

What Optimizing Labs Reveal: Scaling Laws as Production Functions

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Language-model developers trade off quality, training cost and the value of compactness. Optimization cuts both ways: it removes the variation identifying the technology, as curvature information is fourth order in allocation errors, but makes choices informative about objectives. Annealed designed experiments give, without functional-form restrictions, a parameter-data elasticity of substitution near 0.7 at 10^{19} – 3×10^{20} FLOP and about 0.6 at two designs' largest budgets. [TBD-m9 (Amendments 1–2, Q1): outcomes 1–2 add “A controlled experiment at 2×10^{14} – 10^{17} FLOP gives $[x, y]$ for two corpora” with the cuts of M9-14; outcome 3 replaces the σ^ clause by the range across designs, compute levels and the experiment; outcome 4: no change.] Inverted first-order conditions show that most dense open-weight models with disclosed training data released since 2023 were trained beyond training-cost minimization, though not most training compute, and that over-training rose steeply from 2023 to 2025.*

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Training a frontier language model has become one of the largest single investments in the technology sector: frontier training compute has grown four- to fivefold a year since 2018 (Sevilla and Roldán, 2024), and its amortized cost about 2.4-fold a year since 2016 (Cottier et al., 2024). Labs, our term for both the laboratories that run training experiments and the developers that release models, plan this work with neural scaling laws, which relate a model’s loss L on held-out text to its parameters N and training tokens D (Kaplan et al., 2020; Hoffmann et al., 2022). Training costs about $6ND$ floating-point operations (FLOP) and serving about $2N$ FLOP per token, so a scaling law is a production function with

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a known cost side, and the choice of N and D depends on what else a lab values: one that expects heavy use of a model (Sardana et al., 2024), or wants it to run on a laptop (Ben Allal et al., 2025), trains a smaller model on more data. Two objects govern this choice, how easily parameters and data substitute and how much labs value a compact model; both enter forecasts of data demand and of the returns to compute in models of growth. Labs’ optimization cuts both ways for measuring them.

A lab that minimizes cost places its models on the expansion path, where they carry almost no information about the curvature of the technology. The same optimization makes its choices informative about its objectives: the further a model lies from the path that minimizes training cost, the more the lab must value something that training cost leaves out. Designed experiments break the circle: they supply the technology that turns choices into revealed objectives.

Theory. Because training compute is multiplicative in parameters and data, isocost lines are straight in logs, and cost minimization equates the output elasticities of the two inputs, ε_N and ε_D . An interior optimum then requires an elasticity of substitution σ between zero and one (Lemma 1); U-shaped IsoFLOP profiles, loss against model size at fixed compute, are its empirical content. The ratio $w \equiv \varepsilon_N/\varepsilon_D$ measures what else a lab values. A lab stops adding parameters when their marginal product equals their full marginal cost, which includes whatever it loses when a model of given quality is larger; tokens cost only training. Under any objective that rewards lower loss, and absent data costs, caps on tokens and size-dependent FLOP prices, $w - 1$ is therefore the value of compactness at given quality per unit of training cost, including the shadow value of any binding limit on size (Proposition 1); we report it as the compactness share $s \equiv (w - 1)/w$. Compactness can pay through serving cost, adoption, memory tiers or latency; the wedge measures their sum.

On the expansion path, labs’ data identify the path and the loss–compute frontier, but the curvature of the technology only through functional form (Proposition 2). Off the path, information about the compute-optimal ratio of tokens to parameters, M^* , is second order in labs’ allocation errors, and information about curvature is fourth order: curvature affects loss only through the loss from misallocation, which is itself second order. Designed experiments restore identification without a functional form: with σ^* the elasticity on the expansion path, $1/\sigma^* - 1$ equals the curvature of the IsoFLOP profile in log model size at its minimum, divided by twice the slope of the loss–compute frontier (Proposition 3). Beyond the largest designed budget, the sign of $w - 1$ needs even less: for any technology whose IsoFLOP profiles have a single peak and no other critical point, a model with $M \equiv D/N$ trained with compute C is over-trained exactly when $M > M^*(C)$, so bounds on $M^*(C)$ identify the sign (Proposition 3(v)).

The technology. We apply the estimator to the annealed IsoFLOP experiments of three laboratories, Chinchilla (Hoffmann et al., 2022), Llama 3 (Grattafiori et al., 2024) and Marin’s ladders on three corpora (Held and Marin Community,

2026), and to the near-factorial Farseer sweep (Li et al., 2025b). Where the designs overlap, 10^{19} to 3×10^{20} FLOP, σ^* is about 0.7 with no common drift; one estimate per study gives 0.69, with a 95 percent confidence interval of $[0.64, 0.73]$, and FLOP and parameter conventions, estimators and bandwidths move this mean between 0.66 and 0.70. At 6×10^{20} – 3×10^{21} FLOP, in the two IsoFLOP designs that reach those budgets, σ^* is about 0.59 (Farseer’s local path gives 0.66 at 10^{21}), and beyond the designs it is only partially identified. On Llama 3 and Farseer, parametric fits come close to these values only when the outer curvature of the loss function is free. The level of the technology does not travel: at 10^{21} FLOP, M^* differs more than tenfold across the sweeps and published laws we use. Our own experiment, with a pre-analysis plan committed before estimation, trains small transformers at 2×10^{14} to 10^{17} FLOP on two corpora that differ in quality. [TBD-m9: one sentence per Amendments 1 and 2: model-free σ^* by corpus, $[x, y]$; neutrality of the filter; extrapolation slope against its null.]

What over-training reveals. We invert labs’ first-order conditions on a clean sample of 77 dense open-weight base models released in 2023–2025 by developers that disclose training tokens. Family members trained on one token budget reveal one number, so the models form 49 allocation decisions. In the families we fit, $\ln w = (1/\sigma^* - 1) \ln[M/M^*(C)]$, so the inversion needs the curvature and the zero point $M^*(C)$ at each model’s compute. Under our reference technology, the Chinchilla data fitted with the outer curvature free, the median decision was made as if a model one percent smaller at the same loss were worth 2.8 percent of its training cost ($s = 0.73$), with a 95 percent interval for s of $[0.51, 0.81]$ under a cluster bootstrap. The level cuts both ways: a zero-point allowance as large as the shift across DataDecide’s recipes lowers the models’ median lower bound on s to 0.03, a curvature that keeps falling with compute raises the median decision’s s to 0.87 at $\sigma^* = 0.60$ (zero points fixed), and the 20 decisions under a possible cap on tokens are lower bounds; the 29 point estimates have a median of 0.74. The evidence is more consistent with adoption and memory tiers than with serving cost: models built for devices are the most over-trained and open-weight models more so than closed ones at given compute, while too few developers serve their own sizes for a serving test to have power.

The sign and the trend. The sign of over-training is more robust than its size. With M^* extrapolated along the IsoFLOP designs’ own paths from their largest bracketed budgets, with the sampling error of level and slope, $w > 1$ is identified for 78 percent of models and 69 percent of decisions. The share is 53 percent if labs’ M^* may differ by a factor of 3.4, the shift across data recipes in DataDecide (Magnusson et al., 2025), 13 percent if only normal inputs bound the path and 23 percent if any technology in our set may anchor it, and it stays above one half up to a factor of about 3.7. Identified models carry only 24 percent of training compute, because flagships lie close to the path. Inside the designs, parametric forms understate the wedge at high M in all three recipes that allow the check, but the local log wedge is convex in two and concave in one, so the direction

of extrapolation error in M beyond the designs is recipe-dependent. From 2023 to 2025 revealed over-training rose steeply: under the reference the compute-weighted share over decisions rose from 0.24 to 0.81 and the median from 0.49 to 0.74, under every technology in our set and without any single developer, though a bootstrap over developers establishes the rise over the two years but not the step from 2024 to 2025. The trend largely restates the rise in tokens per parameter, and measures a rising value of compactness only if developers optimized against technologies like ours.

Why these objects matter. Which objects matter is itself a finding (Section V). The allocation exponent, among the least portable objects, drives forecasts of data demand: across its estimates, compute-optimal data demand grows 2.0- to 2.9-fold a year at current compute growth. For the median open-weight decision, the rise in over-training since 2023 has multiplied tokens per model at given compute by about 2.1 a year (95 percent interval 0.95 to 4.4). The curvature σ^* scales the revealed value of compactness but barely moves the cost of a binding cap on unique data when repeated data lose value slowly; as such caps bind, measured wedges will understate the value of compactness, so a flattening of the trend would not show that labs value it less. Levels hinge on M^* at frontier compute, which no public design observes; IsoFLOP profiles published at frontier budgets would turn our signs into levels.

The methods travel beyond language models. The identification result gives rates to the observation of [Marschak and Andrews \(1944\)](#) and [Bond and Söderbom \(2005\)](#) that optimizing firms facing common prices do not reveal their technology. The model-free estimator, with a correction for the curvature of the iso-cost itself, applies wherever experimenters vary two inputs at known cost (Online Appendix Corollary A3), as in factorial fertilizer trials, chip-design sweeps and dose-combination trials. The sign result identifies the direction of a first-order-condition wedge from the location of the cost-minimizing input ratio alone, which bears on measuring markups and input distortions ([De Loecker and Warzynski, 2012](#); [Raval, 2023](#)) wherever designed experiments locate that ratio, as in engineering or agronomic trials; production data alone do not.

Related literature. [Villalobos and Atkinson \(2023\)](#), [Erdil \(2024\)](#), [Sardana et al. \(2024\)](#), [Bian, Yan and Venkataraman \(2025\)](#) and [Roberts et al. \(2026\)](#) solve the forward problem: optimal size and data given inference demand, latency or repeated sampling at test time. We solve the inverse problem; their mechanisms, the cost of serving ([Erdil, 2025](#)) and who bears it determine what the recovered value of compactness contains. [Hao and Merrill \(2026\)](#) model a developer that pays for training and serving under a Leontief approximation, which rules out over-training; Proposition 1 inverts such problems when $\sigma > 0$, also for labs that do not bear serving costs. [Kricheli et al. \(2026\)](#) show that fixed-ratio designs leave scaling-law parameters ill-conditioned when the two exponents are close; on a compute-optimal path the degeneracy arises whatever the exponents (Proposition 2), and economic wedges supply the design spread that a curvature parameter

needs (Kiefer and Wolfowitz, 1959; Box and Lucas, 1959, see also Akerberg, Caves and Frazer, 2015; Gandhi, Navarro and Rivers, 2020). The inversion follows the production approach of De Loecker and Warzynski (2012), but w is not a markup, and its elasticities come from experiments, which avoids the circularity stressed by Doraszelski and Jaumandreu (2021) at the cost of external validity; the critiques of Bond et al. (2021) and Raval (2023) carry over (Section IV). A companion paper (Okar, 2026) takes the framework to observational data and algorithmic progress (Mertens, Fischl-Lanzoni and Thompson, 2026; Whitfill, 2025).

I. The Training Problem

This section writes a scaling law as a production technology with a known, multiplicative cost.

A. Technology and Cost

A developer trains a language model with N parameters on D tokens (tokens processed, counting repetitions) and measures its cross-entropy loss L , in nats per token, on a held-out corpus. Following Hoffmann et al. (2022), with productivity terms and an outer exponent added, we write

$$(1) \quad L = E + e^{-\omega} \left[A(e^{\psi_N} N)^{-\alpha} + B(e^{\psi_D} D)^{-\beta} \right]^{\kappa} e^v, \quad A, B, \alpha, \beta, \kappa > 0.$$

Here E is the irreducible loss, and the bracket collects the losses from finite parameters and data. Productivity is Hicks-neutral (ω) or factor-augmenting (ψ_N and ψ_D ; a cleaner corpus raises ψ_D), and v is seed and evaluation noise. The outer exponent scales both output elasticities and leaves the isoquants unchanged, so it separates returns to scale from substitution; $\kappa = 1$ is testable (Section III).

Lower-case letters denote logs, $n \equiv \ln N$, $d \equiv \ln D$ and $c \equiv \ln C$, and $y \equiv -\ln(L - E)$ is the log output index. With $u \equiv A(e^{\psi_N} N)^{-\alpha}$ and $v \equiv B(e^{\psi_D} D)^{-\beta}$, the output elasticities are

$$(2) \quad \varepsilon_N \equiv \frac{\partial y}{\partial n} = \frac{\kappa \alpha u}{u + v}, \quad \varepsilon_D \equiv \frac{\partial y}{\partial d} = \frac{\kappa \beta v}{u + v},$$

and neither depends on ω , v or E . The family (1) is quasi-homothetic: the scaling $(N, D) \mapsto (t^\beta N, t^\alpha D)$ multiplies reducible loss by $t^{-\kappa\alpha\beta}$ for every $t > 0$ (Online Appendix Lemma A4). Isoquants are therefore rescaled copies of one another, the compute-optimal mix drifts with compute as a power law, and all curvature lies across the expansion path (Section II). This restriction of the family, not a property of scaling laws, makes the curvature at the optimum the same at every compute, whereas Section III finds σ^* lower at the largest budgets of the two designs that reach them; Lemma 1, Proposition 1 and the model-free results of Section II do not use it.

Training a dense transformer takes about six floating-point operations (FLOPs) per parameter per token, so training compute is $C \approx 6ND$ (Kaplan et al., 2020). The approximation omits attention and embedding FLOPs, and FLOPs are not dollars, so Proposition 1 allows FLOP prices that vary with size and a cost of data.

In FLOP units the cost is multiplicative: both log-cost weights, $\partial c/\partial n$ and $\partial c/\partial d$, equal one, isocosts are straight lines in logs, $n + d = c - \ln 6$, and the implicit price of a parameter, $6D$, rises with the quantity of the other input. Minimizing compute for a given loss therefore equates the output elasticities, $\varepsilon_N = \varepsilon_D$, without the expenditure-share data that the first-order-condition approaches of De Loecker and Warzynski (2012) and Gandhi, Navarro and Rivers (2020) require: FLOP accounting supplies the cost shares.

In this language, iso-loss curves are isoquants, an IsoFLOP profile traces loss along an isocost, the compute-optimal allocation $(N^*(C), D^*(C))$ is a conditional factor demand, and the loss–compute frontier $L^*(C)$ is the inverse of a cost function (Figure 1, panel a). We write $M \equiv D/N$ for tokens per parameter and $M^*(C)$ for its compute-optimal value. Isoquants, the elasticity of substitution, the compute-optimal allocation and the wedge defined below are ordinal: they are unchanged by any strictly increasing transformation of the same output (loss, bits per byte, or accuracy on a benchmark that is monotone in loss) and do not depend on E , ω or κ (Online Appendix Lemma A1). A capability that pretraining loss does not order, such as quality after quantization (Kumar et al., 2025), is a different output with its own technology (Proposition 1(iii)). The frontier elasticity is cardinal, and M^* depends on the tokenizer and the parameter count. Online Appendix Table D1 gives a full dictionary between scaling-law practice and production economics.

B. Duality

Let $a \equiv \beta/(\alpha + \beta)$, $\gamma \equiv \kappa\alpha\beta/(\alpha + \beta)$ and $G \equiv (\alpha A/\beta B)^{1/(\alpha+\beta)}$, and set productivity and noise to zero. The allocation that minimizes $6ND$ subject to $L \leq \bar{L}$, or maximizes y at given compute, satisfies $\varepsilon_N = \varepsilon_D$. The conditional factor demands are

$$(3) \quad N^*(C) = G\left(\frac{C}{6}\right)^a, \quad D^*(C) = G^{-1}\left(\frac{C}{6}\right)^{1-a}, \quad M^*(C) = G^{-2}\left(\frac{C}{6}\right)^{1-2a},$$

and the loss–compute frontier and the cost function are

$$(4) \quad L^*(C) = E + K\left(\frac{C}{6}\right)^{-\gamma}, \quad C^*(L) = 6\left(\frac{K}{L - E}\right)^{1/\gamma}, \quad K \equiv \left(\frac{\alpha + \beta}{\beta} AG^{-\alpha}\right)^\kappa.$$

Along the expansion path $\varepsilon_N = \varepsilon_D = \gamma$. At given compute, Hicks-neutral productivity leaves the allocation unchanged, and factor-augmenting productivity moves it only through the bias index $\chi \equiv \beta\psi_D - \alpha\psi_N$: $\ln M^*(C)$ falls by $2\chi/(\alpha + \beta)$

(Online Appendix Lemma A3).

This is Shephard–Uzawa duality for the scaling law (Shephard, 1953; Uzawa, 1964): the technology, the factor demands and the cost function carry the same information, linked by a and γ . The outer exponent moves the frontier but not the path. The frontier elasticity γ is the returns-to-scale measure of Nerlove (1963); under the reference technology of Section IV, the κ -free fit to Chinchilla’s 240 runs in total parameters (Table 1, panel B), $\gamma = 0.165$; because compute is quadratic in the scale of the inputs, the technological scale elasticity on the path is 2γ . A fit of (1) to all runs (Approach 3 of Hoffmann et al., 2022) is a primal estimator; a regression of the loss-minimizing $\ln N$ of each IsoFLOP profile on $\ln C$ (their Approach 2) estimates the factor demand, and a frontier through the same minima the cost function (Online Appendix D tests duality by comparing the two; Klump, McAdam and Willman, 2007 instead estimate a production function jointly with its first-order conditions). The minima are outcomes of inputs that an experimenter assigned, not choices, so the endogeneity problems of IO arise for released models, not for experiments.

C. Substitution

The Hicks elasticity of substitution between parameters and data depends on the input mix only through the ratio of the output elasticities, $w \equiv \varepsilon_N/\varepsilon_D = \alpha u/(\beta v)$, which we call the wedge. For every member of the family (1),

$$(5) \quad \sigma(w) = \frac{1+w}{1+\alpha+w(1+\beta)}, \quad \sigma^* \equiv \sigma(1) = \frac{2}{2+\alpha+\beta},$$

which follows from the formula of Hao and Merrill (2026) (Online Appendix Lemma A1). The local elasticity lies between $1/(1+\alpha)$ and $1/(1+\beta)$ and equals σ^* on the compute-optimal path, where $w = 1$. Under the reference technology, $\sigma^* = 0.701$, and σ varies only between 0.699 and 0.702 over all input mixes.

Lemma 1 (Interior compute optima require $0 < \sigma < 1$). *Let $y = F(n, d)$ be any twice continuously differentiable technology with $F_n, F_d > 0$, and let $C = 6ND$. At an allocation with $\varepsilon_N = \varepsilon_D$:*

- (i) *the second-order condition for minimizing compute at given output holds strictly if and only if $0 < \sigma < 1$ there;*
- (ii) *if $\sigma > 1$ or $\sigma < 0$ there, the allocation minimizes rather than maximizes output along its isocost; if $\sigma > 1$ everywhere, compute-optimal training is a corner solution at a bound on N or D .*

Every member of the family (1) has $\sigma < 1$ at every input mix, so the first-order condition $\varepsilon_N = \varepsilon_D$ is sufficient.

The proof is in Online Appendix A (Lemma A2 and Corollary A1). With a linear cost $q_N N + q_D D$, isocosts are straight lines in levels, and any convex isoquant

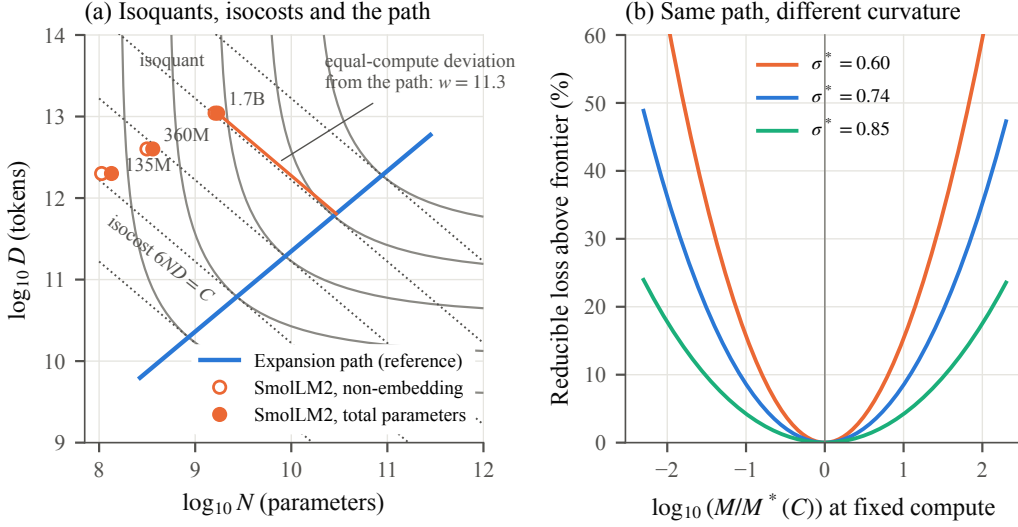


FIGURE 1. THE GEOMETRY OF THE TRAINING PROBLEM

Note: Panel (a): the $(\log_{10} N, \log_{10} D)$ plane under the reference technology of Section IV (the κ -free fit to Chinchilla’s 240 runs, in total parameters; $\sigma^* = 0.701$). Solid gray curves are isoquants at the frontier losses for $C = 10^{20}, \dots, 10^{24}$ FLOP; dotted lines are isocosts $6ND = C$; the blue line is the compute-optimal expansion path. Filled orange dots are the three SmoLLM2 base models (135M, 360M and 1.7B parameters, trained on 2, 4 and 11 trillion tokens) at their total parameter counts, and hollow circles the same models at their non-embedding counts; for the 1.7B model the two markers nearly coincide, and the orange line starts at the total count. It joins SmoLLM2 1.7B to the path at equal compute; its wedge under the reference is $w = 11.3$. Panel (b): three members of the κ family with $\sigma^* = 0.60, 0.74$ and 0.85 , constructed to share the expansion path and the loss–compute frontier of the Besiroglu et al. (2024) parameters (Online Appendix Proposition A1). The vertical axis is reducible loss in excess of the frontier at $C = 10^{22}$ FLOP, as a function of the deviation of tokens per parameter from the compute-optimal ratio. The technologies coincide on the path and differ off it.

($\sigma > 0$) makes a tangency a cost minimum. With $C = 6ND$ they are straight lines in logs, and shifting toward one input lowers its own implicit price and raises the other’s, so cost alone favors a corner; an interior optimum requires the isoquant to be more curved than the isocost, which is complementarity in this sense ($\sigma < 1$).

The lemma’s observable implication is that IsoFLOP profiles are U-shaped with an interior minimum, as every design of Section III shows wherever its grid brackets the minimum. That $\sigma^* < 1$ is therefore not a finding; its level is. That $\sigma > 0$ makes over-training a sensible response to a value of compactness: in the Leontief limit of Hao and Merrill (2026) the optimum sits at the kink whatever that value, whereas for $0 < \sigma < 1$ parameters can be traded for tokens at a cost governed by σ^* (Figure 1, panel b).

D. What Over-Training Reveals

Most open-weight language models released since 2023 lie far from the expansion path, with M well above $M^*(C)$ (Section IV), whereas a developer that minimized training cost would have $w = 1$.

Proposition 1 (What over-training reveals). *A developer chooses (N, D) to maximize $V(L(N, D), N) - X_T - q_D D$. Here V is any differentiable objective that falls with loss and may depend on size directly; $X_T \equiv q_T(N) 6ND$ is training expenditure at the (shadow) price $q_T(N)$ of a training FLOP, whose size elasticity is δ ; and $q_D \geq 0$ is a cost per token. Let $m_N \equiv -(\partial V / \partial \ln N)_L / X_T$ be the objective's own value of compactness: making the model one percent smaller at the same loss is worth m_N percent of training expenditure. Let $m_D \equiv q_D D / X_T$. Let v_N be the shadow value, per unit of training expenditure, of any binding constraint on size (a memory tier, a latency objective or a size menu; $v_N \geq 0$ for a cap, $v_N = 0$ if none binds), and let $v_N \equiv m_N + v_N$ be the value of compactness: the value at which a developer free of it would make the same choice, a virtual price in the sense of [Neary and Roberts \(1980\)](#).*

(i) (The wedge.) *At an interior optimum with no binding cap on tokens,*

$$(6) \quad w = \frac{\varepsilon_N}{\varepsilon_D} = \frac{1 + \delta + v_N}{1 + m_D}.$$

With $\delta = m_D = 0$, $w - 1 = v_N$, and the compactness share $s \equiv (w - 1)/w$ equals $v_N/(1 + v_N)$.

(ii) (Serving.) *Let $\delta = m_D = v_N = 0$ and $V = R(L) - \lambda X_S$, so that serving is the only effect of size on V at given loss, where X_S is the expenditure on serving T tokens over the model's life at $2N$ FLOPs per token and a constant ϱ times the price of a training FLOP, with T independent of size at given loss. Then $w = 1 + \lambda \varrho T / (3D)$, $w - 1 = \lambda X_S / X_T$, and $s = \lambda X_S / (X_T + \lambda X_S)$ is the internalized serving share of lifetime expenditure.*

(iii) (Constraints and misspecification.) *A binding cap on tokens adds $\mu \geq 0$ to the denominator of (6), so with $\delta = m_D = 0$, $w - 1$ bounds v_N from below; a binding cap on size enters v_N , so, if $\mu = 0$, $w - 1$ bounds the objective's own m_N from above. Logit distillation from a teacher with N_T parameters is, if it leaves the student's technology unchanged, a data cost with $m_D = N_T / (3N)$; if it also raises the productivity of tokens ($\chi > 0$), it lowers the measured wedge further. If the developer's first-order conditions involve a technology L' other than the econometrician's L (its own productivity, a law it believes, or the output it values), the measured wedge is $\hat{w} = w_{L'} w_L / w_{L'}$, where $w_{L'}$ is the wedge that the developer's technology assigns to its choice and w_L the one that the econometrician's assigns; it equals $w_{L'} e^{-\chi}$ under factor-augmenting productivity.*

- (iv) (*Family token budgets.*) Let a family train models of sizes $N_1, \dots, N_I$ on one token budget D , with $\delta = m_D = 0$. Member wedges then differ only through the technology; in family (1), $\ln w_i - \ln w_j = -\alpha(\ln N_i - \ln N_j)$. Let $\bar{w}_H$ be the compute-weighted harmonic mean of the member wedges, and let the weight π_i be proportional to member i 's share of the family's training compute divided by w_i .
- (a) If D is chosen for the family, the family's first-order conditions identify one number, $\sum_i \pi_i v_{N,i} = \bar{w}_H - 1$, whether the sizes are chosen at the margin or set by binding size constraints.
- (b) If a cap on tokens binds for the family, $\bar{w}_H - 1 \leq \sum_i \pi_i v_{N,i}$.

The proof is in Online Appendix A (Propositions A3 and A4); part (iv)(b) is Proposition A4(vi). The developer adds parameters until their marginal value in loss equals their marginal cost: training expenditure plus whatever it loses when the model is larger at the same quality (serving cost, adoption, latency, a memory tier). Tokens cost only training and data. An econometrician who sees only training cost therefore observes a wedge: the developer acts as if parameters were w times as expensive, relative to tokens, as training accounting says.

The value of compactness. What v_N measures depends on the developer's objective (Online Appendix Table E1). For a developer that serves this model, bears the cost ($\lambda = 1$) and faces no binding size constraint, s is planned serving expenditure as a share of lifetime expenditure; a count of tokens, $T = 3D(w - 1)/(\lambda \varrho)$, also needs the price ratio ϱ , likely above one because decoding is often bound by memory bandwidth (Erdil, 2025). For an open-weight developer whose users bear the serving cost, m_N is the value of compactness through adoption; under a binding tier, memory or latency cap (Bian, Yan and Venkataraman, 2025), the value of compactness includes the cap's shadow value. Section IV tests these readings as a problem of conduct (Berry and Haile, 2014; Backus, Conlon and Sinkinson, 2021; Duarte et al., 2024), and the evidence is more consistent with adoption and memory tiers than with the internalization of serving cost. We therefore call v_N the value of compactness and s the compactness share throughout, and read s as planned serving expenditure only where the developer served the size and neither a common token budget nor a tier window intervenes (Section IV.E).

Other sources of a wedge. Their signs are known (Online Appendix Proposition A3(iv)). A price of a FLOP that rises with size ($\delta > 0$) and parameter-augmenting productivity ($\chi < 0$) raise the measured wedge. So does post-training compute that scales with N and is planned when N is chosen: counting its expenditure X_P as a cost alongside X_T , with $x_P \equiv X_P/X_T$, $w = 1 + x_P + m_N$, where m_N excludes post-training, and Section IV bounds x_P with disclosed budgets. Data costs, caps on tokens, distillation (Busbridge et al., 2025) and data-augmenting productivity ($\chi > 0$) lower the wedge. Serving demand cannot produce a wedge below one as long as serving cost rises with size. Gopher's $w = 0.28$ under the reference technology (Rae et al., 2021) is thus consistent with a binding data

constraint or a data cost, better data, a size target (a minimum-size floor), or optimization against another technology, plausibly the law of Kaplan et al. (2020).

Family budgets. On one token budget, member wedges are fixed by the technology and are not revealed preferences. A binding cap, as when every size trains on the whole curated corpus, makes the family share $1 - 1/\bar{w}_H$ a lower bound, and a member’s own wedge one only if that member would have trained on more tokens at its own size, which a cap on the family’s budget does not imply.

A closed form. In the κ family the wedge of a model with compute C and tokens per parameter M is

$$(7) \quad \ln w = \left(\frac{1}{\sigma^*} - 1 \right) \ln \frac{M}{M^*(C)},$$

where $M^*(C)$ is the compute-optimal ratio at the model’s own compute (Online Appendix Lemma A4 and Corollary A2). The log wedge is the distance from the expansion path at equal compute, scaled by curvature (Figure 1, panel a); the technology supplies the zero point $M^*(C)$ and the scale $1/\sigma^* - 1$. SmolLM2 1.7B (Ben Allal et al., 2025), trained on 11 trillion tokens, has $M \approx 6,400$, about 290 times the reference technology’s compute-optimal ratio at its compute. With the reference scale, $1/\sigma^* - 1 = 0.427$, its wedge is $w \approx 11.3$: its developer behaves as if making it one percent smaller at the same loss were worth about 10 percent of the training bill, and $s \approx 0.91$. Such numbers are only as good as the technology’s two inputs, neither of which any public design observes at SmolLM2’s compute. The sign needs only the zero point: for any technology with single-peaked IsoFLOP profiles, $w > 1$ if and only if $M > M^*(C)$, whatever the scale, so bounds on $M^*(C)$ identify the sign (Proposition 3(v)).

II. What Optimizing Labs’ Data Identify

Reading developers’ choices requires σ^* and $M^*(C)$, the scale and the zero point of the wedge in equation (7). This section asks which data identify them; proofs are in Online Appendix A.

A. Optimization Removes the Variation That Identifies Curvature

A sample is *on-path* if its runs share one technology, each run minimizes training compute for its loss (every wedge equals one), and compute varies over an interval.

Proposition 2 (What optimizing labs’ data identify). *Let runs follow (1) with $\psi_N = \psi_D = 0$.*

- (i) *(On the path.) In an on-path sample, n_i and d_i are affine in c_i , with slopes a and $1 - a$: choices reveal the path, that is, a and $M^*(C)$, and nothing else. Outcomes satisfy $\mathbb{E}[y \mid c] = \gamma c + \text{const.} + \mathbb{E}[\omega \mid c]$, so within the model γ is identified if and only if $\mathbb{E}[\omega \mid c]$ is known up to a constant, as in a single-lab sweep. Curvature is not identified: for every $\sigma^* \in (0, 1)$ some member of*

family (1) generates the same joint distribution of (n, d, y) , and by Lemma 1 no value $\sigma^* > 1$ does (in family (1), none with $\sigma^* \geq 1$).

- (ii) *(Global versus local identification.)* Suppose also that $\mathbb{E}[\omega \mid c]$ is constant and $\kappa = 1$. Globally, with $A, B > 0$, $\alpha = \gamma/a$ and $\beta = \gamma/(1-a)$ are identified by the functional form, and A/B , hence the technology's own cost-minimizing mix, only by imposing that the observed path is optimal. Locally, identification fails: every perturbation that moves the decay rates of the two loss terms along the path, αa and $\beta(1-a)$, in opposite directions leaves on-path outcomes unchanged to first order once the common level is re-split between the two terms (at a given split only one such direction is flat). The least-squares criterion, profiled over that split, is therefore of fourth order along such directions, and σ^* is not identified at first order, whether or not $\alpha = \beta$. Imposing $\beta/(\alpha + \beta) = a$ restores first-order identification.
- (iii) *(Off the path.)* Let runs deviate from the path with log wedges $\ln w_i$, with E known and Gaussian noise. To leading order, the information about the path's location ($\ln G$, and so $\ln M^*$ at a given compute) is proportional to the residual dispersion of the $\ln w_i$. The information about σ^* is bounded by a multiple of $\sum_i \Phi_i^2$, where $\Phi_i \equiv \ln(C_i/C_{\min}(L_i))$ is run i 's Farrell allocative loss: the log of its compute relative to the least compute that attains its loss. If the deviations of a fixed design are scaled by $\varsigma \rightarrow 0$, the two informations are $O(\varsigma^2)$ and $O(\varsigma^4)$, so the standard error of $\ln \hat{M}^*$ grows at least like ς^{-1} and that of $\hat{\sigma}^*$ at least like ς^{-2} , exactly so for designs that cross several compute levels with symmetric transverse offsets.

A ray has no curvature. Facing the same log-cost weights, cost-minimizing labs that share a technology place every run on the expansion path, where reducible loss is log-linear in compute. Choices trace the path and outcomes the frontier, but neither reveals how fast loss rises off the path, which is what σ^* measures (Figure 1, panel b): at five times the compute-optimal ratio, reducible loss lies 2.0, 4.1 and 7.6 percent above the frontier for three technologies that share it and the path, with $\sigma^* = 0.85, 0.74$ and 0.60 .

Remark 1 (The rate on the path). Part (ii) makes the on-path model the analogue of an over-fitted mixture, whose parameters can be learned at best at the rate $n^{-1/4}$ (Chen, 1995). We conjecture the same bound here for estimators of σ^* from on-path data that leave $\beta/(\alpha + \beta)$ free, such as unrestricted least-squares fits of the Chinchilla form, whose Wald intervals, built on the near-singular Hessian, are then invalid. Imposing $\beta/(\alpha + \beta) = a$, which the optimality that defines an on-path sample implies under $\kappa = 1$, removes the singularity: $\sigma^* = 2/\{2 + \gamma/[a(1-a)]\}$ is then estimated at the usual root- n rate. The argument is heuristic: the results of Rotnitzky et al. (2000) for singular information do not apply when A/B is unidentified at the truth (see also Sargan, 1983). In simulations with 40 on-path compute levels (holding the design fixed and letting the noise fall, which is equivalent to replicating it n times; Online Appendix Remark A3), the 75th

percentile of the error of unrestricted least-squares estimates of σ^* shrinks as $n^{-0.24}$, and that of restricted ones as $n^{-0.50}$.

Real ladders and labs deviate from the path. Because the frontier is a power law, output decomposes exactly as

$$(8) \quad y_i = \gamma c_i + \text{const.} - \gamma \Phi_i - v_i,$$

with ω normalized to zero. Curvature enters output only through the Farrell losses, which are Harberger triangles, $\Phi_i \approx \sigma^*(\ln w_i)^2/[4(1 - \sigma^*)]$. A run reveals where the path lies to the extent that it is off the path, first order in its log wedge, and the path's curvature only to the extent that it wastes compute, second order. Information squares these: halving the dispersion of log wedges quarters the information about M^* and divides the information about σ^* by sixteen (the orders are unchanged when E is estimated; Online Appendix A).

Proposition 2 is the point of [Marschak and Andrews \(1944\)](#) in its cleanest form: identification requires the variation in relative prices, adjustment frictions or optimization errors of [Bond and Söderbom \(2005\)](#) (see also [Gandhi, Navarro and Rivers, 2020](#)); here it comes from heterogeneous wedges (Proposition 1) and belief errors such as allocation by the [Kaplan et al. \(2020\)](#) law. The rates in part (iii) are the standard design rates for a curvature parameter, whose information grows with the fourth power of the design's spread ([Kiefer and Wolfowitz 1959](#) on the highest coefficient of a polynomial regression; [Box and Lucas 1959](#) on nonlinear models); what is specific here is that economic wedges generate the design's spread.

The Chinchilla data show part (i) directly. On the 41 runs within 0.15 log points of the fitted compute-optimal size, the profile likelihood for σ^* with κ free is flat: its 95 percent confidence set is $[0.20, 0.99]$, the edge of the grid searched, with χ_1^2 critical values (about $[0.15, 0.99]$ with bootstrap ones), against $[0.69, 0.71]$ ($[0.68, 0.72]$) on the full 240-run design, which adds transverse variation (Online Appendix D).

Figure 2 shows what the rates mean at realistic noise and design sizes, in simulated designs of 90 runs with equal total compute that differ in where the runs sit. With $\varsigma = 0$, a Huber fit of the Chinchilla form ends on a corner, with one power-law term switched off and its exponent not identified at all, in 87 percent of replications (56 percent with 64 random starts), each fitting the data at least as well as the truth. As ς rises from 0.05 to 0.2, the root mean squared error of $\hat{\sigma}^*$ falls from 0.089 to 0.015, against 0.0036 for an IsoFLOP design with the same compute (panel A); for $\ln \hat{M}^*$ it is 4.6 at $\varsigma = 0.1$ and 0.59 at $\varsigma = 1$, against 0.16.

With κ free (panel B), this precision proves to be functional-form information, as part (ii) predicts: the 95 percent profile confidence set for σ^* covers the entire grid searched, $[0.05, 0.99]$, in 91 percent of on-path replications, and at $\varsigma = 0.3$ it excludes low elasticities but still contains every value above 0.9 in 76 percent of replications, whereas the IsoFLOP and factorial designs give sets 0.016 and 0.019 wide. On the path, standard inference also fails, and heteroskedastic noise

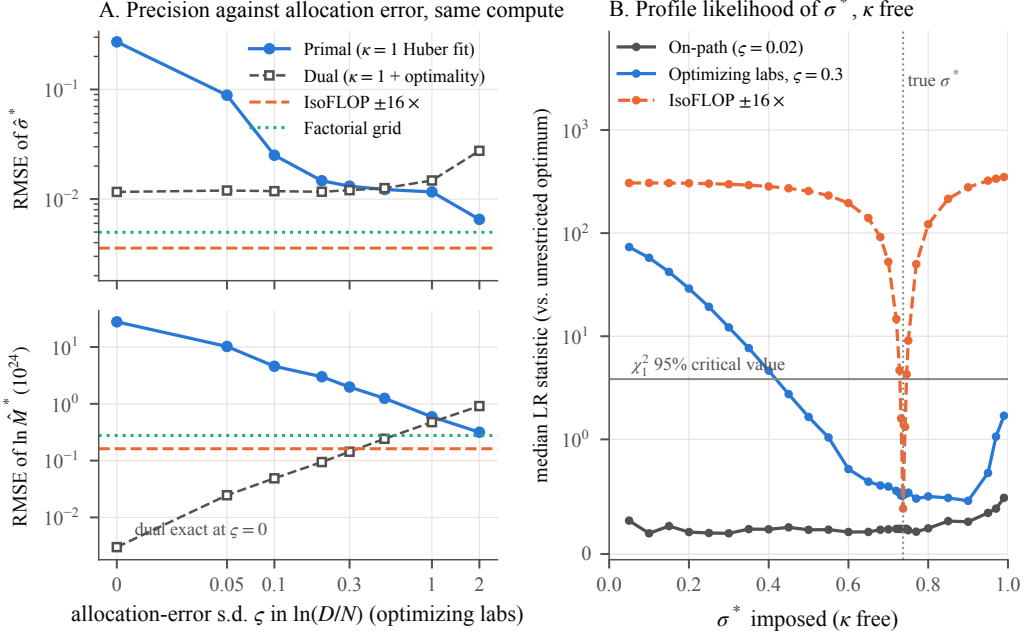


FIGURE 2. IDENTIFICATION BY DESIGN: MONTE CARLO EVIDENCE

Notes: Simulations of the Besiroglu et al. (2024) technology ($\sigma^* = 0.737$; $M^* = 18.1$ at 10^{24} FLOP) with normal noise of standard deviation 0.0075 in $\ln L$ (the Besiroglu fit's residual s.d. is 0.0076). Every design has 90 runs and the same total compute, 5.1×10^{22} FLOP; except in the factorial design, the runs sit ten on each of nine budgets matching Chinchilla's nominal IsoFLOP budgets (6×10^{18} to 3×10^{21} FLOP). Optimizing labs set $\ln(D/N)$ at its compute-optimal value plus a normal allocation error with standard deviation ζ ; the IsoFLOP design places ten sizes log-spaced over $[N^*/16, 16N^*]$ at each budget, and the factorial design crosses nine values of N with ten of D , each spanning a factor of 256 around the expansion path, so that its runs range from 1.8×10^{17} to 1.2×10^{22} FLOP. Panel A: root mean squared error of $\hat{\sigma}^*$ and of $\ln \hat{M}^* (10^{24})$ for optimizing labs, 500 replications per design. Primal: fit of the Hoffmann et al. (2022) form with $\kappa = 1$ by the Huber loss with threshold 10^{-3} , started from nine values drawn at random within the region of Hoffmann et al.'s initialization grid (exponents on $[0.1, 1.2]$ rather than $[0, 2]$), none at the truth. Dual: exponents from the path of the optima and the frontier under $\kappa = 1$ and optimality. Horizontal lines: primal estimator under the IsoFLOP and factorial designs. Panel B: median profile likelihood-ratio statistic for σ^* with κ free, relative to the unrestricted optimum, on a 29-point grid over $[0.05, 0.99]$ (150 replications), for on-path labs ($\zeta = 0.02$), optimizing labs with $\zeta = 0.3$ and the IsoFLOP design; horizontal line, the 95 percent χ^2_1 critical value; dotted line, the truth. *Source:* Author's simulations; design, estimators and full tables in Online Appendix C.

correlated within budgets or training trunks leaves the ranking of the designs unchanged (Online Appendix C).

B. Designed Variation Identifies Curvature, and What Survives beyond the Design

The next result reads the technology off movement across the path without a functional form, and says what survives where no design reaches. Let $x \equiv \ln M$, so that an IsoFLOP profile is loss as a function of x at fixed c , with minimum at $x^*(c) \equiv \ln M^*(C)$; subscripts denote partial derivatives. A design budget is *bracketed* if at least two of its IsoFLOP runs lie on each side of the profile's fitted minimum, within the estimation window (Online Appendix D3).

Proposition 3 (Model-free σ^* and w , and the sign beyond the design). *Let L be any measure of output that falls as quality rises (loss, bits per byte, or $\ln(L - E)$ for any E), and let the technology be twice continuously differentiable with positive marginal products.*

(i) (Wedge.) *At any point, $w = (L_c - L_x)/(L_c + L_x)$.*

(ii) (Curvature.) *At a compute-optimal point,*

$$(9) \quad \frac{1}{\sigma^*} - 1 = \frac{L_{nn}|_C}{2|dL^*/dc|},$$

where $L_{nn}|_C$ is the curvature of the IsoFLOP profile in $\ln N$ at its minimum and $L^*(c)$ is the loss-compute frontier. The ratio is invariant to strictly monotone transformations of output.

(iii) (Slope of the log wedge.) *At $M^*(C)$, $\partial \ln w / \partial \ln M|_C = 1/\sigma^* - 1$. In family (1), $\ln w$ is exactly linear in $\ln(M/M^*(C))$ with this slope; in general the slope varies with M .*

(iv) (What $\kappa = 1$ imposes.) *Under $\kappa = 1$ the curvature of every IsoFLOP profile is pinned by the slopes of the frontier and the path, $-y_{nn}|_C = \gamma^2/[a(1-a)]$; with κ free it is a separate parameter.*

(v) (Beyond the design.) *Let c_J be the log of the largest bracketed design budget, and consider a model with log compute $c > c_J$ and log ratio x . Suppose that (a) every IsoFLOP profile of the developer's technology is single-peaked in x (its only critical point is the peak), which $0 < \sigma < 1$ everywhere guarantees (Lemma 1); (b) the anchor $x^*(c_J)$ lies in $[\underline{x}_J, \bar{x}_J]$; (c) the path slope $e_p \equiv d \ln M^* / d \ln C$ lies in $[e_L, e_U]$ between c_J and c ; and (d) at every compute, the developer's own compute-optimal ratio lies within a factor $\tau \geq 1$ of the anchored technology's. Then the wedge in the developer's first-order conditions satisfies $\text{sign}(w - 1) = \text{sign}(x - x_{\text{dev}}^*(c))$, where the developer's compute-optimal log ratio $x_{\text{dev}}^*(c)$ lies in*

$$\mathcal{X}^*(c) \equiv [\underline{x}_J + e_L(c - c_J) - \ln \tau, \bar{x}_J + e_U(c - c_J) + \ln \tau].$$

Hence $w > 1$ is identified if $x > \sup \mathcal{X}^(c)$ and $w < 1$ if $x < \inf \mathcal{X}^*(c)$; when the anchor is a point and the anchored technology is in family (1), these conditions are also necessary, and otherwise the sign is not identified. Neither σ^* nor a functional form is needed.*

Part (i) is a change of coordinates: slopes along the isocost and along a ray of constant M give the two marginal products, whose ratio is the wedge. Part (ii) writes σ^* in observables: at the minimum of an IsoFLOP profile the marginal products of parameters and tokens are equal, the profile’s curvature measures how fast they diverge as the mix moves, and the frontier slope, the marginal product of compute by the envelope theorem, measures their common level. Because both terms are in the same output units, the irreducible loss, the transformation of output and κ cancel. Both are familiar: the quadratic coefficient of the parabola fitted to each profile, as in Approach 2 of Hoffmann et al. (2022), and the slope of the frontier through the minima; quadratics fitted over wide grids overstate the curvature slightly, which narrow windows avoid (Online Appendix Remark A6; on the corresponding shift of the fitted minimum, see Czech et al., 2026). With a known cost that is not multiplicative, the same curvature and frontier slope, weighted by the cost’s elasticities, identify $1/\sigma^*$ less the inverse elasticity of substitution of the isocost itself (Online Appendix Corollary A3), so the estimator applies to any design that varies two inputs along known isocosts. Part (iii) gives a test of the family.

Parts (i) to (iv) hold only inside a design’s support, but nearly all released models used more compute than 3×10^{21} FLOP, the largest bracketed budget of any public design (Section IV); part (v) says what extrapolating the zero point delivers.

The sign result needs only the location of the minimum: on a single-peaked profile, a model with more tokens per parameter than is optimal at its compute has a higher marginal product of parameters than of tokens, so $w > 1$ whatever the curvature. Assumption (a) extends the U-shaped profiles of Lemma 1 and every bracketing design. The anchor in (b) is a compute-optimal ratio at a bracketed budget, for Llama 3 at 10^{21} FLOP rather than at its unbracketed 3×10^{21} or 10^{22} , and the path-slope range in (c) is the substantive extrapolation; Section IV.D takes both, with their sampling error, from the designs’ own paths. Normal inputs alone, factor demands that do not fall with compute, imply only $|e_p| \leq 1$. The allowance τ in (d) admits cross-lab differences in technology (Section III): a factor-augmenting bias χ moves $\ln M^*$ by $2\chi/(\alpha + \beta)$, and the largest tilt across DataDecide’s recipes, 0.26, corresponds to $\tau \approx 1.8$ under the reference exponents (we use 1.84) and to 3.4 under DataDecide’s own.

Given (b) to (d), a point anchor and an anchored technology in family (1), the set is sharp: every point in it is the compute-optimal ratio of a technology with positive marginal products and single-peaked profiles that reproduces the designs up to c_J , shifted by a constant factor-augmenting tilt of at most $\ln \tau$ (Online Appendix Proposition A6(ii) and (vii); Manski, 2003). With sampling error, a

confidence set replaces the anchor and the slope range; if it is a joint band for the level and slope of a linear path, the set covers $x_{\text{dev}}^*(c)$ with at least the band’s coverage probability, and with several anchored paths the union does so if the developer’s technology lies within τ of at least one of them (Proposition A6(viii)).

What the data identify at frontier scale is therefore a sign, not a level. SmolLM2 1.7B, trained on about 6,400 tokens per parameter with 1.1×10^{23} FLOP, has a ratio about 47 times the upper end of the set under the anchors of Section IV, so its over-training is identified for any tilt allowance below that factor. Llama 3.1 405B, at 38 tokens per parameter and 3.8×10^{25} FLOP, lies inside the set: even the sign of its wedge is not identified. Magnitudes need the curvature as well: if $\partial \ln w / \partial x|_c \in [\theta_L, \theta_U]$ with $\theta_L > 0$, then for $x > \sup \mathcal{X}^*(c)$, $\ln w \in [\theta_L(x - \sup \mathcal{X}^*(c)), \theta_U(x - \inf \mathcal{X}^*(c))]$ (Online Appendix Proposition A6(iv)); in family (1) a range for σ^* at the model’s own compute gives $[\theta_L, \theta_U]$, and that range is itself an extrapolation.

Where the results bind. Proposition 2 binds for compute-optimal ladders and for releases of the compute-optimal era; by the same mechanism, fixed tokens-per-parameter designs run by one lab are ill-conditioned when $\alpha \approx \beta$ (Kricheli et al., 2026), as under the reference technology (DataDecide trains every size to about 100 tokens per parameter; Magnusson et al., 2025). Released models since 2023 lie far off the path, but using their values of compactness as transverse variation requires that these values move the mix without moving the technology, which factor-biased technology would violate. We therefore take the technology from designed sweeps and read objectives from released models’ wedges.

III. The Technology from Designed Variation

This section takes Proposition 3 to the data. Throughout, σ^* is the elasticity of substitution on the expansion path between FLOP-effective parameters, $N_F = C/(6D)$, and training tokens. On every public IsoFLOP design with annealed runs and on Farseeer’s factorial grid, σ^* is about 0.70 where the designs overlap, 10^{19} to 3×10^{20} FLOP. It falls to about 0.59 at the largest budgets of the two IsoFLOP designs that go beyond (Farseeer’s local path gives 0.66 at 10^{21} FLOP), and past the designs it is only partially identified. Where the same-run comparison discriminates, on Llama 3 and Farseeer, only the κ -free form agrees with the model-free benchmark. Neither a nor M^* is portable across designs, conventions and data.

A. Designs

Table 1 lists the designs. Four studies provide IsoFLOP profiles, models of different sizes trained on the isocost line $6ND = C$ at each of several budgets. Epoch AI’s digitization of the runs of Hoffmann et al. (2022), the sample of Besiroglu et al. (2024), has 137 runs on nine IsoFLOP profiles from 6×10^{18} to 3×10^{21} FLOP and 108 from 35 fixed-size trunks; parametric fits drop the

five with the highest loss ($n = 240$). Meta’s Llama 3 profiles (Grattafiori et al., 2024), digitized by Czech (2026), cover ten budgets up to 10^{22} FLOP, of which the eight up to 10^{21} bracket their minimum; their unstated loss units do not matter, because the estimator is invariant to monotone transformations of output. Marin’s ladders (Held and Marin Community, 2026) repeat the design on three corpora up to 3×10^{20} FLOP. Porian et al. (2024) release loss curves of architectures trained at a constant learning rate, from which we read profiles at 12 budgets up to 2.6×10^{19} FLOP; these runs are not annealed.

The 404 released runs of Farseeer (Li et al., 2025b) form a near-factorial grid dense enough to locate the expansion path. Three smaller sweeps have no profiles: the three web corpora of Gadre et al. (2025), the OLMo ladder (Bhagia et al., 2025) and the single-epoch runs of Muennighoff et al. (2023); with Chinchilla’s full design and Farseeer, they form seven sweep–corpus technologies. DataDecide (Magnusson et al., 2025) serves only to compare data recipes. The model-free estimator needs exact isocosts, so N is counted so that $C = 6ND$ holds along each profile: $N_F = C/(6D)$, with C each study’s own FLOP count, for Chinchilla, Llama 3 and Marin, and the non-embedding count for Farseeer. For Chinchilla we rebuild FLOPs per token from Hoffmann et al.’s architecture table (Online Appendix Table D2). The off-path spread, the counterpart of ς in Proposition 2, is 1.4–2.5 in the IsoFLOP designs and 0.94 on the OLMo ladder.

B. The Elasticity without a Functional Form

To apply equation (9), a quadratic in $\ln N$ fitted to the runs within one log point of the path gives the curvature at each budget, a cubic in $\ln C$ through the logged minima gives the slope of the IsoFLOP-minima frontier, and a design’s estimate is the random-effects mean over budgets (DerSimonian and Laird, 1986), with standard errors from a design-conditional wild bootstrap that re-estimates every step. On Farseeer we fit a kernel-weighted local quadratic of $\ln L$ in $(\ln N, \ln D)$, locate the points where the local wedge equals one, and read σ^* off the slope of $\ln w$ in $\ln M$ there, which equals $1/\sigma^* - 1$ for any smooth technology (Proposition 3). Online Appendix D gives details and robustness.

Levels. In FLOP-effective parameters, σ^* is 0.660 (standard error 0.023) on Chinchilla’s profiles, 0.660 (0.023) on Meta’s and 0.700, 0.713 and 0.705 on Marin’s three corpora (Figure 3, panel a); Farseeer’s local expansion path gives 0.708. Marin’s corpora share code, ladder and FLOP accounting, so we treat them as one study. With one estimate per study, the random-effects mean is 0.687, with a 95 percent confidence interval of [0.640, 0.734] (Table 1, panel C). Chinchilla and Meta give 0.66 and Marin and Farseeer 0.71, but the evidence that they differ is borderline: Cochran’s statistic is 8.3 on three degrees of freedom ($p = 0.04$) and ranges from 4.4 to 14.8 across Chinchilla’s windows, samples and profile memberships.

The interval reflects sampling error only. FLOP accounting, parameter conventions, Chinchilla’s windows and samples, and estimators move the mean between

TABLE 1. DESIGNS AND THE ELASTICITY OF SUBSTITUTION ON THE EXPANSION PATH

Design	Runs	Budgets	N	$M = D/N$	Off-path		σ^*		
					spread	Conv.	Model-free	κ free	$\kappa = 1$
<i>Panel A. IsoFLOP designs and Farseer’s factorial grid</i>									
Chinchilla, profiles ^a	137	9/9	57M–16B	0.04–294	1.65	F	0.660 (0.023)	0.656 (0.028)	0.690 (0.015)
Llama 3	133	8/10	59M–17B	0.44–643	1.38	F	0.660 (0.023)	0.693 (0.020)	0.769 (0.006)
Marin, Comma	85	5/7	0.12–13B	0.09–3,706	2.46	F	0.700 (0.027)	0.678 (0.015)	0.690 (0.015)
Marin, DCLM	85	7/7	0.12–13B	0.09–3,706	2.46	F	0.713 (0.027)	0.663 (0.013)	0.673 (0.013)
Marin, Nemotron-CC	88	7/8	0.12–13B	0.09–3,706	2.42	F	0.705 (0.023)	0.677 (0.014)	0.684 (0.014)
Porian, RefinedWeb ^b	121	12/12	5.2M–0.90B	0.02–346	2.29	NH	0.518 (0.017)	0.345 (0.141)	0.599 (0.032)
Porian, OWT2 ^b	116	12/12	5.2M–0.90B	0.02–194	2.20	NH	0.505 (0.014)	0.347 (0.160)	0.609 (0.039)
Farseer	404	8 ^c	0.10–6.4B	0.31–2,570	1.97	NE	0.708 ^c (0.005)	0.710 (0.003)	0.772 (0.010)
<i>Panel B. Other sweep-corpus technologies (parametric fits to all runs)^d</i>									
Chinchilla, all runs	240	–	57M–16B	0.46–341	1.35	T	–	0.701	0.737
Gadre, C4	34	–	11M–6.9B	5–640	1.46	T	–	0.607	0.795
Gadre, RedPajama	35	–	11M–6.9B	5–640	1.47	T	–	0.623	0.795
Gadre, RefinedWeb	35	–	11M–6.9B	5–640	1.47	T	–	0.594	0.796
OLMo ladder	30	–	0.19–3.2B	10–200	0.94	NI	–	0.544	0.825
Muennighoff	33	–	7.1M–8.7B	0.04–303	1.97	A	–	0.511	0.735

Note: Budgets: bracketed budgets used by the model-free estimator, of all. Conv.: parameter count that makes each profile an exact isocost; T, total; F, FLOP-implied $C/(6D)$ in the study’s own FLOP count (Chinchilla’s rebuilt, Online Appendix Table D2); NH, non-embedding plus output head; NE, non-embedding; NI, without the input embedding; A, the authors’ count. Off-path spread: residual s.d. of $\ln M$ given $\ln C$. Model-free: equation (9) per budget, random-effects mean. Parametric: Huber loss, same runs in Panel A. Standard errors: design-conditional wild bootstrap (Farseeer’s parametric columns: pairs or cell). ^aFive high-loss runs lie outside every model-free window; without them, 132 runs, M from 0.44, spread 1.48, $\sigma^* = 0.681$. ^bConstant-learning-rate runs without cooldown; OWT2, OpenWebText2. ^cLocal-path compute levels; first-derivative estimate (0.701–0.711 across bandwidths), s.e. of its mean; Panel C’s slopes use level-wise Hessian estimates (simple mean 0.707; random-effects mean 0.703). ^dWild-bootstrap s.e. of the κ -free fits: Online Appendix Table D8. Panel C and notes *e* to *k* follow on the next page.

Source: Studies in Panels A and B (subsection III.A); author’s calculations.

TABLE 1 (CONTINUED). DESIGNS AND THE ELASTICITY OF SUBSTITUTION ON THE EXPANSION PATH

Panel C. Model-free σ^ : study-level summary and dependence on compute*

	Estimate	95% CI	Diagnostics
Study-level mean ^e	0.687	[0.640, 0.734]	$k = 4$, $\hat{\varphi} = 0.023$, $Q = 8.3$ ($p = 0.04$)
Budgets $\leq 3 \times 10^{20}$ FLOP ⁱ	0.708	[0.690, 0.726]	$Q = 1.0$
Other parameter conventions	0.668–0.673		Marin, Farseer
FLOP accounting, $\eta = \pm 0.1$	0.673–0.700	[0.598, 0.748] ^f	Meta
Chinchilla’s variants ^l	0.674–0.702		$Q = 4.4$ –14.9
Farseer’s bandwidths ^j	0.680	[0.643, 0.717]	Farseer
First-derivative estimator	0.670	[0.603, 0.737]	Marin, Meta
Slope per decade of compute ^g	−0.033	[−0.081, 0.015]	44 budgets, wild $p = 0.049$
Budgets $\leq 3 \times 10^{20}$ FLOP	−0.008	[−0.050, 0.035]	37 budgets, wild $p = 0.87$
Chinchilla and Llama 3 ^h	−0.058	(0.014)	17 budgets
σ^* at 10^{21} FLOP, pooled line	0.647	[0.554, 0.741]	
σ^* , budgets $\geq 6 \times 10^{20}$ FLOP ^k	0.594	[0.561, 0.627]	Chinchilla, Llama 3; $Q = 2.5$
With Farseer’s 10^{21} level	0.606	[0.558, 0.655]	$Q = 21.1$ ($p < 0.001$)

Note: ^eOne estimate per study (Marin one; not Porian) in Panel A’s conventions; DerSimonian–Laird mean, modified Hartung–Knapp–Sidik–Jonkman interval (Knapp and Hartung, 2003). k : number of studies; $\hat{\varphi}$: between-study s.d.; Q : Cochran’s statistic. All variants give 0.663–0.702, and several variants’ intervals reach about 0.60. Within Marin the first-derivative estimator gives 0.666–0.676. ^f $\eta = -0.1$ for Meta; η , elasticity of FLOP per token in the parameter count. ^gThree-level random-effects meta-regression on $\log_{10} C$ (Farseer a sixth design); CR2 interval, about 3 degrees of freedom, and wild cluster bootstrap- t (Webb weights, enumerated), both with Marin as one cluster. The CR2 interval and the wild p -value disagree about significance, and p is 0.03–0.49 across weighting, scale, test and design fixed effects; no p -value is reliable with four to six clusters. ^hDesign fixed effects; s.e. from the joint design-conditional bootstrap draws, scaled for the budgets’ excess dispersion (model-based 0.013). ⁱEach IsoFLOP design’s random-effects mean over its budgets up to 3×10^{20} FLOP; Farseer’s first-derivative estimate on its path up to 3×10^{20} , which carries 84 percent of the weight. With the uncertainty of Farseer’s bandwidth range (s.e. 0.019; note j) the mean is 0.701 [0.665, 0.738]. ^jFarseer at the midpoint of its Hessian path mean over six bandwidths (0.664–0.727), the range treated as uniform uncertainty. ^kInverse-variance mean of Chinchilla’s three and Llama 3’s two budgets (Llama 3 85 percent of the weight); modified Hartung–Knapp–Sidik–Jonkman interval with the variance of the joint bootstrap draws. Farseer’s local path gives 0.722 (0.008) at 5×10^{20} and 0.658 (0.011) at 10^{21} FLOP; next row: random-effects mean with the latter. ^lChinchilla’s FLOP counts, parameter conventions, windows, samples and profile memberships (38 variants).

Source: Studies in Panel A (subsection III.A); author’s calculations.

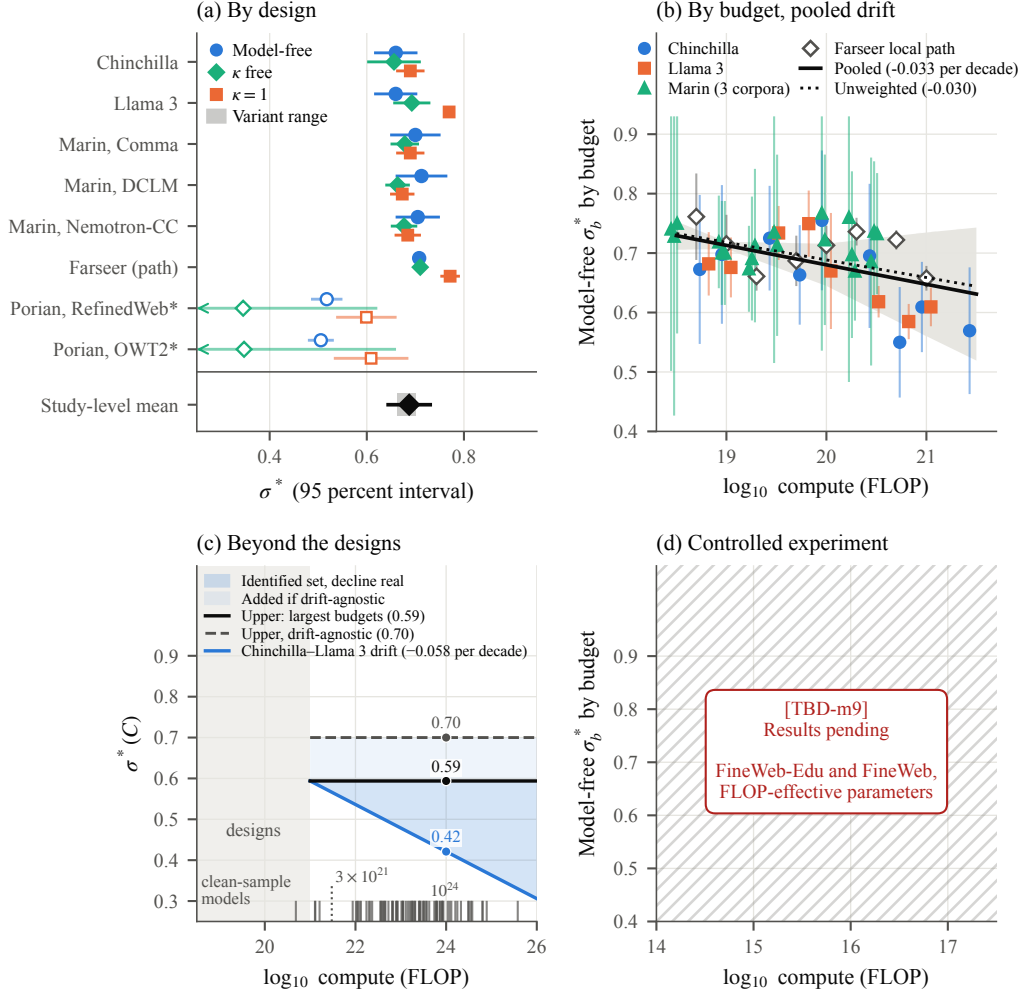


FIGURE 3. THE ELASTICITY OF SUBSTITUTION BY DESIGN, BY COMPUTE BUDGET, BEYOND THE DESIGNS, AND IN OUR EXPERIMENT

Note: Panel a: σ^* with 95 percent intervals. Circles: model-free estimate, random-effects mean over budgets (Farseeer: first-derivative estimate on the local path, with the standard error of its mean over compute levels). Diamonds and squares: κ family and $\kappa = 1$ fitted by Huber loss to the same runs. Open markers: Porian et al.'s unannealed runs. Study-level mean: one estimate per study (Marin one study) with its modified Hartung–Knapp–Sidik–Jonkman interval; the gray band spans the means of all variants (Table 1, panel C). Panel b: budget-level model-free estimates with 95 percent intervals, clipped at 0.40 and 0.93 (Farseeer: Hessian-based, eight compute levels). Solid line and band: three-level random-effects meta-regression with cluster-robust 95 percent band; dotted: unweighted. Panel c: identified set from 10^{21} FLOP, the largest budget bracketed by more than one design (Chinchilla's profiles reach 3×10^{21}): between a linear continuation of the Chinchilla–Llama 3 drift and 0.70 if σ^* does not rise with compute and falls no faster than that drift, and 0.59 if the top-budget decline is real; values at 10^{24} FLOP marked; ticks: training compute of the clean sample of Section IV. Panel d: [TBD-m9: model-free σ_b^* by budget for both corpora in FLOP-effective parameters, plan and CR2-scheme intervals; the drift is exploratory.] *Source:* Designs of Table 1; author's calculations.

0.663 and 0.702, and several of these variants have intervals that reach about 0.60 (Table 1, panel C). Without noise, the estimator’s bias on each design’s fitted technologies is at most 0.006; with noise, at the fitted κ -family technologies of three designs, its interval covers the truth 93 to 95 percent of the time, and 93 to 94 percent when half the noise variance is shared within a budget (Online Appendix D3).

Porian et al.’s profiles are the exception, at 0.518 (0.017) and 0.505 (0.014), and we keep them out of the summary. If the loss gain from annealing, which depends on the whole learning-rate path (Hägele et al., 2024; Tissue, Wang and Wang, 2025), varies with D/N at fixed compute, it bends the profile. Their estimates move between 0.45 and 0.55 across windows of 0.8 to 1.5 log points (0.33–0.34 with a global quadratic), and unlike the annealed designs they rise with compute (+0.040 per decade, standard error 0.006). The exclusion rests on a hypothesis about schedules, which our experiment tests.

Scale. Over the 44 budget-level estimates of Figure 3, panel b, a three-level random-effects meta-regression (Konstantopoulos, 2011) gives a slope of -0.033 per decade of compute, but the evidence of a common drift is weak: with four design clusters (Marin as one study) or six, no cluster-robust p -value is reliable (Bell and McCaffrey, 2002; Pustejovsky and Tipton, 2018), and across weighting, scale, test and design fixed effects the p -value lies between 0.03 and 0.49. The pooled line gives $\sigma^* = 0.713$ at 10^{19} FLOP, 0.680 at 10^{20} and 0.647 at 10^{21} .

At or below 3×10^{20} FLOP, where every design has budgets, the pooled slope is -0.008 , although the designs’ own slopes differ: Llama 3 already falls by 0.050 per decade and gives 0.618 (0.013) at 3×10^{20} FLOP, the budget where Marin stops, not a break the data locate. Above it only Chinchilla and Llama 3 have bracketed budgets. Across their 17 budgets σ^* falls by 0.058 per decade (standard error 0.014, from joint bootstrap draws scaled for excess dispersion), and their five budgets at 6×10^{20} FLOP and above agree ($Q = 2.5$, $p = 0.65$) and give $\sigma^* = 0.594$, with a 95 percent confidence interval of $[0.561, 0.627]$, 85 percent of the weight from Llama 3. Farseer’s local path gives 0.658 (0.011) at 10^{21} FLOP; with it the six top estimates are heterogeneous ($Q = 21.1$, $p < 0.001$). The pooled drift loses significance without either design or without each design’s largest budget, and both designs are digitized from figures. Marin shows no drift, but a design of its size detects a drift of -0.06 per decade only 31 percent of the time. The drift does not look like an artifact of the estimator: in simulations on three designs’ grids it creates none and attenuates an imposed one by at most 20 percent, and symmetric windows or a power-law frontier leave the Chinchilla–Llama 3 drift between -0.049 and -0.064 per decade (Online Appendix D3).

Beyond the designs. All but one of the 77 models in the clean sample of Section IV were trained with more than 10^{21} FLOP, and 73 with more than 3×10^{21} , Chinchilla’s largest bracketed budget; there σ^* is at best partially identified (Manski, 2003). If σ^* does not rise with compute and falls no faster than the drift within the designs, $\sigma^*(C)$ lies between a linear continuation of the Chinchilla–Llama 3

drift and 0.70, the level where the designs overlap (0.59, the top-budget value, if the decline there is real; Figure 3, panel c). The lower end is 0.48 at 10^{23} FLOP and 0.42 at 10^{24} with the top-budget value placed at 10^{21} FLOP (0.49 and 0.43 at the budgets’ compute-weighted center). Neither assumption follows from the data: a hinge at 3×10^{20} FLOP gives a decline of 0.15 per decade above the knot, and nothing rules out a reversal.

The curvature prices departures from the compute-optimal mix. In the κ family with equal exponents, a model trained on ten times the compute-optimal tokens per parameter needs 73 percent more compute than the compute-optimal model of the same loss at $\sigma^* = 0.70$, and 124 percent more at 0.60 (72–75 and 122–132 percent with the published splits of the exponents; Online Appendix D5).

C. Parametric Forms

Scaling laws are usually fitted in the Chinchilla form, the $\kappa = 1$ member of family (1), which ties the curvature of IsoFLOP profiles to the slopes of the frontier and the path (Proposition 3). The restriction is rejected wherever it can be tested. On Chinchilla’s full design $\hat{\kappa} = 0.774$, and $\kappa = 1$ is rejected at 1 percent under five of six inference schemes; it is rejected in all seven sweep-corpus technologies and on every IsoFLOP design, on Chinchilla’s profile runs only in FLOP-effective parameters (quasi-likelihood ratio 12.4, against 7.8 in total parameters and a critical value of 8.6 on the full design). Where the same-run comparison discriminates, only the κ -free form agrees with the model-free benchmark (Figure 3, panel a): on Llama 3, $\kappa = 1$ gives 0.769, 4.8 standard errors above the model-free 0.660, and κ free 0.693; on FarSeer, $\kappa = 1$ gives 0.772, against 0.708 on the local path and 0.710 with κ free. On Chinchilla’s full design (240 runs, total parameters) the two forms give 0.737 and 0.701, and the bootstrap-calibrated profile-likelihood set with κ free, $[0.68, 0.72]$, excludes 0.74; on Marin the Chinchilla form is slightly the closer.

With κ free, the three sweeps without profiles give lower and heterogeneous values (Table 1, panel B). With 30–35 runs, E and κ are hard to separate (Online Appendix Table D8), but the OLMo ladder is precise and still low, so a difference in recipe cannot be ruled out. Section IV therefore uses as its reference the κ -free Chinchilla technology, whose σ^* of 0.701 (0.700 refitted in N_F) lies inside the study-level interval.

D. What Is Not Portable

The inversion of Section IV also needs a zero point, $M^*(C)$, and this is where designs disagree. Under the Chinchilla form, the allocation exponent a ranges from 0.37 (OLMo ladder) to 0.57 (Gadre et al., RefinedWeb) across the seven sweep-corpus technologies, and $M^*(10^{21})$ from 3.4 to 60 tokens per parameter; even the model-free minima paths have allocation exponents of 0.49 on Chinchilla, 0.50 on Llama 3 and 0.35–0.43 on Marin’s corpora. M^* is a level: it depends on

the ratio A/B , which absorbs every difference in tokenizers, parameter counts and recipes. Counting Farseeer’s embedding parameters moves a from 0.53 to 0.41 and $M^*(10^{23})$ from 19.5 to 45. On the 975 released runs of [Porian et al. \(2024\)](#), the allocation exponent falls from 0.84 in Kaplan et al.’s setup to 0.50 through two familiar problems of production-function estimation: input measurement, here the counting of the output head ([Collard-Wexler and De Loecker, 2016](#)), and the re-tuning of flexible inputs ([Akerberg, Caves and Frazer, 2015](#); [Gandhi, Navarro and Rivers, 2020](#)) (Online Appendix D6).

Data quality moves allocation through one channel. Better data can lower the irreducible loss, scale reducible loss neutrally, or tilt the ratio A/B ; only the tilt changes M^* and the wedge. Across Gadre et al.’s three web corpora, all evaluated on C4, a neutral scale with corpus-specific asymptotes is not rejected ($p = 0.90$), but with little power. DataDecide’s 25 recipes reject every restriction at the resolution of the bootstrap, yet the tilt adds only 0.05 points to the 99.7 percent of cross-recipe variation in fit that asymptotes and a neutral scale explain. It is small in loss but not in allocation: it spans 0.22–0.26 across recipes, carried by the heavily filtered DCLM recipes, which moves M^* 2.9–3.4-fold and the wedge at given inputs 1.25–1.29-fold. DataDecide measures it before the schedule ends, and on a common validation set part of any corpus difference is distribution match (Online Appendix D4). In DeepSeek’s IsoFLOP experiments, too, the allocation exponent rose from 0.45 to 0.58 as data quality improved ([DeepSeek-AI et al., 2024b](#)), and in image-text pretraining the value of filtering depends on compute ([Goyal et al., 2024](#)). The allocation exponent and M^* are thus properties of a recipe, a tokenizer and a counting convention as much as of the technology.

E. A Controlled Experiment

No public design compares two data recipes on a common output with annealed runs, so we ran our own experiment (Online Appendix B4). Two corpora, FineWeb-Edu and FineWeb ([Penedo et al., 2024](#)), draw on the same web crawl and differ in filtering: FineWeb-Edu keeps the documents that an educational-quality classifier rates highly. Both share one 8,192-token tokenizer, and losses are in bits per byte. The main grid crosses eight transformers of width 128 to 640 (0.39M–49M non-embedding parameters) with token budgets from 25M to 800M: 44 endpoints per corpus, from 2×10^{14} to 7×10^{16} FLOP. M runs from 0.51 to 2,031 with non-embedding parameters. Each width has one constant-learning-rate trunk, from which a copy branches off for each budget and cools down ([Hägele et al., 2024](#)), so every endpoint is annealed. Four-layer models trained to 3.2 billion tokens (width 128 on both corpora, width 256 on FineWeb-Edu) raise the largest M to 4,063.

The largest run uses about 10^{17} FLOP, five orders of magnitude below Chinchilla’s largest, so the experiment tests functional form, neutrality and the direction of extrapolation errors, not production-scale levels. Nor does it extend Farseeer’s range in M : above about 1,000 tokens per non-embedding parameter

nearly every run is a width-128 model, whose embedding is 57 to 73 percent of its parameters. What it adds is a second recipe at those ratios, a controlled contrast in data quality, and annealed runs at the scale of Porian et al.’s.

Pre-analysis plan. We fixed the analysis before fitting any model (on such plans, see [Olken, 2015](#)). The plan was committed to the project’s working repository at 03:12 (UTC+3) on 24 September 2026 (bd5c0ad), when part of the FineWeb-Edu grid had finished and its raw endpoint losses had appeared in run logs, but no model had been fitted. It fixes the output; two parameter conventions, non-embedding (primary for the extrapolation test) and total (primary for comparisons with public designs); the estimators of subsections [III.B](#) and [III.C](#); a wild cluster bootstrap by width ([Webb, 2023](#)) and a seed-covariance parametric bootstrap; and six questions. Only the question on data quality has a decision rule: the filter is “factor-biased” if the 95 percent interval of the tilt $\Delta \ln(A/B)$ between corpora excludes zero with the same sign on both validation sets, “neutral” if it contains zero on both with a half-width below 0.10, and inconclusive otherwise. The others are a test of equal σ^* across corpora, a signed statistic (the slope in $\ln M$ of the log ratio of the wedge extrapolated from runs with $M \leq 100$ to the model-free local wedge) and descriptive checks of noise, tuning and on-path identification.

Power and a declared deviation. A power study from the design alone, run before the first estimation at 17:12, changed how the experiment can be read. At a noise standard deviation of 0.005 in log loss, the smallest tilt detected with 80 percent power is 0.05, against DataDecide’s 0.22. But with eight width clusters the plan’s bootstrap is anti-conservative ([MacKinnon and Webb, 2018](#)): its nominal 95 percent intervals for the tilt cover the truth 73 to 76 percent of the time (Online Appendix Table B5), and no scheme reaches nominal coverage for the model-free σ^* or for the extrapolation slope, whose null value is not zero. We therefore declared a deviation, D8, which we call the CR2 scheme: every table reports the plan’s intervals beside intervals from leverage-adjusted (CR2) cluster residuals ([Bell and McCaffrey, 2002](#)), widened for σ^* and the slope by calibration factors, and the slope is read against its design-specific null. A conclusion is robust only if it holds under both schemes and the seed-covariance bootstrap. D8 was written into the analysis memo at about 17:05 but committed only at 17:44 (a5f47f8); its timing rests on our account, and the coverage outputs that fix its factors carry file times of 16:55 to 17:02.

Amendments. The same commit adds Amendment 1, which maps each outcome of the curvature, neutrality and extrapolation questions to what the abstract, the introduction and Sections [III–IV](#) will say, and moves the comparison with Table 1 from total to FLOP-effective parameters. It was written before any FineWeb technology estimate or tilt, but after the FineWeb learning-rate calibration, matched-cell loss differences between the corpora, and preliminary FineWeb-Edu estimates of the model-free σ^* , 0.666 in non-embedding and 0.607 in total parameters. As FLOP-effective parameters are within 3 to 7 percent

of the total count here, its band for σ^* , $[0.60, 0.76]$, was drawn with FineWeb-Edu’s estimate effectively known: FineWeb-Edu’s arms of the curvature and extrapolation questions are therefore exploratory, and the confirmatory content is FineWeb’s arms, the test of equal σ^* and the neutrality question. Three mapped statements were later changed on public-design evidence alone (Online Appendix B4). Amendment 2, committed and pushed to the public repository at [[X1: time]], before any FineWeb technology estimate, restates the map, defines when an estimate lies outside the band and fixes the calibration factors (1.74 for σ^* , 1.75 for the slope). It also replaces D8’s restricted bootstrap test of the tilt, which its second power study found oversized (9.2 percent on one validation set) and weak (power 0.65 against 0.22), by the CR2 interval widened by 1.25. The public repository (<https://github.com/yigitokar/scaling-laws-as-production-functions>), first pushed at 18:10 on 24 September (59c5707), after the private commits but before any FineWeb technology estimate, carries the plan and Amendment 1 unchanged.

Two further deviations bear on the questions. The high- M runs were trimmed for time: FineWeb trains only the width-128 four-layer shape, and the tuning checks at high M exist for FineWeb-Edu only and stop at 1.6 billion tokens. And the main grid’s job lists predate the neutral sets: FineWeb’s widths 320 to 640 are evaluated on them afterwards from saved weights (deviation D13) and FineWeb-Edu’s widths 384 and 512 are retrained with them (Amendment 2), but FineWeb-Edu’s original grid has no saved weights, so the main-grid tilt rests on the two in-distribution validation sets. Online Appendix Table B3 lists every deviation with its time, reason and effect.

Exploratory additions. Trunks started after the trainer was changed (a5f47f8) are evaluated before each cooldown except the last (FineWeb widths 320 to 640, FineWeb-Edu’s retrained widths, the seed replicates and most high- M runs), which gives unannealed and annealed losses at the same (N, D) and tests the reason for excluding Porian et al.’s estimates, on the main grid only up to $M \approx 65$. Learning-rate and weight-decay variants at the high- M shape check tuning (Bergsma et al., 2025; Lourie et al., 2026). [TBD-m9: completed endpoints by corpus and tag, and which shapes have trunk evaluations and neutral sets.]

Results. [TBD-m9: each result under the plan’s scheme and the CR2 scheme (and the seed-covariance bootstrap), labeled robust only if it holds under all three; outcomes mapped to headline claims as in Amendment 1. (Q1) Model-free and κ -free σ^* by corpus in FLOP-effective parameters (primary; FineWeb-Edu exploratory), with both other conventions, against the study-level 0.687 and Porian et al.’s 0.51; test of equality across corpora. (Q2) Tilt between corpora on each in-distribution validation set, the decision under the plan’s rule and under the CR2 scheme, under the robustness conditions of Amendment 2, the implied factors for M^* and $\hat{w}$, the sign on the neutral sets where available, and the tilt with D in bytes. (Q3) Slope of $\ln(w_{M \leq 100}/w_{\text{local}})$ in $\ln M$ against the design-specific null, non-embedding convention; level gap as descriptive. (Q4) Seed noise

against residual s.d. (Q5) Learning-rate excess loss at the corners and the high- M checks; bound on the bias in a . (Q6) On-path against full-grid profile likelihood. Exploratory: trunk (unannealed) against endpoint (annealed) σ^* on the same grid; drift of σ_b^* across budgets (Figure 3, panel d).]

IV. What Over-Training Reveals

This section uses the technology of Section III to read the choices of the developers that release models. Two findings stand out. With the zero point extrapolated from the designs’ own paths and their sampling error, over-training is identified beyond the designs for most allocation decisions, though not for most training compute; weaker restrictions on the zero point identify far less (Section IV.D). And revealed over-training rose steeply from 2023 to 2025: the compute-weighted share rose under every technology we consider, though the step from 2024 to 2025 is not statistically established (Section IV.F).

A. The Inversion

By Proposition 1, when the price of a training FLOP does not vary with model size, data cost nothing beyond compute and no cap on tokens binds, $w - 1 = v_N$, the value of compactness (including the shadow value of any binding limit on size). In the κ family the wedge depends on a model’s inputs only through $M = D/N$ and $C = 6ND$ (equation (7)):

$$(10) \quad \ln w = \left(\frac{1}{\sigma^*} - 1 \right) \ln \frac{M}{M^*(C)}, \quad s \equiv \frac{w - 1}{w} = 1 - \left(\frac{M}{M^*(C)} \right)^{-(1/\sigma^* - 1)}.$$

At given compute every technology in the family ranks models by M , and in the sample below a regression of $\ln \hat{w}$ on $\ln M$ alone has an R^2 of 0.9999: what the technology adds to the model cards is the translation into an economic quantity.

We report the compactness share $s = v_N/(1 + v_N)$, which needs no conversion of FLOPs into dollars; what the value consists of depends on conduct (Section I.D). Released models lie outside every design, so equation (10) is an extrapolation; Section IV.D checks its direction in M and replaces it with a sign result in C .

B. The Clean Sample and Decision Units

Models. The clean sample starts from the 173 general-purpose open-weight models of the verified sample, whose parameter and token counts we audited (Online Appendix B). Nine steps drop models whose (N, D) is not a choice of the kind Proposition 1 describes, such as research suites and replications whose D was fixed by design, mixture-of-experts and distilled models, and token counts without a primary source. The result is 77 dense models from 18 developers, released between 2023 and 2025 (Online Appendix E2); an earlier version reported wedges for all 173 under 12 of the 32 technologies below.

Decision units. When members of a family share one token budget, their wedges differ only through the technology (Proposition 1(iv)), and a budget chosen for them reveals one number, the family share, a compute-weighted average of member shares. Our unit of analysis is therefore the allocation decision, one choice of (N, D) : members of a family whose budgets are within 10 percent of each other form one decision, and the other models are decisions of their own, 49 in all. Under a written protocol, checked by a second coder blind to the wedges (Online Appendix E5), we read each common budget in the reports as a family *choice*, a size *menu*, a *cap* (the developer’s whole curated corpus), a cap and a menu, or, where only the corpus size is stated, *cap or choice*. A cap on tokens, for a family or for a member that repeated its data, makes the share a lower bound on the value of compactness; under a choice or a menu it is a point estimate, since a binding limit on size is part of that value (Proposition 1(iii)). Of the 49 decisions, 29 are point estimates and 20 lower bounds. Separately, 49 of the 77 models sit in five size windows around quantized memory tiers, where a binding tier’s shadow value is part of the value of compactness. Table 2 flags both.

Technologies. A fixed rule admits 32 technologies: every technology estimated on a designed sweep in Section III, with $\kappa = 1$ and with κ free, the path of each IsoFLOP design with its model-free curvature, the literature’s estimates on the Chinchilla data, and the published laws of DeepSeek LLM (DeepSeek-AI et al., 2024b) and MiniCPM (Hu et al., 2024). The reference is the Chinchilla technology with κ free, fitted to all runs in total parameters like the models’ M ($\sigma^* = 0.701$, $M^*(10^{21}) = 22.7$; 0.700 and 20.6 in FLOP-effective parameters). Its intervals come from a design-conditional wild bootstrap (Feng, He and Hu, 2011), joint across the sample, the narrowest of six schemes we compared; we also report a cluster bootstrap that resamples the Chinchilla runs by IsoFLOP budget and model-size trunk (44 clusters).

C. What Allocation Decisions Reveal

Figure 4 and Table 2 report the 49 decision units. Under the reference technology the median decision has $\hat{w} = 3.8$ and $s = 0.73$, with a 95 percent joint wild-bootstrap interval of $[0.66, 0.78]$: it was made as if a model one percent smaller at the same quality were worth 2.8 percent of its training cost. The cluster bootstrap widens it to $[0.51, 0.81]$. Ninety-eight percent of decisions are over-trained ($\hat{w} > 1$). The unit of analysis hardly moves the median: it is 0.75 over the 77 models, 0.74 over 56 units that group only whole families on one budget, 0.75 over 66 units that enter members of cap families as lower bounds, and 0.74 over the 29 point-identified decisions.

That interval measures sampling error within one technology; the choice of technology matters more. Across the 32 technologies the median decision share ranges from 0.04 to 0.80, and from 0.17 to 0.80 without MiniCPM’s law, whose zero point lies outside its own design; the κ -free fits of the large designs and the developers’ paths with model-free curvature give 0.62 to 0.75. All 32 technologies

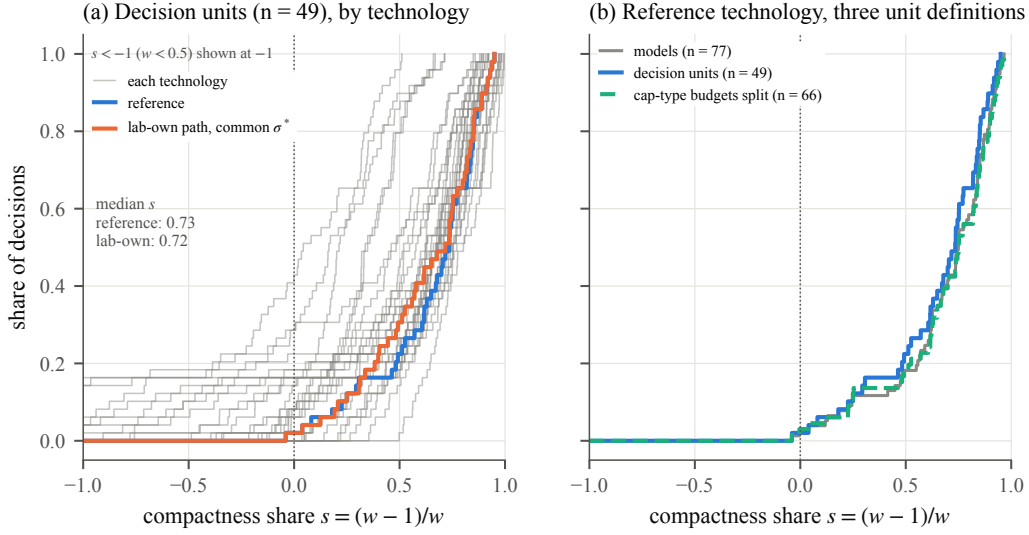


FIGURE 4. THE VALUE OF COMPACTNESS REVEALED BY ALLOCATION DECISIONS

Note: Empirical distribution functions of the compactness share $s = (w - 1)/w$. Panel a: the 49 decision units of the clean sample, in which members of a family that share one token budget enter once, at their family share. Gray: each of the 32 technologies. Blue: the reference technology (Chinchilla, κ free). Orange: the developer's own expansion path with the common model-free σ^* where available, otherwise the reference. Panel b: the reference technology over the 77 models, the 49 decision units and 66 units that split the eight cap-type budgets into their members (members of cap-reading families as lower bounds, assuming each would have trained longer at its own size). Values below -1 ($w < 0.5$) are drawn at -1 . s can be read as planned serving expenditure only where the developer served the size and neither a common budget nor a tier window intervenes (Section IV.E).

TABLE 2. THE REVEALED VALUE OF COMPACTNESS, BY ALLOCATION DECISION

Decision unit	Budget	Id.	Served (model)	Tier window	N_{emb}/N	M/M^*	s	Reference [95%]	Lab-own s	Band of s	Sign id.
<i>Panel A. Members of a family that share one token budget: one decision each</i>											
LLaMA (7B, 13B)	choice	pt	0/2	2/2	0.03–0.04	4.4	0.46	[0.37, 0.55]	0.56	[-0.35, 0.63]	N/N
StableLM-Alpha (3B, 7B)	choice	pt	0/2	1/2	0.08–0.11	5.9	0.53	[0.44, 0.59]	–	[-0.20, 0.65]	Y/N
MPT (7B, 30B)	menu	pt	2/2	2/2	0.01–0.03	2.1	0.25	[0.11, 0.36]	–	[-1.27, 0.56]	N/N
Llama 2 (7B, 13B, 70B)	choice	pt	0/3	3/3	0.01–0.04	2.0	0.23	[0.06, 0.36]	0.34	[-2.00, 0.58]	N/N
Yi (6B, 34B)	menu	pt	0/2	0/2	0.03–0.09	5.2	0.49	[0.38, 0.59]	–	[-0.39, 0.67]	N/N
Qwen2 (1.5B, 7B, 72B) [†]	menu	pt	3/3	2/3	0.03–0.15	6.0	0.51	[0.38, 0.61]	–	[-0.68, 0.69]	N/N
OLMo 2 (1B, 7B) [†]	choice	pt	0/2	1/2	0.11–0.28	33	0.77	[0.71, 0.81]	0.58	[0.20, 0.86]	Y/Y
Apertus (8B, 70B) [†]	choice	pt	0/2	2/2	0.03–0.13	12	0.65	[0.54, 0.73]	–	[-0.28, 0.75]	N/N
OLMo 3 (7B, 32B) [†]	choice	pt	0/2	2/2	0.03–0.11	11	0.63	[0.53, 0.70]	0.19	[-0.16, 0.73]	N/N
LLaMA (30B, 65B)	cap or choice	lb ^c	0/2	2/2	0.01	1.2	0.08	[-0.11, 0.23]	0.21	[-2.79, 0.52]	N/N
Qwen (14B, 72B)	cap or choice	lb ^c	2/2	2/2	0.03–0.11	2.5	0.31	[0.15, 0.43]	–	[-1.47, 0.61]	N/N
Llama 3 herd (8B, 70B, 3.1 405B)	cap or choice	lb ^c	1/3	2/3	0.01–0.13	2.5	0.29	[0.08, 0.46]	0.38	[-2.94, 0.64]	N/N
Qwen2.5 (0.5B–72B) [†]	cap or choice	lb ^c	7/7	5/7	0.03–0.28	22	0.72	[0.63, 0.78]	–	[-0.03, 0.78]	N/N
Granite 3.0 (2B, 8B) [†]	cap or choice	lb ^c	2/2	2/2	0.02–0.04	89	0.85	[0.81, 0.89]	–	[0.32, 0.94]	Y/Y
Qwen3 (0.6B–14B) [†]	cap or choice	lb ^c	5/5	2/5	0.10–0.26	194	0.89	[0.85, 0.92]	–	[0.35, 0.96]	Y/Y
DeepSeek LLM (7B, 67B)	cap+menu	lb	1/2	2/2	0.03–0.12	1.7	0.18	[0.00, 0.32]	0.31	[-2.25, 0.56]	N/N
SmolLM (135M, 360M) [†]	cap or choice	lb	0/2	0/2	0.13–0.21	96	0.86	[0.83, 0.88]	–	[0.41, 0.97]	Y/Y
<i>Panel B. Families with size-specific budgets (selected): each member is a decision</i>											
SmolLM2 135M [†]	size-specific	pt	N	N	0.21	658	0.94	[0.92, 0.95]	–	[0.51, 1.00]	Y/Y
SmolLM2 360M [†]	size-specific	pt	N	N	0.13	495	0.93	[0.91, 0.95]	–	[0.48, 0.99]	Y/Y
SmolLM2 1.7B [†]	size-specific	lb	N	N	0.06	293	0.91	[0.88, 0.93]	–	[0.42, 0.99]	Y/Y
OLMo 2 13B [†]	size-specific	lb	N	Y	0.07	17	0.70	[0.63, 0.76]	0.40	[0.02, 0.77]	Y/Y
OLMo 2 32B [†]	size-specific	lb	N	Y	0.03	8.9	0.61	[0.50, 0.69]	0.12	[-0.23, 0.72]	N/N
<i>Panel C. Single-model releases: the 5 with the largest training compute of 15</i>											
Gemma 2 27B	singleton	pt	Y	Y	0.04	22	0.73	[0.66, 0.79]	–	[0.02, 0.80]	Y/N
Falcon3 7B	singleton	pt	N	Y	0.11	87	0.85	[0.81, 0.89]	–	[0.31, 0.94]	Y/Y
Marin 8B [†]	singleton	pt	N	Y	0.13	73	0.84	[0.79, 0.88]	0.81	[0.30, 0.93]	Y/Y
Falcon 2 11B	singleton	pt	N	N	0.05	23	0.74	[0.67, 0.79]	–	[0.08, 0.81]	Y/Y
MAP-Neo 7B	singleton	pt	N	Y	0.05	26	0.75	[0.69, 0.80]	–	[0.13, 0.84]	Y/Y
<i>Panel D. Over decision units</i>											
Median s , 49 decision units							0.73	[0.66, 0.78]	0.72		
interval under the budget-plus-trunk cluster bootstrap								[0.51, 0.81]			
29 point-identified (pt)							0.74	[0.67, 0.79]			
interval under the budget-plus-trunk cluster bootstrap								[0.54, 0.82]			
20 lower bounds (lb, lb ^c)							0.71	[0.63, 0.77]			
Median s , 56 family-label units							0.74	[0.67, 0.78]	0.73		
Median s , 66 units, cap-type budgets split into members							0.75	[0.69, 0.80]	0.75		
Median s , 77 models							0.75	[0.69, 0.79]	0.74		
Share of decision units with $w > 1$							0.98	[0.94, 1.00]	0.98		
Share of decision units with $w > 1$ identified, $\tau = 1$ and $\tau = 1.84$: unweighted 0.69 and 0.61; compute-weighted 0.14 and 0.11											

Note: A decision unit is one choice of (N, D) : members of a family whose token budgets are within 10 percent of each other ($\max/\min D \leq 1.10$) form one decision; its entry is the family share $1 - 1/\bar{w}_H = \sum_j h_j s_j$ (h_j : training-compute shares; Proposition 1(iv) and Online Appendix Proposition A4), and M/M^* is compute-weighted. Budget: the reading of a common budget from the developers' reports, coded under a written protocol and checked by a second coder (Online Appendix E5): cap, the whole curated corpus binds, so the family share bounds the value of compactness from below, and a member's wedge does so only if that member would have trained longer at its own size; choice, the family share is revealed; menu, sizes set by a hardware or configuration menu, whose shadow value is part of the value of compactness, so the share is revealed; cap+menu, a lower bound; cap or choice, the reports state only the corpus or budget size, or give evidence of both. Id.: pt, point estimate of the value of compactness; lb, lower bound (a binding cap on tokens, or a member that repeated its corpus); lb^c, lower bound that is a point estimate if the budget was chosen. Served: this size (any post-trained variant) was offered through the developer's own per-token API or consumer product within 180 days of release; families show members served/members. Tier window: 2.4–3.3, 6.5–9.5, 11.5–14.9, 26–32.9 or 65–72.9B parameters, where a binding tier's shadow value is part of the value of compactness. N_{emb}/N : embedding share of parameters (range over members). $s = (w - 1)/w$. Reference: Chinchilla with κ free, in total parameters like the models' M ; brackets: 95 percent joint wild-bootstrap intervals of one technology, and every bracketed interval is the narrowest of the six bootstrap schemes compared (Panel D also gives the cluster bootstrap, 44 clusters). Lab-own: the lab's own expansion path with the common model-free σ^* (Meta: 8 bracketed IsoFLOP budgets; AI2: OLMo ladder; DeepSeek: published law; Marin: Nemotron-CC ladder); in Panel D, the reference technology where no lab-own path exists. Band: range of point estimates over the 32 technologies. Sign id.: $w > 1$ identified beyond the designs (Proposition 3(v); Online Appendix E7) with no tilt allowance ($\tau = 1$) and with $\tau = 1.84$ on M^* ; a decision is identified only if every member is. [†]Trained partly on teacher-generated synthetic data (the reference technology is estimated on natural text). Token counts corrected: StableLM-Alpha 0.8T, MPT-30B 1.05T.

Source: Author's calculations; units, readings, serving codes and their evidence are in the replication package.

agree that $w > 1$ for 28 of the 49 decisions.

A size-specific family. SmolLM2 (Ben Allal et al., 2025), the only family whose three members have their own budgets and lie outside the tier windows, trained 135 million parameters on 2 trillion tokens, 360 million on 4 trillion and 1.7 billion on 11 trillion. Under the reference the shares are 0.94, 0.93 and 0.91 (a lower bound: the 1.7B model trained for about two epochs on its data), and each member’s over-training is identified beyond the designs even with a tilt allowance of 1.84. Hugging Face released the sizes for on-device use and did not serve them, so s is a value of compactness to users who run the models; trained partly on synthetic data, SmolLM2 has a more secure sign than level. OLMo 2 (Team OLMo et al., 2024) is less clean: its 1B and 7B models share a budget, and its 13B and 32B repeated part of the stage-1 mix (1.2 and 1.5 epochs), a binding cap on unique tokens that makes their shares lower bounds. With D varying 1.5-fold across a 22-fold range of N , the fall of its shares with size, from 0.87 to 0.61, is largely the technology’s (Online Appendix E5).

Common budgets. Meta trained the Llama 3 herd, the 8B and 70B models and the 405B flagship of Llama 3.1, on 15 to 15.6 trillion tokens. The herd reveals one share: 0.29 under the reference, with an interval of $[0.08, 0.46]$, and 0.38 under Meta’s own path; the flagship’s own sign is not identified beyond the designs. Meta’s sources are consistent with a choice and with a cap: the report describes a curated corpus of about 15 trillion tokens, a flagship sized as compute-optimal for its budget, and smaller models trained far beyond that point for inference efficiency (Grattafiori et al., 2024), and the release post says that the 8B and 70B were still improving when training stopped (Meta AI, 2024). Under a choice 0.29 is the herd’s value of compactness; under a cap it is a lower bound. Trained on one corpus, the herd cannot reveal the contrast Meta describes between a flagship sized for training efficiency and siblings trained for inference. Across the 17 common budgets the shares run from 0.08 for LLaMA 30B and 65B (Touvron et al., 2023b) to 0.89 for Qwen3 (Yang et al., 2025), both read as cap or choice and so lower bounds.

D. Beyond the Designs

Released models lie beyond the designs in two directions. Forty-six of the 77 clean models have M above 341, the largest ratio in the Chinchilla design, and 73 used more compute than 3×10^{21} FLOP, the largest budget at which any public design brackets a training optimum (Section II.B).

Tokens per parameter. Figure 5 compares the local wedge of Proposition 3(i), which needs no functional form, with those of parametric forms fitted to the same runs in three recipes: Farseeer’s design (Li et al., 2025b), Marin’s IsoFLOP ladders on three corpora (Held and Marin Community, 2026) and Meta’s Llama 3 IsoFLOP profiles (Grattafiori et al., 2024; Czech, 2026). In all three, parametric forms understate the wedge at high M , and the gap grows with M : at 256 to 1,024 tokens per parameter the κ -free form fitted to all runs gives 0.65 to 0.87 of

the local wedge, and the Chinchilla form fitted to runs with $M \leq 100$ about 0.4.

The direction replicates; the shape does not. Measured against a linear extrapolation from the local slope at the path, the form of the κ family (Proposition 3(iii)), $\ln w$ is convex in Farseeer and Marin, by 0.17 and 0.43 log points at M of 256 to 1,024. In Llama 3 it is concave at the primary bandwidth, though not at every one (-0.13 at M of 64 to 256), and a steeper local slope at the path (a local σ^* of 0.63, against 0.69 for the κ -free fit) makes the κ -free fit overstate the local wedge at M below 16. Inside the designs, then, parametric forms understate the wedge at high M , but the direction of the error beyond them is recipe-dependent. The evidence comes from small models, and at high M from few points (Online Appendix E7). [TBD-m9: Our experiment repeats the check under question Q3 of its pre-analysis plan and Amendments 1 and 2, up to $M \approx X$ in non-embedding parameters. The extrapolation slope lies below / above / is not distinguishable from its design-specific null, so [sentence from Amendment 2’s map] (Section III.E).]

Compute. Proposition 3(v) replaces a point extrapolation of the zero point with an identified set. For each design we fit a line to the logged IsoFLOP minima at its bracketed budgets and extrapolate it with a simultaneous 95 percent band for its level and slope (a wild bootstrap- t): Chinchilla, anchored at 3×10^{21} FLOP in total parameters, the convention of the models’ M ; Llama 3, at 10^{21} ; and Marin, at 3×10^{20} , with one slope for its three corpora, as in Section III. DeepSeek’s published law is a fourth path. The identified set for $M^*(10^{24})$ is then [5.7, 248]; Marin’s path, the steepest, sets its upper end. The minima curve inside the designs; continuing Chinchilla’s and Llama 3’s paths at their top-budget slopes, beside the bands, identifies 75 percent of models (Online Appendix E7).

The sign. Over-training is identified when M exceeds every ratio in the identified set, whatever the curvature and the functional form (Proposition 3(v)): for 78 percent of models and 69 percent of decisions. No model is identified as under-trained, but that is mechanical: every clean model has $M \geq 19$, while the lower end of the set is 4 to 11 at their compute. The 17 models whose sign is not identified all have M below 250. They include Llama 2 70B, Llama 3.1 405B, Qwen2 72B, Falcon 180B and DeepSeek LLM 67B, and they carry most of the training compute: weighted by compute, the share identified is 24 percent over models and 14 percent over decisions. With the slopes bounded by the designs’ point estimates alone, the share is 87 percent of models.

The bands assume that a developer’s recipe shares the designs’ zero points, but recipes, tokenizers and data quality move M^* (Section III.D); the tilt allowance τ of Proposition 3(v) admits such differences. The share of models identified is 74 percent at $\tau = 1.3$, for tokenizer differences; 66 percent at 1.84, DataDecide’s largest tilt under the reference exponents; and 53 percent at 3.4, the same tilt under DataDecide’s own exponents. Weighted by compute, it is 17 and 16 percent at the last two. [TBD-m9: The experiment’s tilt between its two corpora, $\hat{\chi} = X$ (Amendment 1, Q2), implies an M^* factor of Y at 10^{15} – 10^{17} FLOP, a magnitude

at small scale beside DataDecide’s (Online Appendix Figure E1).] Weaker assumptions identify less: 13 percent of models with normal inputs alone ($|e_p| \leq 1$), and 23 percent with every technology’s in-support zero point as an anchor. The share of models identified stays above one half up to a factor of 3.7 on M^* , and for models below 15 billion parameters above 40 percent even at 10; for the flagships that carry most compute the sign is open (Online Appendix Figure E1).

Magnitudes. With $1/\sigma^* - 1$ between 0.40 and 0.68, from the in-design estimates to the top-budget value, the medians of the models’ lower and upper bounds on s are 0.41 and 0.95 (0.99 with σ^* at the lower end of its identified set). Tilt allowances of 1.84 and 3.4 lower the median lower bound to 0.24 and 0.03. With the zero points fixed, the reference levels are lower bounds if σ^* keeps falling with compute: a constant σ^* of 0.60, close to the top-budget value, raises the median decision share from 0.73 to 0.87, and one of 0.62 to 0.85, while the share of decisions over-trained, which depends only on $M^*(C)$, does not move. [TBD-m9: If the experiment’s σ^* lies outside the range from 0.60 to 0.76 for either corpus (Amendment 1, Q1.3), the curvature range widens to X–Y and the median bounds become A and B.]

E. Conduct and Rival Wedges

The objectives that Proposition 1 admits predict different patterns (Online Appendix Remark A4). Serving cost predicts more over-training for sizes a developer serves and for closed models; adoption by users who bear the cost, the most for models meant for users’ devices; memory tiers, bunching at tier boundaries. Inference uses a restricted wild cluster bootstrap by developer (Cameron, Gelbach and Miller, 2008; Webb, 2023).

Serving. Coded by model (this size offered through the developer’s own per-token interface or consumer product within 180 days of release), 26 of the 77 clean models were served, 19 of them by Alibaba, and 25 of the 26 share a budget with other sizes or sit in tier windows. Given compute and release year, the coefficient of $\ln \hat{w}$ on the serving indicator is 0.56 (standard error 0.24), the sign that serving-cost internalization predicts, but with six treated developers the test cannot detect a relation: the bootstrap p -value is 0.41. Randomization inference gives $p < 0.001$ when it permutes the code across models, treating one developer’s sizes as separate assignments, but 0.10 when it permutes it across developers (MacKinnon and Webb, 2020), and 0.94 without Alibaba. Without Alibaba the coefficient is -0.09 ; over decisions it is 0.28 (p -value 0.67).

Open weights, tiers and devices. The one test with power points away from serving cost. Among 159 production-scale decisions from 67 developers, in a broader universe than the clean sample that includes unverified Epoch rows, open-weight decisions are more over-trained than closed ones by 0.45 log points given compute and release year (standard error 0.17, p -value 0.018), whereas a developer bearing all of its serving cost should over-train at least as much. Closed developers, however, disclosed only eight allocations in 2024 and 2025, and none for their small,

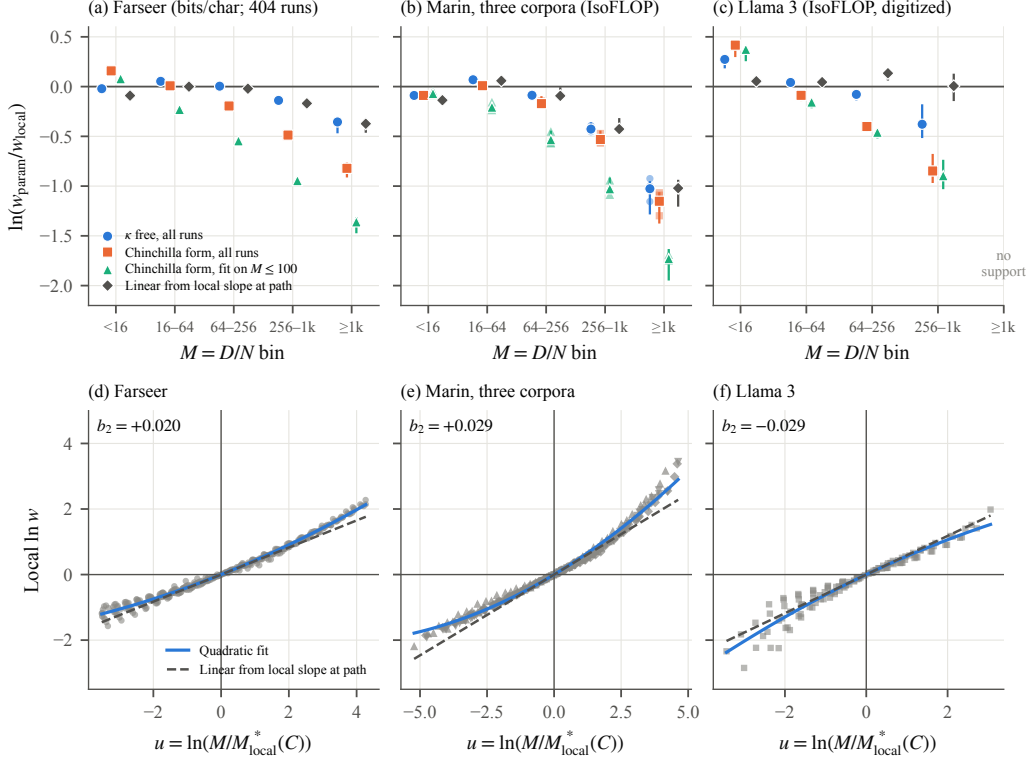


FIGURE 5. PARAMETRIC EXTRAPOLATION AGAINST THE MODEL-FREE WEDGE IN THREE RECIPES

Note: Panels a–c: mean of $\ln(w_{\text{param}}/w_{\text{local}})$ by bin of $M = D/N$ for parametric forms fitted to the same runs: the κ family with κ free and the Chinchilla form ($\kappa = 1$), each fitted to all runs, and the Chinchilla form fitted only to runs with $M \leq 100$. Gray diamonds: a linear extrapolation of $\ln w$ from its local slope at the path, whose gap is minus the convex part of the local wedge. Negative values mean that the form understates the wedge. The local wedge is $w = (f_c - f_x)/(f_c + f_x)$ with $f = \ln L$, $x = \ln M$ and $c = \ln C$ (Proposition 3(i)), from kernel-weighted local quadratic fits. Farseer: 404 runs, bits per character, non-embedding N . Marin: IsoFLOP ladders on three corpora, pooled with equal weights (the corpora are shown faintly). Llama 3: digitized IsoFLOP profiles. Bars: 95 percent percentile intervals from a wild cluster bootstrap (by model size for Farseer and by budget otherwise; Webb weights, 999 draws), which exclude smoothing bias; 7 of the 51 intervals of the parametric forms exclude their own estimate. In Llama 3 the gaps are positive at M below 16 (+0.27 for the κ -free form, +0.42 for the Chinchilla form): the fitted forms rotate against the local wedge. The local wedge is evaluated where the kernel’s effective sample size is at least 15 (Farseer) or 8 (otherwise): up to $M \approx 2,570$ in Farseer, 1,110 in Marin and 290 in Llama 3. Marin’s highest bin rests on one or two corner points per corpus, and Llama 3’s 256–1,024 bin on two points. Panels d–f: local $\ln w$ against $u = \ln(M/M_{\text{local}}^*(C))$, where $M_{\text{local}}^*(C)$ is the root of the local wedge on each budget line, with the quadratic fit $\ln w = b_1 u + b_2 u^2$ and its tangent at $u = 0$: the linear extrapolation from the local slope at the path, which is also the linear form of the κ family.

Source: Li et al. (2025b); Held and Marin Community (2026); Grattafiori et al. (2024); Czech (2026); author’s calculations.

typically distilled models, so the closed sample is selected on flagships. Sizes bunch: 51 percent of the 220 open-weight production-scale models of 2023–2026 sit in the tier windows, against 13 percent under a smooth counterfactual (Chetty et al., 2011; Kleven, 2016). And the nine on-device models, from two developers, have a median share of 0.90, against 0.70 for other targets.

The data are thus more consistent with adoption and tier mechanisms than with the internalization of serving cost, and we read $w - 1$ as the revealed value of compactness. Planned serving expenditure is fully licensed only where the developer served the size and neither a common budget nor a tier window intervenes: Qwen2 0.5B alone (Qwen2 1.5B shares its budget with the 7B and 72B). The cost side matters little (Online Appendix E9).

Recipes. A developer’s own recipe may place M^* away from the reference, the factor-bias critique of markup estimation (Demirer, 2026; Raval, 2023). Where a developer published its own scaling evidence we apply one rule: its expansion path, in its own parameter convention, with the common model-free $\sigma^* = 0.687$. Llama 1 and 2, allocated before Meta’s 2024 path existed, are left out. On the 15 models covered, the median share is 0.53 under developers’ own technologies, against 0.70 under the reference; AI2’s ladder (Bhagia et al., 2025), which puts M^* at 3.7 to 6.7 times the reference zero point, drives the gap, and without AI2 the order reverses (0.71 against 0.65). Over all 49 decisions, with developers’ own technologies where they exist, the median is 0.72; like the reference’s, these levels are lower bounds if σ^* falls with compute (Online Appendix E5).

A second output. A developer that values an output other than validation loss equates a different wedge, $\hat{w} = w_{L'} w_L / w_{L'}$ (Proposition 1(iii)), the output critique of Bond et al. (2021). On the OLMo ladder’s 30 runs, the median of $\ln(w_{\text{task}}/w_{\text{C4}})$ over the 77 clean models, task bits per byte against C4 loss, is -0.11 with $\kappa = 1$ and -0.37 with κ free, neither distinguishable from zero; with κ free, a task-valuing developer’s wedge could be a third of the loss-based one or three and a half times it. The only second-output design available has no power, and the output critique remains unbounded.

Post-training and capability. Post-training compute, which scales with N but not with D , raises the wedge to $w = 1 + x_P + m_N$, with x_P its ratio to pretraining compute. Disclosed budgets put x_P at 0.2 percent for DeepSeek-V3 (DeepSeek-AI et al., 2024a), below 0.1 percent for the supervised stage of Tülu 3 (Lambert et al., 2025) (converting disclosed GPU-hours at an assumed 40 percent utilization) and at most 32 percent for Olmo 3.1 Think 32B (Team Olmo, Ettinger et al., 2025). At 32 percent the median model share falls from 0.75 to 0.73, and the median decision share by 0.025. Loss, moreover, is not capability: over-trained models lose more quality under quantization (Kumar et al., 2025), so $\hat{w} - 1$ understates their value to users who run quantized models, whereas the second-output estimates point the other way; neither direction is measured at production scale.

F. The Trend since 2023

Under the reference technology, the compute-weighted share over decisions, $\sum_i (w_i^+ - 1)C_i / \sum_i w_i^+ C_i$ with $w_i^+ = \max(w_i, 1)$, rose from 0.24 among 2023 releases to 0.54 in 2024 and 0.81 in 2025, and over models from 0.27 to 0.83. A bootstrap over the 18 developers establishes the rise from 2023 to 2025, with a 95 percent interval for the change of $[0.19, 0.72]$, and from 2023 to 2024. It does not establish the step from 2024 to 2025 $([-0.12, 0.44])$. The weighted shares rise in both steps under all 32 technologies and in each of the 18 samples that leave out one developer, but the median decision share rose from 0.49 to 0.77 and then fell to 0.74. The level does not travel: the 2025 weighted share over decisions ranges from 0.23 to 0.92 across technologies (Online Appendix E10).

Five qualifications apply. First, it largely restates the rise in tokens per parameter, whose median rose from 146 in 2023 to 1,217 in 2024 and 2,438 in 2025. Second, part of the weighted rise is composition: Llama 3.1 405B, whose sign is not identified, carries 55 percent of 2024 compute, and the sign is least identified in the base year, 2023 (12 percent of its compute). Third, 2025 is thin, with 13 dense models in six decisions; without any one of Alibaba, Hugging Face or Marin the median decision share falls further. Fourth, at x_P of 32 percent the 2025 weighted share over models falls from 0.83 to 0.82, or to 0.77 with post-training compute in the cost base. Fifth, 92 percent of 2025 clean-sample compute sits in tier windows, so the rise may reflect tiers' shadow values, and the excluded mixture-of-experts models carried more 2025 open-weight compute than the clean sample; with them the 2025 share is 0.73 at total and 0.82 at active parameters. With developer fixed effects, identified by the nine developers with decisions at more than one date, the wedge grew 1.6-fold a year (95 percent interval $[0.91, 2.61]$). Most technologies, the reference among them, date from 2022 to 2024, and better data shift the optimum toward parameters (DeepSeek-AI et al., 2024b), so improving data would make them understate the rise, and parameter-augmenting architectural change overstate it.

The rise is a rise in revealed over-training; it is a rise in the value of compactness only if developers optimized against technologies like ours. Among decisions whose developers state a deployment rationale the median share rose from 0.48 in 2023 to 0.89 in 2025, while the share of decisions citing a compute-optimal target fell from 0.12 to zero (Online Appendix Table E15).

We do not extend the series before 2023. OPT-175B and GLM-130B, with 92 percent of the compute of the eight open-weight production-scale releases of 2019–2022, were allocated when the law of Kaplan et al. (2020) guided practice; with Kaplan's path as the believed technology (Proposition 1(iii)), the period's compute-weighted share is 0.19 to 0.29 rather than 0.04 (0.43 to 0.57 with Kaplan's curvature as well).

Public data cannot validate levels (Online Appendices E6 and E10). The level, a marginal value of compactness of about 2.8 percent of training cost per percent of size for the median decision, remains conditional on a zero point that no public

design observes at the relevant compute.

V. Economic Implications

Three economic questions turn on the objects of Sections III and IV: how fast frontier training’s demand for data grows, what a limit on unique text costs, and how the returns to compute should enter models of growth. The answers load on different objects, and the object on which the designs agree most closely matters least. Compute-optimal data demand depends on the allocation exponent a and the compute-optimal ratio $M^*(C)$, not on σ^* ; the cost of a data wall depends mainly on how fast repeated data lose value. The curvature sets the scale of the revealed value of compactness, and it converts that value into data: at given compute $D/D^*(C) = [M/M^*(C)]^{1/2}$, so by equation (7) a model with wedge w processes

$$(11) \quad \frac{D}{D^*(C)} = w^{\sigma^*/[2(1-\sigma^*)]}$$

times the compute-optimal tokens.

A. The Growth of Data Demand

Along the expansion path, compute-optimal data grow as $D^*(C) \propto C^{1-a}$. Frontier training compute grew 5.06-fold a year from 2018 to 2026 on the current Epoch AI database (95 percent confidence interval 4.27 to 6.02), faster than the 4.2-fold rate that Sevilla and Roldán (2024) published for 2018 to May 2024; the difference is a vintage effect, and we report both rates.¹ Across technologies, a runs from 0.36 (Farsee’s own form, beyond its design) to 0.57 (the RefinedWeb sweep of Gadre et al. 2025), so compute-optimal data demand grows 2.0-fold to 2.8-fold a year at 5.06-fold compute growth and 1.9-fold to 2.5-fold at the published rate (Table 3, panel A). The top of these ranges rests on two imprecise slopes, Farsee’s and the OLMo ladder’s; the model-free IsoFLOP paths of Section III give 2.3-fold to 2.9-fold. Levels are not portable either: at the fitted frontier compute of September 2026, 9.4×10^{26} FLOP, compute-optimal data range from 9 to 519 trillion tokens.

We therefore anchor each technology at the observed data use of the four disclosed dense training runs above 10^{25} FLOP, whose median lies close to the expansion path ($w = 1.21$ under the reference). Frontier runs then reach the quality-adjusted stock of public text, about 100 trillion tokens (Villalobos et al., 2024), between 2026.6 and 2027.8 across technologies at 5.06-fold compute growth, and

¹We regress $\log_{10} C$ on release date for the running top ten of training compute at release in Epoch AI (2026) (92 runs by 27 developers), with a wild cluster bootstrap- t interval by developer (Webb, 2023). On Epoch’s public file of May 31, 2024 (Epoch AI, 2024), the same estimator gives 4.13-fold for January 2018 to May 2024, against 5.23-fold on the current file for the same window; revised compute estimates explain 64 percent of that gap and entries added later the rest (Online Appendix Table F1).

the repetition-adjusted stock of 320 trillion tokens between 2027.8 and 2029.6 (2028.0 and 2030.0 at the published rate). The stock’s size matters more than α : at its lower bound, today’s frontier runs already process 2.2 to 4.9 times the stock.

Rising wedges. These forecasts hold the frontier’s data multiple D/D^* fixed, but the wedges revealed by open-weight releases rose. Under the reference technology, a regression of $\ln w$ on release date over the 49 decision units gives growth of 1.86-fold a year from 2023 to 2025 (95 percent confidence interval 0.96 to 3.54). If the frontier adopts this trend, a scenario across populations since the trend comes mostly from small models, equation (11) makes its data multiple grow 2.06-fold a year at $\sigma^* = 0.70$, and frontier data demand 4.6-fold a year rather than 2.2-fold (Table 3, panel B). At the point estimate the wedge trend adds about as much to the growth of data demand as compute growth does, but its interval includes no growth, as do those of the trends within developers, above 10^{24} FLOP and weighted by compute (1.59-fold, 1.76-fold and 2.21-fold a year). Extrapolating the trend in $M/M^*(C)$ instead needs no curvature and gives 1.97-fold to 2.38-fold a year for the data multiple over the 77 models under the 32 technologies (2.24 under the reference); the σ^* rows of panel B convert a projected value of compactness into data and are not a forecast interval. If the rise stops at the 2025 compute-weighted wedge of the decision units, 5.38, which rests largely on Alibaba’s Qwen3, five-year frontier data demand is 171 to 468 times today’s (114 to 215 without Alibaba), against 56 times with the wedge held.

B. The Data Wall

A data wall concerns unique tokens U , whereas D counts tokens processed, including repetitions. We combine the technology with the repetition model of Muennighoff et al. (2023), in which effective data are $D' = U + UR_D^*(1 - e^{-R/R_D^*})$ with $R = D/U - 1$ repetitions: the data-decay scale $R_D^* = 15.4$ alone is the most favorable case, their data-only fit has $R_D^* = 2.9$, and in their full model excess parameters also lose value. The cost of the wall is the increase in lifetime cost, training plus a value of compactness proportional to N that reproduces the developer’s wedge, at the loss of its unconstrained choice; for a compute-optimal run it depends on scarcity $r \equiv D^*(C)/U$ alone. At $r = 4$ it is 7.7, 7.4 and 7.2 percent for $\sigma^* = 0.60, 0.70$ and 0.74 if repeated data decay slowly, but 51, 40 and 35 percent under the data-only fit (Table 3, panel C). The repetition technology, not the curvature, drives the cost of the wall.

Over-training brings the wall forward. A model with the 2025 aggregate wedge processes about 7.2 times the compute-optimal tokens at given compute, and at $r = 4$ it pays 16 percent more lifetime cost rather than 7.4 (Online Appendix Table F3). Today’s frontier runs, at their observed data use, process 0.5 to 1.1 times the 100-trillion-token stock, so the wall costs them less than 0.1 percent. It binds within the forecast horizon: with the wedge held and 5.06-fold compute growth, frontier runs process 3.5 times the stock in 2029, and the wall costs

4.6 percent of lifetime cost if repeated data decay slowly (about \$2 billion at 10^{-18} per FLOP, the median amortized cost of 2024–2025 frontier runs; Epoch AI, 2026) and 23 percent under the data-only fit. With the wedge rising, they process 18 times the stock, at a cost of 10.4 or 48 percent. At the 2026 frontier, the shadow value of a unique token exceeds posted prices of generated tokens (DeepSeek, 2026) once compute-optimal data exceed 1.5 to 2.9 times the stock, a lower bound because synthetic tokens are imperfect substitutes (Online Appendix Table F4).

Feedback on measured wedges. A binding cap on tokens adds a term to the denominator of the wedge (Proposition 1(iii)), so measured wedges understate the value of compactness as the wall approaches. When a developer with the 2025 aggregate wedge would process four times the unique stock, its measured wedge falls short of the wedge it would reveal without the wall by 4 percent if repeated data decay slowly, by 15 percent under the data-only fit and by 53 percent under a hard cap (Online Appendix Table F3, panel B). As releases approach the stock, measured wedges should therefore flatten even if the value of compactness keeps rising, so a flattening among 2026 and 2027 releases would not by itself show that compactness has become less valuable (except under the full repetition model; Online Appendix F).

C. Growth Models

For models of AI and growth that map compute into capability, the relevant elasticity is γ , the compute elasticity of reducible loss along the frontier; near the expansion path compute is a sufficient statistic for the input bundle, and σ^* enters only when the input mix is constrained, as behind a data wall. For calibrations at 10^{25} to 10^{27} FLOP we recommend the reference value, $\gamma = 0.165$ (standard error 0.006). Halving reducible loss then takes 66 times the compute (95 percent confidence interval 50 to 90), or 2.6 years of frontier compute growth at 5.06-fold a year and 2.9 years at 4.2-fold. Because γ is cardinal in reducible loss, it needs the irreducible loss E , and the Chinchilla design (Hoffmann et al., 2022) alone has known loss units (it defines its loss as per-token log loss on MassiveText, that is, nats per token) and identifies E twice, from off-path runs and from its IsoFLOP minima (Online Appendix F); unlike σ^* , its local γ does not drift with compute. Other fits, of Chinchilla with $\kappa = 1$ and of the Llama 3 and Farseeer designs, give 0.14 to 0.18, or 49 to 161 times the compute per halving (Online Appendix Table F4). The recommendation extrapolates three to five decades beyond the design, and loss is not capability: at 10^{26} FLOP reducible loss is about 4 percent of total loss in Chinchilla’s units, so a calibration must say which loss it maps into output. Integrated-assessment models such as GATE (Erdil et al., 2025) can use these numbers only on their input side.

VI. Conclusion

Optimization cuts both ways for measurement. Developers who minimize cost generate data that say little about the technology they optimize over (Proposition 2) and choices that say a good deal about what they optimize for (Proposition 1). Designed experiments break the circle. Wherever the cost of two inputs is known and experiments vary them along isocosts (with a correction when the cost is not multiplicative), as in agronomic factorial trials, three results carry over: a quantified version of the argument of Marschak and Andrews (1944) and Bond and Söderbom (2005); an estimator of the elasticity of substitution that needs no functional form; and a revealed-preference result, of use for the production approach to markups (De Loecker and Warzynski, 2012), that signs a first-order-condition wedge from the location of the optimum alone, where experiments locate it (Proposition 3).

Applied to language models, they deliver three findings. The elasticity of substitution between parameters and data is about 0.7 at 10^{19} to 3×10^{20} FLOP, where the designs overlap, and about 0.6 at the largest budgets of the two IsoFLOP designs that go beyond. Most dense open-weight models released since 2023 are trained beyond training-cost minimization, though not most training compute. And the over-training their developers reveal rose steeply from 2023 to 2025, weighted by compute under every technology we consider; its level depends on the technology. [TBD-m9: one sentence on what our experiment adds: σ^* by corpus and the FineWeb-Edu versus FineWeb neutrality verdict.] The object on which the designs agree most closely, σ^* , does not enter compute-optimal data demand and barely moves the cost of a data wall unless repeated data decay fast; it matters by scaling the revealed value of compactness. The least portable objects, the allocation exponent and the compute-optimal ratio at frontier scale, matter most; laboratories could resolve them by publishing bracketed IsoFLOP minima at frontier compute.

What economists should use. For substitution between FLOP-effective parameters and tokens, 0.68 at 10^{20} FLOP from the pooled meta-regression (95 percent interval 0.64 to 0.72), and 0.59 at 6×10^{20} to 3×10^{21} FLOP in Chinchilla and Llama 3. As a summary over the designs' budgets, one estimate per study gives 0.69 (0.64 to 0.73). Beyond the designs σ^* is partially identified: if it neither rises with compute nor falls faster than inside the designs, it lies between 0.42 and 0.70 at 10^{24} FLOP (at most 0.59 if the top-budget decline is real). For growth calibrations at 10^{25} to 10^{27} FLOP, $\gamma = 0.165$, or 66 times the compute per halving of reducible loss. For data demand, the range across technologies, not one path. For the value of compactness, its sign and its rise rather than its level: zero-point allowances lower the level, a curvature that keeps falling with compute raises it at fixed zero points, and binding caps on tokens make it a lower bound. The sign is identified beyond the designs for 78 percent of clean-sample models, given the designs' paths extrapolated with their sampling error, but for only 24 percent of their training compute.

Implications for scaling studies. Studies that aim to measure the technology should place runs off the expansion path, bracket the loss minimum at every budget and test rather than impose $\kappa = 1$. They should anneal learning-rate schedules, tune hyperparameters by size and token budget (Tissue, Wang and Wang, 2025; Bergsma et al., 2025; Lourie et al., 2026), and state their FLOP and parameter conventions.

Limitations. First, units: only unit-free objects compare across designs. Second, extrapolation: 76 of the 77 clean-sample models were trained above 10^{21} FLOP. Parametric forms understate the wedge at high tokens per parameter inside three recipes’ designs, but the direction beyond them is recipe-dependent and nothing is observed at frontier compute. [TBD-m9: whether our high- M runs confirm this direction.] A curvature that falls with compute also breaks the quasi-homotheticity the inversion assumes. Third, developers that do not disclose training tokens, most closed ones among them, are absent. Fourth, the wedge leaves out post-training compute, which moves the median share by about 0.02 at the largest disclosed amount, and the gap between loss and capability: over-trained models degrade more under quantization (Kumar et al., 2025). Fifth, the data-wall results rest on repetition estimates from models of at most nine billion parameters.

Designed training runs let production-function econometrics check what optimizing agents reveal, and what they conceal, against experimental variation.

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ONLINE APPENDIX

APPENDIX A. FORMAL RESULTS AND PROOFS

This appendix states and proves the formal results that the main text uses. All statements are exact unless an order of approximation is given. Lemma 1 of the main text combines Lemmas A1 and A2 with Corollary A1. The duality statements of Section I are Lemma A3. Proposition 1 is Proposition A3, with Proposition A4 for family token budgets and binding data caps, Lemmas A4 and A5, and Corollary A2. Proposition 3(i)–(iv) is Proposition A5(i)–(iv), and its part (v) is Proposition A6(ii)–(iv) and (vii)–(viii); Proposition 2(i) combines Proposition A1(i), (ii) and (iv), part (ii) is Proposition A1(iii), and part (iii) is Proposition A2; the rate statement is Remark A3. The statement of Section II for costs that are not multiplicative is Corollary A3. Every algebraic identity below was also verified symbolically, and every comparative-static, bias and information statement numerically, by brute-force optimization, simulation or direct computation of Fisher information; the replication package lists each check and its tolerance (`output/docs/verification.md`). Results on observational production functions and technical change over time are in the companion paper (Okar, 2026). Throughout, E denotes irreducible loss and $\mathbb{E}$ the expectation operator.

A1. Setup and Notation

Definition A1 (Technology). A training run uses N parameters and D training tokens (tokens processed, including repetitions) and attains a held-out cross-entropy

$$(A1) \quad L = E + e^{-\omega} \left[A(e^{\psi_N} N)^{-\alpha} + B(e^{\psi_D} D)^{-\beta} \right] e^v, \quad A, B, \alpha, \beta > 0.$$

Lower-case letters are logs: $n = \ln N$, $d = \ln D$, $c = \ln C$ with training compute $C = 6ND$, so $c = \ln 6 + n + d$; $M \equiv D/N$. Hicks-neutral productivity ω and factor-augmenting productivity (ψ_N, ψ_D) are known to the lab when it chooses inputs; the seed and evaluation noise v is not. Write $u \equiv Ae^{-\alpha(n+\psi_N)}$, $v \equiv Be^{-\beta(d+\psi_D)}$ and $R \equiv u + v$, so that $L - E = e^{-\omega+v}R$. The log output index is $y \equiv -\ln(L - E)$, and $F(n, d) \equiv -\ln(Ae^{-\alpha n} + Be^{-\beta d})$. The output elasticities are $\varepsilon_N \equiv -\partial \ln(L - E)/\partial n$ and $\varepsilon_D \equiv -\partial \ln(L - E)/\partial d$, and the wedge is $w \equiv \varepsilon_N/\varepsilon_D$. Expansion-path objects are $a \equiv \beta/(\alpha + \beta)$ (so that $1 - a = \alpha/(\alpha + \beta)$), $\gamma \equiv \alpha\beta/(\alpha + \beta)$, $G \equiv (\alpha A/\beta B)^{1/(\alpha+\beta)}$, $K \equiv \frac{\alpha+\beta}{\beta} AG^{-\alpha}$ and $\sigma^* \equiv 2/(2 + \alpha + \beta)$. Finally $\chi \equiv \beta\psi_D - \alpha\psi_N$ is the bias index of productivity and $\Omega \equiv \omega + \gamma(\psi_N + \psi_D)$ is productivity in compute-equivalent units.

Remark A1 (Exact log form). Taking logs of (A1) gives, exactly and for any known E , $y = \omega + F(n + \psi_N, d + \psi_D) - v$, $\varepsilon_N = \alpha u/(u + v)$ and $\varepsilon_D = \beta v/(u + v)$. None of the three depends on ω , v or E ; the elasticities use the ω -free sum $u + v$, not the observed $L - E$.

Definition A2 (Generalized family). The generalized family replaces the bracket in (A1) by $[Ae^{-a_1(n+\psi_N)} + Be^{-b_1(d+\psi_D)}]^\kappa$ with $a_1, b_1, \kappa > 0$. Chinchilla is $\kappa = 1$ ($a_1 = \alpha, b_1 = \beta$). The joint law of Kaplan et al. (2020, eq. 1.5), fitted to early-stopped runs with D unique tokens and non-embedding N , is the member with $E = 0$, $a_1 = \alpha_N/\alpha_D \approx 0.738$, $b_1 = 1$ and $\kappa = \alpha_D \approx 0.103$. For this family write $Q \equiv u + v$ with $u = Ae^{-a_1n}$ and $v = Be^{-b_1d}$, $S \equiv a_1 + b_1$, $a \equiv b_1/S$ (so $1 - a = a_1/S$), $\gamma \equiv \kappa a_1 b_1 / S$, $G \equiv (a_1 A / b_1 B)^{1/S}$, $K \equiv (\frac{S}{b_1} A G^{-a_1})^\kappa$ and $\sigma^* \equiv 2/(2 + S)$; with $\kappa = 1$ these coincide with Definition A1. Because $R = Q^\kappa$ and the path does not depend on κ , the outer exponent leaves the expansion path unchanged, multiplies the inner frontier elasticity $a_1 b_1 / S$ by κ and raises the inner frontier constant to the power κ (Lemma A3(ii)).

A2. The Elasticity of Substitution and the Second-Order Condition

Lemma A1 (Elasticity of substitution; Lemma 1). *For the technology (A1), and for the generalized family with (a_1, b_1) in place of (α, β) :*

- (i) *The Hicks elasticity of substitution between N and D is $\sigma = (\alpha u + \beta v)/[\alpha u(1 + \beta) + \beta v(1 + \alpha)]$.*
- (ii) *σ depends on (N, D) only through the wedge,*

$$\sigma(w) = \frac{1 + w}{1 + \alpha + w(1 + \beta)}, \quad \text{equivalently} \quad \frac{1}{\sigma} = 1 + \frac{\alpha + \beta w}{1 + w},$$

which is monotone in w between $1/(1 + \alpha)$ ($w \rightarrow 0$) and $1/(1 + \beta)$ ($w \rightarrow \infty$).

- (iii) *On the compute-optimal expansion path ($w = 1$), $\sigma = \sigma^* = 2/(2 + \alpha + \beta)$.*
- (iv) *σ is invariant to any strictly increasing transformation of output; in particular it does not depend on E, ω, v or κ , or on whether output is measured by loss, bits per byte or a monotone benchmark link.*
- (v) *For the Kaplan et al. (2020) law, $\sigma \in [1/2, 1/1.738] = [0.500, 0.575]$ and $\sigma^* = 2/3.738 = 0.535$.*

With two inputs the Hicks, Allen–Uzawa and Morishima elasticities coincide (Blackorby and Russell, 1989).

Proof. (i) With output $f(N, D) = -(AN^{-\alpha} + BD^{-\beta})$ in levels, $f_N = \alpha u/N$, $f_D = \beta v/D$, $f_{NN} = -\alpha(1 + \alpha)u/N^2$, $f_{DD} = -\beta(1 + \beta)v/D^2$ and $f_{ND} = 0$. The Hicks elasticity for two inputs, $\sigma = -f_N f_D (N f_N + D f_D) / \{N D (f_{NN} f_D^2 - 2 f_{ND} f_N f_D + f_{DD} f_N^2)\}$, then equals the stated expression. (ii) Substitute $\alpha u = w \beta v$; the derivative of $\sigma(w)$ has the sign of $\alpha - \beta$. (iii) is (ii) at $w = 1$. (iv) For g strictly increasing, the first derivatives of $g \circ f$ are g' times those of f , and the second derivatives add terms proportional to $g'' f_i f_j$; in the denominator these sum to $g'' g'^2 (f_N^2 f_D^2 - 2 f_N^2 f_D^2 + f_D^2 f_N^2) = 0$, so numerator and denominator both scale by g'^3 . (v) Apply (ii) with $(a_1, b_1) = (0.738, 1)$. $\square$

Lemma A2 (Log coordinates and the second-order condition; Lemma 1). *Let $y = F(n, d)$ be twice continuously differentiable with $F_n, F_d > 0$, and define $\mathcal{P} \equiv F_n F_d (F_n + F_d) > 0$ and $\mathcal{Q} \equiv -(F_{nn} F_d^2 - 2F_{nd} F_n F_d + F_{dd} F_n^2)$. Then*

- (i) $\sigma = \mathcal{P}/(\mathcal{P} + \mathcal{Q})$; hence $\sigma \in (0, 1)$ if and only if $\mathcal{Q} > 0$, that is, if and only if the isoquant is strictly convex in $(\ln N, \ln D)$ coordinates.
- (ii) Consider $\max y$ subject to $n + d = c - \ln 6$ (equivalently $\min 6ND$ subject to $y \geq \bar{y}$). At a point with $F_n = F_d$, the second derivative of y along the isocost is $F_{nn} - 2F_{nd} + F_{dd} = -\mathcal{Q}/F_n^2$. The point is a strict local maximum of output (minimum of compute) if $\sigma \in (0, 1)$, and a strict local minimum of output if $\mathcal{Q} < 0$, which requires $\sigma > 1$ or $\sigma < 0$.

Proof. (i) With $f(N, D) = F(\ln N, \ln D)$: $f_N = F_n/N$, $f_D = F_d/D$, $f_{NN} = (F_{nn} - F_n)/N^2$, $f_{DD} = (F_{dd} - F_d)/D^2$ and $f_{ND} = F_{nd}/(ND)$. Substituting into the Hicks formula, the powers of N and D cancel and $\sigma = -F_n F_d (F_n + F_d) / [(F_{nn} - F_n) F_d^2 - 2F_{nd} F_n F_d + (F_{dd} - F_d) F_n^2] = \mathcal{P}/(\mathcal{P} + \mathcal{Q})$. The curvature of the log-isoquant $d = \phi(n)$ is $\phi'' = \mathcal{Q}/F_d^3$, so $\mathcal{Q} > 0$ is strict convexity. (ii) Along $(n, d) = (n_0 - t, d_0 + t)$, $d^2 y / dt^2 = F_{nn} - 2F_{nd} + F_{dd}$; at $F_n = F_d$ the bracket in $\mathcal{Q}$ equals $F_n^2 (F_{nn} - 2F_{nd} + F_{dd})$. If $\mathcal{Q} < 0$, then $\sigma = \mathcal{P}/(\mathcal{P} + \mathcal{Q})$ lies outside $[0, 1]$. $\square$

Corollary A1 (Complementarity, $\sigma < 1$; Lemma 1). *Under the multiplicative cost $C = 6ND$, every interior compute-optimal allocation that satisfies the second-order condition strictly has $\sigma < 1$ at the optimum. If $\sigma > 1$ everywhere, compute-optimal training puts all compute on one input, subject to bounds: for the CES technology $Y = (N^r + D^r)^{1/r}$ with $r \in (0, 1)$, output on the isocost $ND = X$ is $\sqrt{X} (2 \cosh(rt))^{1/r}$ with $t = \frac{1}{2} \ln(D/N)$, which is minimized, not maximized, at the symmetric point. Every member of the generalized family has $\sigma < 1$ everywhere.*

Proof. The first two statements are Lemma A2(ii); the CES computation is direct. For the generalized family, Lemma A1(ii) gives $\sigma \leq 1/(1 + \min\{a_1, b_1\}) < 1$. $\square$

Remark A2. With a linear cost the second-order condition requires only $\sigma > 0$; with the multiplicative cost the isocost is a line in logs, so the isoquant must be convex in logs, $\sigma < 1$; for any known cost the condition is $\sigma < \sigma_C$, the elasticity of substitution of the isocost itself (Corollary A3). The empirical content of Corollary A1 is that IsoFLOP profiles are U-shaped in $\ln N$; Proposition A5 turns the curvature of the U into the level of σ^* .

A3. Duality and Allocation

Lemma A3 (Cost function and conditional factor demands; the duality statements of Section I). *Consider the problem of minimizing $6ND$ subject to $L \leq \bar{L}$, or equivalently of maximizing y at given c , for the technology (A1).*

- (i) The first-order condition is $\alpha u = \beta v$ ($\varepsilon_N = \varepsilon_D$), and it is sufficient because $\sigma < 1$ everywhere.
- (ii) With $\psi_N = \psi_D = 0$: $N^* = G(C/6)^a$, $D^* = G^{-1}(C/6)^{1-a}$ and $R^*(C) = K(C/6)^{-\gamma}$ with $K = \frac{\alpha+\beta}{\beta}AG^{-\alpha} = (\alpha+\beta)(A/\beta)^a(B/\alpha)^{1-a}$. The cost function is $C^*(R) = 6(K/R)^{1/\gamma}$, with $\partial \ln C^*/\partial \ln R = -1/\gamma$; along the path $\varepsilon_N = \varepsilon_D = \gamma$. For the generalized family the same statements hold with (a_1, b_1) in place of (α, β) and with γ and K of Definition A2: $R^*(C) = K(C/6)^{-\gamma}$.
- (iii) In general,

$$\begin{aligned} n^* &= \ln G + a(c - \ln 6) + \chi/(\alpha + \beta), & y^* &= \Omega + \gamma(c - \ln 6) - \ln K, \\ d^* &= -\ln G + (1 - a)(c - \ln 6) - \chi/(\alpha + \beta), \\ \ln M^*(C) &= -2\ln G + (1 - 2a)(c - \ln 6) - 2\chi/(\alpha + \beta). \end{aligned}$$

- (iv) At given compute, ω does not enter (n^*, d^*) ; $\partial n^*/\partial \psi_D = a > 0$, $\partial d^*/\partial \psi_D = -a < 0$ and $\partial \ln M^*/\partial \psi_D = -2a < 0$: better data raise parameters and lower both tokens and tokens per parameter. Only the bias index χ moves the mix; productivity with $\chi = 0$ is equivalent to a Hicks-neutral shift $\Delta\omega = \alpha\psi_N$.

Proof. (i) The Lagrangian of $\max\{-\ln(u+v)\}$ subject to $n+d = c - \ln 6$ gives $\alpha u = \beta v$. The upper contour sets of y are convex in (n, d) (log-sum-exp is convex) and the constraint is linear. (ii) Substituting $n = \ln G + a \ln(C/6)$ and $d = -\ln G + (1-a) \ln(C/6)$ gives $\alpha u/(\beta v) = (\alpha A/\beta B)G^{-(\alpha+\beta)}(C/6)^{-\alpha a + \beta(1-a)} = 1$, because $\alpha a = \beta(1-a) = \gamma$. At the optimum $v = \alpha u/\beta$, so $R^* = u(\alpha + \beta)/\beta$; inverting $R^*(C)$ gives $C^*(R)$, and on the path $\varepsilon_N = \alpha u/R^* = \gamma$. For the generalized family, apply this to the inner aggregate Q , whose path does not involve κ , and raise its frontier to the power κ . (iii) Effective inputs $n + \psi_N$ and $d + \psi_D$ enter R exactly as (n, d) do when $\psi = 0$ and satisfy $(n + \psi_N) + (d + \psi_D) = c - \ln 6 + \psi_N + \psi_D$. Apply (ii) with effective compute, subtract ψ_N and ψ_D , and use $a\psi_D - (1-a)\psi_N = \chi/(\alpha + \beta)$. (iv) Differentiate (iii). If $\alpha\psi_N = \beta\psi_D$ then $u + v = e^{-\alpha\psi_N}(Ae^{-\alpha n} + Be^{-\beta d})$. $\square$

Part (ii) links the frontier $R^*(C)$ (a cost function), the demands $N^*(C), D^*(C)$ and the primal $L(N, D)$: Hoffmann et al. (2022)'s Approaches 1 and 2 estimate the first two, and Approach 3 the primal.

A4. Geometry and the Wedge as a Sufficient Statistic

Lemma A4 (Geometry of the technology; used in Propositions 1, 2 and 3). *For the generalized family (in effective inputs):*

- (i) $\nabla_{(n,d)}^2 \ln R = \kappa \frac{uv}{Q^2} \begin{pmatrix} a_1^2 & -a_1 b_1 \\ -a_1 b_1 & b_1^2 \end{pmatrix}$ has rank one, and its null vector (b_1, a_1) is the direction of the expansion path, $(\partial n^*/\partial c, \partial d^*/\partial c) = (a, 1-a)$.
- (ii) With the transverse coordinate $z \equiv a_1 n - b_1 d$, $\ln R = \kappa [-\frac{1}{2}(a_1 n + b_1 d) + g(z)]$ with $g(z) = \ln(Ae^{-z/2} + Be^{z/2})$ and $g''(z) = s_u(1-s_u) \in (0, \frac{1}{4}]$, $s_u = u/Q$. Thus $\ln R$ is affine along every line with direction (b_1, a_1) , and w and σ are constant along such lines.
- (iii) $\ln w = z^* - z$ with $z^* = \ln(a_1 A/b_1 B) = S \ln G$; the expansion path is the level set $\{z = z^*\}$; and for any run with compute C and ratio M ,

$$(A2) \quad \ln w = \frac{S}{2} \ln(M/M^*(C)) = \left(\frac{1}{\sigma^*} - 1\right) \ln(M/M^*(C)),$$

where $M^*(C) = G^{-2}(C/6)^{1-2a}$ is the training-optimal ratio at the run's own compute.

- (iv) (Quasi-homotheticity.) $R(\lambda^{b_1} N, \lambda^{a_1} D) = \lambda^{-\kappa a_1 b_1} R(N, D)$ for every $\lambda > 0$. More generally, for any C^2 technology, translation along (v_n, v_d) maps isoquants onto isoquants if and only if $\ln w$ is constant along it; then, with a unique optimum at each compute, the expansion path is a straight line in logs with slope $a = v_n/(v_n + v_d)$, and all information about σ is transverse to the path.

Proof. (i) $\partial \ln Q / \partial n = -a_1 u / Q$, so $\partial^2 \ln Q / \partial n^2 = a_1^2 uv / Q^2$; the other entries follow in the same way, and $\ln R = \kappa \ln Q$. The determinant is zero and (b_1, a_1) is annihilated. (ii) Since $-\frac{1}{2}(a_1 n + b_1 d) \mp \frac{1}{2}z$ equals $-a_1 n$ and $-b_1 d$ respectively, $Q = e^{-(a_1 n + b_1 d)/2} (Ae^{-z/2} + Be^{z/2})$; along $(n_0 + b_1 t, d_0 + a_1 t)$, z is constant and $a_1 n + b_1 d$ is affine in t . (iii) $w = a_1 u / (b_1 v) = (a_1 A / b_1 B) e^{-z}$. At given compute write $n = n^* - x/2$ and $d = d^* + x/2$ with $x = \ln(M/M^*(C))$; then $z - z^* = -Sx/2$. (iv) The first statement is direct. The log-isoquant has slope $-F_n/F_d$, so translates of isoquants are isoquants if and only if F_n/F_d is constant along the translation; with $w = 1$ on the whole line through an optimum, which crosses every isocost once, that line contains the optimum at every compute. $\square$

Lemma A5 (Allocative loss is a function of the wedge; used in Propositions 1 and 2). *Let $C_{\min}(R)$ denote the least training compute $6ND$ that attains reducible loss R . For any run with wedge w ,*

$$(A3) \quad \frac{C}{C_{\min}(R)} = \left(\frac{a_1 + b_1 w}{a_1 + b_1}\right)^{S/(a_1 b_1)} w^{-1/a_1} \stackrel{\kappa=1}{=} \left(\frac{\alpha + \beta w}{\alpha + \beta}\right)^{1/\gamma} w^{-1/\alpha},$$

which depends neither on (A, B, E, κ) nor on productivity. To second order, $\ln(C/C_{\min}) = (\ln w)^2 / (2S) + O(|\ln w|^3) = \sigma^*(\ln w)^2 / [4(1 - \sigma^*)] + O(|\ln w|^3)$: the allocative loss is a Harberger triangle whose area is governed by the elasticity of substitution.

Proof. Let $Q_0 = u + v$ at the run. The cheapest bundle producing Q_0 has $a_1 u^* = b_1 v^*$, so $u^* = aQ_0$ and $v^* = (1 - a)Q_0$. Because $u \propto N^{-a_1}$ and $v \propto D^{-b_1}$, $C/C_{\min} = (u^*/u)^{1/a_1} (v^*/v)^{1/b_1}$ with $u/Q_0 = b_1 w / (a_1 + b_1 w)$, which gives (A3). For the expansion, let $x = \ln w$: $\ln((a_1 + b_1 e^x)/S) = ax + \frac{1}{2}a(1 - a)x^2 + O(x^3)$; multiplying by $S/(a_1 b_1) = 1/[a(1 - a)S]$ and subtracting $x/a_1 = x/[(1 - a)S]$ leaves $x^2/(2S)$. Finally $S = 2(1 - \sigma^*)/\sigma^*$. $\square$

A5. What On-Path Data Identify

Definition A3 (On-path sample). A sample of runs is on-path if (a) all runs share the technology (generalized family, common parameters); (b) productivity is Hicks-neutral ($\psi \equiv 0$); (c) every run minimizes training compute for its loss (in the notation of Proposition A3, $\delta = m_D = v_N = 0$ and no constraint binds, so that $w = 1$); and (d) compute varies over an interval (or at least five distinct values).

Proposition A1 (What on-path data identify; Proposition 2). *In an on-path sample:*

- (i) $n_i = \ln G + a(c_i - \ln 6)$ and $d_i = -\ln G + (1 - a)(c_i - \ln 6)$. The inputs are deterministic affine functions of compute, and the choice data reveal a and G (the observed compute-optimal ratio $M^*(C)$).
- (ii) $\mathbb{E}[y \mid c] = \gamma(c - \ln 6) - \ln K + \mathbb{E}[\omega \mid c]$. Without instruments for compute or panel restrictions on ω , the frontier elasticity γ is identified if and only if $\mathbb{E}[\omega \mid c]$ is known up to an additive constant, as in a single-lab sweep.
- (iii) Suppose in addition $\mathbb{E}[\omega \mid c]$ is constant and $\kappa = 1$. Write $c' \equiv c - \ln 6$, $r_N \equiv \alpha a$ and $r_D \equiv \beta(1 - a)$ for the rates at which the two terms decay along the observed path, and $U_0 \equiv Ae^{-\alpha n_0}$, $V_0 \equiv Be^{-\beta d_0}$ (with K normalized by $\mathbb{E}[e^{-\omega + v}] = 1$), so that on the path $\mathbb{E}[L \mid c] = E + U_0 e^{-r_N c'} + V_0 e^{-r_D c'}$, a two-rate exponential mixture whose rates coincide at the truth.
 - (a) (Global identification, by functional form.) E , γ and K are identified, and so are $\alpha = \gamma/a$ and $\beta = \gamma/(1 - a)$: every parameter with $A, B > 0$ that generates the same on-path outcomes has $r_N = r_D = \gamma$. The ratio A/B is identified only by imposing that the observed path is optimal ($G^{\alpha+\beta} = \alpha A/\beta B$); outcomes identify only $U_0 + V_0$.
 - (b) (No local identification.) The Jacobian of the on-path mean of $\ln L$ with respect to $(E, \ln A, \ln B, \alpha, \beta)$ has rank three. Because A/B is not identified, every perturbation that moves r_N and r_D in opposite directions leaves the on-path mean unchanged to first order (the change is absorbed by re-splitting $U_0 + V_0$; at a given split only one such direction is flat), so along each such direction the least-squares or likelihood criterion is of fourth order in the step, and σ^* is identified only at second order whether or not $\alpha = \beta$. The twin map $(\alpha, \beta, A, B) \mapsto$

$(\beta(1-a)/a, \alpha a/(1-a), Be^{-\beta d_0 + \beta(1-a)n_0/a}, Ae^{-\alpha n_0 + \alpha a d_0/(1-a)})$ leaves the on-path outcome distribution exactly unchanged, so every parameter value off the ray $\alpha a = \beta(1-a)$ has an observationally equivalent twin whose σ^* differs unless $a = \frac{1}{2}$. On the boundary of the parameter space identification fails altogether: as $A \rightarrow 0$ ($B \rightarrow 0$) the on-path distribution converges to the true one for any α (β).

(iv) Without the restriction $\kappa = 1$: for every $S > 0$, the member

$$(A4) \quad \begin{aligned} a_1 &= (1-a)S, & b_1 &= aS, & \kappa_S &= \frac{\gamma}{a(1-a)S}, \\ A_S &= \frac{b_1}{S} K^{1/\kappa_S} G^{a_1}, & B_S &= \frac{a_1 A_S}{b_1 G^S} \end{aligned}$$

generates exactly the same joint distribution of (n, d, y) as the true technology. Hence $\sigma^* = 2/(2+S)$ is not identified: every value in $(0, 1)$ is consistent with the data, whereas by Lemma A2 no value $\sigma^* > 1$ is (in the generalized family, none with $\sigma^* \geq 1$).

Proof. (i) is Lemma A3(ii). (ii) On the path $y = \omega + \gamma(c - \ln 6) - \ln K - v$; if $\mathbb{E}[\omega | c] = h(c)$ with h unknown, then $\gamma(c - \ln 6) + h(c)$ and $\tilde{\gamma}(c - \ln 6) + h(c) + (\gamma - \tilde{\gamma})(c - \ln 6)$ are indistinguishable. (iii)(a) With ω independent of c , $\mathbb{E}[L | c] = E + \tilde{K}e^{-\gamma c'}$; a three-parameter exponential is identified from an interval of c . If $(\tilde{\alpha}, \tilde{\beta}, \tilde{A}, \tilde{B})$ with $\tilde{A}, \tilde{B} > 0$ fits the same curve on the path $n = n_0 + ac'$, $d = d_0 + (1-a)c'$, then $\tilde{U}_0 e^{-\tilde{\alpha} ac'} + \tilde{V}_0 e^{-\tilde{\beta}(1-a)c'} = \tilde{K}e^{-\gamma c'}$ on an interval; exponentials with distinct rates are linearly independent, so $\tilde{\alpha}a = \tilde{\beta}(1-a) = \gamma$. The remaining condition, $\tilde{U}_0 + \tilde{V}_0 = \tilde{K}$, is one equation in two unknowns. (b) On the path $u = u_0 e^{-\gamma c'}$ and $v = v_0 e^{-\gamma c'}$. The scores of $\ln L$ are $1/L$ times: 1 for E ; u and v for $\ln A$ and $\ln B$; $-(n_0 + ac')u$ for α ; and $-(d_0 + (1-a)c')v$ for β . They span $\{1, e^{-\gamma c'}, c'e^{-\gamma c'}\}$, so the rank is three. Every (U_0, V_0) with $U_0 + V_0 = U_0^\circ + V_0^\circ$ (the true values) and equal rates generates the true distribution, and at such a point the first-order change of the mean from $(\delta r_N, \delta r_D)$ is $-c'e^{-\gamma c'}(U_0 \delta r_N + V_0 \delta r_D)$, which vanishes for $U_0/V_0 = -\delta r_D/\delta r_N > 0$ whenever $\delta r_N \delta r_D < 0$; at a given split U_0/V_0 this holds for one direction only. The second-order term, $\frac{1}{2}c'^2 e^{-\gamma c'}(U_0 \delta r_N^2 + V_0 \delta r_D^2)$, is not in the span of $\{1, e^{-\gamma c'}, c'e^{-\gamma c'}\}$, so the criterion is exactly of fourth order along such directions. Since $\delta(\alpha + \beta) = \delta r_N/a + \delta r_D/(1-a)$, generic opposite-sign directions move σ^* at first order. The twin claim follows by substitution: the twin's two terms are $V_0 e^{-r_D c'}$ and $U_0 e^{-r_N c'}$ with the roles of the inputs exchanged. (iv) Apply Lemma A3(ii) to the inner aggregator of member S : its path slope is $b_1/S = a$, its path intercept is G by construction of B_S , and its inner frontier is $K^{1/\kappa_S}(C/6)^{-a_1 b_1/S}$ by construction of A_S . Hence $R^* = K(C/6)^{-\kappa_S a_1 b_1/S} = K(C/6)^{-\gamma}$, and inputs and outputs coincide run by run for every (ω, v) . $\square$

Remark A3 (Rates and inference on the path). The on-path model of Proposition A1(iii) is an over-fitted mixture: writing its mean as $E + (U_0 +$

$V_0) \int e^{-rc'} d\mathcal{G}(r)$, with $\mathcal{G}$ the two-point distribution on $\{r_N, r_D\}$, the data identify the mean of $\mathcal{G}$ through the coefficient on $c'e^{-\gamma c'}$ and its second moment about γ through the coefficient on $c'^2 e^{-\gamma c'}$, each at best at rate $n^{-1/2}$, so the spread $|r_N - r_D|$, and with it σ^* , at best at rate $n^{-1/4}$, as in the bound for over-fitted mixtures of [Chen \(1995\)](#). The bound concerns estimators that leave $\beta/(\alpha + \beta)$ free, such as unrestricted least-squares fits of the Chinchilla form. Imposing $\beta/(\alpha + \beta) = a$, which the optimality that defines an on-path sample implies under $\kappa = 1$, forces $r_N = r_D$ and removes the singularity: $\sigma^* = 2/\{2 + \gamma/[a(1 - a)]\}$ combines $\hat{a}$ from the choices with $\hat{\gamma}$ from a regular three-parameter exponential and is estimated at the usual root- n rate. This is a heuristic argument, not a theorem for this model; the results of [Sargan \(1983\)](#) and [Rotnitzky et al. \(2000\)](#) on singular information do not apply directly because U_0/V_0 is not identified at the truth.

A Monte Carlo supports it. It holds the design fixed, 40 on-path compute levels spanning four decades, and lets the standard deviation of Gaussian noise on L fall from 10^{-3} to 10^{-7} in five steps, with 100 replications at each; for least squares this is equivalent to replicating the design n times, with n proportional to the inverse noise variance. The unrestricted fit is least squares over (E, U_0, V_0, r_N, r_D) , equivalently over $(E, \ln A, \ln B, \alpha, \beta)$, with U_0 and V_0 bounded below at 10 percent of the smaller true value, which excludes the boundary where an exponent is not identified; the error recorded is the larger of the errors of the two exact twins of Proposition [A1\(iii\)\(b\)](#). The Hoffmann et al. exponents ($\alpha \neq \beta$) and equal exponents give the same picture. The 75th and 90th percentiles of $|\hat{\sigma}^* - \sigma^*|$ scale as $n^{-0.24}$ and as $n^{-0.27}$ to $n^{-0.28}$, and as $n^{-0.50}$ when the restriction $\beta/(\alpha + \beta) = a$ is imposed; the 25th percentile scales as $n^{-1/2}$, because in about 40 percent of replications the fitted rates coincide. Wald and delta-method intervals based on the near-singular Hessian are then invalid ([Online Appendix C](#)).

Proposition A2 (Information rates; Proposition [2](#)). *Consider a sweep with $y_i = F(n_i, d_i; \phi) - v_i$, v_i i.i.d. $N(0, V_v)$, E known, and the generalized family parameterized by $\phi = (S, a, \gamma, \ln G, \ln K)$. For run i let w_i be its wedge, $\Delta_i \equiv (1 - a)n_i - ad_i - \ln G = -(\ln w_i)/S$ its transverse deviation ([Lemma A4](#)), and $\Phi_i \equiv \ln(C_i/C_{\min}(R_i))$ its allocative loss ([Lemma A5](#)).*

- (i) *(Exact decomposition.)* $y_i = \gamma(c_i - \ln 6) - \ln K - \gamma \Phi_S(\Delta_i) - v_i$, where $\Phi_S(\Delta) = [\ln(1 - a + ae^{-S\Delta}) + aS\Delta]/[a(1 - a)S] = \Phi_i \geq 0$ and $\Phi_S(0) = \Phi'_S(0) = 0$. The curvature parameter S , and hence σ^* , enters output only through the runs' allocative losses.
- (ii) *(Scores.)* $\partial y_i / \partial \ln G = \gamma(1 - w_i)/(1 - a + aw_i) = -\gamma \ln w_i + O((\ln w_i)^2)$ and $\partial y_i / \partial S = -(\gamma/S) \Phi_i [1 + O(|\ln w_i|)] = -\gamma(\ln w_i)^2/(2S^2) + O(|\ln w_i|^3)$.
- (iii) *(Efficient information.)* Let ζ_k be the vector of scores for parameter k and M_{-k} the residual-maker for the other four score vectors. Then $\mathcal{I}^*(\ln G) = (\gamma^2/V_v) \|M_{-G} \ln w\|^2 [1 + O(\max_i |\ln w_i|)]$ and

$\mathcal{I}^*(S) \leq (\gamma^2/(S^2 V_v)) \sum_i \Phi_i^2 [1 + O(\max_i |\ln w_i|)] = O(\sum_i (\ln w_i)^4)$. If every run lies on the expansion path, both informations are zero. If the transverse deviations of a fixed design are scaled by $\varsigma \rightarrow 0$, then $\mathcal{I}^*(\ln G) = O(\varsigma^2)$ and $\mathcal{I}^*(\sigma^*) = O(\varsigma^4)$, so the standard error of $\hat{\sigma}^*$ grows at least like ς^{-2} and that of $\ln \hat{G}$ at least like ς^{-1} . The same order holds for the path's location at any given compute C_0 , $\ln M^*(C_0) = -2 \ln G + (1 - 2a) \ln(C_0/6)$, whose efficient information combines ζ_G with the score for a ; both are $O(\varsigma)$, but the level of the information differs from $\mathcal{I}^*(\ln G)/4$. Both rates are attained, so that the standard errors grow exactly like ς^{-2} and ς^{-1} , by designs that cross several compute levels with symmetric transverse offsets, such as factorial designs.

(iv) If E is estimated and the noise is Gaussian on $\ln L$, the score for S is multiplied by $R_i/L_i \in (0, 1)$, so the bound in (iii) still holds.

Proof. (i) Because the frontier is a power law, $R = K(C_{\min}(R)/6)^{-\gamma}$ for every R , so $\ln R = \ln K - \gamma(c - \ln 6) + \gamma \ln(C/C_{\min}(R))$. By Lemmas A5 and A4(iii), $\ln(C/C_{\min}) = \ln((a_1 + b_1 w)/S)/[a(1 - a)S] - \ln w/a_1$ with $w = e^{-S\Delta}$, $a_1 = (1 - a)S$ and $b_1 = aS$, which is $\Phi_S(\Delta)$; it is nonnegative because $C \geq C_{\min}$, and $\Phi'_S(\Delta) = (1 - e^{-S\Delta})/(1 - a + ae^{-S\Delta})$ vanishes at zero. (ii) Holding (n, d) fixed, Δ depends on $\ln G$ (with derivative -1) but not on S , so $\partial y/\partial \ln G = \gamma \Phi'_S(\Delta)$. Write $\Phi_S = h(S\Delta)/[a(1 - a)S]$ with $h(x) = \ln(1 - a + ae^{-x}) + ax$; then $\partial \Phi_S/\partial S = [xh'(x) - h(x)]/[a(1 - a)S^2]$, and $h(x)$ and $xh'(x) - h(x)$ both equal $\frac{1}{2}a(1 - a)x^2 + O(x^3)$. (iii) Under Gaussian errors the efficient information for one parameter is the Schur complement $V_v^{-1} \zeta'_k M_{-k} \zeta_k$; the inequality uses $\|M_{-S} \zeta_S\| \leq \|\zeta_S\|$ and (ii), and the orders follow from $\zeta_G = O(\varsigma)$ and $\zeta_S = O(\varsigma^2)$. For a design that crosses several compute levels with symmetric transverse offsets, Δ_i^2 is not in the span of the leading-order nuisance scores $\{1, c_i - \ln 6 - \Phi_i, \Delta_i, \Delta_i(c_i - \ln 6)\}$ and Δ_i is not in the span of the others, which gives attainment; in a nine-budget, five-offset factorial design the numerical log-log slopes are 4.00 and 2.00, also with E estimated, and 2.00 for $\ln M^*(C_0)$ at a fixed C_0 . (iv) $\partial \ln L/\partial S = -(R/L) \partial y/\partial S$. $\square$

A6. What Over-Training Reveals: The Generalized Wedge

Proposition A3 (Generalized wedge; Proposition 1). *A developer chooses (N, D) to maximize*

$$(A5) \quad \begin{aligned} \Pi(N, D) &= V(L(N, D), N) - X_T(N, D) - q_D D \\ \text{subject to } g_k(n, d) &\leq 0, \quad k \in \mathcal{K}, \end{aligned}$$

where V is differentiable with $\partial V/\partial L < 0$; $X_T \equiv q_T(N) 6ND$ is training expenditure at the (shadow) price $q_T(N)$ of a training FLOP, whose size elasticity is $\delta \equiv d \ln q_T/d \ln N$; $q_D \geq 0$ is a cost per training token; L is given by (A1) or by any technology with $\varepsilon_N, \varepsilon_D > 0$; and the g_k are differentiable constraints.

Let $m_N \equiv -(\partial V / \partial \ln N)_L / X_T$ be the objective's own value of compactness at fixed quality per unit of training expenditure, $m_D \equiv q_D D / X_T$, and $\xi_k \geq 0$ the multipliers (value per log point). Let

$$\nu_N \equiv \sum_{k \in \mathcal{K}: g_{k,d}=0} \xi_k g_{k,n} / X_T, \quad v_N \equiv m_N + \nu_N :$$

ν_N is the shadow value of the constraints on size alone per unit of training expenditure (a cap has $g_{k,n} > 0$ and a minimum-size floor $g_{k,n} < 0$, so floors enter with a negative sign), and v_N is the value of compactness, a virtual price in the sense of [Neary and Roberts \(1980\)](#): the value of m_N at which a developer free of those constraints would make the same choice. Without binding constraints on size, $v_N = m_N$.

(i) At any interior Karush–Kuhn–Tucker point,

$$(A6) \quad w = \frac{1 + \delta + m_N + \sum_k \xi_k g_{k,n} / X_T}{1 + m_D + \sum_k \xi_k g_{k,d} / X_T}.$$

With no binding constraint that involves d , $w = (1 + \delta + v_N) / (1 + m_D)$; if also $\delta = m_D = 0$, $w - 1 = v_N$, and the compactness share $s \equiv (w - 1) / w$ is $v_N / (1 + v_N)$.

(ii) (Invariance and contamination.) w is computed from (N, D) and the technology alone; it is invariant to ω , v , E and κ . If the developer's first-order conditions involve a technology L' (its own productivity, a law it believes, or its own output concept) while the econometrician uses L , then $\hat{w} = w_{L'} \cdot w_L / w_{L'}$, where $w_{L'}$ is the wedge that (A6) determines at (N, D) under L' and w_L the one that L assigns to the same inputs. With factor-augmenting productivity $w_L / w_{L'} = e^{-\chi}$, so $\hat{w} = w_{L'} e^{-\chi}$.

(iii) (Special cases.) Let T be lifetime inference tokens in present value (discounted at the interest rate plus an obsolescence hazard), $q_S(N)$ the price of a serving FLOP, $X_S \equiv q_S 2NT$ serving expenditure and $\varrho \equiv q_S / q_T$ the price ratio.

(a) (Serving cost borne at rate λ .) If $V = R(L) - \lambda X_S$ with $T = T(L, N)$, $q_D = 0$ and no binding constraint, then $w = 1 + \delta + \lambda \varrho (\phi + \eta) T / (3D)$, where $\phi \equiv 1 + d \ln q_S / d \ln N$ and $\eta \equiv (\partial \ln T / \partial \ln N)_L$; a direct effect of size on revenue at fixed quality adds $-(\partial R / \partial \ln N)_L / X_T$. With $\delta = \eta = 0$ and $\phi = 1$: $w = 1 + \lambda \varrho T / (3D)$, $w - 1 = \lambda X_S / X_T$ and $s = \lambda X_S / (X_T + \lambda X_S)$; no conversion of FLOPs into dollars is needed. Compute proportional to N that is planned at training time, such as post-training, reinforcement-learning rollouts or synthetic-data generation, enters in the same way. Counting post-training expenditure X_P

as a cost alongside X_T (so that V excludes it), with $x_P \equiv X_P/X_T$ and $\delta = m_D = 0$, gives $w = 1 + x_P + m_N$, where m_N is net of post-training.

- (a') (Adoption.) If the developer bears no serving cost and $V = v(\mathcal{A})$, where adoption $\mathcal{A}(L, P_u)$ falls with users' serving cost per token $P_u \propto N$, then $m_N = v'(\mathcal{A})\mathcal{A}|_{\varepsilon_{\mathcal{A},P}}/X_T$: $w - 1$ is the adoption value of compactness, not a token count.
- (b) (Tier or memory cap.) If $N \leq \bar{N}$ binds, $w = (1 + \delta + v_N)/(1 + m_D)$ with $\nu_N = \xi/X_T \geq 0$; $w - 1$ identifies v_N and bounds m_N from above (when $\delta = m_D = 0$), but does not identify m_N : any $w \geq (1 + \delta + m_N)/(1 + m_D)$ is rationalizable, including with $m_N = 0$. A menu of sizes is this case at the chosen size, with ν_N of either sign (negative when the menu binds as a floor).
- (c) (Latency.) A binding service-level objective $Z(N) \leq \bar{Z}$, with elasticity $\ell \equiv d \ln Z / d \ln N > 0$, is case (b) with $\nu_N = \ell \xi_Z / X_T$. A latency cost inside V instead enters m_N .
- (d) (Data cost and data cap.) $w = (1 + \delta + v_N)/(1 + m_D)$, so a data cost lowers w for given v_N . A binding cap $D \leq \bar{D}$ adds $\mu \equiv \xi_D / X_T \geq 0$ to the denominator, $w = (1 + \delta + v_N)/(1 + m_D + \mu)$, so with $\delta = m_D = 0$, $w - 1$ bounds v_N from below; w can fall below one with $v_N = 0$.
- (e) (Logit distillation.) If distillation leaves the student's technology unchanged, teacher forward passes cost $2N_T D$ FLOPs at the training price, a per-token cost with $m_D = N_T/(3N)$: with $\delta = 0$ and $\phi = 1$, $w = (1 + \lambda q_T/(3D))/(1 + N_T/(3N))$. If it also raises the productivity of tokens ($\chi > 0$), the measured wedge falls further, by (ii).
- (f) (Utilization varying with size.) If the prices of training and serving FLOPs have size elasticities δ and δ_S ($\phi = 1 + \delta_S$), then $w = 1 + \delta + \lambda(1 + \delta_S)q_T/(3D)$, which equals $(1 + \delta)(1 + \lambda q_T/(3D))$ if and only if $\delta_S = \delta$.
- (iv) (Signs, on m_N .) Let $w_0 \equiv (1 + \delta + m_N)/(1 + m_D)$. A single binding constraint raises w above w_0 if and only if $g_n > w_0 g_d$. Relative to the objective's own m_N , memory or tier caps and latency objectives raise w (they raise v_N); data caps and deadlines that bind on tokens lower it; deadlines proportional to compute pull it toward one; minimum-size floors lower it. Hence $\hat{T} \equiv 3D(\hat{w} - 1)$ overstates λq_T under caps, latency objectives, $\delta > 0$ (with $\delta_S \geq 0$), post-training compute and $\chi < 0$, and understates it under data costs, data caps, distillation, minimum-size floors, $\delta < 0$ (with $\delta_S \leq 0$) and $\chi > 0$. A wedge below one cannot be generated by serving demand as long as serving expenditure rises with size at given quality ($\phi + \eta \geq 0$).

Proof. In (n, d) the Lagrangian is $\Pi - \sum_k \xi_k g_k$. Since $\partial L / \partial n = -(L - E)\varepsilon_N$ and $\partial L / \partial d = -(L - E)\varepsilon_D$, with $\Lambda \equiv -(\partial V / \partial L)(L - E) > 0$, $\partial X_T / \partial n = (1 + \delta)X_T$ and $\partial X_T / \partial d = X_T$, the first-order conditions are $\Lambda \varepsilon_N + (\partial V / \partial \ln N)_L -$

$(1 + \delta)X_T - \sum_k \xi_k g_{k,n} = 0$ and $\Lambda \varepsilon_D - X_T - q_D D - \sum_k \xi_k g_{k,d} = 0$; dividing gives (A6). Constraints with $g_{k,d} = 0$ enter only the numerator, where they add ν_N to m_N . (ii) Remark A1; in the generalized family $w = a_1 u / (b_1 v)$ does not involve κ . If the first-order conditions involve L' , (A6) determines $w_{L'}(N, D)$, and the econometrician computes $w_L(N, D) = w_{L'}(N, D) \cdot w_L / w_{L'}$; with factor-augmenting productivity $w_{L'} / w_L = e^{\beta \psi_D - \alpha \psi_N} = e^\chi$. (iii)(a) $-(\partial V / \partial \ln N)_L = \lambda X_S (\phi + \eta)$ because $\ln X_S = \ln q_S + \ln 2 + \ln N + \ln T$; divide by X_T and use $X_S / X_T = \varrho T / (3D)$. Post-training compute has elasticity one in N and zero in D . (a') $(\partial V / \partial \ln N)_L = v'(\mathcal{A}) \partial \mathcal{A} / \partial \ln P_u$. (b) $g = n - \bar{n}$, so $g_n = 1$ and $g_d = 0$, and $\nu_N = \xi / X_T$. A menu fixes n at the chosen size; with $\xi \equiv \partial \Pi / \partial n$ there, of either sign, the condition in n holds with $\nu_N = \xi / X_T$. (c) $g = \ln Z(n) - \ln \bar{Z}$, so $g_n = \ell$ and $g_d = 0$. (d) The cap is $g = d - \bar{d}$. (e) The teacher's cost $2N_T D q_T$ has elasticity zero in N and one in D , so $m_D = 2N_T D / (6ND)$; a rise in the productivity of tokens is $\chi > 0$ in (ii). (f) The elasticity of X_T in N is $1 + \delta$ and that of X_S is $1 + \delta_S$. (iv) With $A_0 \equiv 1 + \delta + m_N$ and $B_0 \equiv 1 + m_D$, $\partial_\xi [(A_0 + \xi g_n / X_T) / (B_0 + \xi g_d / X_T)]$ has the sign of $g_n B_0 - g_d A_0$ for every ξ . The first-order conditions are necessary, and the inversion uses nothing else; they are also sufficient when Π is concave in (n, d) , as in the lifetime-compute problem. Every case was checked by brute-force maximization of the objective, including binding size caps, floors, menus and latency objectives with and without a cap on tokens (`code/analysis/rb5_theory/verify_virtual_value.py`). $\square$

Remark A4 (Conduct and the parallel with markup estimation). Cases (a), (a') and (b) are three models of conduct: a developer that bears the cost of serving its own model; an open-weight developer that internalizes users' costs only through adoption; and size-class competition with N set by a menu of deployment tiers (compare the Leontief model of [Hao and Merrill, 2026](#), in which over-training does not arise). Under (a), $w - 1$ rises with the developer's own serving footprint; under (a'), with the sensitivity of adoption to users' serving cost; under (b), sizes bunch at tier boundaries and $w - 1$ is unrelated to demand given the tier. The wedge is the ratio statistic of [Raval \(2023\)](#) with FLOP-accounting weights, and part (ii) contains the two classic critiques of the production approach to markups, factor-augmenting heterogeneity ([Demirer, 2026](#)) and a mismatch between measured and valued output ([Bond et al., 2021](#)). Here the elasticities come from designed experiments rather than from proxy estimation on the same choices ([Doraszelski and Jaumandreu, 2021](#)). As in [De Ridder, Grassi and Morzenti \(2026\)](#), levels are fragile and ranks robust: at given compute the rank of $\hat{w}$ across models is the rank of M for every member of the generalized family.

Corollary A2 (Sufficient statistics; used in Proposition 1). *In the generalized family, for a model with compute C and ratio M at an interior optimum of Proposition A3 with $\delta = m_D = 0$ and no binding constraint on tokens,*

$$v_N = w - 1 = \left(\frac{M}{M^*(C)} \right)^{1/\sigma^* - 1} - 1, \quad s = 1 - \left(\frac{M}{M^*(C)} \right)^{-(1/\sigma^* - 1)},$$

and $C/C_{\min}$ is given by Lemma A5. $M^*(C)$ is the training-optimal ratio at the model's own compute. Under Proposition A3(iii)(a) with $\phi = 1$, $\eta = 0$ and no binding size constraint, $\lambda \varrho T/D = 3[(M/M^*(C))^{1/\sigma^*-1} - 1]$. In the homothetic case $\alpha = \beta = \rho$, M^* is constant and $C/C_{\min} = \cosh(\frac{\rho}{2} \ln(M/M^*))^{2/\rho}$. With unequal exponents, $\ln w = \frac{\alpha+\beta}{2} \ln(M/M^*(C))$ still holds, with the mean exponent $(\alpha + \beta)/2 = 1/\sigma^* - 1$ in place of ρ and $M^*(C)$ in place of M^* , but $C/C_{\min}$ is then given by Lemma A5: the cosh form holds only to second order in $\ln w$ (with the exponents of Hoffmann et al. (2022), it gives 78.0 against an exact 101.2 at $w = 13.5$). This justifies evaluating a technology's compute-optimal law at each model's own compute.

Proof. Combine Proposition A3(i), equation (A2) and Lemma A5. Both $\ln(C/C_{\min})$ from Lemma A5 and $\frac{2}{\rho} \ln \cosh(\frac{\rho}{2} \ln(M/M^*(C)))$ with $\rho = (\alpha + \beta)/2$ equal $(\ln w)^2/[2(\alpha + \beta)] + O(|\ln w|^3)$. $\square$

Proposition A4 (Family token budgets and binding data caps; Proposition 1). *A developer trains I models of sizes $N_1, \dots, N_I$ on a common token budget D and chooses D (and, where stated, the sizes) to maximize $\sum_i V_i(L(N_i, D), N_i) - \sum_i X_{T,i} - q_D D$ subject to $\underline{n}_i \leq n_i \leq \bar{n}_i$ and $d \leq \bar{d}$, where $X_{T,i} = q_T 6N_i D$ and the data cost is incurred once. Let w_i be member i 's wedge computed from the technology, $h_i \equiv N_i / \sum_j N_j$ its compute share, $s_i \equiv (w_i - 1)/w_i$, $m_{N,i} \equiv -(\partial V_i / \partial \ln N_i)_L / X_{T,i}$, $m_D \equiv q_D D / \sum_j X_{T,j}$, $\Lambda_i \equiv -(\partial V_i / \partial L)(L_i - E)$, and $\rho_i \equiv \Lambda_i \varepsilon_{D,i} / X_{T,i}$, the marginal value of data to member i per unit of its marginal training cost. Let ξ_i be the net multiplier on member i 's size constraints (zero if they are slack, $\xi_i \geq 0$ at a cap, $\xi_i \leq 0$ at a floor) and $v_{N,i} \equiv m_{N,i} + \xi_i / X_{T,i}$ its value of compactness (Proposition A3). Let $\bar{w}_H \equiv (\sum_i h_i / w_i)^{-1}$ be the compute-weighted harmonic mean of the member wedges and $\pi_i \equiv (h_i / w_i) / \sum_j (h_j / w_j)$.*

- (i) (Member wedges are technological.) Given D , $\ln w_i - \ln w_j = \ln w(n_i, d) - \ln w(n_j, d)$, which equals $-a_1(n_i - n_j)$ in the generalized family: the profile of $\hat{T}_i \equiv 3D(w_i - 1)$ across sizes is fixed by the technology.
- (ii) (Family first-order condition.) At an interior optimum in D , $\sum_i h_i \rho_i = 1 + m_D$. If the sizes are set by a menu, this is the only first-order condition: the w_i carry no information about the $m_{N,i}$, and (iii) applies to the virtual values $v_{N,i}$.
- (iii) (Sizes chosen or constrained.) If each n_i is at an interior optimum or at a binding size constraint, $\sum_i \pi_i (1 + v_{N,i}) = (1 + m_D) \bar{w}_H$ with $v_{N,i} \equiv m_{N,i} + \xi_i / X_{T,i}$, where $\bar{w}_H$ is computed from the technology alone and satisfies $1 - 1/\bar{w}_H = \sum_i h_i s_i$: the family's first-order condition pins the π -weighted average of the members' values of compactness, whether the sizes are chosen at the margin or set by binding size constraints. Under Proposition A3(iii)(a) with common (λ, ϱ) , $\delta = \eta = 0$, $\phi = 1$, $m_D = 0$ and no binding size constraint: $\sum_i \pi_i \hat{T}_i = \lambda \varrho \sum_i \pi_i T_i$. Only this average is a

revealed-preference object; the member values are not identified (one equation for I unknowns).

- (iv) (Binding size caps and floors.) Binding size caps make $\xi_i \geq 0$ and binding minimum-size floors make $\xi_i \leq 0$. If every size is at a binding cap, $\sum_i \pi_i(1+m_{N,i}) \leq (1+m_D)\bar{w}_H$: $\bar{w}_H-1$ bounds the π -weighted average of the objective's own $m_{N,i}$ from above (with $m_D = 0$). Binding floors reverse the inequality.
- (v) (A common D under member-level choice is a knife-edge.) If instead each member chooses its own D_i with $\delta = m_D = 0$ and no binding constraint, equal token counts $D_i = \bar{D}$ occur only if $1+m_{N,i} = w(n_i, \bar{d})$ for every i ; in the generalized family, only if $(1+m_{N,i})/(1+m_{N,j}) = (N_i/N_j)^{-a_1}$ exactly.
- (vi) (Binding data cap.) If the token budget is at a binding cap $d = \bar{d}$ with multiplier $\xi_D \geq 0$ and each n_i is at an interior optimum or at a binding size constraint, then $\sum_i \pi_i(1+v_{N,i}) = (1+m_D+\mu)\bar{w}_H \geq (1+m_D)\bar{w}_H$ with $\mu \equiv \xi_D/\sum_j X_{T,j}$: with $m_D = 0$, $\bar{w}_H - 1$ bounds the family's π -weighted value of compactness from below. Member i has $w_i \leq (1+v_{N,i})/(1+m_D)$ if and only if $\rho_i \geq 1+m_D$, that is, if it would train on more tokens at its own size when it bears the data cost in proportion to its compute share, which a binding cap on the family's budget does not imply. If every member would train on more tokens at its own size ($\rho_i \geq 1+m_D$ for each i), each member's $w_i - 1$ bounds its own $v_{N,i}$ from below (with $m_D = 0$).

Proof. (i) By Lemma A4(iii), $\ln w = \ln(a_1 A/b_1 B) - a_1 n + b_1 d$ in the generalized family. (ii) The first-order condition in d is $\sum_i \Lambda_i \varepsilon_{D,i} = \sum_i X_{T,i} + q_D D$. Divide it by $\sum_j X_{T,j}$.

(iii) The condition in n_i is $\Lambda_i \varepsilon_{N,i} = X_{T,i}(1+m_{N,i}) + \xi_i$, with $\xi_i = 0$ at an interior optimum, so $\rho_i = (1+v_{N,i})/w_i$; substitute into (ii) and multiply by $\bar{w}_H$. For the shares, $1 - \sum_i h_i/w_i = \sum_i h_i(1-1/w_i)$. With $m_D = 0$, the π -weighted mean of $v_{N,i}$ is $\bar{w}_H - 1$, which equals the π -weighted mean of $w_i - 1$ because $\sum_i \pi_i w_i = \bar{w}_H$; without binding size constraints $v_{N,i} = m_{N,i} = \lambda \varrho T_i/(3D)$, and multiplying by $3D$ gives the statement on $\hat{T}_i$. (iv) Substitute $v_{N,i} = m_{N,i} + \xi_i/X_{T,i}$ in (iii), with the sign of ξ_i . (v) Proposition A3(i) member by member, and (i).

(vi) With the cap binding, the condition in d becomes $\sum_i h_i \rho_i = 1 + m_D + \mu$; substituting $\rho_i = (1+v_{N,i})/w_i$ from (iii) and multiplying by $\bar{w}_H$ gives the equality. Since $\rho_i = (1+v_{N,i})/w_i$, $\rho_i \geq 1+m_D$ if and only if $w_i \leq (1+v_{N,i})/(1+m_D)$. A member that bears the share h_i of the data cost gains $\Lambda_i \varepsilon_{D,i} = \rho_i X_{T,i}$ and pays $X_{T,i} + h_i q_D D = X_{T,i}(1+m_D)$ per log point of tokens at its own size, so $\rho_i \geq 1+m_D$ says that it would train on more tokens. $\square$

Remark A5. In two brute-force examples with three members on a common budget, the member inversions $\hat{T}_i/T_i$ are 0.86, 0.91 and 1.06 in one and 6.9, 2.3 and 0.65 in the other, while the π -weighted average is exact in both: member wedges can be off by an order of magnitude. With the budget of the second example

capped at 30, 50 and 80 percent of the family's optimum, the equality in (vi) holds to numerical precision; one of the three members, the smallest, violates the member bound at 30 and 50 percent, and two do at 80 percent, exactly those with $\rho_i < 1$. With a binding size cap on one member and a floor on another, the equalities of (iii) and (vi) in virtual values hold to numerical precision, with and without a data cost and for the reference technology with κ free. For the objective's own $m_{N,i}$, a size cap and a data cap point in opposite directions: a capped member with $\rho_i \geq 1$ has $w_i - 1 \leq v_{N,i}$ but can have $w_i - 1 > m_{N,i}$ (code/analysis/rb5_theory/verify_virtual_value.py).

A7. *Model-Free Identification of the Elasticity of Substitution and the Wedge*

Proposition A5 (Model-free σ^* and w ; Proposition 3). *Let $y = F(n, d)$ be C^2 with $F_n, F_d > 0$, where y is any strictly increasing transformation of $-L$. Write $x \equiv \ln M = d - n$ and $c \equiv \ln C = \ln 6 + n + d$, let $Y(x, c) \equiv F((c - \ln 6 - x)/2, (c - \ln 6 + x)/2)$, let L also denote loss as a function of (x, c) , and let $L^*(c) \equiv \min_x L(x, c)$ be the loss-compute frontier with $\arg\min x^*(c) = \ln M^*(C)$.*

(i) (Wedge.) *At any point, $F_n = Y_c - Y_x$, $F_d = Y_c + Y_x$ and*

$$w = \frac{Y_c - Y_x}{Y_c + Y_x} = \frac{L_c - L_x}{L_c + L_x}.$$

w is invariant to strictly monotone transformations of output, so it requires neither E nor κ nor a functional form: only the slopes of any output measure along the isocost and along the ray of constant M .

(ii) (Elasticity of substitution at the compute optimum.) *At $x = x^*(c)$, where $L_x = 0$,*

$$(A7) \quad \frac{1}{\sigma^*} - 1 = \frac{2(-Y_{xx})}{Y_c} = \frac{2L_{xx}}{|dL^*/dc|} = \frac{L_{nn}|_C}{2|dL^*/dc|},$$

where $L_{nn}|_C$ is the second derivative of the IsoFLOP profile with respect to $\ln N$ at its minimum and $dL^/dc = L_c$ by the envelope theorem. The ratio is invariant to strictly monotone transformations of output (with a nonzero derivative) at the argmin, but not elsewhere: the curvature of the IsoFLOP profile divided by twice the slope of the loss-compute frontier, in any common output units, identifies σ^* .*

(iii) (Slope of the log wedge.) *At $x^*(c)$, $\partial \ln w / \partial \ln M|_C = 1/\sigma^* - 1$. In the generalized family $\ln w$ is exactly linear in $\ln(M/M^*(C))$ with this slope at every compute (Lemma A4(iii)); in general the slope varies with M , which gives a test of the family.*

(iv) (What $\kappa = 1$ imposes.) *In the generalized family, with $y = -\ln(L - E)$, $dy^*/dc = \gamma$ and $-y_{nn}|_C = S\gamma$ at the argmin, so (A7) returns $S/2$ for every*

κ and E . The restriction $\kappa = 1$ imposes $S = \gamma/[a(1-a)]$: the curvature of every IsoFLOP profile in log reducible loss is pinned by the slopes of the frontier and of the path. With κ free the curvature is a separate parameter, which (A7) measures.

(v) (Local σ anywhere.) $F_{nn} = Y_{cc} - 2Y_{cx} + Y_{xx}$, $F_{dd} = Y_{cc} + 2Y_{cx} + Y_{xx}$ and $F_{nd} = Y_{cc} - Y_{xx}$, so $\sigma = \mathcal{P}/(\mathcal{P} + \mathcal{Q})$ (Lemma A2) is identified at any point from the second-order expansion of any output in (x, c) .

(vi) (Designs.) σ^* is identified by IsoFLOP profiles: the curvature of a profile at its minimum, and the frontier slope from the minima at two or more budgets (or locally). On-path designs identify the frontier slope but not the curvature. The wedge at a point is identified by local variation in two non-collinear directions around it, and only inside the design's support; its precision falls as $\varepsilon_D \rightarrow 0$, since $w \rightarrow \infty$ when $L_c + L_x \rightarrow 0$.

Proof. (i) $\partial/\partial n = \partial_c - \partial_x$ and $\partial/\partial d = \partial_c + \partial_x$. With $L = h(Y)$ and $h' < 0$, $L_x = h'Y_x$ and $L_c = h'Y_c$, and h' cancels. (ii) At $F_n = F_d \equiv f$, $\mathcal{P} = 2f^3$ and $\mathcal{Q} = -f^2(F_{nn} - 2F_{nd} + F_{dd}) = -4f^2Y_{xx}$, since $Y_{xx} = (F_{nn} - 2F_{nd} + F_{dd})/4$. By Lemma A2, $1/\sigma - 1 = \mathcal{Q}/\mathcal{P} = -2Y_{xx}/f$ with $f = Y_c$. For $L = h(Y)$, $L_{xx} = h'Y_{xx}$ at $Y_x = 0$ and $dL^*/dc = h'Y_c$; at fixed c , $dn = -dx/2$, so $L_{nn}|_C = 4L_{xx}$. For a monotone g , $(g \circ L)_{xx} = g''L_x^2 + g'L_{xx} = g'L_{xx}$ only where $L_x = 0$, and the ratio is preserved where $g' \neq 0$. (iii) $\partial_x \ln w = (Y_{cx} - Y_{xx})/(Y_c - Y_x) - (Y_{cx} + Y_{xx})/(Y_c + Y_x)$, which is $-2Y_{xx}/Y_c$ at $Y_x = 0$. (iv) Along the path $y^* = \gamma(c - \ln 6) - \ln K$, so $dy^*/dc = \gamma$; by (ii), $-y_{nn}|_C = -4Y_{xx} = S\gamma$; with $\kappa = 1$, $\gamma = a(1-a)S$. (v) Second-order chain rule. (vi) Part (ii) uses only derivatives of a profile at its minimum and of the frontier; part (i) uses two directional derivatives at the point. $\square$

Corollary A3 (Known costs that are not multiplicative; Section II). *Let $y = F(n, d)$ be C^2 with $F_n, F_d > 0$, where y is any strictly increasing transformation of $-L$, and let the cost $C(N, D)$ be any known twice-differentiable function, with $\mathcal{C}(n, d) \equiv \ln C(e^n, e^d)$ and log-cost elasticities $g_n \equiv \mathcal{C}_n > 0$ and $g_d \equiv \mathcal{C}_d > 0$. At a point that maximizes output on its isocost (equivalently, minimizes cost at its output), let σ^* be the technology's Hicks elasticity of substitution there, let $r \equiv g_n/g_d$, let σ_C be the Hicks elasticity of substitution of the isocost itself (of C read as a production function), let $y_{nn}|_C$ be the second derivative of output along the isocost with respect to n , and let $y^*(c)$ be the output-cost frontier. Then*

$$(A8) \quad \frac{1}{\sigma^*} - \frac{1}{\sigma_C} = \frac{|y_{nn}|_C}{r(1+r)g_d dy^*/dc}.$$

(i) (Multiplicative cost.) With $C = 6ND$, $\sigma_C = 1$ and $r = g_d = 1$, and (A8) is (A7).

- (ii) (Linear cost.) With $C = q_N N + q_D D$, $\sigma_C = \infty$ and (g_n, g_d) are the cost shares (s_N, s_D) , so $1/\sigma^* = |y_{nn}|_C s_D / (s_N dy^*/dc)$.
- (iii) (Interior optima.) The second-order condition for a cost minimum holds strictly if and only if $1/\sigma^* > 1/\sigma_C$; for positive elasticities, $\sigma^* < \sigma_C$. This generalizes Lemma 1 ($\sigma_C = 1$) and the condition $\sigma^* > 0$ under a linear cost ($\sigma_C = \infty$).
- (iv) (Invariance.) The right side of (A8) is invariant to strictly monotone transformations of output with a nonzero derivative at the point, because the first derivative of output along the isocost vanishes there; it may be computed from loss, with $L_{nn}|_C$ and $|dL^*/dc|$ in place of $|y_{nn}|_C$ and dy^*/dc .

Proof. Parameterize the isocost as $d = \phi(n)$. Differentiating $\mathcal{C}(n, \phi(n)) = c$ twice gives $\phi' = -r$ and $\phi'' = \mathcal{Q}_C/g_d^3$, with $\mathcal{Q}_C \equiv -(\mathcal{C}_{nn}g_d^2 - 2\mathcal{C}_{nd}g_n g_d + \mathcal{C}_{dd}g_n^2)$; by Lemma A2(i) applied to $\mathcal{C}$, $\mathcal{Q}_C = \mathcal{P}_C(1/\sigma_C - 1)$ with $\mathcal{P}_C \equiv g_n g_d(g_n + g_d) = r(1+r)g_d^3$. Output along the isocost, $Y(n) \equiv F(n, \phi(n))$, has $Y' = F_n - rF_d = 0$ at the point and $Y'' = F_{nn} - 2rF_{nd} + r^2F_{dd} + F_d\phi''$. With $F_n = rF_d$, $F_{nn} - 2rF_{nd} + r^2F_{dd} = -\mathcal{Q}/F_d^2$, and Lemma A2(i) gives $\mathcal{Q} = \mathcal{P}(1/\sigma^* - 1)$ with $\mathcal{P} = r(1+r)F_d^3$. Hence $y_{nn}|_C = Y'' = -r(1+r)F_d(1/\sigma^* - 1/\sigma_C)$. By the envelope theorem dy^*/dc is the Lagrange multiplier, F_d/g_d , and dividing gives (A8). (i) $\mathcal{C}$ is linear, so $\mathcal{Q}_C = 0$. (ii) With $\mathcal{C} = \ln(q_N e^n + q_D e^d)$, $g_n = s_N$, $g_d = s_D$ and $\mathcal{C}_{nn} = \mathcal{C}_{dd} = -\mathcal{C}_{nd} = s_N s_D$, so $\mathcal{Q}_C = -s_N s_D = -\mathcal{P}_C$ and $1/\sigma_C = 0$; and $r(1+r)g_d = s_N/s_D$. (iii) $Y'' < 0$ if and only if $1/\sigma^* > 1/\sigma_C$, since $r, F_d > 0$. (iv) For h strictly monotone, $(h \circ Y)'' = h''Y'^2 + h'Y'' = h'Y''$ at $Y' = 0$ and $dh(y^*)/dc = h'dy^*/dc$. The identity was checked symbolically for generic derivatives of F and $\mathcal{C}$, and numerically, to at least five digits, for CES technologies with $\sigma^* = 0.40, 0.67$ and 0.83 under $C = ND$ and with $\sigma^* = 0.40, 0.67, 0.83$ and 1.43 under a linear cost, for the reference technology under multiplicative, linear and size-dependent-price costs, for a CES technology under a CES cost, and for a non-separable technology with a data cost (`code/analysis/rb5_theory/verify_general_isocost.py`). $\square$

Remark A6 (Finite grids). IsoFLOP profiles are sampled on finite grids and fitted by low-order polynomials (Czech et al., 2026). If the profile $f(t)$, with t the distance from its minimum in $\ln N$, is sampled at points symmetric about the minimum and fitted by least squares on $(1, t, t^2)$, the quadratic coefficient is $f''/2 + (f'''/24)\kappa_4 + O(h^4)$, with $\kappa_4 \equiv (m_6 - m_2 m_4)/(m_4 - m_2^2)$, m_k the k -th moment of the grid and h its half-width, and the fitted minimum is displaced by $-(f'''/6)(m_4/m_2)/f'' + O(h^3)$. For the Besiroglu et al. (2024) technology at 10^{21} FLOP and nine equally spaced sizes, a grid spanning a factor of 9 (100) in N overstates the curvature by 1.3 (5.9) percent, so the model-free estimate is $\hat{\sigma}^* = 0.734$ (0.726) against 0.737; a quartic fit removes the leading term. Quadratic fits on grids narrower than a factor of about 3 in N keep the bias below 0.001. These are noise-free calculations of bias only: a narrower grid or a quartic fit raises the variance of the estimated curvature.

A8. Partial Identification beyond the Design

Proposition A6 (Partial identification of $M^*(C)$ and of the wedge; Proposition 3, beyond the design). *Let $x^*(c) \equiv \ln M^*(C)$ be the unique argmin of the IsoFLOP profile at compute C , absolutely continuous in c , with path slope $e_p(c) \equiv dx^*/dc = 1 - 2a(c)$, where $a(c) \equiv d \ln N^*/d \ln C$. Suppose $x^*(c_J)$ is identified at the largest design budget c_J , and consider a model with compute $c > c_J$ and log ratio $x = \ln M$.*

(i) (Normal inputs.) *If $N^*(C)$ and $D^*(C)$ are nondecreasing, then $e_p \in [-1, 1]$ and $x^*(c) \in [x^*(c_J) - (c - c_J), x^*(c_J) + (c - c_J)]$.*

(ii) (Bounded path slope.) *If $e_p(c') \in [e_L, e_U]$ for $c' \in [c_J, c]$, the sharp identified set is*

$$\mathcal{X}^*(c) = [x^*(c_J) + e_L(c - c_J), x^*(c_J) + e_U(c - c_J)],$$

whose width grows linearly in the log extrapolation distance; monotonicity of M^ is the case $e_L \geq 0$. When the true technology is in the generalized family, every point of $\mathcal{X}^*(c)$ is generated by a technology with positive marginal products and single-peaked IsoFLOP profiles that coincides with the true one for $c \leq c_J$. If confidence intervals CI_j for $x^*(c_j)$ at budgets $c_1 < \dots < c_J$ hold jointly with probability $1 - p$ and the slope bounds hold from c_1 on, then $\bigcap_j [\inf \text{CI}_j + e_L(c - c_j), \sup \text{CI}_j + e_U(c - c_j)]$ covers $x^*(c)$ with probability at least $1 - p$.*

(iii) (Sign of the wedge.) *If each IsoFLOP profile is single-peaked in x , that is, $Y_x > 0$ for $x < x^*(c)$ and $Y_x < 0$ for $x > x^*(c)$, which holds whenever $0 < \sigma < 1$ throughout (Lemma A2), then $\text{sign}(w - 1) = \text{sign}(x - x^*(c))$ at every (c, x) , for any technology. Hence $w > 1$ is identified if $x > \sup \mathcal{X}^*(c)$ and $w < 1$ if $x < \inf \mathcal{X}^*(c)$; when $\mathcal{X}^*(c)$ is sharp, as in (ii) for a technology in the generalized family, these conditions are also necessary, and the sign is not identified otherwise. Neither σ^* nor the functional form is needed.*

(iv) (Magnitude.) *Suppose $\partial \ln w / \partial x|_c \in [\theta_L, \theta_U]$ with $\theta_L > 0$ between $x^*(c)$ and x ; in the generalized family $\theta_L = \theta_U = 1/\sigma^* - 1$, so a range for σ^* gives $[\theta_L, \theta_U]$. With $\mathcal{X}^*(c) = [\ell, u]$: $\ln w \in [\theta_L(x - u), \theta_U(x - \ell)]$ if $x > u$; $\ln w \in [\theta_U(x - u), \theta_L(x - \ell)]$ if $x < \ell$; and $\ln w \in [\theta_U(x - u), \theta_U(x - \ell)]$ if $\ell \leq x \leq u$.*

(v) (Compactness share and T .) *Since $s = 1 - 1/w$ is increasing in w , bounds $[w_L, w_U]$ give $s \in [1 - 1/w_L, 1 - 1/w_U]$. Under Proposition A3(iii)(a) with $\phi = 1$ and $\eta = \delta = 0$, $\lambda \varrho T / (3D) \in [w_L - 1, w_U - 1]$.*

(vi) (A bound inside the support.) *Let $F_{nn}, F_{dd} \leq 0$ and $F_{nd} \geq 0$, which hold throughout the generalized family. Then $\ln w$ is nonincreasing in n and nondecreasing in d , so for any support point with $n_b \geq n_0$ and $d_b \leq d_0$, $w(n_0, d_0) \geq w(n_b, d_b)$, where $w(n_b, d_b)$ is identified without extrapolation.*

No finite upper bound follows from these shape restrictions, and without $F_{nd} \geq 0$ the bound can fail.

- (vii) (Interval anchor, joint path bands and tilt allowance.) Let the anchored technology have $x^*(c_J)$ in $[\underline{x}_J, \bar{x}_J]$ and path slopes in $[e_L, e_U]$ on $[c_J, c]$, and let the developer's compute-optimal log ratio lie within $\ln \tau$ of the anchored technology's at every compute, $|x_{\text{dev}}^*(c') - x^*(c')| \leq \ln \tau$ with $\tau \geq 1$. Then

$$x_{\text{dev}}^*(c) \in \mathcal{X}_\tau^*(c) \equiv [\underline{x}_J + e_L(c - c_J) - \ln \tau, \bar{x}_J + e_U(c - c_J) + \ln \tau],$$

and, if the developer's IsoFLOP profiles are single-peaked, (iii) and (iv) hold for the wedge in its first-order conditions with $\mathcal{X}_\tau^*(c)$ in place of $\mathcal{X}^*(c)$ and x_{dev}^* in place of x^* . If the anchor is a point ($\underline{x}_J = \bar{x}_J$) and the anchored technology is in the generalized family, $\mathcal{X}_\tau^*(c)$ is sharp: every point of it is $x_{\text{dev}}^*(c)$ for a developer technology with positive marginal products and single-peaked IsoFLOP profiles that lies within $\ln \tau$, at every compute, of a technology that coincides with the anchored one for $c' \leq c_J$ and has path slopes in $[e_L, e_U]$. With sampling error, let $\mathcal{B}(c') = [b_L(c'), b_U(c')]$ be a simultaneous confidence band for the anchored path over $[c_J, \bar{c}]$ at level $1 - p$, $\Pr\{x^*(c') \in \mathcal{B}(c') \text{ for all } c' \in [c_J, \bar{c}]\} \geq 1 - p$, for instance, for a path linear in c , one built from a joint confidence set for its level and slope. Then $[b_L(c) - \ln \tau, b_U(c) + \ln \tau]$ covers $x_{\text{dev}}^*(c)$ at every $c \in [c_J, \bar{c}]$ simultaneously with probability at least $1 - p$.

- (viii) (Several anchored paths.) If the developer's compute-optimal log ratio lies within $\ln \tau$ of at least one of finitely many anchored paths j , each with its own anchor budget and its own set $\mathcal{X}_j^*(c)$ from (vii) with $\tau = 1$, it lies in $\bigcup_j \mathcal{X}_{j,\tau}^*(c)$, and (iii) and (iv) hold with the union, with ℓ and u its infimum and supremum. If each $\mathcal{X}_j^*$ is a simultaneous confidence band for path j over its range of compute up to $\bar{c}$ at level $1 - p_j$, the widened union covers $x_{\text{dev}}^*(c)$ with probability at least $1 - \max_j p_j$, because it covers whenever the band of the path that the developer's technology lies within τ of covers.

Proof. (i) $d \ln N^* + d \ln D^* = d \ln C$ with both terms nonnegative gives $a \in [0, 1]$; integrate $e_p = 1 - 2a$. (ii) Integrate e_p over $[c_J, c]$; the linear continuations attain every point. For the technology: in the generalized family let $\tilde{L}(n, d) = E + [Ae^{-a_1(n+\psi_N(c))} + Be^{-b_1(d+\psi_D(c))}]^\kappa$ with $c = \ln 6 + n + d$ and nondecreasing ψ_N, ψ_D that vanish for $c \leq c_J$. At fixed c the profile is a translate of a profile of the original technology, hence single-peaked, with $\arg\min x^*(c) + 2(1 - a)\psi_N(c) - 2a\psi_D(c)$ (Lemma A3(iii)); $\tilde{F}_n = F_n(1 + \psi'_N) + F_d\psi'_D > 0$ and $\tilde{F}_d = F_n\psi'_N + F_d(1 + \psi'_D) > 0$. Choosing $\psi'_N = \Delta'_+/[2(1 - a)]$ and $\psi'_D = \Delta'_-/(2a)$, with $\Delta'_\pm$ the positive and negative parts of Δ' , generates any continuation $x^*(c) + \Delta(c)$ with $\Delta(c_J) = 0$. On the event that every CI_j covers, every cone contains $x^*(c)$. (iii) By Proposition A5(i), $w - 1 = -2Y_x/(Y_c + Y_x) = -2Y_x/F_d$ with $F_d > 0$, and single-peakedness gives the sign of Y_x . If $0 < \sigma < 1$ throughout, every critical

point of a profile is a strict local maximum of y (Lemma A2), so there is at most one. Sharpness makes every x inside the set consistent with both signs. (iv) $\ln w(x) = \int_{x^*}^x \partial_x \ln w$, with $\ln w(x^*) = 0$ and $x^* \in [\ell, u]$. (v) Monotonicity. (vi) $\partial \ln w / \partial n = F_{nn}/F_n - F_{nd}/F_d \leq 0$ and $\partial \ln w / \partial d = F_{nd}/F_n - F_{dd}/F_d \geq 0$; move from (n_b, d_b) to (n_b, d_0) and then to (n_0, d_0) . A saturating continuation of the data term beyond the support satisfies the shape restrictions and sends $w \rightarrow \infty$; the concave quadratic $F = 0.05n + 0.3d - 0.025n^2 - 0.005d^2 - 0.02nd$ has $F_{nd} < 0$ and $w(0, 0.5) < w(0, 0)$.

(vii) Integrating e_p from a point of $[\underline{x}_J, \bar{x}_J]$ puts the anchored $x^*(c)$ in $[\underline{x}_J + e_L(c - c_J), \bar{x}_J + e_U(c - c_J)]$, and the allowance widens it by $\ln \tau$ on each side; (iii) and (iv) use only the developer's own technology. For sharpness, write a point of $\mathcal{X}_\tau^*(c)$ as $y_0 + t$ with y_0 in the set of (ii) and $|t| \leq \ln \tau$. By (ii), y_0 is the compute-optimal log ratio at c of a technology $\tilde{F}$ with positive marginal products and single-peaked profiles that coincides with the anchored one for $c' \leq c_J$ and has path slopes in $[e_L, e_U]$. The constant factor-augmenting tilt $\psi_N = -\psi_D = t/2$, whose bias index is $\chi = -(a_1 + b_1)t/2$, maps (n, d) into $(n + t/2, d - t/2)$ and leaves compute unchanged, so at every compute the tilted technology's IsoFLOP profile is that of $\tilde{F}$ translated by t in x : its argmin moves by $t = -2\chi/(a_1 + b_1)$ at every compute, as in Lemma A3(iii), its profiles stay single-peaked, and its marginal products are those of $\tilde{F}$ at shifted inputs, hence positive. It lies within $\ln \tau$ of $\tilde{F}$ at every compute and has $x_{\text{dev}}^*(c) = y_0 + t$. On the event that $\mathcal{B}$ covers the anchored path on $[c_J, \bar{c}]$, the widened band covers x_{dev}^* there. (viii) Let j^* be a path that the developer's technology lies within τ of. By (vii), $x_{\text{dev}}^*(c) \in \mathcal{X}_{j^*, \tau}^*(c)$, which is contained in the union, and the event that band j^* covers has probability at least $1 - p_{j^*} \geq 1 - \max_j p_j$. The tilt construction, the containment and sign statements (20,000 random draws) and the band coverage (a Monte Carlo with three anchored paths) were also checked numerically (code/analysis/rb5_theory/verify_prop_pi.py). $\square$

Remark A7 (Parametric projection and the direction of extrapolation errors). A confidence set for $(\ln(A/B), \alpha, \beta)$ projects into one for the wedge whose width grows linearly in the log extrapolation distance, since $\text{Var}(\ln \hat{w})$ is exactly quadratic in $\ln M$ at fixed compute. For the Chinchilla extraction a pairs bootstrap gives $\text{sd}(\ln \hat{w}) = 0.161$ at $M = 1,875$ and $C = 7.2 \times 10^{23}$, which is sampling error within one sweep and one functional form; the union of the designs' path bands of Proposition A6(viii), which carries each path's slope and its sampling error, puts $M^*(10^{24})$ in a set about 3.8 log points wide (Online Appendix E7), and the 32 technologies in the set put $M^*(10^{24})$ between 1.4 and 426, or 5.7 log points. Evidence that fitted laws misstate losses at extreme ratios (Sardana et al., 2024) does not sign the error in $\hat{w}$: revealed preference recovers the technology the developer believed (Proposition A3(ii)), so if developers planned with fitted laws, correcting toward the true technology moves $\hat{w}$ away from their first-order condition.

TABLE B1. SAMPLES

Sample	Units	N	$M = D/N$	Off-path spread	Use
<i>Panel A. Public designed sweeps (unit: training run)</i>					
Chinchilla (digitized)	245 ^a	57M–16B	0.46–341	1.35	III; IV; D
Llama 3 (digitized)	133	59M–17B	0.44–643	1.38	III; IV; D
Marin, 3 corpora	258	0.12–13B	0.09–3,706	2.42–2.46	III; IV; D
Porian et al.	975 ^b	5.2M–0.90B	0.02–346	2.20–2.29	III; D
Farseeer	404	0.10–6.4B ^c	0.31–2,570	1.97	III; IV; D
Gadre et al., 3 corpora	104	11M–6.9B	5–640	1.44	III; D
OLMo ladder	30	0.19–3.2B ^d	10–200	0.94	III; IV; E
Muennighoff et al.	33	7.1M–8.7B	0.04–303	1.97	III; D
DataDecide, 25 recipes	1,100 ^e	4M–1.2B ^d	5–110	0.73	III; D
Step Law	1,911 ^f	0.21–1.07B ^c	19–466	–	D
<i>Panel B. Our controlled experiment, FineWeb-Edu and FineWeb (unit: annealed endpoint)</i>					
Main grid, 2 corpora	2×44	0.39–49M ^c	0.51–2,031	1.96	III; E
High- M runs	10	0.79–3.2M ^c	254–4,063	–	III; E
<i>Panel C. Released models (unit: model)</i>					
Verified sample	173	70M–406B ^g	2.1–60,398	–	IV; E
Clean sample	77	0.13–406B	19–60,398	–	IV; E
Production-scale universe	335	0.13–600B ^g	0.2–60,398	–	IV; V; E

Note: N is parameters (total, including embeddings, unless marked), D is tokens processed, $M = D/N$ and $C = 6ND$; ranges are over the estimation sample. Off-path spread: residual standard deviation of an OLS regression of $\ln M$ on $\ln C$, zero for compute-optimal runs; “–”: not computed. Roman numerals refer to sections of the main text, letters to Online Appendices. The benchmark panel and the algorithmic-progress samples are documented in the companion paper. Llama 3 and Marin: $N = C/(6D)$ at the nominal budget; Porian et al.: non-embedding plus output head. ^a137 runs on nine IsoFLOP profiles and 108 off-profile runs; parametric fits drop the five highest-loss runs ($n = 240$), model-free estimates use the 137 profile runs. M and spread are over the 240 runs of the parametric fits; the model-free sample (137 profile runs, FLOP-effective N_F) has M 0.04–294 and spread 1.65 (Table 1). ^b16 architectures; losses read along each run’s curve at 12 IsoFLOP budgets (121 and 116 profile points). ^cNon-embedding parameters. In our experiment N including the tied embedding is 1.4–54M and M is 0.46–555 in the main grid (up to 1,743 in the high- M runs). High- M runs: new endpoints only; the width-256 shape at 200 and 800 million tokens repeats main-grid cells. ^dExcludes the input embedding only. ^e25 recipes $\times$ 14 sizes $\times$ 3 seeds (5 at 1B); 21,888 intermediate checkpoints enter estimation. ^f17 (N, D) cells, each with a grid of learning rates and batch sizes. ^gActive parameters for mixture-of-experts models.

Source: Author’s calculations from the sources cited in the text of this appendix; Panel B: this paper.

APPENDIX B. DATA CONSTRUCTION

This appendix documents the samples in Table B1: sources, conventions, corrections to source data and sample restrictions. A single script per module regenerates every sample from the raw files with fixed seeds, and the replication package lists the download commands. Snapshots of living databases are from September 23 and 24, 2026.

B1. Public Sweeps

For every sweep we build a run-level panel of (N, D, L) with a cluster identifier, $M = D/N$ and $C = 6ND$. We use $C = 6ND$ even when a sweep reports a more detailed FLOP count, so that compute means the same thing across

sweeps; the model-free estimator, which needs exact isocosts, is the exception. The sweeps share neither a parameter-count convention, nor a tokenizer, nor an output. We keep each sweep’s conventions and compare only unit-free objects across sweeps: α , β , a , γ and σ^* are invariant to the unit of loss but not to the parameter-count convention, while E , A , B and M^* are not comparable across rows. The off-path spread in Table B1 is the residual standard deviation of an OLS regression of $\ln M$ on $\ln(6ND)$. Farseer is a near-factorial grid in $(\ln N, \ln D)$ with $\text{corr}(\ln N, \ln D) = -0.04$; the Chinchilla runs combine nine IsoFLOP profiles with off-profile streaks at fixed size; Gadre et al. and the OLMo ladder are grids in size and token multiplier.

Farseer. The released file of Li et al. (2025b) contains 404 single-recipe rows: 400 distinct runs of 25 sizes, each on up to 18 token budgets, plus four runs listed twice with identical records (the copies differ only in a mislabeled D/N column); we keep all 404 rows, and dropping the four copies leaves the statistics of Table B1 unchanged. Hyperparameters are set by the Step Law rules. N excludes both untied 65,536-row embedding matrices (the reported total exceeds it by exactly 131,072 h for hidden size h). The output is English bits per character on the authors’ validation set. The column labeled D/N takes five values unrelated to D/N , so we recompute M (0.31–2,570); the file’s compute column includes attention FLOPs, and we use $6ND$ instead.

Gadre et al., the OLMo ladder and Muennighoff et al. The 104 runs of Gadre et al. (2025) (34 on C4, 35 on RedPajama and 35 on RefinedWeb) count all parameters and have $D = 20 \times \text{multiplier} \times N$; the output is C4 validation loss, on which every model is evaluated, because tests of neutral data quality need a common output. Their token-level intervals imply a sampling standard error of $\ln L$ of 0.0016, against residual standard deviations of 0.023–0.030 for a fitted Chinchilla form, so residuals reflect misspecification rather than evaluation noise. For the OLMo ladder we take the final logged step of the 30 runs released with Bhagia et al. (2025) in the OLMo-ladder repository, a superset of the 16-model ladder of their paper: five sizes at 0.5–10 times the Chinchilla budget plus five reruns, clustered with their originals. N excludes only the input embedding, the output is C4-en validation cross-entropy, and task bits per byte on the same runs are the second output of Online Appendix E8. From Muennighoff et al. (2023) we parse 229 final losses with the authors’ code, rebuild N , D and unique tokens from the model names (the aggregated third-party file mislabels two columns), and keep the 33 single-epoch runs.

DataDecide. The release (Magnusson et al., 2025) crosses 25 recipes with 14 sizes (4M to 1.2B parameters without the input embedding) and three seeds (five at 1B), for 1,100 runs in the release; the two auxiliary seeds of the 530M, 750M and 1B models stop early, at about 89, 49 and 21 tokens per parameter, and we keep them as truncated runs on the common schedule. We keep 21,888 checkpoints with $D/N \geq 5$. Full-length runs end at $D \approx 85$ –110 N , and even with checkpoints $\text{corr}(\ln N, \ln D) = 0.91$; the checkpoints that break the collinearity come before

the cosine schedule ends, and schedule progress is a function of D/N . We therefore use DataDecide only for comparisons across recipes, which share sizes, steps and schedules, and exclude it by rule from the technologies of Section IV. We did not use the release of Kricheli et al. (2026), which contains only per-epoch training losses.

B2. Chinchilla Extraction

The Epoch AI digitization of Figure 4 of Hoffmann et al. (2022) contains 245 runs (Besiroglu et al., 2024), from 1.4×10^{18} to 1.3×10^{22} FLOP and 57 million to 16 billion parameters. N is the total parameter count and C is read off the figure’s horizontal axis, so $D = C/(6N)$ is constructed: digitization error in C passes one for one into $\ln D$. We rebuild FLOPs per token for all 50 models of Hoffmann et al.’s Table A9. Their parameter counts are reproduced, to 0.8 percent, by attention, MLP and one relative-position projection per layer plus one $32,000 \times d$ vocabulary matrix. Their Appendix F formula as printed does not reproduce their Table A4; without its embedding and final-logit terms it reproduces all six ratios. In that count, FLOPs per token are 0.98–1.15 times $6N$ for the models in the profiles. The within-budget pattern of the digitized FLOP coordinates is flat once the budgets are read in that count (slope -0.002 , s.e. 0.007) and not in the printed count (-0.065), in executed FLOPs (-0.036) or in $6ND$ ($+0.020$). This is what one expects if Figure 4 plots runs at $6ND$ with token counts set in Hoffmann et al.’s count. Then the digitization’s $D = C/(6N)$ is the true token count, each profile is an exact isocost in that count, and $N_F = C/(6D)$ is FLOPs per token over six, whatever D (Online Appendix Table D2). The 44M model of Table A9 does not appear in the digitization. The digitized offset of $\log_{10} C$ below the nominal budgets (-0.018 ; below) is mostly this accounting: net of it, -0.006 remains. Drawn on the corrected coordinates, the membership rule below gives 147 profile runs instead of 137, and $\sigma^* = 0.668$ (Online Appendix D3). The loss is Hoffmann et al.’s smoothed final training loss on MassiveText, in nats per token (each token is seen once, so it is an unbiased estimate of held-out loss), read from a color map and quantized to about ± 0.01 . Besiroglu et al. drop the five highest-loss runs, all from the 10^{19} budget with 2.0 to 6.8 billion parameters, and so do we ($n = 240$).

Digitized $\log_{10} C$ sits systematically 0.018 below the nine nominal budgets, 6×10^{18} to 3×10^{21} FLOP; we estimate this offset as the median deviation of runs within 0.1 of a budget and remove it. A run belongs to an IsoFLOP profile if its corrected $\log_{10} C$ is within 0.045 of a budget, which gives 137 profile runs of 245 (132 of 240). The other 108 runs form 35 streaks of fixed model size, the training-horizon runs that Hoffmann et al. pool into their Approach 3. In the 240-run sample, 33 percent of the variance of centered $(\ln N, \ln D)$ lies across the expansion path. Assigning every run to its nearest budget, as a nine-cluster bootstrap requires, puts off-profile runs from a 10^{22} -FLOP trunk into the 3×10^{21} IsoFLOP experiment; our cluster schemes therefore group profile runs by budget and off-

profile runs by their model-size trunk. Both Chinchilla series in the compilation of [Open Athena \(2026\)](#), associated with [Czech et al. \(2026\)](#) (123 and 124 points), are snapped subsets of the same digitization, not independent sources.

B3. Laboratory IsoFLOP Profiles and Hyperparameter Sweeps

Llama 3. The 133 points of Meta’s IsoFLOP figure ([Grattafiori et al., 2024](#)) come from the digitization of [Czech \(2026\)](#), distributed in the compilation of [Open Athena \(2026\)](#), associated with [Czech et al. \(2026\)](#); the two agree row by row. There are ten budgets from 6×10^{18} to 10^{22} FLOP. Two of them, 3×10^{21} and 10^{22} , have six runs each and do not bracket their minimum with two runs on each side; the model-free estimator and the anchors of Online Appendix E7 use the other eight, and Meta’s ten-budget planning law is a sensitivity. D is read off the figure and $N = C/(6D)$ uses the nominal budget. The loss units are not stated, but the model-free estimator and the wedge are invariant to monotone transformations of the output.

Marin. The scaling ladders of [Held and Marin Community \(2026\)](#), in the March-2026 W&B export distributed in the compilation of [Open Athena \(2026\)](#) (experiments `marin_202603`), train a Llama-2-style architecture on three corpora (Comma, DCLM and Nemotron-CC): 85, 85 and 88 runs on seven or eight budgets from 1.8×10^{18} to 3×10^{20} FLOP, with the macro-averaged Paloma loss as output. The nominal budgets are three times the forward FLOPs and differ from $6ND$ with the configuration parameter count by -7 to $+35$ percent, a difference strongly negatively correlated with N within a budget. Model-free estimates use $N = C_b/(6D)$, which makes each profile an exact isocost; with the configuration count they are 0.04 to 0.07 lower. The checkpoint-interpolated profiles that [Czech et al. \(2026\)](#) construct from the training logs of [Li, Kudugunta and Zettlemoyer \(2025\)](#), whose points come before the learning-rate schedule ends, enter only the duality tests.

Porian et al. and Step Law. The released results of [Porian et al. \(2024\)](#) contain 975 runs of 16 architectures on RefinedWeb and OpenWebText2. We count parameters and FLOPs without the output head (Kaplan), with it (standard), with attention FLOPs and with all embeddings, and our re-implementation of their IsoFLOP pipeline reproduces their path estimates to within 0.003. The 264 runs that reproduce Kaplan et al.’s setup use cosine decay, either to zero at a fixed horizon of about 131B tokens or to 1 percent at each target budget; the other 711, the 61 with tuned hyperparameters (among them the 32 of their final configuration, whose IsoFLOP profiles we use), their hyperparameter sweeps and their seed replicates, use a constant learning rate without cooldown. The cosine runs enter only the decomposition of Online Appendix D6. The dense-model file of [Li et al. \(2025a\)](#) contains 1,911 runs in 17 (N, D) cells (N 215M–1.07B non-embedding parameters, D 4B–100B, M 19–466), each with a grid of up to 12 learning rates and 10 batch sizes; the output is the smoothed final training loss, and we classify 9.5 percent of runs as diverged (log loss more than 0.30 above the

cell minimum).

B4. The Controlled Experiment

Corpora and tokenizer. The corpora are FineWeb-Edu and FineWeb (Penedo et al., 2024); FineWeb-Edu keeps the documents of the same crawl that an educational-quality classifier rates highly. The first 20,000 documents of each sample-10BT release are held out for validation, and the training tokens exceed every budget, so no run repeats data. A BPE tokenizer (Sennrich, Haddow and Birch, 2016) over raw bytes with 8,192 tokens, trained on both corpora and shared, puts losses in the same units; we report bits per byte (3.89 and 3.71 bytes per token in the scored tokens of the FineWeb-Edu and FineWeb validation sets).

Architecture and training. The models are decoder-only transformers with RMSNorm pre-normalization (Zhang and Sennrich, 2019), rotary position embeddings (Su et al., 2024), a four-times-wide GELU feed-forward block and an input embedding tied to the output projection; width d_{model} sets $d_{\text{model}}/64$ layers and heads (Table B2). The tied embedding is 73 percent of all parameters at width 128 and 10 percent at 640, so the counting convention matters far more than in production models (Pearce and Song, 2024; Porian et al., 2024). All runs but one (described below) use AdamW (Loshchilov and Hutter, 2019) with weight decay 0.1, which the implementation (MLX) multiplies by the learning rate, so its averaging timescale relative to run length varies across the design. The learning rate warms up over 250 steps and stays constant on a trunk; for each budget D_k the state is copied at $0.8D_k$ and the copy is cooled to zero (Hägele et al., 2024), so one trunk delivers a column of the design and every endpoint is annealed. Endpoints of one shape share the trunk, so inference clusters by width. The peak learning rate follows $\text{lr}^*(d_{\text{model}}) = 3.07 \times 10^{-3} (d_{\text{model}}/256)^{-0.90}$, calibrated at widths 128, 320 and 512 on 50 million FineWeb-Edu tokens. All models were trained in MLX (Hannun et al., 2023) on one Apple M5 Max. [TBD-m9: completed endpoints by corpus and tag, diverged or failed runs, and total wall-clock hours.]

Additions. Four further sets of runs measure noise and tuning: the learning-rate calibration repeated on FineWeb; peak rates of 0.5 and 2 times the rule at width 128 over the full budget ladder and at width 640 at 25 and 50 million tokens, on both corpora; seed replicates (widths 128, 256 and 384 with two further seeds at 50–200 million tokens, and, at width 128, the first replicate seed continued to 200–800 million tokens), each changing initialization and data order; and, at the four-layer width-128 high- M shape on FineWeb-Edu, rates of 0.5 and 2 times the rule and one run whose weight decay is scaled to hold the timescale-to-run-length ratio at its main-grid median. The high- M tuning checks exist for FineWeb-Edu only and stop at 1.6 billion tokens. Every endpoint except the 12 FineWeb-Edu learning-rate calibration runs, which were evaluated on FineWeb-Edu only, is evaluated on the first 1,048,576 tokens of both validation sets, and the added runs other than the FineWeb-Edu seed replicates also on WikiText-103 (Merity et al., 2017) during training.

TABLE B2. DESIGN OF THE CONTROLLED EXPERIMENT

Width	Layers	N (millions)		Embedding share	Peak LR (10^{-3})	Budgets D_k (M)	$M = D/N$		FLOP per $6N_{\text{ne}}D$
		Non-emb.	Total				Non-emb.	Total	
128	2	0.394	1.44	0.73	5.72	25–800	63–2,031	17–555	3.83
192	3	1.33	2.90	0.54	3.97	25–800	19–602	8.6–276	2.29
256	4	3.15	5.25	0.40	3.07	25–800	7.9–254	4.8–153	1.75
320	5	6.15	8.77	0.30	2.51	25–800	4.1–130	2.9–91	1.49
384	6	10.6	13.8	0.23	2.13	25–800	2.4–75	1.8–58	1.35
448	7	16.9	20.5	0.18	1.85	25–400	1.5–24	1.2–19	1.27
512	8	25.2	29.4	0.14	1.64	25–400	0.99–16	0.85–14	1.21
640	10	49.2	54.4	0.10	1.35	25–200	0.51–4.1	0.46–3.7	1.14
<i>High-M runs with a four-layer floor (extension data)</i>									
128	4	0.788	1.84	0.57	5.72	200–3,200	254–4,063	109–1,743	2.50
256	4	3.15	5.25	0.40	3.07	200–3,200	64–1,017	38–610	1.75

Note: Main grid: each width is trained once per corpus (FineWeb-Edu and FineWeb) with a constant-learning-rate trunk. The token budgets D_k are taken from $\{25, 50, 100, 200, 400, 800\}$ million; each budget below the largest is a cooldown branch started at $0.8D_k$, and the largest is the trunk’s own cooldown, which gives 44 endpoints per corpus. There are $d_{\text{model}}/64$ attention heads of dimension 64; sequence length 256; batch 64×256 tokens. Non-embedding parameters are $n_{\text{layer}}(12d_{\text{model}}^2 + 2d_{\text{model}}) + d_{\text{model}}$ for width d_{model} and n_{layer} layers; the tied embedding adds $8,192 d_{\text{model}}$. The peak learning rate is $3.07 \times 10^{-3}(d_{\text{model}}/256)^{-0.90}$. The last column is training FLOP per token, including attention and the output projection, relative to $6N_{\text{ne}}$; relative to $6N$ with the total count it is 1.03–1.07 at every shape. High- M runs: budgets $\{200, 800, 1,600, 3,200\}$ million on a longer data stream (the width-256 shape on FineWeb-Edu only). Further runs, all on both corpora unless stated: learning-rate calibration for FineWeb (widths 128, 320 and 512 at four peak rates, 50 million tokens); peak rates of 0.5 and 2 times the rule at width 128 over the full budget ladder and at width 640 at 25 and 50 million tokens; two further seeds at widths 128, 256 and 384 with 50–200 million tokens, and at width 128 the first replicate seed continued to 200–800 million tokens; and the FineWeb-Edu main-grid widths 384 and 512 retrained with the same seed and data order and every evaluation set.

Neutral sets and trunk evaluations. Runs launched after the trainer update of 17:44 on 24 September 2026 log the unannealed trunk’s validation loss at every branch point except the last and save their annealed endpoint weights. The evaluation sets are fixed when a queue’s job list is generated. The lists for the FineWeb main grid (15:37) and the FineWeb-Edu high- M runs (17:33) predate the update, so during training the FineWeb main grid was evaluated only on the two in-distribution sets and those high- M runs also on WikiText-103; the exception is FineWeb’s width-640 endpoint at 200 million tokens, whose trunk was rerun after a machine reboot from a list generated after the update. Runs from later job lists, other than the FineWeb-Edu seed replicates, are evaluated on C4 and PG-19 as well. Where an endpoint’s weights were saved but a neutral set is missing (FineWeb main-grid widths 320 to 640, the FineWeb-Edu (256, 4) high- M shape and the FineWeb-Edu seed replicates), the missing losses are computed after training from those weights with the same evaluation code (deviation D13); each such evaluation recomputes the recorded losses as a check on the reload, and the first check reproduced them exactly. The FineWeb-Edu main grid and the FineWeb main grid at widths 128 to 256 have no saved weights and no neutral-set losses, and the FineWeb-Edu (128, 4) high- M shape has WikiText-103 losses but no C4 or PG-19 losses. A between-corpus tilt on the neutral sets is therefore not identified on the main grid: the main-grid verdict on neutrality rests on the two in-distribution validation sets, and the neutral sets enter through matched-cell sign checks. C4 is lightly filtered web text, closer to FineWeb than to FineWeb-Edu, so it is not neutral between the corpora [TBD-m9: endpoints with post hoc neutral-set losses]. To supply matched cells at the rule’s learning rate beyond width 128, FineWeb-Edu’s main-grid trunks at widths 384 and 512 are retrained, with the same seed and data order, with every evaluation set, trunk evaluations and saved weights (Amendment 2); the original endpoints stay in the fits. Unannealed trunk losses thus exist for FineWeb widths 320 to 640 (up to 400 million tokens, so $M \leq 65$ in convention P), for FineWeb-Edu widths 384 and 512 from the retrained trunks, for the seed replicates, and for the high- M shapes other than FineWeb-Edu’s (128, 4) run at the rule, never at a trunk’s last budget.

Pre-analysis plan, deviations and power. The pre-analysis plan, which is in the replication package, was committed at 03:12 (UTC+3) on 24 September 2026 (commit bd5c0ad); its header gives 03:15 as the time of writing, three minutes after the commit. At that moment part of the FineWeb-Edu main grid had finished and its raw endpoint losses had appeared in run logs, but no model had been fitted; the plan’s primary choices, including the non-embedding convention for the extrapolation question, were therefore made after those logs existed. The FineWeb main grid and the high- M runs had not started, and no addition had produced a result: the first addition, the FineWeb learning-rate calibration, was launched 46 seconds before the commit, and its first endpoint finished more than two minutes after it. The plan fixes the output, both parameter conventions, the robustness samples, the estimators, the inference (a wild cluster bootstrap

by width and a parametric bootstrap with the within-trunk covariance from the seed replicates), six questions and a commitment to report every answer. Only the neutrality question (Q2) has a decision rule; Q1 is a test of equal curvature across corpora, Q3 a signed statistic, and Q4–Q6 are descriptive.

The power, size and coverage of the plan’s inference were computed from the design alone between 15:52 and 17:03, before the first estimation at 17:12 (Table B4). The plan’s wild cluster bootstrap with eight width clusters is anti-conservative on this design: its intervals cover 40–83 percent at a nominal 95, and its test of a zero tilt rejects a true null about 20 percent of the time. That study used 40 to 150 replications. A second study from the design alone, run on 25 September for Amendment 2 before any FineWeb technology estimate, uses 500 replications for the tilt test and for coverage at a noise s.d. of 0.005 (150 to 200 in sensitivity cells), adds the FLOP-effective convention and κ -family truths, and gives a Monte Carlo standard error for every rate (Table B5). It shows that the restricted test of D8 is less reliable than 40 replications suggested: on one validation set it rejects a true zero tilt 9.2 percent of the time (Monte Carlo s.e. 1.3 points) and a tilt of 0.22 only 65 percent of the time, because the residuals of the fit that imposes a zero tilt carry the tilt; its two-set rule calls a tilt of 0.22 neutral 15 percent of the time. Amendment 2 therefore uses, in the CR2 scheme, the CR2 interval with its half-width multiplied by 1.25. Its two-set rule calls a tilt of 0.22 factor-biased in every replication, and a true zero tilt factor-biased 5.4 percent of the time, against 3.0 percent for D8’s rule. At that noise level the model-free σ^* in FLOP-effective parameters has a Monte Carlo standard deviation of 0.019 on FineWeb-Edu’s design and 0.014 on FineWeb’s. The CR2 scheme’s intervals for the model-free σ^* in FLOP-effective parameters cover 74 percent at a noise s.d. of 0.005, and 92 percent once widened by 1.74, the largest calibration factor across noise levels, within-trunk correlations, conventions and truths (85 percent at a within-trunk share of 0.9); the factor is 1.75 for the Q3 slope and 1.64 for the difference in σ^* between corpora, whose smallest value detectable with 80 percent power is then 0.075. The null value of the Q3 slope is not zero: at noise s.d. 0.002 to 0.010 it is -0.01 to -0.09 in convention P, and in convention T it is $+0.11$ with the high- M runs, or -0.08 on the main grid without noise (-0.08 to -0.12 at noise s.d. 0.002–0.005), because the Chinchilla form is misspecified in total-parameter units when it holds in non-embedding units. Q3 is therefore read against a design-specific null, recomputed at the noise measured in Q4 (Amendment 2).

Table B3 lists every deviation from the plan with its time, reason and effect; five bear most on the questions: the trimmed high- M runs, the duplicated cells, the missing neutral sets, D8 and the change of comparison convention (D15). D8 is a size-corrected inference that we call the CR2 scheme. Every table reports the plan’s intervals beside intervals from CR2 cluster residuals with the bandwidth re-selected in each draw; the tilt test becomes a restricted wild cluster bootstrap on CR2 residuals, so that “excludes zero” in Q2 means a p -value below 0.05 and “neutral” requires $p \geq 0.05$ and a CR2 half-width below 0.10 on both validation

sets; a conclusion is called robust only under the plan’s scheme, the CR2 scheme and the seed-based bootstrap; and model-free and Q3 intervals are widened by calibration factors before a statement is made. D8 was written into the analysis memo at about 17:05, before the first estimation, and first committed at 17:44 (a5f47f8); its timing rests on our account, and the coverage outputs that fix its calibration factors carry file times of 16:55 to 17:02. Amendment 2 replaces its restricted test by the calibrated CR2 interval and its calibration factors by those of the second study.

Amendment 1 to the plan, committed at 17:44 (a5f47f8), maps each outcome of Q1–Q3 to the sentences of the abstract, the introduction and Sections III–IV that it would change; its committed header misstated the time of writing as about 18:10, after the commit itself, and b2e9bf8, eight seconds later, changed it to 17:40. It was written before any FineWeb technology estimate or tilt, but after the FineWeb learning-rate calibration and the matched-cell loss differences between corpora had been computed, and after preliminary FineWeb-Edu estimates: a model-free σ^* of 0.666 in convention P (0.665 in P6) and 0.607 in T, κ -family values of 0.662 and 0.605, and a Q3 slope of -0.29 with a level gap of -0.69 , not decided under D8 (files of 17:26–17:28). At a referee’s request it also replaced the plan’s convention for comparisons with public designs that count all parameters, the total count, by FLOP-effective parameters, $N_F = C/(6D)$; in this design N_F is 1.03 to 1.07 times the total count, so FineWeb-Edu’s position relative to the amendment’s band of $[0.60, 0.76]$ was effectively known when the band was drawn. Amendment 2 therefore declares FineWeb-Edu’s arms of Q1 and Q3 exploratory; the confirmatory content is the FineWeb arms of Q1 and Q3, the test of equal σ^* and Q2. The power and coverage outputs, the pipeline and the analysis memo were committed with the amendment. It gives about 17:05 as the start of estimation; the run logs record the first estimation at 17:12, and 17:05 is when the CR2 scheme was written. The public repository of the replication package (<https://github.com/yigitokar/scaling-laws-as-production-functions>), first pushed at 18:10 on 24 September 2026, provides an external timestamp for the plan, the amendment and the power study. It is a snapshot whose first commit, 59c5707, contains these files unchanged; it does not carry the working repository’s history, so the earlier commit hashes cannot be verified from it. That history is not public because it contains third-party files without redistribution licenses; the plan and Amendment 1 carry the commit hashes and times given here, and the public repository timestamps Amendment 2 and the replication snapshot.

Three mapped statements were changed after Amendment 1, each on the evidence of the public designs alone. The headline’s conditioning of σ^* on compute (about 0.7 at 10^{19} – 3×10^{20} FLOP, rather than the 10^{19} – 10^{21} FLOP named in Amendment 1) was narrowed (Section III.B). Chinchilla’s parameter convention was changed: Amendment 1 fixes the comparison with the public designs in FLOP-effective parameters, but when it was written Chinchilla’s estimate was in

total parameters (0.673, and a study-level mean of 0.693); it is now in FLOP-effective parameters, with FLOPs per token rebuilt from Hoffmann et al.’s architecture table (0.660 and 0.687; Online Appendix B2 and Table D2), and the amendment’s thresholds of 0.60 and 0.76 are unchanged. The number of recipes in the extrapolation statement was also changed afterwards. Amendment 1’s map for Q3 was written when the check covered Farseeer alone (“in one recipe (Farseeer)” if the slope is undetermined, “two recipes (Farseeer and our experiment)” if it lies below its null). The same check on Marin’s and Llama 3’s public IsoFLOP profiles now covers three recipes (Section IV.D), and the introduction’s statement was then revised to a recipe-dependent direction; no estimate from this experiment enters any of the three changes. Amendment 2, committed at [[X1: time]] and pushed to the public repository at [[X1: time]] on [[date]], before any FineWeb technology estimate, restates the map against the sentences of this version, defines when an estimate lies inside or outside the band, fixes the calibration factors (1.74 for the model-free σ^* , 1.75 for the Q3 slope, 1.64 for the difference in σ^* between corpora and 1.25 for the tilt), replaces D8’s restricted tilt test and sets the conditions for a robust factor-biased verdict; it also declares the retrained FineWeb-Edu runs and their analyses, and lists what had been seen when it was written.

B5. Released Models

The Epoch universe. From the 2,070 language-model rows of the Epoch AI database (Epoch AI, 2026) (snapshot of September 23, 2026) we exclude 1,728, among them fine-tunes and other derived models, rows without N or a recoverable token count, multimodal and non-text models, encoders, benchmark-scale research models and 29 manual exclusions with recorded reasons; the replication package lists every count. D is dataset size times epochs; seven word counts are converted to tokens, 67 rows with operation-counted compute but no dataset size get $D = C/(6N)$, and six token counts are corrected from primary sources (GPT-3 175B, 300B tokens; GLM-130B, 400B).

The verified sample. It starts from the 148 base models of the ObsScaling evaluation file (Ruan, Maddison and Hashimoto, 2024), with nine documented corrections (for example, OPT’s D is 300B tokens, not its 180B-token corpus), drops 17 models with undisclosed or ambiguous token counts or that are upcycled, duplicated or instruction-tuned, replaces the deduplicated Pythia suite with the standard one (Biderman et al., 2023), and adds 57 models released between September 2023 and November 2025 whose token counts come from model cards or technical reports. Excluding 15 code-specialized models leaves 173 general-purpose models. For 143 of the 188 models (131 of the 173 general-purpose ones) N is the exact count of floating-point tensors in the Hugging Face checkpoint, excluding integer buffers such as stored causal masks; mixture-of-experts models enter with active parameters, and configuration files give the non-embedding N . For OLMo 2 we count pretraining tokens plus the mean mid-training budget of

TABLE B3. DEVIATIONS FROM THE PRE-ANALYSIS PLAN OF THE CONTROLLED EXPERIMENT

No.	Deviation (reason)	Time (source)	Effect on the questions
<i>Panel A. Departures from the plan</i>			
D1	High- M runs trimmed (GPU time): FineWeb has only the (128, 4) shape; (128, 2) not continued to 3.2B	07:48 (commit)	Less reach for Q3 on FineWeb; high- M tuning checks on FineWeb-Edu only, to 1.6B
D2	Reboot (hardware); interrupted trunks rerun, same seed and data order	15:32 (system); 15:37 (log)	Endpoints of a trunk from two runs; duplicates checked
D3– D7, D9– D11	Details the plan left open: bytes per scored token (D3); σ^* identity under P, and P6 (D4); bandwidths, compute levels (D5); intervals, joint draws (D6); WikiText-103 via matched cells (D7); power-study truth (D9); Q6 bins (D10); Q5 argmins (D11)	17:44 (commit); D9 16:44 (file)	FineWeb validation levels (D3); quality label from matched cells (D7); else none
D8	CR2 scheme beside the plan’s intervals (which cover 40–83 percent): CR2 residuals, bandwidth re-selected, restricted tilt test, calibration factors	17:05 (account); 16:55–17:02 (file); 17:44 (commit)	Robust only under all three schemes; factors and tilt test replaced by Amendment 2
D12	Clusters are architectures; duplicated cells ((256, 4) at 0.2B and 0.8B; width-128 seed run at 0.2B) left out of the fits	17:44 (commit)	Duplicates check nondeterminism
D13	Neutral-set losses computed after training from saved weights (FineWeb widths 320–640, FineWeb-Edu (256, 4), seed replicates): job lists predated the update of 17:44	23:07 (commit); 00:49 [†] (commit)	No neutral-set tilt on the main grid; matched-cell sign checks
D14	Second reboot (hardware): FineWeb trunks at width 640 (200M) and (128, 4) (1.6B, 3.2B) rerun; the rerun width-640 endpoint has all five sets	03:17 [†] (system); 10:07 [†] (log)	As D2
D15	Amendment 1: $N_F = C/(6D)$, not total parameters, is the comparison convention (referee request)	17:40 (account); 17:44 (commit)	Made after FineWeb-Edu’s total-parameter value was seen: its Q1, Q3 arms exploratory
D16	FineWeb-Edu widths 384 and 512 retrained with every evaluation set and saved weights (matched neutral-set cells)	11:34 [†] (log); 11:35 [†] (commit)	Exploratory; originals stay in the fits

Note: Panel B and the notes follow on the next page.

TABLE B3 (CONTINUED). DEVIATIONS FROM THE PRE-ANALYSIS PLAN OF THE CONTROLLED EXPERIMENT

No.	Deviation (reason)	Time (source)	Effect on the questions
<i>Panel B. Changes to the outcome-to-headline map of Amendment 1</i>			
A1	Window of the headline σ^* narrowed from 10^{19} – 10^{21} FLOP	22:27 (commit)	Public designs only
A2	Chinchilla’s σ^* in N_F : 0.673 to 0.660; study-level mean 0.693 to 0.687	00:21 [†] (commit)	Band unchanged
A3	Q3: three public recipes, then a recipe-dependent direction beyond the designs	22:27 (commit); round 3 [†]	Map restated in Amendment 2
A4	Amendment 2: definitions, restated map, Q2 robustness, calibration factors, CR2-scheme tilt rule, added runs	11:45 [†] (account); [[X1: time]] (commit)	Before any FineWeb technology estimate

Note: Times are UTC+3 on 24 September 2026, or 25 September ([†]). Time sources: commit, the git time of the first commit with the change; log, a queue-script timestamp; file, a file time; system, the reboot record; account, our account. Seen before Amendment 1 (files of 17:26–17:28): FineWeb-Edu’s model-free σ^* of 0.666 (P), 0.665 (P6) and 0.607 (T), κ -family values of 0.662 and 0.605, a Q3 slope of -0.29 and a level gap of -0.69 ; FineWeb’s learning-rate calibration; between-corpus loss differences at matched cells. Before Amendment 2, also raw losses in the queue logs. No FineWeb technology estimate or tilt existed before either. Source: analysis memo, section 6.

the runs averaged into each released model (the report’s totals, which add every averaged run, are 5.6 and 6.6 trillion tokens for the 13B and 32B). MPT-30B is 1.05 trillion tokens (1 trillion at 2k context and 50 billion at 8k). Yi uses 3 trillion tokens, as in its report’s over-training statement and model card; the report also gives 3.1 trillion.

The clean sample and decision units. Nine steps, listed with their sources in Online Appendix E2, reduce the verified sample to 77 models from 36 families and 18 developers, released 2023–2025. An audit of the clean sample’s token counts against primary sources corrected StableLM-Alpha: 800 billion training tokens, not the 1.5-trillion-token dataset size, and 3.64 and 7.87 billion parameters, not the nominal 3 and 7 billion; no headline median moves. Qwen3’s report documents the tokens of every size, but Qwen2.5 states only “up to 18 trillion” tokens for all sizes, so its members’ D is flagged as unaudited and robustness rows drop both families. A decision unit is one choice of (N, D) : the 11 families that train every size on token budgets within 10 percent of each other (32 models) are one unit each, and the 30 members of families with size-specific budgets and the 15 singletons are one unit each, for 56 units. For each common-budget family the technical reports and release posts are read for whether the budget was chosen, set by a binding corpus cap, or accompanied by sizes from a hardware menu (Online Appendix E5); every quote is saved with its URL and retrieval time and checked against the saved source.

Serving at the model level. A model is coded as served if a first-party per-token API or consumer product offered this size (any post-trained variant of the

TABLE B4. POWER, SIZE AND COVERAGE IN THE CONTROLLED EXPERIMENT, STUDY 1 (COMPUTED BEFORE ESTIMATION)

Statistic	Log-loss noise s.d.			
	0.002	0.005	0.010	
<i>Panel A. Expected standard errors (Monte Carlo s.d.), FineWeb-Edu design</i>				
σ_κ^* , convention P / T	0.003 / 0.004	0.007 / 0.007	0.014 / 0.014	
Model-free σ^* , P, FLOP-optimal path / $w = 1$ path	0.009 / 0.005	0.019 / 0.013	0.028 / 0.020	
Model-free σ^* , T	0.008	0.018	0.034	
Q3 slope, P, with high- M runs / main grid only	0.030 / 0.030	0.062 / 0.073	0.107 / 0.119	
Q3 slope, T, with high- M runs	0.018	0.044	0.088	
<i>Panel B. Between-corpus tilt $\hat{\chi}$, convention P (s.e. and detectable tilt: Monte Carlo s.d. with a normal approximation; probabilities: plan's wild bootstrap and decision rule)</i>				
S.e. of $\hat{\chi}$	0.007	0.018	0.038	
Minimum detectable χ (80% power)	0.020	0.051	0.107	
Pr("factor-biased" $\chi = 0.22$)	1.00	1.00	1.00	
Pr("factor-biased" $\chi = 0$)	0.17	0.08	0.12	
Pr("neutral" $\chi = 0$)	0.57	0.50	0.58	
<i>Panel C. Coverage of nominal 95 percent intervals and size, noise s.d. 0.005</i>				
	Plan (wild)	CR2, fixed h	CR2 scheme	MC s.d./ boot. s.e.
σ_κ^*	0.83	0.88	–	1.12
Model-free σ^* , FLOP-optimal path	0.40	0.58	0.65	1.73
Model-free σ^* , $w = 1$ path	0.45	0.65	0.70	1.57
Q3 slope	0.55	0.75	0.68	1.39–1.75
Tilt $\hat{\chi}$, $\chi = 0.22$ / $\chi = 0$	0.68 / 0.80	0.78 / 0.93	–	1.41 / 0.95
Size of the interval test of $\chi = 0$	0.20	0.075	–	
Size of the restricted test, raw / CR2	0.075 / 0.050			

Note: Monte Carlo from the design alone (design: `code/sweep/run_grid.py`; code: `code/analysis/m9_sweeps/m9_power.py` and `m9_coverage.py`), computed on 24 September 2026 between 15:52 and 17:03, before any estimation on the experiment's results (first estimation at 17:12). Truth: the Chinchilla form with the exponents of Besiroglu et al. (2024) ($\sigma^* = 0.737$) in non-embedding parameters, anchored at $M^* = 20$ and 3.8 nats at the design's central compute; FineWeb-Edu differs from FineWeb by a data-augmenting shift χ . Noise: $\ln L$ error with a component common to each training trunk (share 0.5 of the variance) and a second validation set with noise correlated 0.8. Panels A–B: 150 replications per cell; the decision-rule rows use 60 replications with 199 Webb draws each and the plan's rule on two validation sets. Panel C: 40 replications (Monte Carlo error of a coverage rate about ± 0.07); coverage is measured against each estimator's noise-free value on the design. Plan: wild cluster bootstrap by width on the residuals of the unrestricted fit. CR2: the same draws on bias-reduced (CR2) cluster residuals; CR2 scheme: for model-free statistics and the Q3 slope, with the bandwidth h re-selected in each draw (deviation D8). Restricted test: wild cluster bootstrap that imposes $\chi = 0$. Last column: the factor by which the best scheme's standard error understates the Monte Carlo standard deviation. Table B5 replaces these estimates with more replications. Convention P: non-embedding parameters with compute measured in actual training FLOPs; T: total parameters with $C = 6ND$.

Source: Author's calculations.

TABLE B5. POWER, SIZE AND COVERAGE IN THE CONTROLLED EXPERIMENT, STUDY 2 (COMPUTED FROM THE DESIGN BEFORE ANY FINEWEB TECHNOLOGY ESTIMATE)

	Log-loss noise s.d.			
Statistic	0.002	0.005	0.010	
<i>Panel A. Monte Carlo s.d. and null values, FineWeb-Edu design (FineWeb in brackets)</i>				
Model-free σ^* , N_F	0.009	0.019	0.036	
$\sigma^*_{\text{edu}} - \sigma^*_{\text{web}}$; model-free, N_F	0.010	0.024	0.047	
Null of the Q3 slope, P, with high- M runs	-0.023 [-0.014]	-0.052 [-0.034]	-0.092 [-0.045]	
Null of the Q3 slope, T, main grid only	-0.094 [-0.084]	-0.124 [-0.109]	-0.159 [-0.133]	
<i>Panel B. Coverage of nominal 95 percent intervals, noise s.d. 0.005</i>				
	Plan (wild)	CR2 scheme	MC s.d./ boot. s.e.	Calibrated CR2
Model-free σ^* , N_F	0.52 (0.02)	0.74 (0.01)	1.44 (0.04)	0.92 (0.01)
Model-free σ^* , P, FLOP-optimal path	0.52 (0.02)	0.75 (0.01)	1.38 (0.04)	0.93 (0.01)
Model-free σ^* , P, $w = 1$ path	0.63 (0.02)	0.81 (0.01)	1.30 (0.04)	0.95 (0.01)
Model-free σ^* , T	0.53 (0.02)	0.76 (0.01)	1.42 (0.04)	0.93 (0.01)
Q3 slope, P	0.51 (0.02)	0.71 (0.01)	1.37 (0.05)	0.91 (0.01)
$\sigma^*_{\text{edu}} - \sigma^*_{\text{web}}$, N_F	0.52 (0.02)	0.74 (0.02)	1.43 (0.05)	0.92 (0.01)
rejects $\sigma^*_{\text{edu}} = \sigma^*_{\text{web}}$	0.46 (0.02)	0.26 (0.02)		0.10 (0.01)
<i>Panel C. Tests of $\chi = 0$, noise s.d. 0.005</i>				
	Rate at $\chi = 0$		Rate at $\chi = 0.22$	
Plan's interval excludes zero (one set)	0.270 (0.020)		1.000 (0.000)	
CR2 interval excludes zero (one set)	0.214 (0.018)		1.000 (0.000)	
Plan's interval covers χ (one set)	0.730 (0.020)		0.756 (0.019)	
CR2 interval covers χ (one set)	0.786 (0.018)		0.820 (0.017)	
Restricted test on raw residuals, $p < 0.05$ (one set)	0.124 (0.015)		0.974 (0.007)	
Restricted test on CR2 residuals, $p < 0.05$ (D8; one set)	0.092 (0.013)		0.652 (0.021)	
CR2 interval $\times 1.25$ excludes zero (Amendment 2; one set)	0.118 (0.014)		1.000 (0.000)	
“Factor-biased”: plan's rule / D8's rule	0.134 / 0.030		1.000 / 0.468	
“Factor-biased”: Amendment 2's rule	0.054 (0.010)		1.000 (0.000)	
“Neutral”: plan's rule / D8's rule	0.624 / 0.848		0.000 / 0.148	
“Neutral”: Amendment 2's rule	0.818 (0.017)		0.000 (0.000)	
<i>Panel D. Calibration factors fixed by Amendment 2</i>				
Model-free σ^* (largest over cells, conventions and truths)	1.74 (0.13)			
Q3 slope (largest 1.50 (0.10); floor 1.75)	1.75			
$\sigma^*_{\text{edu}} - \sigma^*_{\text{web}}$, N_F (largest)	1.64 (0.10)			
Tilt, CR2 interval (largest)	1.25			
Detectable difference (80% power): correct size / CR2 / calibrated	0.068 / 0.054 / 0.075			

Note: Monte Carlo from the design alone (code: `m9_power_v2.py` and `m9_coverage_v2.py` in `code/analysis/m9_sweeps`), computed on 25 September 2026 before any FineWeb technology estimate or tilt, reading no endpoint loss. Data-generating process as in Table B4, plus κ -family truths ($\sigma_{\kappa}^* = 0.70$; $\kappa = 0.36$, the FineWeb-Edu dry-run estimate, or 0.6) and $N_F = C/(6D)$. Panels A and B: Chinchilla-form truth, within-trunk share 0.5; 500 replications per corpus design at noise s.d. 0.005 (pooled over the two designs in Panel B) and 150 at 0.002 and 0.010, with 99 bootstrap draws (49 in cells of 150–200 replications); Q3 nulls from 500 replications (Monte Carlo s.e. at most 0.013). Panel C: 500 replications per value of χ , 99 draws, two validation sets with noise correlated 0.8. Panel D: the largest ratio of the Monte Carlo s.d. to the mean CR2-scheme bootstrap s.e. across noise s.d. 0.002–0.010, within-trunk shares 0.5 and 0.9, the $\kappa = 0.36$ truth and conventions N_F , P and T (for Q3 at least 1.75). Coverage is of each estimator's noise-free value; a calibrated interval has its half-width multiplied by the factor of Panel D, and covers less than 0.95 because the CR2-scheme half-width is only about 1.85 bootstrap s.e. (light tails with eight clusters, also with 999 draws). "Factor-biased": zero excluded on both validation sets with the same sign; "neutral": zero included on both with a half-width below 0.10 (D8: restricted-test $p \geq 0.05$ and a CR2 half-width below 0.10). The noise-free $\sigma_{\text{edu}}^* - \sigma_{\text{web}}^*$ on the design is 0.005, which adds to its rejection rates. Detectable differences: normal approximation with the Monte Carlo s.d.

Source: Author's calculations.

same run) within 180 days of its release. The evidence is dated first-party pages, Internet Archive captures of Alibaba’s model catalogs, and dated press. Self-deploy catalogs (Vertex AI Model Garden, the Azure AI Studio catalog), where the customer rents the accelerators, and research demos (Hugging Face Spaces, the Ai2 Playground) are coded 0 and kept as flags. Twenty-six of the 77 models are served, 19 of them Alibaba’s. Of the 77 codes, 48 rest on the model’s own dated evidence, 13 on a sibling’s from the same release, 2 on an undated page of the same developer, and 14, all coded 0, on the developer-level coding of the previous version without a model-level source; every code of 1 has a dated, quote-verified source. Three sizes are ambiguous (Llama 3 8B and 70B behind “Meta AI, built with Llama 3 technology”, and Yi-34B’s unnamed early-access API); they are coded 0, and 1 in a sensitivity. A model has an on-device target if its card names on-device, mobile or phone use.

Production-scale universe. Trends and the open-weight comparison use models trained with $6ND \geq 10^{21}$ FLOP: Epoch rows that are not speculative, not code models and not matched to the verified sample (176), plus the verified general-purpose models above the threshold (159), for 335 models from 94 developers after name variants are merged. The open-closed comparison uses the 186 with confident parameter and token counts, without mixture-of-experts or non-transformer architectures and with the clean-sample exclusions applied to verified models; the bunching test uses the 220 open-weight models released in 2023–2026 with their total parameter counts.

Derivatives and units. From the Hugging Face model tree we count, for 74 clean models, the quantizations, fine-tunes, adapters and merges derived from the base repository and official releases. Wedges are computed in each technology’s own parameter convention, with D in each model’s own tokens; under the reference technology $\partial \ln w / \partial \ln D = 0.43$, so a 20 percent difference in tokenizer efficiency moves $\ln w$ by about 0.08.

B6. Licenses

Epoch AI data are released under CC-BY; ObsScaling, the OLMo ladder, Pythia, DataDecide and the Muennighoff et al. data under Apache-2.0 or ODC-BY; Porian et al. and Gadre et al. under MIT. The FarSeer and Step Law repositories and Epoch’s Chinchilla-extraction repository carry no license; we use them for estimation and redistribute only derived aggregates, with download scripts. The Llama 3 and Marin points we use are taken from the compilation of [Open Athena \(2026\)](#) (Apache-2.0). FineWeb, FineWeb-Edu and C4 (ODC-BY), WikiText-103 (CC BY-SA) and PG-19 (Apache-2.0) are not redistributed. Model cards, Hub metadata, model-tree counts and the saved primary sources for family readings and serving codes were used to verify facts and are not redistributed.

APPENDIX C. MONTE CARLO DESIGN AND RESULTS

This appendix documents the Monte Carlo design behind Section II: what alternative experimental designs identify when they use the same total compute. The truth is the technology of Besiroglu et al. (2024): $E = 1.8172$, $A = 482.01$, $B = 2085.43$, $\alpha = 0.3478$ and $\beta = 0.3658$, so that $a = 0.513$, $\gamma = 0.178$, $\sigma^* = 0.737$ and $\ln M^*(10^{24}) = 2.898$ ($M^* = 18.1$). “Bias” is the Monte Carlo mean error and “RMSE” the root-mean-squared error around the truth. No estimator is started at the true parameters.

Design. Every design has 90 runs and the same total compute, 5.1×10^{22} FLOP, that of ten runs on each of Chinchilla’s nine nominal IsoFLOP budgets (6×10^{18} to 3×10^{21} FLOP). All designs except the factorial grid place ten runs on each of these budgets; the factorial grid crosses nine values of N with ten of D , each spanning a factor of 256 around the expansion path and scaled to the same total compute, so that its runs sit at 89 compute levels from 1.8×10^{17} to 1.2×10^{22} FLOP. Apart from that benchmark, the designs differ only in where the runs sit on their isocost lines (Table C1): near the expansion path, as in a compute-optimal ladder; spread by IsoFLOP profiles or by a factorial grid; spread by allocation errors with standard deviation ς in $\ln(D/N)$ at given compute, as by optimizing labs that err; or tilted by labs that follow Kaplan’s allocation rule. Log loss carries normal noise with standard deviation 0.0075, about the residual standard deviation (0.0076) of the Besiroglu fit on the 240-run Chinchilla sample, an upper value for pure seed noise. Three estimators are compared. The primal is the Huber ($\delta = 10^{-3}$) fit of $\ln L$ with $\kappa = 1$ (Hoffmann et al., 2022; Besiroglu et al., 2024), from nine starting values drawn at random within the region of Hoffmann et al.’s initialization grid ($\ln A, \ln B \in [0, 25]$, $\ln E \in [-1, 1]$, $\alpha, \beta \in [0.1, 1.2]$). The dual reads a from the optima (or IsoFLOP minima) and γ from the frontier and recovers $\alpha = \gamma/a$ and $\beta = \gamma/(1 - a)$, which uses optimality and $\kappa = 1$. The third is a profile likelihood for σ^* with κ free, on a 29-point grid over $[0.05, 0.99]$, started at each grid point from the members of the observationally equivalent family of Proposition A1(iv) that reproduce the primal and dual fits.

On-path data do not identify the technology through the primal estimator. When every run is exactly compute-optimal ($\varsigma = 0$), 87 percent of fits end on a corner, with one power-law term switched off, and the RMSE of $\hat{\sigma}^*$ is 0.27, against 0.004 for an IsoFLOP design spanning $\pm 16\times$ the optimal size and 0.005 for the factorial grid. No corner fit has a higher objective than the true parameters, so the data are fitted as well by a technology with one input switched off as by the truth. Near the path the failure is a ridge: at $\varsigma = 0.02$ the median error of $\hat{\sigma}^*$ is -0.12 and the 5–95 percent range of $\hat{\sigma}^*$ is $[0.34, 0.75]$, and the normalized condition number of the information matrix rises from 1.4×10^3 to 4.0×10^3 in the designed experiments to 1.5×10^7 . As ς rises to 0.05, 0.1 and 0.2, the RMSE of $\hat{\sigma}^*$ falls to 0.089, 0.025 and 0.015. The compute-optimal ratio is more demanding once optimality is not imposed: the RMSE of $\ln \hat{M}^*(10^{24})$ is 28 at $\varsigma = 0$, 4.6 at 0.1, 2.0 at 0.3 and 0.59 at 1, against 0.16 under the IsoFLOP design. The RMSE of

$\ln \hat{M}^*$ falls roughly as $1/\varsigma$, and the κ -free profile sets below narrow roughly as $1/\varsigma^2$ (median widths 0.104 at $\varsigma = 1$ and 0.028 at $\varsigma = 2$): these are the finite-sample counterparts of the information rates of Proposition A2. With $\kappa = 1$ imposed, the RMSE of $\hat{\sigma}^*$ instead flattens, for the reason given next.

With $\kappa = 1$ imposed, apparent precision at intermediate ς is functional form. For $0.1 \leq \varsigma \leq 0.3$ the primal RMSE of $\hat{\sigma}^*$ (0.013 to 0.025) is no better than that of the dual estimator (0.012), which uses only the path: with $\kappa = 1$, $\alpha + \beta = \gamma/[a(1-a)]$, and $a(1-a)$ is flat near $a = \frac{1}{2}$, so $\sigma^* \approx 2/(2 + 4\gamma)$ is pinned down by the frontier elasticity. Freeing κ removes that information (Table C1, Panel B). The 95 percent profile set for σ^* is the whole grid in 91 percent of on-path replications and 95 percent at $\varsigma = 0$. Larger allocation errors first rule out poor substitutes: at $\varsigma = 0.3$ the median bounds are $[0.42, 0.99]$, and the set contains every $\sigma^* \geq 0.9$ in 76 percent of replications. The asymmetry follows from the Harberger triangle of Lemma A5: an error in $\ln M$ costs a Farrell loss proportional to $(1 - \sigma^*)/\sigma^*$, which vanishes as $\sigma^* \rightarrow 1$. The set becomes informative at $\varsigma = 1$, with median bounds of $[0.69, 0.79]$. In the designed experiments it is tight (median bounds 0.729 and 0.745 for IsoFLOP $\pm 16\times$), and it covers the truth in 93 to 100 percent of replications in every design with independent noise. The dual estimator is precise on the path (RMSE 0.011 for σ^*) but inherits any systematic allocation error: when labs optimize against Kaplan’s beliefs ($N \propto C^{0.73}$) it is biased by -3.92 in $\ln M^*$, understating M^* about 50-fold.

Inference, starting values and noise. On the path the Wald covariance is numerically singular, or the fit is at a corner, in 79 percent of replications, and the pairs bootstrap with every draw warm-started at the full-sample estimate covers σ^* in 44 percent of 150 replications; restarting every draw also from the nine starting values raises coverage from 33 to 78 percent in a separate check, with a median interval width of 0.43 against 0.016 under the IsoFLOP design. With designed variation, Wald and bootstrap coverage is 94 to 97 percent. Starting values matter only near the path: with 64 random starts the corner shares on the path and at $\varsigma = 0$ fall from 72 and 87 percent to 18 and 56 percent, with RMSEs of $\hat{\sigma}^*$ of 0.19 and 0.30; for $\varsigma \geq 0.1$ and the designed experiments the start set makes no difference. The RMSE of $\hat{\sigma}^*$ under the IsoFLOP design is 0.0024, 0.0036 and 0.0078 at noise standard deviations of 0.005, 0.0075 and 0.015. With heteroskedastic noise (standard deviation proportional to $(ND)^{-1/4}$) and half of each run’s variance shared within its budget or training trunk, the ranking of designs is unchanged (RMSE 0.0042 for IsoFLOP, 0.0052 for the factorial grid, 0.022 at $\varsigma = 0.3$ and 0.25 on the path), but Wald intervals that assume independence cover σ^* in only 82, 74 and 61 percent of replications under the IsoFLOP, factorial and $\varsigma = 0.3$ designs, and profile sets calibrated as if the noise were independent cover it in 71 to 92 percent. Inference must allow for dependence within budgets or training trunks, as the analysis of our experiment does (Online Appendix B4). The seed standard deviation of $\ln L$ measured in the experiment is [TBD-m9: seed standard deviation of $\ln L$], against the noise standard deviations

of 0.005 to 0.015 simulated here [TBD-m9: below, inside or above that range].

A simulated industry. A second design simulates forty labs that choose compute and allocations over six model generations and asks which production-function estimators recover the technology from released models; it is reported in the companion paper (Okar, 2026). One result bears on Section IV: stacking the loss equation with the training-only first-order condition ($w = 1$) biases $\ln M^*$ by +1.25 to +1.28 under every compute rule, close to the closed form $2 \mathbb{E}[\ln w]/(\alpha + \beta) = 1.28$, because the first-order-condition residual is the wedge of Proposition A3; ignoring it makes over-trained models look like evidence that data are cheap.

Reproducibility. The code is in `code/analysis/m6_montecarlo` (entry point `run_v2.py`). Every design cell seeds its own random-number generator, so results do not depend on the number of processes; on four CPU processes Design A takes 18 minutes.

APPENDIX D. THE TECHNOLOGY: ADDITIONAL RESULTS

This appendix supports Sections I through III: a dictionary between scaling-law practice and production economics; estimators and inference on the Chinchilla data; the model-free estimator, its budget-level estimates and their drift with compute; specification tests across sweeps; the compute cost of over-training; and input measurement and flexible inputs.

D1. A Dictionary

Table D1 lists nineteen correspondences. Three analogies break down: parameters and tokens have no market prices, so size-dependent prices of training and serving FLOPs enter the first-order conditions as wedges; returns to scale measured in loss depend on how output is cardinalized, whereas the elasticity of substitution, the optimal mix and the wedge do not; and the inputs of a designed sweep are assigned, not chosen, so the endogeneity machinery of IO applies to cross-lab data (the companion paper), not to single-lab experiments.

D2. The Chinchilla Data

Estimators and inference. On the 240-run sample (Online Appendix B2; total parameters), Huber loss with $\delta = 10^{-3}$, the estimator of Besiroglu et al. (2024), is least absolute deviations in practice. Least squares, on log loss or in levels, gives a larger data exponent (paired differences in β of 0.039 and 0.060) and compute-optimal ratios at 5.76×10^{23} FLOP lower by 3.7 and 7.6 tokens per parameter. Estimating the loss equation jointly with the nine IsoFLOP argmins under the restrictions of Lemma A3 leaves the estimates unchanged ($p = 0.60$). The 41 runs within 0.15 log points of the expansion path carry less than 1 percent of their input variance across the path, and the condition number of the Jacobian rises from 62 on the full design to 413 on them, against 69 to 95 for random subsamples of the same size. The restriction $\kappa = 1$ is rejected at 1 percent under every bootstrap

TABLE C1. MONTE CARLO DESIGN A: WHAT ALTERNATIVE DESIGNS IDENTIFY AT EQUAL COMPUTE

Design	Primal estimator ($\kappa = 1$)				Dual estimator	
	a	γ	σ^*	$\ln M^*(10^{24})$	σ^*	$\ln M^*(10^{24})$
<i>Panel A. Bias [root-mean-squared error]</i>						
(i) On path, $\varsigma = 0.02$	0.050 [0.296]	0.058 [0.096]	-0.166 [0.222]	-1.08 [23.8]	0.000 [0.011]	0.00 [0.01]
(ii) IsoFLOP, $\pm 16\times$	0.001 [0.007]	0.000 [0.003]	0.000 [0.004]	-0.02 [0.16]	0.000 [0.016]	0.02 [0.14]
(ii') IsoFLOP, $\pm 4\times$	0.000 [0.015]	0.000 [0.008]	0.000 [0.009]	0.00 [0.31]	-0.001 [0.016]	0.02 [0.36]
(iii) Factorial grid	-0.001 [0.012]	0.000 [0.005]	0.000 [0.005]	0.02 [0.28]	-	-
(iv) Optimizing, $\varsigma = 0$	0.140 [0.331]	0.093 [0.113]	-0.240 [0.272]	-11.8 [27.7]	0.000 [0.012]	0.00 [0.00]
(iv) Optimizing, $\varsigma = 0.1$	0.014 [0.225]	-0.032 [0.051]	-0.006 [0.025]	-0.25 [4.58]	0.000 [0.012]	0.00 [0.05]
(iv) Optimizing, $\varsigma = 0.3$	0.000 [0.098]	-0.005 [0.016]	-0.002 [0.013]	0.00 [1.97]	0.000 [0.012]	-0.01 [0.14]
(iv) Optimizing, $\varsigma = 1$	-0.002 [0.029]	0.000 [0.011]	-0.001 [0.012]	0.04 [0.59]	-0.001 [0.015]	-0.04 [0.48]
(v) Kaplan beliefs	-0.028 [0.265]	-0.028 [0.072]	-0.023 [0.102]	0.34 [4.97]	-0.088 [0.088]	-3.92 [3.92]
<i>Panel B. Inference for σ^* and the profile likelihood with κ free</i>						
	Corner	Wald	Bootstrap	95% profile set, κ free		
	fits (%)	coverage (%)	coverage (%)	Median bounds	Flat (%)	Flat' (%)
(i) On path, $\varsigma = 0.02$	72	100 [†]	44 [§]	[0.05, 0.99]	91	94
(ii) IsoFLOP, $\pm 16\times$	0	96	97	[0.729, 0.745]	0	0
(ii') IsoFLOP, $\pm 4\times$	0	94	-	[0.714, 0.760]	0	0
(iii) Factorial grid	0	95	97	[0.727, 0.746]	0	0
(iv) Optimizing, $\varsigma = 0$	87	100 [†]	-	[0.05, 0.99]	95	97
(iv) Optimizing, $\varsigma = 0.1$	0	99 [†]	-	[0.08, 0.99]	21	93
(iv) Optimizing, $\varsigma = 0.3$	0	93	96 [§]	[0.42, 0.99]	0	65
(iv) Optimizing, $\varsigma = 1$	0	95	95	[0.69, 0.79]	0	0
(v) Kaplan beliefs	0	92 [†]	-	[0.08, 0.99]	21	84

Note: Designs, truth ($a = 0.513$, $\gamma = 0.178$, $\sigma^* = 0.737$, $\ln M^*(10^{24}) = 2.90$) and noise as described in the text: 90 runs with equal total compute, on Chinchilla's nine budgets except in the factorial grid, 500 replications per design; (v) labs follow Kaplan's rule $N \propto C^{0.73}$ with errors $\varsigma = 0.1$. Primal: Huber ($\delta = 10^{-3}$) fit of $\ln L$ from nine starting values drawn at random within the region of Hoffmann et al. (2022)'s initialization grid, none at the truth, inside a parameter box. Dual: a from the (noisy) optima or IsoFLOP minima, γ from the frontier (eight starting values on a grid), $\alpha = \gamma/a$, $\beta = \gamma/(1 - a)$; not computed for the factorial grid. Panel B: corner = estimate on the box boundary or one power-law term below 2 percent of reducible loss at every design point; Wald = LAD-sandwich delta-method 95 percent interval ([†]: covariance singular or corner in more than 5 percent of replications, 79 percent on the path; coverage among computable ones); bootstrap = pairs-bootstrap percentile interval (149 draws, first 150 replications), each draw warm-started at the replication's estimate ([§]: in a separate check with 60 replications and 49 draws, restarting every draw also from the nine starting values moves on-path coverage from 33 to 78 percent and the median width from 0.04 to 0.43; at $\varsigma = 0.3$ both give 100 percent); 95 percent profile set = likelihood-ratio confidence set for σ^* in $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ on a 29-point grid over [0.05, 0.99], with the likelihood ratio measured against the unrestricted optimum (first 150 replications): medians of the lower and upper bounds, share of replications whose set is the whole grid (flat) and share whose set contains all of [0.50, 0.95], the range searched in the first version (flat'). Monte Carlo standard errors of coverage rates near 95 percent are 1.0, 1.8 and 2.8 percentage points with 500, 150 and 60 replications.

Source: Author's simulations; replication code `code/analysis/m6_montecarlo`.

TABLE D1. SCALING LAWS IN THE LANGUAGE OF PRODUCTION ECONOMICS: FULL DICTIONARY

Scaling-law concept	Production / IO concept	What the mapping delivers (where used)	Key reference
<i>Panel A. Technology and cost</i>			
Iso-loss contour	Isoquant	σ , optimal mix and w are ordinal (Lemma 1)	Brandfonbrener et al. 2025
Chinchilla law	Additive; CES if $\alpha = \beta$; quasi-homothetic	$\sigma(w)$, $\sigma^* = 2/(2 + \alpha + \beta)$; interior optima imply $\sigma^* < 1$ (Lemma 1)	Hao and Merrill 2026
Outer exponent κ	Two-level aggregator	$\kappa = 1$ ties IsoFLOP curvature to frontier and path slopes (Prop. 3)	Sato 1967
IsoFLOP line, $C \approx 6ND$	Isocost with common log-cost weights	Optimality equates output elasticities; FLOP prices that vary with size are wedges (Prop. 1)	Shephard 1953
Compute-optimal $N^*(C)$, $D^*(C)$	Conditional factor demands	Allocation exponent $a = \beta/(\alpha + \beta)$ (Lemma A3)	Uzawa 1964
IsoFLOP-minima frontier	Inverse cost function	Frontier elasticity γ (Lemma A3)	Nerlove 1963
Hoffmann et al.'s Approaches 1–3	Cost function and factor demands (1, 2); primal (3)	Design-consistent duality tests (Appendix D2)	Hoffmann et al. 2022
<i>Panel B. Identification</i>			
Compute-optimal or fixed-ratio designs	Collinearity under common prices	a and γ identified; σ^* only by functional form, at rate $n^{-1/4}$ (Prop. 2; Remark A3)	Bond and Söderbom 2005
IsoFLOP sweeps; factorial grids	Assigned inputs, not price shocks	σ^* from IsoFLOP curvature and the frontier slope, form-free (Prop. 3)	Box and Lucas 1959
Wedges of released models	Relative-price and constraint variation	Information on $\ln M^*$ second order, on σ^* fourth order in the log wedges (Prop. 2)	Kiefer and Wolfowitz 1959
Lab “secret sauce”	Transmitted Hicks-neutral productivity	Biases γ across labs; given compute, the mix cannot proxy for it (companion paper)	Akerberg, Caves and Frazer 2015
Hyperparameters	Flexible inputs	Scale-correlated mis-tuning biases a (Appendix D6)	Gandhi, Navarro and Rivers 2020
<i>Panel C. Measurement</i>			
Omitted embedding or head parameters	Scale-correlated input error	Upward bias in local a that vanishes at scale (Appendix D6)	Collard-Wexler and De Loecker 2016
Loss per token; tokenizer	Units of output and of the data input	Exponents survive proportional rescaling; E , A , B , M^* do not	Tao et al. 2024
Benchmark accuracy; Elo	Bounded monotone measurement map	Ordinal objects survive the link; γ and returns to compute do not	Gadre et al. 2025
<i>Panel D. Productivity and wedges</i>			
Algorithmic progress	Factor-augmenting technical change	Drift of M^* signs the Hicks bias $\beta g_D - \alpha g_N$ (companion paper)	Ho et al. 2024
Rebalancing after Chinchilla	Farrell allocative efficiency	An allocative, not a technical, gain (companion paper)	Gundlach et al. 2025
Over-training, $M > M^*(C)$	Input wedge (Raval’s ratio statistic)	Value of compactness, $s = (w - 1)/w$; a serving share only for a developer that serves (Prop. 1)	Raval 2023
Cross-lab scaling regressions	Transmission and selection bias	Bias signed by the compute regime; low-power benchmark (companion paper)	Whitfill 2025

Note: Panel A’s entries are identities of duality theory once compute is measured by the FLOP-accounting approximation $C \approx 6ND$; the others are analogies in which the econometric problem is the same but its source can differ. $M^*(C)$: compute-optimal D/N ; g_N , g_D : rates of parameter- and data-augmenting technical change.

Source: Author’s compilation.

scheme except nine budget clusters that assign each off-profile run to the nearest budget ($p = 0.052$); clustering off-profile runs by their model-size trunk gives $p = 0.004$ and a 95 percent interval for $M^*(5.76 \times 10^{23})$ of $[5.2, 106]$. The CES restriction $\alpha = \beta$ is not rejected at 5 percent by tests robust to the heavy tails of the residuals (restricted-bootstrap $p = 0.17$ for the Gaussian likelihood ratio; heteroskedasticity-robust Wald $p = 0.06$ – 0.09). With κ free, the Huber profile over σ^* has its minimum at 0.701, with a bootstrap-calibrated 95 percent set of $[0.68, 0.72]$; on the 41 on-path runs the profile is flat (95 percent set $[0.20, 0.99]$ with χ_1^2 critical values, the edge of the grid; about $[0.15, 0.99]$ with bootstrap critical values). The Chinchilla-70B wedge contains one under every scheme, an internal-consistency check with little power because DeepMind sized that model on these runs.

Exclusion rules, duality and lab designs. As the j highest-loss runs are dropped, β falls from 0.453 with all 245 runs to 0.347 at $j = 15$, while α stays between 0.345 and 0.349; the five runs that Besiroglu et al. drop are data-starved runs whose influence is leverage, not truncation bias (a truncated-Gaussian likelihood, Hausman and Wise, 1977, returns the same β). The Chinchilla-70B wedge is 1.54 with them and 0.95 to 1.06 under ten of twelve rules that remove all or most of them (the j highest-loss runs, the largest residuals, or runs with D/N below 0.4, 1 or 2); dropping the whole 10^{19} budget gives 0.78, and the 132 IsoFLOP-profile runs without them give 1.16. Against design-consistent nulls, which allow for the finite-grid bias of fitted parabolas (Czech et al., 2026), our refit is consistent with the IsoFLOP argmins ($p \geq 0.16$). Our argmin fit reproduces Meta’s Llama 3 law; on its eight bracketed budgets the allocation exponent is 0.50 (a path slope e_p near zero) and $M^*(10^{24})$ is 15 to 16 instead of 31. For the Marin ladders the design-consistent tests do not reject (p between 0.33 and 0.54).

D3. The Model-Free Estimator and the Elasticity by Compute

Specification. The primary estimator of Section III fits a quadratic in $\ln N$ to the runs within one log point of a pooled IsoFLOP-argmin path, on budgets whose minimum is bracketed by at least two runs on each side, and pools budgets by a random-effects mean. By Remark A6, the noise-free finite-grid bias of such a quadratic is about -0.003 in σ^* on a grid spanning a factor of 9 in N and below 0.001 on grids narrower than a factor of about 3; the primary window spans a factor of $e^2 \approx 7.4$, and on each design’s own fitted technologies the bias is at most 0.006. In the specification checks, pooled F tests of a cubic term never reject at 5 percent, and across windows, polynomial orders, frontier smoothers and window centering the random-effects σ^* ranges over 0.656–0.702 for Chinchilla (0.665–0.714 in total parameters; 0.58 with the narrowest window, which keeps seven of nine budgets), 0.642–0.677 for Llama 3 and 0.66–0.72 for Marin (0.64 with DCLM’s narrowest window, which keeps four budgets); an independent re-implementation, with Chinchilla in total parameters, gives 0.669–0.694. In a Monte Carlo at the fitted κ -family technologies of Chinchilla (in total parameters),

Llama 3 and Marin’s DCLM ladder, the random-effects interval covers the truth in 93 to 95 percent of 500 replications with a bias of at most 0.006, and in 93 to 94 percent when half the noise variance is shared within a budget. It covers 90 to 97 percent also when the shared half attaches to model size across budgets or to a smooth distortion in size, but such a distortion, which a digitized figure can produce, makes the within-design drift test reject a true zero drift 14 to 19 percent of the time (200 replications). On a κ -family fit to Porian et al.’s runs the noise-free bias is about +0.05. The estimator needs exact isocosts, so each design is analyzed in FLOP-effective parameters, the convention in which its profiles are exact isocosts. On Marin, whose accounting is observed, the configuration count gives 0.04 to 0.07 less.

FLOP accounting. If FLOPs per token of the measured parameter count vary with size with elasticity η , a design’s σ^* shifts by about 0.44η . Meta’s accounting is unobserved, and $\eta = \pm 0.1$ for Meta moves the study-level mean to 0.673 or 0.700; in Marin, counting configuration parameters lowers the mean to 0.673, and counting Farseer’s embeddings lowers it to 0.670. For Chinchilla, FLOPs per token rebuilt from Hoffmann et al.’s architecture table (Online Appendix B2) have an elasticity with respect to N between -0.04 and 0.00 by budget, and Chinchilla’s σ^* is 0.660 against 0.673 in total parameters (Table D2). It is 0.648–0.660 across Hoffmann et al.’s printed, implemented and executed counts on the nominal budgets, and 0.690 in non-embedding parameters; moved onto each count’s own exact isocosts, every count and convention gives 0.660–0.669. Of Hoffmann et al.’s counts, only the rebuilt one reproduces their Table A4 and fits the FLOP coordinates of their Figure 4. On the 137 profile runs the rebuilt count gives less than the total count in every specification of the grid, by 0.008 to 0.018 in the eleven specifications that use all nine budgets and by 0.022 with the narrowest window. At the primary specification about a third of the difference comes from one run that enters the 3×10^{19} window. The level is less robust than the direction. The five high-loss runs lie outside every window but move the window centers: without them the primary specification gives 0.681 (0.674 in total parameters). With profile membership drawn on the corrected coordinates it gives 0.668, and with N_F read directly off the digitized coordinates, as for Llama 3, 0.676. Put in place of 0.660, these values give study-level means of 0.691 to 0.694 (Table D3), and Cochran’s statistic across studies falls from 8.3 ($p = 0.04$) to 5.8–6.3 (p between 0.10 and 0.12). Across all of Chinchilla’s windows, samples and memberships the study-level mean runs from 0.674 to 0.702 and Q from 4.4 to 14.8, so we build no claim on the heterogeneity across studies. The reference technology of Section IV stays in total parameters; refitted in N_F , its σ^* is 0.700 with κ free, against 0.701.

Budget-level estimates. Tables D4 and D5 list every bracketed budget with its runs on each side of the minimum. Fixed-effect and random-effects pooling agree on Marin (0.70–0.71) and nearly so on Chinchilla (0.666 against 0.660), but not on Llama 3 (0.628 against 0.660), whose precise large budgets pull the fixed-effect

TABLE D2. CHINCHILLA’S FLOP ACCOUNTING REBUILT: THE MODEL-FREE ELASTICITY BY PARAMETER COUNT

<i>Panel A. Which FLOP count set Hoffmann et al.’s budgets?</i>					
	Table A4	$F/(6T)$,	η , mean	Figure 4 coordinates	
FLOP count	ratios	profile models	of budgets	slope in $\log N$	offset
T4: App. F, no vocabulary	6 of 6	0.98–1.15	−0.023	−0.002 (0.007)	−0.006
A: App. F as printed	0 of 6	1.00–1.65	−0.083	−0.065 (0.012)	+0.045
X: executed	0 of 6	0.99–1.34	−0.055	−0.036 (0.009)	+0.020
6T	–	1	0	+0.020 (0.004)	−0.023
<i>Panel B. Chinchilla’s model-free σ^* (137 profile runs, nine budgets)</i>					
	Pooled		Drift per	σ_b^* at	Change vs.
Parameter count	Random	Fixed	decade	10^{21} FLOP	T , digitized
T , digitized (earlier primary)	0.673 (0.027)	0.684 (0.018)	−0.072 (0.026)	0.617	–
N_F , count T4	0.660 (0.023)	0.666 (0.018)	−0.061 (0.025)	0.609	−0.014
N_F , count A	0.648	0.651	−0.047	0.591	−0.025
N_F , count X	0.658	0.662	−0.054	0.600	−0.015
$P = T - Vd$	0.690	0.703	−0.082	0.627	+0.016
T , architecture	0.674	0.684	−0.072	0.617	+0.000
N_F , A, own isocosts ^a	0.663	0.667	−0.053	0.596	−0.011
N_F , X, own isocosts ^a	0.666	0.670	−0.057	0.603	−0.008
P , own isocosts ^a	0.669	0.678	−0.070	0.615	−0.004
T , own isocosts ^a	0.660	0.664	−0.059	0.608	−0.014
N_F , T4, 6T tokens ^b	0.671	0.680	−0.071	0.619	−0.002
Memo: T , digitized, $\eta = -0.1$	0.626				−0.048
Memo: T , digitized, $\eta = +0.1$	0.714				+0.040

Note: Hoffmann et al.’s (2022) 50 architectures (their Table A9) with vocabulary $V = 32,000$ and context 2,048. T : total parameters, $L(4dd_{kv}h + 2df) + Ldd_{kv}h + Vd$, with h heads of size d_{kv} , width d , feed-forward size f and L layers (attention, MLP, one relative-position projection per layer, one vocabulary matrix), which reproduces every count of Table A9 to 0.8 percent; $P = T - Vd$; $N_F = C/(6D) = F/6$, with F the training FLOPs per token (forward and backward). Count A is Appendix F as printed; T4 drops its embedding and final-logit terms; X drops only the embedding term. Table A4 ratios: printed ratios reproduced to two decimals. η : slope of $\ln(F/6T)$ on $\ln T$ over each budget’s window runs, averaged over the nine budgets (by budget in the replication files). Figure 4 coordinates: within-budget regression of $\log_{10}$ of the digitized FLOP coordinate, net of the nominal budget and of $-\log_{10}(F/6T)$, on $\log_{10} N$ (budget fixed effects; standard errors clustered by budget); flat if the figure plots $6ND$ at token counts set by count F . Panel B: runs on the nominal budgets unless marked; for count T4 these are exact isocosts. The estimator and bootstrap of Section III.B, unchanged (quadratic in $\ln N$ within one log point of the path, log-cubic frontier, 999 draws). With the windows of the total-parameter estimate held fixed, count T4 changes the budget estimates by -0.018 to $+0.001$. Same-run parametric σ^* in N_F : κ free 0.656 (0.028), $\kappa = 1$ 0.690 (0.015). ^aLosses moved from the observed (T4) isocost to the count’s own isocost with the κ -free surface fitted to the same runs. ^bThe reading in which tokens were set by $D = C_b/(6T)$, rejected by the coordinate test (row 6T).

Source: Hoffmann et al. (2022), Appendix F and Tables A4 and A9; Epoch AI’s digitization (Besiroglu et al., 2024); author’s calculations.

TABLE D3. CHINCHILLA’S FLOP ACCOUNTING REBUILT: PROPAGATION TO THE STUDY-LEVEL SUMMARY AND THE DRIFT

	Chinchilla in T	Chinchilla in N_F , T4 (primary)
Study-level mean, 95% CI ^a	0.693 [0.653, 0.733]	0.687 [0.640, 0.734]
$\hat{\varphi}$; $Q(p)$ ^f	0.017; 5.7 (0.128)	0.023; 8.3 (0.041)
FLOP accounting, $\eta = \pm 0.1$ ^b	0.665–0.708 [0.578, 0.751]	0.673–0.700 [0.598, 0.748]
Chinchilla’s other counts ^c	–	0.682–0.697
Chinchilla’s windows, samples ^f	–	0.674–0.702
Range of variant means ^d	0.663–0.708	0.663–0.702
Slope per decade of compute ^e	−0.032 [−0.075, 0.010]	−0.033 [−0.076, 0.011]
Wild cluster p ; CR2 p	0.042; 0.096	0.039; 0.097
Chinchilla and Llama 3 (fixed effects) ^g	−0.058 (0.013)	−0.058 (0.013; 0.014)
Budgets $\leq 3 \times 10^{20}$ FLOP	−0.007	−0.008
σ^* at 10^{21} FLOP, pooled line	0.650 [0.565, 0.736]	0.647 [0.560, 0.735]
σ^* , budgets $\geq 6 \times 10^{20}$ FLOP ^g	0.596 [0.567, 0.624]	0.594 [0.565, 0.622]

Note: The study-level summary and the drift of Tables 1 and D6, with Chinchilla in total parameters (left) and in N_F with count T4 of Table D2 (right; the primary). ^aOne estimate per study (Hoffmann, Meta, Marin, Farseeer); DerSimonian–Laird mean, modified Hartung–Knapp–Sidik–Jonkman interval. ^b $\sigma_{\text{eff}}^* = 2/[2 + S/(1 + \eta)^2]$ for Chinchilla and Meta (left) and for Meta only (right); means, and the widest interval. ^cChinchilla in counts A and X and conventions P and T , on nominal budgets or exact isocosts, and T4 under $6T$ budgets; other studies unchanged. ^dAll study-level variants of Table D7 (FLOP-accounting rows as in the row above), and, right, the rows of note c and the variants of note f . ^eThree-level random-effects meta-regression of 44 budget-level estimates on $\log_{10} C$; cluster-robust (CR2) interval by design; restricted wild cluster bootstrap- t (9,999 draws). ^fWith Chinchilla at 0.668 (profile membership drawn on the T4-corrected coordinates; 147 runs), 0.676 (N_F read off the digitized coordinates) or 0.681 (without the five high-loss runs, which move the window centers), the mean is 0.691–0.694 and Q is 5.8–6.3 (p between 0.10 and 0.12). Across the window, order, smoother and centering specifications of Online Appendix D3, N_F (T4) gives 0.656–0.702 against 0.665–0.714 in T in the eleven specifications that keep all nine budgets (0.58 against 0.60 with the narrowest window, which keeps seven), lower in every specification; propagated to the study-level mean, these specifications on the 137 and 132 runs and the three variants above give 0.674–0.702 and Q of 4.4–14.8. ^gModel-based s.e. and independent-budget intervals in both columns, for comparability; in N_F the joint design-conditional bootstrap gives 0.014 (scaled for excess dispersion) and [0.561, 0.627] (Table D6).

Source: Hoffmann et al. (2022), Appendix F and Tables A4 and A9; Epoch AI’s digitization (Besiroglu et al., 2024); author’s calculations.

mean down. Cochran’s Q rejects homogeneity within Llama 3 ($p = 0.001$), but not within Chinchilla ($Q = 12.7$, bootstrap $p = 0.06$) or Marin’s corpora.

Drift with compute. Table D6 pools the 44 budget-level estimates of six designs (Farseeer’s local path counts as a sixth design) in a three-level random-effects meta-regression on $\log_{10} C$ (Konstantopoulos, 2011). The slope is -0.033 per decade. With Marin’s three corpora as one cluster there are four clusters, and six with the corpora apart, so the cluster-robust (CR2) standard errors (Bell and McCaffrey, 2002; Pustejovsky and Tipton, 2018) have about three degrees of freedom and a wild cluster bootstrap- t with Webb weights has only $6^4 = 1,296$ distinct draws with four clusters; we enumerate them. In the primary row with Marin as one cluster the two tests disagree (CR2 $p = 0.11$, wild $p = 0.049$), and across weighting, scale, test, cluster definition and design fixed effects the p -value lies between 0.03 and 0.49. The design fixed-effects rows, which use only variation within designs, give -0.021 (inverse-variance weights; CR2 $p = 0.49$, wild $p = 0.37$) and -0.026 (unweighted; 0.17 and 0.051 with Marin as one cluster, 0.065 and 0.073 with six).

The drift comes from the budgets above 3×10^{20} FLOP, which only Chinchilla and Llama 3 reach. Across all 17 of their budgets σ^* falls by 0.058 per decade, with a standard error of 0.014 from the joint bootstrap draws of the budget-level estimates, which share each design’s frontier derivative, scaled for the budgets’ excess dispersion (0.009 from sampling error alone; model-based 0.013). A hinge at 3×10^{20} gives -0.152 above the knot. At or below 3×10^{20} FLOP the pooled slope is -0.008 (p between 0.56 and 0.87 in the primary row), but the designs’ own slopes differ there (Wald 13.9, $p = 0.016$), and Llama 3’s is -0.050 (s.e. 0.017). Leaving out each design’s largest budget gives -0.030 (p between 0.19 and 0.29). The five budgets at 6×10^{20} FLOP and above have an inverse-variance mean of 0.594, 85 percent of it from Llama 3 (46 percent at 6×10^{20} and 38 percent at 10^{21}); with the variance of the joint draws, its modified Hartung–Knapp–Sidik–Jonkman interval is [0.561, 0.627]. Without Chinchilla’s five highest-loss runs, which move its 6×10^{20} estimate from 0.550 to 0.665, the mean is 0.598. Farseeer’s local path, which reaches 10^{21} FLOP without bracketed budgets, gives 0.722 (0.008) at 5×10^{20} and 0.658 (0.011) at 10^{21} ; with the latter the six top estimates are heterogeneous ($Q = 21.1$, $p < 0.001$; random-effects mean 0.606, fixed-effect mean 0.625). With Marin as one cluster the pooled line gives 0.680 at 10^{20} FLOP, with a CR2 interval of [0.638, 0.722]. At 10^{19} the CR2 standard error is an eighth of the model-based one, a small-sample artifact, so we report only the point estimate there, 0.713.

In a Monte Carlo on the grids of Chinchilla (in total parameters), Llama 3 and Marin’s DCLM ladder with a known path of $\sigma^*(C)$, the estimator creates no drift when there is none and attenuates true drifts by at most 20 percent, but a single design has little power against -0.06 per decade: 0.31 on Marin’s grid and 0.37 on Chinchilla’s, against 0.72 on Llama 3’s. Count-symmetric windows and a power-law frontier with E fitted leave the Chinchilla and Llama 3 drift at -0.049

to -0.064 , and Porian et al.’s unannealed profiles drift the other way ($+0.040$), so the estimator does not force a negative slope.

The level. One estimate per study (Marin one study) gives a mean of 0.687 with a modified Hartung–Knapp–Sidik–Jonkman interval of $[0.640, 0.734]$. Restricted to budgets up to 3×10^{20} FLOP, it is 0.708 , with an interval of $[0.690, 0.726]$ in which Farseeer’s first-derivative estimate carries 84 percent of the weight. With Farseeer’s bandwidth range treated as uncertainty, the restricted mean is 0.701 , with an interval of $[0.665, 0.738]$, and the mean over all budgets 0.680 . Across FLOP accounting, conventions, Chinchilla’s windows, samples and memberships, estimators and Farseeer’s bandwidths the mean ranges over 0.663 – 0.702 , and the lowest interval end is 0.596 (Table D7).

Farseeer’s local expansion path. Farseeer’s grid has no IsoFLOP profiles, so we estimate the loss surface locally: a Gaussian product-kernel local quadratic of $\ln L$ in $(\ln N, \ln D)$, with cross-validated bandwidths, gives the local wedge $w = (L_c - L_x)/(L_c + L_x)$ of Proposition A5(i) everywhere, and the point with $w = 1$ lies inside the design on eight isocosts from 5×10^{18} to 10^{21} FLOP. The elasticity there, from the local Hessian, has a random-effects mean of 0.703 (simple mean 0.707) and moves between 0.66 and 0.73 with the bandwidth. A first-derivative estimator is more robust: by Proposition A5(iii) the slope of $\ln w$ in $u = \ln(M/M^*(C))$ at $u = 0$ equals $1/\sigma^* - 1$, and a regression of the local $\ln w$ on u and u^2 gives $\sigma^* = 0.708$, between 0.701 and 0.711 under every bandwidth and 0.670 with embedding parameters counted. The quadratic coefficient, 0.020 (s.e. 0.001), shows that $\ln w$ is convex in u , which no member of the κ family allows. Porian et al.’s unannealed profiles give σ^* near 0.51 and are window-sensitive (0.45 to 0.55 across windows of 0.8 to 1.5 log points; 0.33 – 0.34 with a global quadratic); the gap between an unannealed and an annealed loss plausibly varies with D/N at fixed compute, so we keep them out of the range reported in Section III.

D4. The Technology across Sweeps

Across the seven sweep–corpus technologies from five studies, the restriction $\kappa = 1$ is rejected in every sweep (Wald $p < 0.001$; $\hat{\kappa}$ from 0.22 on the OLMo ladder to 0.77 on Chinchilla). CES is not rejected by the Huber-based Wald test in any sweep (p between 0.22 and 0.90), and rank-one curvature, which the κ family imposes, is rejected in one to two sweeps depending on the estimate of E . Table D8 reports σ_κ^* under changes of sample, convention and irreducible loss. On Farseeer it is stable across size exclusions (0.694 to 0.716) except for the parameter-count convention (0.668 with both embedding matrices counted). On Chinchilla it falls to 0.690 and 0.648 when the smallest tenth and quarter of the runs are dropped. The small sweeps are fragile: on Gadre et al.’s corpora it moves between 0.48 and 0.70 when one or two of the six sizes are dropped, so the heterogeneity of σ_κ^* across sweeps may partly reflect how the κ family fits small designs without IsoFLOP profiles. In the Chinchilla form, conventions matter most for the allocation exponent: counting Farseeer’s two untied embedding

TABLE D4. BUDGET-LEVEL MODEL-FREE ELASTICITIES WITH WINDOW DIAGNOSTICS: CHINCHILLA AND LLAMA 3

	Budget (FLOP)	σ_b^*	(s.e.)	Runs left/right	M_b^*
Chinchilla	6×10^{18}	0.672	(0.064)	3/6	26.1
	10^{19}	0.698	(0.060)	4/9	21.1
	3×10^{19}	0.725	(0.045)	3/11	15.7
	6×10^{19}	0.663	(0.043)	7/7	14.9
	10^{20}	0.755	(0.060)	7/7	13.9
	3×10^{20}	0.695	(0.062)	5/5	14.9
	6×10^{20}	0.550	(0.047)	5/4	21.7
	10^{21}	0.609	(0.039)	10/4	21.3
	3×10^{21}	0.569	(0.054)	3/5	28.3
pooled, FE / RE		0.666 / 0.660	(0.018 / 0.023)	Q 12.7	p 0.060
Llama 3	6×10^{18}	0.682	(0.027)	4/9	17.9
	10^{19}	0.676	(0.026)	5/11	14.2
	3×10^{19}	0.734	(0.023)	3/9	12.1
	6×10^{19}	0.749	(0.028)	4/10	16.3
	10^{20}	0.670	(0.050)	8/5	11.7
	3×10^{20}	0.618	(0.013)	9/5	12.8
	6×10^{20}	0.585	(0.015)	9/2	16.9
	10^{21}	0.609	(0.017)	5/4	17.5
pooled, FE / RE		0.628 / 0.660	(0.011 / 0.023)	Q 50.7	p 0.001

Note: Model-free $\sigma_b^* = 2/(2 + S_b)$, S_b the curvature of budget b 's IsoFLOP profile in $\ln N$ at its minimum over the slope of the loss-compute frontier (quadratic on the window $|\ln N - x_0| \leq 1$ around the pooled path, log-cubic or log-quadratic frontier). Only bracketed budgets are listed (at least two runs on each side of the fitted minimum inside the window); runs left/right count them. s.e.: design-conditional wild bootstrap. $M_b^* = D^*/N^*$ at the fitted minimum. Pooled: fixed-effect (inverse-variance) and DerSimonian–Laird random-effects means of the budgets on the S scale mapped to σ^* , with Cochran's Q and its bootstrap p -value. Farseeer: Hessian-based local expansion path at eight compute levels (the only level-specific Farseeer estimator); pooled: plain mean with the bootstrap s.e. that allows for overlapping kernels, and the random-effects mean. Parameters: FLOP-implied $N_F = C/(6D)$ (Chinchilla, with FLOPs per token rebuilt from Hoffmann et al.'s architecture table, Table D2; Llama 3; Marin), non-embedding N (Farseeer); M_b^* in tokens per FLOP-effective parameter. Without Chinchilla's five highest-loss runs (132 runs), its estimates at 6×10^{20} , 10^{21} and 3×10^{21} FLOP are 0.665 (0.063), 0.611 (0.040) and 0.575 (0.049), its other budgets move by at most 0.048, and its random-effects mean is 0.681.

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); Grattafiori et al. (2024); Czech (2026); Held and Marin Community (2026); Li et al. (2025b); author's calculations.

TABLE D5. BUDGET-LEVEL MODEL-FREE ELASTICITIES WITH WINDOW DIAGNOSTICS (CONTINUED):
MARIN AND FARSEER

	Budget (FLOP)	σ_b^*	(s.e.)	Runs left/right	M_b^*
Marin, Comma	3×10^{18}	0.740	(0.122)	3/2	6.1
	9×10^{18}	0.719	(0.040)	3/5	9.9
	1.8×10^{19}	0.674	(0.037)	3/6	11.5
	1.8×10^{20}	0.760	(0.141)	2/4	28.2
	3×10^{20}	0.686	(0.089)	2/4	20.5
pooled, FE / RE		0.701 / 0.700	(0.025 / 0.027)	Q 0.9	p 0.906
Marin, DCLM	3×10^{18}	0.728	(0.154)	3/2	5.5
	9×10^{18}	0.700	(0.045)	3/5	7.9
	1.8×10^{19}	0.691	(0.053)	3/6	10.8
	3×10^{19}	0.735	(0.112)	2/3	10.3
	9×10^{19}	0.767	(0.118)	2/3	12.9
	1.8×10^{20}	0.697	(0.072)	2/4	12.4
	3×10^{20}	0.737	(0.060)	3/4	11.6
pooled, FE / RE		0.712 / 0.713	(0.026 / 0.027)	Q 0.7	p 0.996
Marin, Nemotron-CC	3×10^{18}	0.750	(0.095)	3/2	5.7
	9×10^{18}	0.700	(0.045)	3/5	8.5
	1.8×10^{19}	0.713	(0.066)	2/7	11.9
	3×10^{19}	0.713	(0.078)	2/3	10.9
	9×10^{19}	0.722	(0.073)	2/3	12.0
	1.8×10^{20}	0.670	(0.043)	2/4	11.8
	3×10^{20}	0.733	(0.052)	2/5	12.5
pooled, FE / RE		0.702 / 0.705	(0.022 / 0.023)	Q 1.1	p 0.978
Farseer, local path	5×10^{18}	0.761	(0.037)	—	45.2
	10^{19}	0.715	(0.025)	—	27.4
	2×10^{19}	0.661	(0.012)	—	31.3
	5×10^{19}	0.687	(0.022)	—	34.4
	10^{20}	0.713	(0.017)	—	31.4
	2×10^{20}	0.736	(0.012)	—	29.2
	5×10^{20}	0.722	(0.008)	—	25.0
	10^{21}	0.658	(0.011)	—	27.0
pooled, mean / RE		0.707 / 0.703	(0.005 / 0.013)	I^2 0.85	p <0.001

Note: See Table D4.

TABLE D6. THE ELASTICITY OF SUBSTITUTION BY COMPUTE LEVEL

<i>Panel A. Drift: regression of σ_b^* on $\log_{10} C$</i>						
	Budgets	Slope	Model s.e.	CR2 s.e. [df], four clusters	CR2 p , four / six	Wild p , four / six
Six designs, 3-level RE	44	−0.033	0.010	0.014 [2.8]	0.11 / 0.097	0.049 / 0.036
Six designs, RE, unweighted	44	−0.030	0.008	0.014 [2.7]	0.13 / 0.050	0.054 / 0.048
Six designs, balanced weights	44	−0.034	—	0.011 [2.8]	0.064 / 0.050	0.049 / 0.033
Design FE, inverse variance	44	−0.021	0.009	0.024 [1.6]	0.49 / 0.49	0.37 / 0.37
Design FE, unweighted	44	−0.026	0.009	0.014 [2.7]	0.17 / 0.065	0.051 / 0.073
Budgets $\leq 3 \times 10^{20}$, 3-level RE	37	−0.008	0.014	0.012 [2.5]	0.58 / 0.56	0.87 / 0.82
Largest budget of each design left out	38	−0.030	0.013	0.018 [2.8]	0.20 / 0.19	0.29 / 0.27
Hinge, slope above 3×10^{20}	44	−0.152	0.038	—	—	—
IsoFLOP designs, FE ^b	36	−0.050	0.009	—	—	0.23
Chinchilla and Llama 3, FE	17	−0.058	0.013	0.014 ^c	—	—
Marin, FE	19	−0.001	0.023	—	—	—
Porian et al. (not pooled), FE	24	+0.040	0.006	—	—	—
<i>Panel B. σ^* by compute level (pooled line; CR2 95% interval)</i>						
			10 ¹⁹	10 ²⁰	10 ²¹	10 ^{22 a}
3-level RE, Marin as one cluster			0.713	0.680	0.647	0.615
			—	[0.638, 0.722]	[0.554, 0.741]	
RE, unweighted, six designs			0.718	0.688	0.659	0.629
			—	[0.657, 0.720]	[0.596, 0.722]	

Note: Panel A: regression of the budget-level model-free estimates σ_b^* on $\log_{10} C$; slope per decade of compute. Inputs: 36 bracketed IsoFLOP budgets of Chinchilla, Llama 3 and Marin’s three corpora, and Farseeer’s local expansion path at 8 compute levels as a sixth design; Porian et al.’s unannealed profiles are reported separately. 3-level RE: design random intercepts plus a within-design between-budget variance on top of each budget’s bootstrap variance (REML; inverse-variance weights; Konstantopoulos 2011). RE, unweighted: linear mixed model with design random intercepts. Balanced: correlated-effects weights $1/[k_j(\bar{v}_j + \hat{\varphi}^2)]$ (Hedges, Tipton and Johnson, 2010) (its model s.e. is not reported). FE: design fixed effects, weights $1/\text{se}^2$ unless unweighted, s.e. scaled by the residual variance when above one. Hinge: descriptive. CR2: cluster-robust s.e. with the bias-reduced linearization of Bell and McCaffrey (2002) and Satterthwaite degrees of freedom (Pustejovsky and Tipton, 2018; Tipton, 2015), with four clusters (Hoffmann, Meta, Marin, Farseeer) and with six (Marin’s corpora apart). Wild p : restricted wild cluster bootstrap- t (Webb 2023 weights), computed by enumerating all 6^G weight vectors. With four clusters no cluster-robust p -value is reliable; across every row on the 44 budgets, both cluster definitions and study-intercept variants, p lies between 0.03 and 0.49, and in the primary row the CR2 interval and the wild p -value disagree about significance. On the scale $S = 2(1/\sigma^* - 1)$ the 3-level RE slope has CR2 p 0.15 / 0.13 and wild p 0.060 / 0.062. Chinchilla in FLOP-effective parameters (Table D2). ^bMarin’s corpora as five design clusters, Farseeer left out (CR2 degrees of freedom below 2); across this sample’s specifications p lies between 0.013 and 0.23. ^cStandard error from the joint design-conditional bootstrap draws of the 17 budget-level estimates, which share each design’s frontier derivative, scaled for the budgets’ excess dispersion (0.009 from sampling error alone); two designs allow no cluster inference. Panel B: population-average line of Panel A’s first two rows; no interval at 10¹⁹, where the CR2 standard error is a small-sample artifact (an eighth of the model-based one). ^aBeyond the largest bracketed budget of every design (the largest, Chinchilla’s, is 3×10^{21} FLOP). Panel C and its notes follow on the next page.

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); Grattafiori et al. (2024); Czech (2026); Held and Marin Community (2026); Porian et al. (2024); Li et al. (2025b); author’s calculations.

TABLE D6 (CONTINUED). THE ELASTICITY OF SUBSTITUTION BY COMPUTE LEVEL

<i>Panel C. σ^* at the largest budgets</i>				
	Budget	σ_b^*	(s.e.)	Design line
Chinchilla	3×10^{21}	0.569	(0.054)	0.624
Llama 3	10^{21}	0.609	(0.017)	0.636
Marin, Comma	3×10^{20}	0.686	(0.089)	0.664
Marin, DCLM	3×10^{20}	0.737	(0.060)	0.668
Marin, Nemotron-CC	3×10^{20}	0.733	(0.052)	0.669
Farseeer, local path	5×10^{20}	0.722	(0.008)	—
Farseeer, local path	10^{21}	0.658	(0.011)	0.659
Chinchilla and Llama 3, budgets $\geq 6 \times 10^{20d}$	5	0.594	[0.561, 0.627]	
Without Chinchilla’s five highest-loss runs	5	0.598		
With Farseeer’s 10^{21} level, random effects	6	0.606	[0.558, 0.655]	$Q = 21.1$

Note: Panel C: raw estimate at each design’s largest bracketed budget (Farseeer: its two local-path levels above 3×10^{20}); design line: BLUP intercept plus the common slope (3-level RE). ^dInverse-variance mean of Chinchilla’s three and Llama 3’s two budgets at 6×10^{20} FLOP and above (Llama 3 85 percent of the weight; $Q = 2.5$), with the modified Hartung–Knapp–Sidik–Jonkman interval computed with the variance of the joint bootstrap draws ([0.565, 0.622] with independent budgets). Without Chinchilla’s five highest-loss runs its 6×10^{20} estimate moves from 0.550 to 0.665. Study-level means are in Table D7. *Source:* Hoffmann et al. (2022); Besiroglu et al. (2024); Grattafiori et al. (2024); Czech (2026); Held and Marin Community (2026); Porian et al. (2024); Li et al. (2025b); author’s calculations.

matrices moves a from 0.526 to 0.411. With task bits per byte instead of C4 cross-entropy, the OLMo ladder’s σ_κ^* falls from 0.544 to 0.512: two outputs can have different isoquants (Online Appendix E8).

Neutrality of data quality. What matters for allocation and for the wedge is the tilt $\ln(A_r/B_r)$ between recipes, which is ordinal; asymptote shifts and neutral scalings of reducible loss leave the isoquant map unchanged. On a common C4 validation output, across the three web corpora of Gadre et al. (2025) a Hicks-neutral scaling with corpus-specific asymptotes is not rejected ($p = 0.90$), but neither are the factor-augmenting alternatives. Across the 25 DataDecide recipes (Magnusson et al., 2025) every restriction is rejected at the resolution of the bootstrap (p at its floor with 19 or 99 draws); asymptote shifts alone explain 98.1 percent of the cross-recipe variation in fit, and the remaining tilt, 0.22 to 0.26, implies compute-optimal ratios that differ by a factor of 2.9 to 3.4 and moves revealed wedges by a factor of 1.25 to 1.29 at the same inputs. DataDecide’s losses are taken before the learning-rate schedule ends (Online Appendix B), and on a common validation set part of any corpus difference is distribution match. On final checkpoints only, one ray at M of about 100, the recipes’ tilt is not identified: the fit has two optima of almost equal fit that exchange the roles of the two power terms and reverse the ranking of the recipes’ tilts, so we do not report a final-checkpoint allowance. The sign result’s breakdown frontier (Online Appendix Figure E1) does not depend on DataDecide. DeepSeek-AI et al. (2024b) report that higher-quality data shift the compute-optimal allocation toward model size.

TABLE D7. THE ELASTICITY OF SUBSTITUTION: ONE ESTIMATE PER STUDY (MARIN ONE STUDY), RANDOM-EFFECTS MEAN

	k	Mean	95% CI	$\hat{\varphi}$	Q (p)
Headline ^b	4	0.687	[0.640, 0.734]	0.023	8.3 (0.041)
Budgets $\leq 3 \times 10^{20}$ FLOP ^c	4	0.708	[0.690, 0.726]	0.000	1.0 (0.797)
Farseer’s bandwidth range as uncertainty ^d	4	0.701	[0.665, 0.738]	0.000	0.6 (0.895)
Farseer’s Hessian path	4	0.702	[0.668, 0.736]	0.000	0.6 (0.898)
IsoFLOP studies only	3	0.697	[0.636, 0.758]	0.000	0.3 (0.857)
Marin’s corpora independent	4	0.691	[0.649, 0.732]	0.019	8.3 (0.041)
IsoFLOP studies only	3	0.673	[0.609, 0.737]	0.010	2.3 (0.311)
Six designs, Marin separate	6	0.691	[0.668, 0.715]	0.008	5.7 (0.336)
Marin in configuration N	4	0.673	[0.615, 0.732]	0.032	14.3 (0.003)
$\eta = -0.1$ (Meta)	4	0.673	[0.598, 0.748]	0.043	21.1 (<0.001)
$\eta = +0.1$ (Meta)	4	0.700	[0.669, 0.730]	0.011	4.3 (0.230)
Farseer, Hessian path, extended-grid	4	0.665	[0.650, 0.679]	0.000	2.7 (0.439)
CV h					
Farseer, total N	4	0.670	[0.656, 0.685]	0.000	2.4 (0.500)
Marin config. N , Farseer total N	4	0.668	[0.654, 0.683]	0.000	1.1 (0.783)
Farseer’s bandwidth range as uncertainty ^d	4	0.680	[0.643, 0.717]	0.007	3.3 (0.349)
Fixed-effect means (Chin., Llama) ^e	4	0.676	[0.600, 0.752]	0.045	46.5 (<0.001)
Local first-derivative (Marin, Llama) ^f	4	0.670	[0.603, 0.737]	0.037	16.8 (<0.001)
Chinchilla’s other counts, conventions ^g	4	0.682–0.697			4.6–14.9 (0.002–0.201)
Chinchilla’s windows, samples, memberships ^h	4	0.674–0.702			4.4–14.8 (0.002–0.220)

Note: DerSimonian–Laird mean (DerSimonian and Laird, 1986), modified Hartung–Knapp–Sidik–Jonkman interval (t_{k-1} , variance factor truncated at one; Hartung and Knapp 2001; Sidik and Jonkman 2002; Knapp and Hartung 2003; Röver, Knapp and Friede 2015); k : number of studies; $\hat{\varphi}$: between-study s.d.; Q : Cochran’s statistic. ^bChinchilla 0.660 (0.023), FLOP-effective $N_F = F/6$ with Hoffmann et al.’s FLOPs per token F rebuilt from their architecture table (Table D2); Meta 0.660 (0.023), FLOP-implied $N_F = C/(6D)$; Marin, FLOP-implied, the equal-weight mean of its three corpora with the s.e. of perfectly correlated errors (shared code and FLOP accounting); Farseer, non-embedding N , the first-derivative path estimate 0.708 with the s.e. of its path mean (0.005), which does not treat compute levels as independent. η : elasticity of FLOPs per token with respect to the measured parameter count, $\sigma_{\text{eff}}^* = 2/[2 + S/(1 + \eta)^2]$, applied to Meta only (Chinchilla’s accounting is rebuilt). ^cEach IsoFLOP design’s random-effects mean over its budgets up to 3×10^{20} FLOP (Chinchilla 0.698, Meta 0.686, Marin 0.706); Farseer’s first-derivative estimate on its path up to 3×10^{20} (0.710, s.e. 0.006), which carries 84 percent of the weight. The next rows replace Farseer’s entry by its bandwidth-robust version (note d) or by its Hessian path over those levels, or drop it. ^dFarseer’s Hessian path mean ranges over 0.664–0.727 across six bandwidths; the range is treated as uniform uncertainty (s.e. 0.019) around Farseer’s estimate on budgets up to 3×10^{20} (0.710) in the first such row and around the range’s midpoint (0.695) in the second. ^eInverse-variance (fixed-effect) pooling of the budgets within Chinchilla and Llama 3 instead of the random-effects means (Table D4). ^fMarin’s three corpora and Llama 3 at the local first-derivative estimate $1/(1 + b_1)$ of the local surfaces of Table E9 (0.669 and 0.629), the estimator used for Farseer. ^gChinchilla alone replaced by: Hoffmann et al.’s printed, executed and $6N$ counts, total and non-embedding parameters, on nominal budgets or each count’s own isocosts (nine variants). ^hChinchilla alone replaced by each specification of the window, polynomial-order, frontier and centering grid with at least eight bracketed budgets, on its 137 profile runs and on the 132 without the five highest-loss runs (23 variants), and by its count-free N_F , its profile membership drawn on the corrected coordinates, and the 132-run primary (three variants); with the narrowest window, which keeps seven budgets, 0.674 ($Q = 14.1$). Range of means; Q (p). Across all study-level variants (the rows of this table and those in the replication files) the mean ranges over 0.663–0.702, and the lowest interval end is 0.596.

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); Grattafiori et al. (2024); Czech (2026); Held and Marin Community (2026); Porian et al. (2024); Li et al. (2025b); author’s calculations.

TABLE D8. ROBUSTNESS OF THE ELASTICITY OF SUBSTITUTION WITH THE OUTER EXPONENT FREE

Technology	Variant	n	σ_κ^* (s.e.)	$\hat{\kappa}$	σ^* ,		σ_κ^* with E	
					$\kappa = 1$	at $\hat{E}_{\kappa=1}$	over E set	
Chinchilla	Baseline	240	0.701 (0.012)	0.77	0.737	0.703	0.697–0.703	
	Drop smallest 10% of N	216	0.690 (0.014)	0.75	0.729	0.690		
	Drop smallest 25% of N	180	0.648 (0.017)	0.69	0.702	0.654		
	All 245 runs	245	0.664 (0.016)	0.70	0.714	0.671		
Farseer	Baseline	404	0.710 (0.004)	0.47	0.772	0.696	0.709–0.711	
	Drop smallest size	389	0.710 (0.004)	0.47	0.773	0.694		
	Drop 2 smallest sizes	374	0.709 (0.004)	0.48	0.770	0.692		
	Drop 5 smallest sizes	324	0.710 (0.005)	0.46	0.774	0.687		
	N with embeddings	404	0.668 (0.004)	0.52	0.724	0.648		
	Drop $M > 1,000$	389	0.716 (0.004)	0.48	0.785	0.705		
	Drop $M < 2$	355	0.694 (0.002)	0.43	0.797	0.690		
Gadre, C4	Baseline	34	0.607 (0.045)	0.35	0.795	0.602	0.600–0.614	
	Drop smallest size	26	0.528 (0.050)	0.41	0.720	0.535		
	Drop 2 smallest sizes	18	0.493 (0.093)	0.30	0.729	0.542		
Gadre, RedPajama	Baseline	35	0.623 (0.052)	0.39	0.795	0.679	0.606–0.685	
	Drop smallest size	27	0.574 (0.071)	0.38	0.771	0.568		
	Drop 2 smallest sizes	19	0.696 (0.097)	0.57	0.783	0.658		
Gadre, RefinedWeb	Baseline	35	0.594 (0.051)	0.36	0.796	0.595	0.580–0.605	
	Drop smallest size	27	0.545 (0.082)	0.38	0.758	0.545		
	Drop 2 smallest sizes	19	0.484 (0.080)	0.26	0.764	0.480		
OLMo ladder	Baseline	30	0.544 (0.019)	0.22	0.825	0.541	0.541–0.545	
	Drop smallest size	24	0.548 (0.027)	0.24	0.810	0.526		
	Drop 2 smallest sizes	18	0.518 (0.018)	0.32	0.709	0.506		
	Task bits per byte	30	0.512 (0.059)	0.33	0.763	0.487		
Muennighoff	Baseline	33	0.511 (0.047)	0.42	0.735	0.558	0.503–0.555	
	Drop smallest size	32	0.527 (0.051)	0.50	0.701	0.527		
	Drop 2 smallest sizes	31	0.536 (0.050)	0.52	0.693	0.514		
	Drop $D/N < 0.4$	31	0.566 (0.058)	0.51	0.726	0.612		

Note: $\sigma_\kappa^* = 2/(2 + a_1 + b_1)$ from $L = E + (AN^{-a_1} + BD^{-b_1})^\kappa$ fitted by Huber loss on $\ln L$. Standard errors: design-conditional wild bootstrap (Feng–He–Hu weights on leverage-adjusted residuals; weights of Webb (2023) by model size for Farseer and by (size, multiplier) cell for the OLMo ladder), 499 draws for baselines and 199 for variants. σ^* , $\kappa = 1$: the Chinchilla form on the same sample. At $\hat{E}_{\kappa=1}$: E fixed at the Chinchilla-form estimate and the rest refitted. Over E set: range of σ_κ^* over the values of E whose profile Laplace quasi-LR is below the χ_1^2 95 percent critical value (descriptive; a 36-point grid in E refined with 25 points around the set). Farseer’s N with embeddings counts both untied embedding matrices. Chinchilla: total parameters (digitized N), as for the reference technology of Section IV; in N_F the baseline gives 0.700 with κ free and 0.740 with $\kappa = 1$.
Source: Hoffmann et al. (2022); Li et al. (2025b); Gadre et al. (2025); Bhagia et al. (2025); Muennighoff et al. (2023); author’s calculations.

TABLE D9. THE COMPUTE COST OF OVER-TRAINING

σ^*	Training compute $C/C_{\min}$ at $M = k M^*$				Implied wedge $w = k^{1/\sigma^* - 1}$				Share s
	$k = 2$	$k = 10$	$k = 100$	$k = 1,000$	$k = 2$	$k = 10$	$k = 100$	$k = 1,000$	$k = 100$
0.60	1.08	2.24	14.3	129	1.59	4.64	21.5	100	0.95
0.70	1.05	1.73	7.23	49.8	1.35	2.68	7.20	19.3	0.86
0.74	1.04	1.57	5.42	31.3	1.28	2.25	5.04	11.3	0.80
0.80	1.03	1.39	3.52	14.5	1.19	1.78	3.16	5.62	0.68

Note: Compute needed to reach the loss of the compute-optimal model when training at k times the compute-optimal tokens-per-parameter ratio, relative to the minimum, in the Chinchilla family: $C/C_{\min} = ((\alpha + \beta w)/(\alpha + \beta))^{1/\gamma} w^{-1/\alpha}$ with $\ln w = (1/\sigma^* - 1) \ln k$. Entries are for the homothetic case $\alpha = \beta = \rho = 1/\sigma^* - 1$, where $C/C_{\min} = \cosh(\rho \ln k/2)^{2/\rho}$; every entry was checked by direct minimization. At the same σ^* , the α/β split of Besiroglu et al. (2024) changes the entries by at most 4.5 percent and that of Hoffmann et al. (2022) by at most 19 percent for $k \leq 100$ (k relative to M^* at the model's own compute). $s = (w - 1)/w$ is the expenditure share of the value of compactness that rationalizes over-training by k (Proposition 1); for a developer that serves its model it is the planned serving share of lifetime cost. With the parameters of de Vries (2023) ($\alpha = 0.32$, $\beta = 0.28$, $\sigma^* = 0.77$), a model at 30 percent of the compute-optimal size needs 6.6 times the tokens and 99 percent more compute ($k = 21.1$); the formula gives 99 percent.

Source: Author's calculations; de Vries (2023).

D5. The Compute Cost of Over-Training

Practitioners use the elasticity of substitution through one quantity: the extra training compute needed to reach the loss of the compute-optimal model while training at k times the compute-optimal ratio M^* . Table D9 reports it. Over-training by a factor of 10 costs 39 percent ($\sigma^* = 0.8$) to 124 percent ($\sigma^* = 0.6$) more compute; by a factor of 100 it costs 3.5 to 14 times the minimum, and by 1,000 it costs 14 to 129 times. At $k = 100$, $\sigma^* = 0.70$ implies 7.2 times the minimum compute and $\sigma^* = 0.74$, the value that the Chinchilla form imposes on the Chinchilla data, 5.4 times. The entries assume a homothetic split of the exponents; the splits of Besiroglu et al. (2024) and Hoffmann et al. (2022) change them by at most 4.5 and 19 percent for $k \leq 100$. The compute-overhead curve of de Vries (2023) lies between our curves for $\sigma^* = 0.74$ and 0.80. The implied wedge $w = k^{1/\sigma^* - 1}$ links the table to Section IV: a developer that trains 100 times beyond M^* reveals a share $s = (w - 1)/w$ of 0.68 to 0.95, depending on σ^* .

D6. Measurement and Flexible Inputs

The largest disagreement in the scaling literature concerns the allocation exponent: $N^* \propto C^{0.73}$ in Kaplan et al. (2020) against about 0.5 in Hoffmann et al. (2022). Counting the parameters that cost FLOPs is input measurement (Collard-Wexler and De Loecker, 2016), and warmup, schedule, learning rate and batch size are flexible inputs, which are not concentrated out of the production function unless re-optimized at each scale (Akerberg, Caves and Frazer, 2015; Gandhi, Navarro and Rivers, 2020). On the 975 released runs of Porian et al. (2024), whose ten published exponents we reproduce to within 0.003, the exponent falls

TABLE D10. HOW MUCH OF FARSEER’S CONVEXITY COULD BE MIS-TUNING

Policy whose inefficiency gradients are applied	ι_n	ι_d	b_2 corrected	Tuning share	Path $a_{\text{obs}} - a$
Step Law rule, in sample	−0.48	+0.07	0.0200	+0.02	+0.0009
Step Law rule, 95% corner ($\iota_n -$, $\iota_d +$)	−1.24	+0.55	0.0188	+0.08	+0.0031
Step Law rule, 95% corner ($\iota_n +$, $\iota_d -$)	+0.28	−0.41	0.0213	−0.04	−0.0011
Random configuration (best of 1): D-biased	−4.38	−6.95	0.0381	−0.87	−0.0036
Random configurations (best of 4)	−1.60	−2.02	0.0235	−0.15	−0.0007
Random configurations (best of 16)	−0.40	−0.63	0.0213	−0.05	−0.0004
One fixed (LR, batch)	−0.79	−0.26	0.0204	−0.00	+0.0009
Porian et al. base (fixed)	+3.17	−2.31	0.0262	−0.29	−0.0051
Porian et al. N -rule	+7.24	−5.73	0.0367	−0.80	−0.0059
DeepSeek C -rule	−1.55	−0.19	0.0199	+0.02	+0.0023
Bjorck et al. (N, D)-rule	−5.69	+0.37	0.0158	+0.22	+0.0038
Stochastic frontier shape, Step Law level	varies	varies	0.0205	−0.01	+0.0003
Stochastic frontier shape, random-configuration level	varies	varies	0.0249	−0.22	+0.0199
Observed (no correction)	0	0	0.0204	0	0
Breakdown: ι_d that removes all convexity ($\iota_n = 0$)	0	+26.0	0	+1.00	

Note: Farseer’s runs follow Step Law’s hyperparameter rules. With observed loss $L^* e^{\iota(N,D)}$, the technology’s elasticities are the observed ones plus the inefficiency gradients, $\varepsilon_N = \varepsilon_N^{\text{obs}} + \iota_n$ and $\varepsilon_D = \varepsilon_D^{\text{obs}} + \iota_d$ ($\iota_n = \partial \iota / \partial \ln N$, in units of 10^{-3} of log loss per log point). Each row corrects the local wedge at every point of Farseer’s grid (the local quadratic of Online Appendix D3, primary bandwidth), re-locates the expansion path as the root of the corrected $\ln w$ and re-estimates $\ln w = b_1 u + b_2 u^2$ ($u = \ln(M/M^*(C))$; 328 points, 309 for the Bjorck et al. rule, $n_{\text{eff}} \geq 15$). Tuning share = $(b_2^{\text{obs}} - b_2)/b_2^{\text{obs}}$: the part of the convexity attributable to the policy’s inefficiency; negative values mean correcting for it makes $\ln w$ more convex. Gradients: estimates on the Step Law grid (17 cells; N 0.21–1.07B non-embedding, D 4–100B, M 19–466); the 95% corners use their cluster-robust standard errors; the stochastic-frontier rows scale ι with elasticities -0.063 (N) and -0.260 (D) around the grid’s centre. Path $a_{\text{obs}} - a$: slope in $\ln C$ of the shift in the located $\ln N^*(C)$, the pointwise counterpart of equation (D2); its constant-gradient form $-(\iota_n - \iota_d)E/[(\alpha + \beta)R^*(C)]$ on Farseer’s Chinchilla-form fit gives values of the same sign in every row, two to seven times larger, because the fitted form is flatter than the local surface. Farseer’s convexity is driven by M up to 2,570, beyond Step Law’s M range, where the rule’s inefficiency is unmeasured.

Source: Li et al. (2025b); Li et al. (2025a); author’s calculations.

from 0.837 to 0.496 on RefinedWeb; counting the head accounts for 39 percent of the drop (47 on OpenWebText2) and flexible inputs for the rest.

Input measurement. Let the true technology be $L = E + AN^{-\alpha} + BD^{-\beta}$ and let the researcher count N_m parameters when the true count is $N = N_m + X(N_m)$, with omitted share $q \equiv X/N$ and a constant scale elasticity $\theta_X \equiv d \ln X / d \ln N_m < 1$. Taking the IsoFLOP argmin in measured units, with $C_m = 6N_mD$, $\vartheta \equiv d \ln N / dn_m = 1 - q(1 - \theta_X)$ and $\zeta \equiv d \ln \vartheta / dn_m$, the local measured allocation exponent is

$$(D1) \quad a_m = \frac{\beta}{\beta + \alpha\vartheta - \zeta}, \quad \zeta = \frac{(1 - \theta_X)q(\vartheta - \theta_X)}{\vartheta} \geq 0.$$

Since $\vartheta \leq 1$ and $\zeta \geq 0$, $a_m \geq a$, with equality when $q = 0$ or $\theta_X = 1$: only the part of the error that is correlated with scale biases the exponent, and the bias vanishes at scale as the omitted share falls, the functional-form lesson of [Nerlove \(1963\)](#). As $q \rightarrow 1$, $a_m \rightarrow \beta/(\beta + \alpha\theta_X)$, the small-scale limit of [Pearce and Song \(2024\)](#); with their embedding map ($\theta_X = 1/3$) the formula predicts 0.792 over Kaplan et al.’s size range under Besiroglu et al.’s parameters, against the 0.78 they obtain by simulation.

Flexible inputs. Let observed loss be $L_{\text{obs}} = L^*e^{\iota(N,D)}$, where L^* is the technology with the hyperparameters concentrated out and $\iota \geq 0$ is the inefficiency of the researcher’s policy. Along the IsoFLOP curve at compute C the observed argmin solves $f'(n) + \Delta(n) = 0$, with f the true profile, $f'' = (\alpha + \beta)\gamma R^*$ at the minimizer and $\Delta \equiv \partial(L_{\text{obs}} - L^*)/\partial n$ at fixed compute. To first order the log argmin shifts by $-\Delta(C)/f''(C)$, so

$$(D2) \quad a_{\text{obs}} - a = -\frac{\partial[\Delta/f'']}{\partial \ln C},$$

with $\Delta = L^*e^{\iota}(\iota_n - \iota_d)$ at the minimizer. Inefficiency that depends on the inputs only through compute ($\iota_n = \iota_d$) leaves a unbiased; with small constant gradients, $a_{\text{obs}} - a = -(\iota_n - \iota_d)E/[(\alpha + \beta)R^*(C)]$, first order in the gradient although the effect on the loss level is second order. On Porian et al.’s runs it explains 47 to 114 percent of the three flexible-input steps.

On the Step Law grid of [Li et al. \(2025a\)](#), inefficiency falls with training tokens (elasticity -0.23 to -0.26) but not detectably with parameters, and one random configuration per cell biases σ^* by -0.051 . Holding the batch fixed shrinks the token elasticity of the optimal learning rate to 0.02 – 0.05 (the conditional demand of [Milgrom and Roberts, 1996](#)), which closes 26 to 43 percent of the gap to the estimate of -0.32 of [Bjorck et al. \(2025\)](#), and 17 to 40 percent of the gap to their steeper estimates for 125M and 350M parameters.

How much of Farseeer’s convexity could be mis-tuning? Farseeer’s runs follow the Step Law rules, so the measured inefficiency gradients bound what tuning could produce (Table D10). Step Law’s own rule changes the convexity of $\ln w$ by

−4 to +8 percent; untuned configurations, whose inefficiency falls with D , would make it larger; and producing all of it would take an inefficiency that rises with D at 47 times the upper bound of the rule’s measured gradient. The gradients are measured only at $M \leq 466$; Farseer’s grid reaches 2,570.

APPENDIX E. THE REVEALED VALUE OF COMPACTNESS: ROBUSTNESS

This appendix supports Section IV. Unless stated otherwise, wedges are evaluated under the reference technology, the Chinchilla form with the outer exponent free, fitted in total parameters like the models’ M ($\sigma_\kappa^* = 0.701$; 0.700 refitted in FLOP-effective parameters, Online Appendix D2), on the 77 models of the clean sample, which form 49 decision units. Intervals for sample statistics come from the joint distribution across bootstrap draws of the technology: each draw recomputes the whole sample, including the family shares.

E1. What the Wedge Identifies

Table E1 lists what the wedge $w = \varepsilon_N/\varepsilon_D$ and the share $s = (w - 1)/w$ measure under the objectives of Propositions A3 and A4. With no binding cap on tokens, $w - 1 = v_N$, the value of compactness: the objective’s own value m_N plus the shadow value of any binding constraint on size, per unit of training expenditure. When a developer bears the cost of serving its model and no size constraint binds, $w - 1$ is internalized serving expenditure relative to training expenditure, which needs no conversion of FLOPs into dollars. Binding memory or tier caps and latency objectives are part of the value of compactness and make the wedge an upper bound on the objective’s own m_N ; data costs, data caps, distillation and binding corpus caps make it a lower bound. The last row is the factor-bias critique of markup estimation, which the developers’ own technologies, the spread across technologies and a second output on the same runs address (Appendices E5, E3 and E8).

E2. The Clean Sample

The clean sample starts from the 173 general-purpose open-weight models of the verified sample (Online Appendix B5). Nine steps, applied in order, drop models whose (N, D) is not the choice that Proposition 1 describes: (a) repeated observations of one pretraining run (8 models); (b1) research suites whose D is fixed by design, such as Pythia, OPT and BLOOM (45); (b2) replications and data benchmarks (8); (c) mixture-of-experts models (12, bounded in Appendix E4); (d) distilled, pruned or derived initializations, such as Gemma 2 2B and 9B, Gemma 3 and Llama 3.2 (10); (e) multimodal pretraining and (g) non-transformers, already removed; (f) models without a base checkpoint (7); and (h) token counts without a primary source (6). The replication package lists every flag with its source. The result is 77 models from 36 families and 18 developers, released 2023–2025. An audit of token counts corrected StableLM-Alpha (800 billion tokens, not the

TABLE E1. WHAT THE WEDGE IDENTIFIES UNDER ALTERNATIVE DEVELOPER OBJECTIVES

Developer objective or cost structure	Wedge $w = \varepsilon_N / \varepsilon_D$	$s = (w - 1)/w$ measures	$\hat{T}$ vs. $\lambda \varrho T$
Lifetime compute, developer serves ($\lambda = \varrho = \phi = 1$)	$1 + T/(3D)$	Serving share of lifetime compute	Equal
(a) Serving cost borne at rate λ , price ratio ϱ , cost elasticity ϕ	$1 + \lambda \phi \varrho T/(3D)$	$\lambda \phi X_S / (X_T + \lambda \phi X_S)$	$\times \phi$
(a') Open release, adoption value $v(\mathcal{A})$, users bear serving	$1 + v' \mathcal{A} \varepsilon_{\mathcal{A}, P} / X_T$	Adoption value of compactness	Not T
(b) Binding memory or tier cap, shadow value $\nu_N \geq 0$	$1 + v_N = 1 + m_N + \nu_N$	Identifies $v_N / (1 + v_N)$; upper bound on $m_N / (1 + m_N)$	Overstates
(b') Menu of sizes, shadow value ν_N at the chosen size	$1 + v_N = 1 + m_N + \nu_N$	Identifies $v_N / (1 + v_N)$; upper bound on $m_N / (1 + m_N)$ if the menu binds as a cap	Overstates if it binds as a cap
(c) Binding latency objective, $Z(N) \leq \bar{Z}$	$1 + v_N, \nu_N = \ell \xi_Z / X_T$	Identifies $v_N / (1 + v_N)$; upper bound on $m_N / (1 + m_N)$	Overstates

1.5-trillion-token dataset) and MPT-30B (1.05 trillion: 1 trillion at a context of 2,048 tokens and 50 billion at 8,192); every table of this appendix uses the audited counts.

The rules implement comments on an earlier draft of this paper, which had reported wedges for all 173 verified models under 12 of the present 32 technologies (marked in Table E3); neither the rules nor the set of technologies was therefore fixed before any wedge had been computed. Table E2, Panel A, follows the headline through the steps: the median wedge stays between 3.50 and 4.46. The sign is less robust than the point estimates suggest: all 32 technologies agree on $w > 1$ for 62 percent of clean models (64 percent with the token counts of the previous version), but the union of their 95 percent intervals lies above one for only 18 percent. Panel B splits the sample. Dropping the nine models trained partly on teacher-generated synthetic data (phi-1.5, phi-2 and SmolLM 1–3) gives a median share of 0.74. The reading protocol’s rule is broader: it flags every decision whose report documents teacher-generated data in pretraining or mid-training, 21 of the 49 decisions with 38 models, including all six decisions of 2025, and the 39 unflagged models have a median share of 0.62. Models whose size was served by their developer have a median share of 0.83, against 0.74 for the others; the nine models with an on-device target, from two developers, have 0.90; and models inside memory-tier windows have 0.68, against 0.86 outside. Inside Chinchilla’s range of M ($M \leq 341$) the median share is 0.56, and only 2 of the 77 models lie inside Chinchilla’s design in both M and compute.

TABLE E1 (CONTINUED). WHAT THE WEDGE IDENTIFIES UNDER ALTERNATIVE DEVELOPER OBJECTIVES

Developer objective or cost structure	Wedge $w = \varepsilon_N/\varepsilon_D$	$s = (w - 1)/w$ measures	$\hat{T}$ vs. λgT
(d) Data cost q_D per token	$(1 + v_N)/(1 + m_D)$	$(X_S - X_D)/(X_T + X_S)$	Understates
(d') Binding data cap, multiplier μ	$(1 + v_N)/(1 + m_D + \mu)$	Lower bound on $v_N/(1 + v_N)$; $w < 1$ possible	Understates
(e) Logit distillation, teacher N_T , student technology unchanged	$\frac{1 + T/(3D)}{1 + N_T/(3N)}$	$(X_S - X_{\text{teach}})/(X_T + X_S)$	Understates
(f) Dollar cost $\propto N^{1+\delta}D$; serving $\propto N^{1+\delta_S}$	$1 + \delta + \lambda(1 + \delta_S)gT/(3D)$	Mixes δ with serving	$+3D\delta + \lambda\delta_S gT$
(g) Family token budget, sizes chosen or set by binding size constraints	$\sum_i \pi_i(1 + v_{N,i}) = (1 + m_D)\bar{w}_H$	$1 - 1/\bar{w}_H = \sum_i h_i s_i$	Only $\sum_i \pi_i \hat{T}_i$ identified
(g') Family budget at a binding corpus cap	$\sum_i \pi_i(1 + v_{N,i}) = (1 + m_D + \mu)\bar{w}_H$	Family share a lower bound; with $m_D = 0$, member $w_i - 1 \leq v_{N,i}$ if and only if the member would train on more tokens at its own size	Understates
Post-training compute $x_P = X_P/X_T$	$1 + x_P + m_N$ (m_N net of X_P)	Mixes x_P with m_N	Overstates
Lab's technology differs (factor bias, beliefs, output)	$\hat{w} = w_{L'} w_L/w_{L'}$	Contaminated	Over iff $w_L > w_{L'}$

Note: The developer chooses (N, D) to maximize $V(L, N)$ minus training expenditure $X_T = q_T 6ND$ (and data cost $q_D D$). $m_N \equiv -(\partial V/\partial \ln N)_L/X_T$ is the objective's own value of compactness at fixed quality per unit of training expenditure; v_N is the shadow value of binding constraints on size alone (memory or tier caps, latency objectives, menus of sizes) per unit of training expenditure, nonnegative for a cap and negative for a menu that binds as a floor; $v_N \equiv m_N + \nu_N$ is the value of compactness, which equals m_N when no size constraint binds; rows other than (d), (d'), (e) and (g)–(g') take $m_D = 0$, and rows other than (f) take $\delta = 0$. $m_D \equiv q_D D/X_T$; X_S is serving expenditure, λ the share of it the developer internalizes, g the price of a serving FLOP relative to a training FLOP, ϕ the size elasticity of serving cost; $\varepsilon_{A,P}$ is the elasticity of adoption with respect to users' serving cost; ξ_Z and $\mu \geq 0$ are the multipliers of the latency objective and the data cap, ℓ the size elasticity of latency; in the post-training row, post-training expenditure X_P is counted as a cost alongside X_T , so that m_N is net of it; δ (δ_S) is the size elasticity of the price of a training (serving) FLOP; in rows (g) and (g'), $v_{N,i}$ is member i 's value of compactness, $\pi_i \propto h_i/w_i$ with h_i the member's compute share, and $\bar{w}_H$ the compute-weighted harmonic mean of member wedges; in the last row, $w_{L'}$ is the wedge that the lab's technology or output assigns to its choice and w_L the one that the econometrician's assigns to the same inputs. The last column compares $\hat{T} \equiv 3D(\hat{w} - 1)$ with λgT , the internalized serving demand in training-FLOP units (rows other than (a) and (f) take $\phi = 1$); "Over iff $w_L > w_{L'}$," means that $\hat{T}$ overstates, which under factor bias is $\chi < 0$. If distillation also raises the productivity of tokens ($\chi > 0$), it lowers the measured wedge further. Every row was verified by brute-force maximization of the objective (maximum relative error 2×10^{-7} ; Propositions A3 and A4).

Source: Author's derivations.

TABLE E2. THE CLEAN SAMPLE: CONSTRUCTION AND SUBSAMPLES

<i>Panel A. Sequential cleaning steps (reference technology: Chinchilla, κ free)</i>								
Step (models dropped)	n	Groups	Median w	Share $w > 1$	All points > 1	Band > 1	Median s	Median w , $\kappa = 1$
Verified sample	173	67	3.50	0.93	0.61	0.17	0.71	3.09
(a) One per pretraining run (−8)	165	65	3.77	0.93	0.61	0.18	0.73	3.19
(b1) Research suites (−45)	120	56	4.20	0.98	0.69	0.16	0.76	3.51
(b2) Replications, benchmarks (−8)	112	51	4.46	0.98	0.71	0.17	0.78	3.70
(c) Mixture of experts (−12)	100	43	4.30	0.98	0.68	0.18	0.77	3.61
(d) Distilled, pruned, derived (−10)	90	40	4.03	0.98	0.64	0.17	0.75	3.42
(e) Multimodal (−0)	90	40	4.03	0.98	0.64	0.17	0.75	3.42
(f) No base checkpoint (−7)	83	37	3.99	0.98	0.63	0.18	0.75	3.34
(g) Non-transformer (−0)	83	37	3.99	0.98	0.63	0.18	0.75	3.34
(h) D undocumented (−6): clean	77	36	3.99	0.97	0.62	0.18	0.75	3.34
<i>Panel B. Subsamples of the clean sample</i>								
Common-budget members	45	16	3.79	0.98	0.53	0.18	0.74	
Size-specific members	17	9	4.42	0.94	0.76	0.29	0.77	
Single-model releases	15	15	3.93	1.00	0.73	0.07	0.75	
Decisions of one model	32	24	4.03	0.97	0.75	0.19	0.75	
No synthetic data, earlier flag	68	31	3.80	0.97	0.59	0.15	0.74	
No synthetic data, protocol rule ^b	39	22	2.66	0.95	0.41	0.05	0.62	
This size served by its developer ^a	26	6	5.84	1.00			0.83	
Not served	51	15	3.80	0.96			0.74	
On-device target	9	2	9.54	1.00			0.90	
In a memory-tier window	49	15	3.11	0.98			0.68	
Outside the tier windows	28	10	7.08	0.96			0.86	
$M \leq 341$ (Chinchilla’s range)	31		2.26	0.94			0.56	
$M > 341$	46		6.64	1.00			0.85	

Note: Panel A: sequential steps from the 173 verified open-weight base models, with the audited token counts (StableLM-Alpha at 0.8 trillion tokens and MPT-30B at 1.05 trillion; Online Appendix B5); each row describes the sample that remains after the step, and a model is dropped at the first step that applies. Reference technology: the Chinchilla form with the outer exponent free (inner exponents 0.424 and 0.431, $\sigma_{\kappa}^* = 0.701$), in total parameters. $s = (w - 1)/w$. All points > 1 : every one of the 32 technologies of Table E3 puts the point estimate above one. Band > 1 : the union of the 95 percent intervals over those technologies lies above one. With the token counts of the previous version the all-points share of the clean sample was 0.64; the StableLM-Alpha correction puts StableLM-Alpha 3B below one under MiniCPM’s law, which lowers it to 0.62. Last column: the median under the Chinchilla form with $\kappa = 1$. Groups: families, except developers in the serving, target and tier rows of Panel B. Panel B, first four rows: members of a family whose token budgets are within 10 percent of each other form one decision (Online Appendix E5). Synthetic data: the earlier flag drops the nine models of phi-1.5, phi-2 and SmolLM 1–3; ^bthe protocol’s rule flags every decision whose report documents teacher-generated data in pretraining or mid-training (21 of the 49 decisions, with 38 models, including all six decisions of 2025). ^aA first-party per-token API or consumer product offered this size within 180 days of release (Online Appendix B5). Tier windows: 2.4–3.3, 6.5–9.5, 11.5–14.9, 26–32.9 and 65–72.9 billion parameters, where a binding tier’s shadow value is part of the value of compactness.

Source: Model cards, technical reports and Hugging Face configuration files; author’s calculations.

E3. Technologies

The inclusion rule. The set of technologies contains (i) every technology estimated in Section III and Online Appendix D from final-checkpoint losses of a designed sweep with the primary estimator, each corpus and parameter-count convention as its own row, under $\kappa = 1$ and, where estimated, with κ free; (ii) for each IsoFLOP design, the argmin path with its $\kappa = 1$ curvature, the primal fit, and the path combined with the design’s model-free curvature; and (iii) the literature’s estimates on the Chinchilla data and the allocation laws published by Meta, DeepSeek (DeepSeek-AI et al., 2024b) and MiniCPM (Hu et al., 2024). DataDecide, the unannealed profiles of Porian et al. (2024) and the checkpoint-interpolated profiles that Czech et al. (2026) build from the runs of Li, Kudugunta and Zettlemoyer (2025), whose losses come before the learning-rate schedule ends, are excluded, as are estimator or sample variants of a design already in the set. When a path and a curvature come from different sources, the inner exponents are $\alpha = S(1 - a)$ and $\beta = Sa$ with $S = 2(1/\sigma^* - 1)$; the wedge depends on the technology only through S , the path slope and the path level (Corollary A2). MiniCPM’s compute-optimal ratio of 192 at 10^{21} FLOP lies outside its own design (ratios of 10 to 60), so it is itself an extrapolation.

What drives the dispersion. Table E3 lists the 32 technologies. The median share ranges from 0.14 to 0.84 over models and from 0.04 to 0.80 over the 49 decision units. The reference, Farseer with κ free, DeepSeek’s law and the developers’ paths with model-free curvature give 0.67 to 0.79 over models (0.62 to 0.75 over decision units); the κ -free fits of the smaller designs of Gadre et al., the OLMo ladder and Muennighoff et al. give 0.74 to 0.84 (0.66 to 0.80). The low medians come from $\kappa = 1$ curvatures far above the model-free estimates, as in Marin’s argmin laws (σ^* of 0.87 to 0.90), from the OLMo ladder with $\kappa = 1$, whose $M^*(10^{24})$ of 291 puts a quarter of the sample below one, and from MiniCPM’s zero point. Estimation choices within the reference design matter as much as the functional form: imposing $\kappa = 1$ lowers the median model wedge from 3.99 to 3.34, and within that form keeping the five outlying Chinchilla runs raises it to 5.50. Curvature governs levels, not signs: holding the reference zero point fixed, lowering σ^* from 0.80 to 0.60 raises the median wedge from 2.25 to 8.64, while the share of models with $w > 1$, 0.974, does not move. With the curvature of the κ -free fit to Chinchilla’s 137 profile runs (σ^* of 0.656 in FLOP-effective parameters) and the reference zero points held fixed, the median decision share is 0.80 rather than 0.73: that fit lowers σ^* and raises the levels. Only σ^* is available for that fit, so this is a curvature sensitivity rather than a refitted technology. For the reference, the 95 percent interval for the median wedge is [3.21, 4.86] under the design-conditional wild bootstrap. Pairs resampling, which changes the design, widens its upper end to 6.04.

TABLE E3. THE REVEALED WEDGE UNDER EVERY TECHNOLOGY IN THE SET

Technology	σ^*	$M^*(10^{24})$	Median w	$w > 1$	Median s	Units
<i>Chinchilla data</i>						
κ free (reference)	0.701	21.6	3.99 [3.21, 4.86]	0.97	0.75 [0.69, 0.79]	0.73
$\kappa = 1^\dagger$	0.737	17.6	3.34 [2.64, 4.81]	1.00	0.70 [0.62, 0.79]	0.69
Besiroglu et al. parameters [†]	0.737	18.1	3.31	1.00	0.70	0.68
Hoffmann et al., Approach 3 [†]	0.762	62.0	2.08	0.88	0.52	0.46
Non-embedding N , $\kappa = 1$	0.753	9.7	3.65 [2.83, 4.71]	1.00	0.73 [0.65, 0.79]	0.70
<i>Farseeer</i>						
Non-embedding N , $\kappa = 1^\dagger$	0.772	17.3	2.74 [2.37, 3.20]	1.00	0.63 [0.58, 0.69]	0.62
Total N , $\kappa = 1^\dagger$	0.724	67.7	2.50 [2.14, 2.92]	0.88	0.60 [0.53, 0.66]	0.55
Non-embedding N , κ free [†]	0.710	18.3	3.91 [3.77, 4.06]	1.00	0.74 [0.73, 0.75]	0.74
<i>Gadre et al.</i>						
RefinedWeb, $\kappa = 1^\dagger$	0.796	1.4	4.41 [2.18, 7.67]	1.00	0.77 [0.54, 0.87]	0.75
C4, $\kappa = 1^\dagger$	0.795	6.0	3.18 [1.11, 7.66]	1.00	0.69 [0.10, 0.87]	0.67
RedPajama, $\kappa = 1^\dagger$	0.795	3.9	3.50 [2.52, 6.94]	1.00	0.71 [0.60, 0.86]	0.70
RefinedWeb, κ free	0.594	61.0	4.76 [3.39, 15.39]	0.88	0.79 [0.70, 0.94]	0.75
C4, κ free	0.607	78.5	3.89 [1.86, 10.60]	0.88	0.74 [0.46, 0.91]	0.68
RedPajama, κ free	0.623	32.3	5.64 [3.48, 12.88]	0.94	0.82 [0.71, 0.92]	0.80
<i>OLMo ladder</i>						
$\kappa = 1^\dagger$	0.825	291	1.28 [0.45, 2.66]	0.75	0.22 [-1.22, 0.62]	0.17
κ free	0.544	142	4.05 [3.22, 5.32]	0.84	0.75 [0.69, 0.81]	0.66
<i>Muennighoff et al.</i>						
$\kappa = 1$	0.735	36.7	2.55 [0.54, 10.02]	0.92	0.61 [-0.85, 0.90]	0.58
κ free	0.511	100	6.24 [1.11, 35.51]	0.86	0.84 [0.10, 0.97]	0.77
<i>Meta (Llama 3 IsoFLOPs)</i>						
Meta’s law (A2, $\kappa = 1$) [†]	0.785	31.3	2.28 [1.93, 2.65]	0.92	0.56 [0.48, 0.62]	0.52
Primal, $\kappa = 1^\dagger$	0.769	9.6	3.26 [2.32, 4.04]	1.00	0.69 [0.57, 0.75]	0.68
A2 path, MF σ^*	0.660	31.3	4.74 [3.55, 6.45]	0.92	0.79 [0.72, 0.84]	0.75
<i>Marin ladders</i>						
Nemotron-CC, primal	0.658	81.8	3.32 [2.36, 6.46]	0.87	0.70 [0.58, 0.85]	0.65
DCLM, primal	0.647	106	3.17 [2.37, 5.87]	0.86	0.68 [0.58, 0.83]	0.63
Comma, primal	0.663	228	2.10 [1.47, 3.71]	0.78	0.52 [0.32, 0.73]	0.48
Nemotron-CC, A2, $\kappa = 1$	0.867	35.1	1.59 [1.34, 2.09]	0.92	0.37 [0.26, 0.52]	0.34
DCLM, A2, $\kappa = 1$	0.866	32.5	1.62 [1.26, 2.40]	0.92	0.38 [0.21, 0.58]	0.35
Comma, A2, $\kappa = 1$	0.904	58.0	1.31 [1.12, 1.54]	0.88	0.24 [0.11, 0.35]	0.21
Nemotron-CC, MF σ^*	0.705	35.1	3.55 [2.38, 5.52]	0.92	0.72 [0.58, 0.82]	0.68
DCLM, MF σ^*	0.713	32.5	3.49 [2.28, 6.01]	0.92	0.71 [0.56, 0.83]	0.67
Comma, MF σ^*	0.700	58.0	3.03 [2.30, 4.94]	0.88	0.67 [0.57, 0.80]	0.62
<i>Published laws</i>						
DeepSeek LLM, ref. σ^*	0.701	13.9	4.57 [3.96, 5.31]	1.00	0.78 [0.75, 0.81]	0.75
MiniCPM	0.794	426	1.17	0.62	0.14	0.04
Range, 32 technologies		1.4–426	1.17–6.24	0.62–1.00	0.14–0.84	0.04–0.80

Note: Clean sample (77 models, audited token counts; 49 decision units in the last column). Brackets (median w and s): 2.5 and 97.5 percentiles over the technology’s bootstrap draws, recomputing the sample statistic in every draw (one technology draw shared by all models); no bracket: published point estimate; the share with $w > 1$ under the reference has the interval [0.92, 1.00]. A2: IsoFLOP-argmin (Approach 2) path; MF: model-free. $M^*(10^{24})$: compute-optimal tokens per parameter at 10^{24} FLOP in the technology’s own convention, an extrapolation for every row. MF rows: the design’s IsoFLOP path with its model-free σ^* . Non-embedding and OLMo-convention technologies are evaluated at each model’s own count. [†]One of the 12 technologies under which an earlier version of this paper reported wedges for all 173 verified models; the other 20 were added in revision. $w > 1$: share of models. Units: median s over the 49 decision units, with members of a family that share one token budget entered once at their family share.

Source: Hoffmann et al. (2022); Besiroglu et al. (2024); Li et al. (2025b); Gadre et al. (2025); Bhagia et al. (2025); Muennighoff et al. (2023); Grattafiori et al. (2024); Held and Marin Community (2026); DeepSeek-AI et al. (2024b); Hu et al. (2024); author’s calculations.

E4. Parameter Conventions and Mixture-of-Experts Models

Each technology is evaluated at the model’s own parameter count in the technology’s convention. A non-embedding technology ignores the output head, which costs $2N_{\text{head}}$ FLOPs per token in training and serving alike, so counting it turns the lifetime-to-training ratio into $w(1 + N_{\text{head}}/N_{\text{ne}})$. Table E4, Panel A, shows that the sample median moves little across conventions, because embeddings are at most 28 percent of parameters in the clean sample. For the eight clean models below one billion parameters, comparing like with like, the non-embedding count lowers the median wedge by 8 percent under the Chinchilla form and by 36 percent under the Farseeer form. The Farseeer change exceeds, and the Chinchilla change matches, the effect of tokenizer differences on D : under the reference $\partial \ln w / \partial \ln D = 0.43$, so a 20 percent difference in tokens per byte moves the wedge by about 8 percent. Panel B bounds the wedges of the 12 excluded mixture-of-experts models. A dense technology evaluated at active parameters treats a mixture of experts as parameter-augmenting and overstates its wedge; at total parameters it ignores the loss advantage of sparsity (Clark et al., 2022; Ludziejewski et al., 2024; Abnar et al., 2025). The median is 5.14 at active and 2.07 at total parameters; DeepSeek-V3 has 3.50 or 1.01.

E5. Decision Units, Family Readings and the Developers’ Own Technologies

Decision units. Members of a family whose token budgets are within 10 percent of each other form one decision: the 77 models form 17 common-budget decisions with 45 models, 17 members of families with size-specific budgets and 15 single-model releases, 49 in all. By Proposition A4, a budget chosen for its members reveals one number, the family share $1 - 1/\bar{w}_H$ with $\bar{w}_H$ the compute-weighted harmonic mean of the member wedges, whether the sizes are chosen at the margin or set by binding size constraints; member wedges differ only through the technology.

Readings. Each common budget is read from the developers’ reports, release posts and model cards under a coding protocol written before any decision was coded; the protocol, its three dated amendments (made during coding and before any class count or median was computed) and every quote, with its URL and retrieval date, are in the replication package. A *cap* requires evidence that the budget is the developer’s whole available corpus, that the corpus was repeated, or that data limited the budget; a *choice*, a stated cost, compute or schedule reason, a larger usable pool, or other models trained longer on the same data; a *menu*, sizes set by a hardware or configuration menu. Where the reports state only the corpus or budget size, or give evidence of both, the reading is *cap or choice*. Statements that a model was still improving, data expansion and training beyond the compute-optimal point are not evidence on the token margin. A second coder, who saw the saved sources but not the wedges, coded every decision independently. The readings agree for 15 of the 17 common budgets (Cohen’s kappa 0.83), the

TABLE E4. PARAMETER-COUNT CONVENTIONS AND MIXTURE-OF-EXPERTS MODELS

Panel A. Clean sample (77 models) under technologies in different parameter conventions							
Technology	Convention of N	Median w	Median w , head FLOPs	Share $w > 1$	Median s	Median w , $N < 1\text{B}$ ($n = 8$)	
Chinchilla, κ free (reference)	Total	3.99		0.97	0.75	15.10	
Chinchilla, $\kappa = 1$	Total	3.34		1.00	0.70	9.96	
Farseer, $\kappa = 1$	Total	2.50		0.88	0.60	10.67	
Chinchilla, $\kappa = 1$ (refit)	Non-embedding	3.65	3.74	1.00	0.73	9.13	
Farseer, $\kappa = 1$	Non-embedding	2.74	2.86	1.00	0.63	6.85	
Farseer, κ free	Non-embedding	3.91	4.15	1.00	0.74	14.26	
MiniCPM (published law)	Non-embedding	1.17	1.24	0.62	0.14	3.04	
Panel B. Mixture-of-experts models (excluded from the clean sample), reference technology							
Model (year)	N (B), D		M ,	w		s	
	active / total	(T)	active / total	Active N	Total N	Active N	Total N
OLMoE 1B-7B (2024)	1.3 / 7	5.1	3,946 / 741	9.18	4.51	0.89	0.78
DeepSeek-V2-Lite (2024)	2.4 / 16	5.7	2,375 / 363	7.40	3.34	0.86	0.70
DeepSeekMoE 16B (2024)	2.8 / 16	2.0	714 / 122	4.42	2.09	0.77	0.52
Phi-3.5-MoE (2024)	6.6 / 42	4.9	742 / 117	4.52	2.06	0.78	0.51
DeepSeek-V2 (2024)	21.0 / 236	8.1	386 / 34	3.43	1.23	0.71	0.19
DeepSeek-V3 (2024)	37.0 / 685	14.8	400 / 22	3.50	1.01	0.71	0.01
Qwen3 30B-A3B (2025)	3.3 / 31	36.0	10,909 / 1,179	14.30	5.56	0.93	0.82
Llama 4 Scout (2025)	17.0 / 109	40.0	2,353 / 368	7.46	3.40	0.87	0.71
Qwen3 235B-A22B (2025)	22.0 / 235	36.0	1,636 / 153	6.39	2.34	0.84	0.57
GLM-4.5 (2025)	32.0 / 358	23.0	719 / 64	4.50	1.61	0.78	0.38
Llama 4 Maverick (2025)	17.0 / 402	22.0	1,294 / 55	5.77	1.51	0.83	0.34
Kimi K2 (2025)	32.0 / 1,026	15.5	484 / 15	3.79	0.87	0.74	−0.15
Median (12 models)				5.14	2.07	0.80	0.52

Note: Panel A: medians over the clean sample (audited token counts), each technology evaluated at the model’s own count in the technology’s convention (non-embedding counts from each model’s configuration file). With the output head counted in FLOPs, the lifetime-to-training ratio for a non-embedding technology is $w(1 + N_{\text{head}}/N_{\text{ne}})$. Last column: the eight clean models below one billion parameters; comparing like with like, the non-embedding count lowers their median wedge by 8 percent under the Chinchilla form (9.13 against 9.96) and by 36 percent under the Farseer form (6.85 against 10.67). Panel B: w and $s = (w - 1)/w$ under the reference technology (Chinchilla, κ free), with N equal to active or to total parameters. A dense technology evaluated at active parameters treats a mixture-of-experts model as parameter-augmenting and overstates its wedge; at total parameters it ignores the loss advantage of sparsity. The two columns bound the wedge under a dense technology; neither is an estimate from a mixture-of-experts scaling law.

Source: Model cards, technical reports and Hugging Face configuration files; author’s calculations.

flags for a member’s own binding cap for 96 percent of the decisions (kappa 0.90) and the stated sizing rationales with kappas of 0.85 to 1.00. Disagreements were resolved under the protocol’s rules by a reviewer who had seen the wedges; both disagreements on a reading (Granite 3.0 and the Llama 3 herd) were resolved to the more conservative class, cap or choice.

Table E5 lists the readings. Choice readings (LLaMA 7B and 13B, Llama 2, StableLM-Alpha, OLMo 2 1B and 7B, Apertus, Olmo 3) and menu readings (MPT, whose 30B size was chosen to fit one GPU; Yi, sized for consumer GPUs; Qwen2 1.5B, 7B and 72B) make the family share a point estimate of the value of compactness, since a binding limit on size is part of that value. A binding cap on tokens makes it a lower bound (Proposition A4(vi)): so do the cap-or-choice readings (LLaMA 30B and 65B, Qwen 14B and 72B, the Llama 3 herd, Qwen2.5, Granite 3.0, Qwen3), which are point estimates only if the budget was chosen, and DeepSeek LLM’s cap and menu. Twelve members of size-specific families and single-model releases with their own binding cap on unique tokens are also lower bounds: OLMo 2 13B and 32B (1.2 and 1.5 epochs of the stage-1 mix), OLMo 1B and 7B, SmolLM 1.7B, SmolLM2 1.7B (about two epochs), phi-1.5, phi-2, StableLM-3B-4E1T (four epochs), TinyLlama v1.1, and Falcon 40B and 180B. SmolLM 135M and 360M repeated their 252-billion-token corpus about 2.4 times and are lower bounds under either reading. In all, 29 decisions are point estimates and 20 are lower bounds, 6 of which would be point estimates if their budget was chosen. A member’s own $w - 1$ bounds its own value of compactness from below only if that member would have trained longer at its own size (Proposition A4(vi)), which a binding cap on the family’s budget does not imply.

The Llama 3 herd and OLMo 2. Family shares run from 0.08 (LLaMA 30B and 65B) to 0.89 (Qwen3). Because the flagship carries most of a family’s compute, they lie far below the members’ own shares: the Llama 3 herd’s member wedges run from 1.29 to 6.73, but its weighted condition gives $\bar{w}_H = 1.42$, a share of 0.29, and 0.38 under Meta’s own eight-budget path. The herd’s 95 percent interval is [0.08, 0.46]. The report and the release post (Grattafiori et al., 2024; Meta AI, 2024) support a cap reading, and the report attributes the smaller models’ long training to inference efficiency; the protocol codes the herd as cap or choice because neither source states that the corpus bound. Under a cap, 0.29 bounds the herd’s value of compactness from below, and the 8B model’s share is at least 0.85 under the reference (0.89 on Meta’s path) if the 8B would have trained longer at its own size ($v_N \geq w - 1 = 5.73$). The flagship’s sign is not identified beyond the designs (Appendix E7). OLMo 2’s 1B and 7B models share a budget of 4.05 trillion tokens, and its 13B and 32B repeated part of the stage-1 mix. Its member shares fall from 0.87 to 0.61 under the reference and from 0.80 to 0.12 under AI2’s own ladder path with the common curvature; the ladder’s allocation exponent is imprecise (a bootstrap interval of [0.11, 0.82] with $\kappa = 1$). OLMo 2’s tokens are counted as pretraining plus the mean mid-training budget of the averaged runs

(Online Appendix B5).

Unit definitions and classes. Over the 49 decisions the median share is 0.735, with a joint wild-bootstrap interval of [0.657, 0.782]. Under a bootstrap that re-samples the Chinchilla runs by budget and model-size trunk (44 clusters) the interval widens to [0.512, 0.806], so the statement that the value of compactness exceeded training cost at the margin for the median decision ($s > 0.5$) survives only barely. The 29 point-identified decisions have a median of 0.737, with an interval of [0.672, 0.790], and the 20 lower bounds 0.710. Under the cluster bootstrap the point-identified interval is [0.539, 0.824]. The point-identified median is 0.735 over 35 decisions if every cap-or-choice budget without repetition was chosen, 0.737 under the second coder’s readings (31 decisions) and 0.745 without MPT and Yi, whose agreed codes rest on the weakest evidence (27). Without synthetic-flagged decisions it is 0.675 under the protocol’s rule (18) and 0.678 without SmoLM 1–3, phi-1.5, phi-2, Qwen2.5, Qwen3 and OLMo 2 (25). Over the 17 decisions outside tier windows, an interpretation check rather than a condition for identification, the median is 0.837. Other unit definitions give 0.736 over the 56 units of the earlier version, which grouped only families trained entirely on one budget; 0.751 when the eight cap-type budgets are split into their members (66 units; 0.752 for the earlier version’s 66 reading-consistent units); 0.736 with OLMo 2 1B as a decision of its own; and 0.749 over the 77 models. The median is 0.737 without Alibaba’s models (43 decisions), 0.735 without Qwen2.5 and Qwen3, whose per-member token counts are not audited, and 0.735 with StableLM-3B-4E1T and TinyLlama at the effective data of Muennighoff et al. (2023) (their wedges move from 5.96 to 5.78 and from 6.57 to 6.52). The token audit moves the StableLM-Alpha member wedges from 3.79 to 2.66 and from 2.65 to 1.92, and MPT-30B’s from 1.20 to 1.22 (the MPT family share from 0.24 to 0.25), but not the median, which is 0.735 with the uncorrected counts.

The developers’ own technologies under one rule. Where a developer has published scaling evidence, one rule builds its technology: the developer’s own expansion path, in its own parameter convention (the OLMo ladder without the input embedding, DeepSeek’s law in non-embedding FLOPs per token), combined with the common model-free $\sigma^* = 0.687$. For Meta the path is fitted to the eight IsoFLOP budgets that bracket their minimum; for Marin it is the Nemotron-CC ladder, for AI2 the OLMo ladder with κ free, and for DeepSeek its published law. Llama 1 and 2 are left out of the lab-own set, whose 2024 path their developers could not have used; 15 models remain. Table E6 shows that the median share is 0.53 under the one rule against 0.70 under the reference, and 0.71 against 0.65 without AI2’s models, so the order reverses. AI2 drives the gap: its ladder path puts M^* at 82 to 145 at these models’ compute, 3.7 to 6.7 times the reference, and its models’ median share is 0.44 with the common curvature against 0.66 with the ladder’s own. On Meta’s bracketed path, Llama 3 8B has $w = 8.75$, with a 95 percent interval of [4.58, 18.13], and Llama 3.1 405B has 1.47; the 405B is on Meta’s path ($w = 0.97$) only under the ten-budget planning law, which uses two

TABLE E5. READINGS OF COMMON TOKEN BUDGETS

Decision unit	D (T)	Reading	Id.	s_f	Evidence (one quote; all quotes, URLs and retrieval dates in the replication package)
LLaMA (7B, 13B)	1.00	choice	pt	0.46	“LLaMA-33B and LLaMA-65B were trained on 1.4T tokens. The smaller models were trained on 1.0T tokens.”
LLaMA (30B, 65B)	1.40	cap or choice	lb ^c	0.08	“Overall, our entire training dataset contains roughly 1.4T tokens after tokenization.”
Llama 2 (7B, 13B, 70B)	2.00	choice	pt	0.23	“We trained on 2 trillion tokens of data as this provides a good performance–cost trade-off”
MPT (7B, 30B)	1.00, 1.05	menu	pt	0.25	“The size of MPT-30B was also specifically chosen to make it easy to deploy on a single GPU”
StableLM-Alpha (3B, 7B)	0.80	choice	pt	0.53	“StableLM-Alpha models are trained on a new dataset that builds on The Pile, which contains 1.5 trillion tokens, roughly 3x the size of The ...”
Qwen (14B, 72B)	3.00	cap or choice	lb ^c	0.31	“Finally, we have built a dataset of up to 3 trillion tokens.”
Yi (6B, 34B)	3.00	menu	pt	0.49	“when choosing model scale, the desiderata is to have small enough model that is feasible for inference on consumer-grade hardware like the RTX 4090 where the bounding ...”
DeepSeek LLM (7B, 67B)	2.00	cap+menu	lb	0.18	“we have developed a dataset that currently consists of 2 trillion tokens and is continuously expanding.”
Llama 3 herd (8B, 70B, 3.1 405B)	15.00, 15.00, 15.60	cap or choice	lb ^c	0.29	“We pre-train Llama 3 on a corpus of about 15T multilingual tokens, compared to 1.8T tokens for Llama 2.”

unbracketed budgets. Over the 49 decisions, with developers’ own technologies where they exist and the reference elsewhere, the median is 0.717. As for the reference, the lab-own levels are lower bounds if σ^* falls with compute.

E6. Serving, Conduct and Tiers

Serving coded at the model level. Online Appendix B5 gives the coding rule. Twenty-six of the 77 models were served by their developer at release, 19 of them Alibaba’s (every Qwen size was in Alibaba’s own catalog, the smallest Qwen2.5 sizes at a zero price). Seven models that the previous, developer-level code counted as served are not: Gemma 2B and 7B (self-deploy catalog only), DeepSeek LLM 7B (the chat product served only the 67B), phi-1.5, phi-2, and Llama 3 8B and 70B, whose consumer product did not name a size. Twenty-five of the 26 served models share a budget with other sizes or sit in a tier window, which makes their wedges family-level or partly a tier’s shadow value; only Qwen2 0.5B is neither (Qwen2 1.5B shares its 7T budget with the 7B and 72B).

Tests. Table E7 reports restricted wild cluster bootstrap tests by developer (Cameron, Gelbach and Miller, 2008; Webb, 2023). Coded by model, serving has no detectable relation to the wedge at given compute and year: the coefficient is 0.56 (s.e. 0.24), with a p -value of 0.41 from six treated developers, and it is -0.09 without Alibaba. At the decision level it is 0.28 ($p = 0.67$). Randomization in-

TABLE E5 (CONTINUED). READINGS OF COMMON TOKEN BUDGETS

Decision unit	D (T)	Reading	Id.	s_f	Evidence (one quote; all quotes, URLs and retrieval dates in the replication package)
Qwen2 (1.5B, 7B, 72B)	7.00	menu	pt	0.51	“Considering training costs, we opted to use the higher-quality 7 trillion token dataset for training larger models, leaving further exploration for future model ...”
SmolLM (135M, 360M)	0.60	cap or choice	lb	0.86	“Therefore, we decided to train the 1.7B model on 1 trillion tokens and the 135M and 360M models on 600B tokens, as the performance gains after 400B tokens begin to slow ...”
Qwen2.5 (0.5B–72B)	18.00	cap or choice	lb ^c	0.72	“all models are pretrained on our latest large-scale dataset, encompassing up to 18 trillion tokens.”
Granite 3.0 (2B, 8B)	12.00	cap or choice	lb ^c	0.85	“Dense Models: 2B and 8B parameter models, trained on 12 trillion tokens in total.”
OLMo 2 (1B, 7B)	4.05	choice	pt	0.77	“Similar to OLMo 2 7B, we use 2000 steps of warmup, set the schedule to 5 trillion tokens but truncate at the 4 trillion mark.”
Qwen3 (0.6B–14B)	36.00	cap or choice	lb ^c	0.89	“All Qwen3 models are trained on a large and diverse dataset consisting of 119 languages and dialects, with a total of 36 trillion tokens.”
Apertus (8B, 70B)	15.00	choice	pt	0.65	“For all other languages, we subsample documents at random.”
Olmo 3 (7B, 32B)	6.03, 5.60	choice	pt	0.63	“Composition of Dolma 3 Mix including our 9T pool of data, the 6T mix we used for final model training, and the 150B mix we used for experimentation.”

Note: Readings, coded under a written protocol from the saved reports (Online Appendix E5; a second coder who did not see the wedges agreed on 15 of the 17 readings, Cohen’s kappa 0.83): cap, the budget is the developer’s whole available corpus, or the corpus was repeated, or data limited the budget; choice, the budget is stated as set by cost, compute or schedule, or lies below a larger usable pool, or other models trained longer on the same data; menu, a size set by a hardware or configuration menu; cap or choice, only the corpus or budget size is stated, or evidence for both is present (SmolLM trained 600B tokens on a 252B-token corpus, a repetition that makes it a lower bound under either reading). Statements that a model was still improving are not evidence for either reading. Id.: pt, point estimate of the value of compactness; lb, lower bound; lb^c, lower bound that is a point estimate if the budget was chosen. D : the unit’s token budgets in the order of the listed members; one value if shared. s_f : family share under the reference technology. Sources: Touvron et al. (2023b) (both LLaMA budgets), Touvron et al. (2023a), Databricks AI Research Team (MosaicML) (2023a,b), Stability AI (2023), Bai et al. (2023), 01.AI, Young et al. (2024), DeepSeek-AI et al. (2024b), Grattafiori et al. (2024); Meta AI (2024), Yang et al. (2024), Ben Allal, Lozhkov and Bakouch (2024), Qwen Team et al. (2024), IBM Granite Team (2024), Team OLMo et al. (2024), Yang et al. (2025), Project Apertus et al. (2025), Team Olmo, Ettinger et al. (2025). Every quote is verified against the saved source.

TABLE E6. LAB-OWN TECHNOLOGIES UNDER ONE RULE

Models	n	Reference	One rule	Lab curvature	Meta 10-budget law	OLMo $\kappa = 1$ path	Earlier rule
Lab-own models without Llama 1 and 2 (primary)	15	0.70	0.53	0.73	0.53	0.41	0.66
without AI2	6	0.65	0.71	0.73	0.64	0.71	0.65
Meta: the Llama 3 herd	3	0.63	0.69	0.74	0.55	0.69	0.60
All lab-own models (earlier set, with Llama 1 and 2)	22	0.65	0.52	0.67	0.49	0.45	0.61
without AI2	13	0.56	0.64	0.69	0.54	0.64	0.59
Meta, with Llama 1 and 2	10	0.49	0.58	0.62	0.47	0.58	0.51
AI2	9	0.70	0.44	0.66	0.44	0.32	0.66

Note: Median s over the models for which a developer published scaling evidence. Llama 1 and 2 were allocated before Meta’s 2024 path existed and are left out of the primary set; the earlier set, which applied Meta’s path to them as well, is shown for comparison. One rule: the lab’s own expansion path, in its own parameter convention (the OLMo ladder without the input embedding, DeepSeek’s law in non-embedding FLOPs per token), plus the common model-free $\sigma^* = 0.687$ (the study-level mean of Table D7). Meta: path of the 8 IsoFLOP budgets whose minimum is bracketed; AI2: OLMo ladder path (κ free); DeepSeek: published law; Marin: Nemotron-CC ladder path. Lab curvature: Meta 0.660, Marin 0.705, AI2 0.544 (κ -free ladder). Meta 10-budget law: the path that reproduces Meta’s published planning law (it uses two unbracketed budgets). OLMo $\kappa = 1$ path: AI2’s ladder with its $\kappa = 1$ fit. In the lab-curvature, Meta 10-budget and OLMo $\kappa = 1$ columns, models without the variant keep the one-rule wedge. Earlier rule: the assignments of an earlier version (Meta’s ten-budget path and Marin’s path with their model-free curvatures; AI2’s κ -free ladder with its own curvature; DeepSeek with the reference curvature).

ference that permutes the model-level code within year-by-compute strata rejects ($p < 0.001$ for the coefficient, 0.06 for its t -statistic), but it treats each size as a separate assignment, although one developer decides for all its sizes and 19 of the 26 served models are Alibaba’s. Permuting the developers’ shares of served models across developers (MacKinnon and Webb, 2020) gives 0.10, and 0.94 without Alibaba. Open weights carry a premium: at the decision level, with the clean sample’s common budgets entered once, open-weight decisions have log wedges 0.45 higher ($p = 0.018$), and 0.63 higher among releases since 2023 ($p = 0.001$). Over-training is highest, by deployment target, for models built for devices their developers do not serve (median share 0.90 against 0.70), although the single most over-trained decision, Qwen2 0.5B, was served; the contrast is descriptive and rests on two developers.

Tiers and derivatives. Windows around quantized memory boundaries hold 51 percent of the 220 open-weight production-scale models released in 2023–2026, against 13 percent under a smooth counterfactual (Chetty et al., 2011; Kleven, 2016), but position relative to the tiers does not predict the wedge beyond the mechanical relation between N and $\ln w$ when compared with randomly shifted tier menus. Models with higher M at given compute have more derivative repositories on Hugging Face (elasticity 0.80, p -value 0.031), as planned demand predicts but also as accessibility predicts; the elasticity with respect to planned T/D is only 0.60 (p -value 0.086), and the verified sample selects on popularity. These two tests use the token counts and serving code of the previous version.

The reading the data favor. A developer that bears all of its serving cost would over-train more the more it serves, and a developer that serves no one would over-train less. The data show no detectable serving effect once Alibaba is set

aside, over-training is highest, by deployment target, for the on-device models that their developers do not serve, and open weights carry a premium of about half a log point. They are more consistent with the adoption and tier mechanisms (Proposition A3(iii)(a') and (b)) than with serving-cost internalization.

E7. Extrapolation and Partial Identification

Extrapolation in M . Table E9 compares the local wedge of Proposition A5(i), from a kernel-weighted local quadratic of the loss surface, with the wedges of parametric forms fitted to the same runs, in three recipes. In each, parametric forms fitted to the whole design understate the local wedge at M of 256 to 1,024, and the gap grows with M . The κ -free form's log gap there is -0.14 on Farseeer, -0.43 on Marin's three corpora pooled (s.e. 0.03) and -0.38 on Llama 3 (two evaluation points); a Chinchilla form fitted to runs with $M \leq 100$ gives -0.95 , -1.03 and -0.90 . Measured against a linear extrapolation from the local slope at the path, the form of the κ family, the local $\ln w$ is convex on Farseeer and Marin, by 0.17 and 0.43 log points at M of 256 to 1,024. On Marin that is the whole of the κ -free form's gap; leaving out the corner runs of each budget gives 0.38 to 0.46, but the bin then stops at M of about 320 or 595, so the change reflects which points remain. On Llama 3 the local $\ln w$ is concave and bandwidth-sensitive (-0.13 at M of 64 to 256, about zero at its two points near $M = 290$), and its gap comes from a steeper local slope at the path. The local wedge is evaluated only up to $M \approx 2,570$ on Farseeer, 1,110 on Marin and 290 on Llama 3, so the direction of the extrapolation error is established in three recipes at 10^{18} to 10^{22} FLOP, not in general. Twenty clean models lie beyond Farseeer's range of M in its non-embedding convention, and the measured inefficiency gradients of Farseeer's tuning rule change its convexity by only -4 to $+8$ percent, but they are measured at M up to 466 only, while Farseeer's grid reaches 2,570 (Online Appendix D6). Revealed preference recovers the technology a developer believed, which may be the parametric one (Remark A7). [TBD-m9: the controlled experiment's local-wedge check under Amendments 1 and 2.]

Extrapolation in compute: the identified set. No design observes $M^*(C)$ above 10^{22} FLOP. For each study we fit a line to the logged IsoFLOP minima at its bracketed budgets, $\ln M_b^* = a_s + e_s(c_b - c_J)$, and extrapolate it with a simultaneous 95 percent band that carries the joint sampling error of level and slope: a wild bootstrap- t with Webb weights and 9,999 draws, studentized by the within-study pooled residual variance, with a sup- t critical value at least as large as the normal-theory one (notes to Table E10). Chinchilla has nine bracketed budgets up to 3×10^{21} FLOP, in total parameters as digitized, the convention of the models' M ; Llama 3 has eight up to 10^{21} , in FLOP-implied parameters; and Marin's three corpora, which share code and ladder, have seven, seven and five up to 3×10^{20} FLOP, with corpus-specific levels and one slope, 0.198 (95 percent interval $[0.132, 0.265]$). DeepSeek's published law is a fourth, point path. By Proposition A6(vii) and (viii), the union of the bands, widened by $\ln \tau$, is an

TABLE E7. CONDUCT TESTS WITH MODEL-LEVEL SERVING AND DECISION UNITS

	n	Clusters	Treated	Coef.	(s.e.)	WCR p
<i>Panel A. Serving, outcome $\ln \hat{w}$ given $\ln C$ and year effects</i>						
Models: developer-level code	77	18	7	0.43	(0.27)	0.481
Models: model-level code	77	18	6	0.56	(0.24)	0.413
Models: model-level, ambiguous sizes served	77	18	7	0.62	(0.24)	0.220
Models without Alibaba	58	17	5	-0.09	(0.28)	0.751
Decisions: any member served	49	18	6	0.28	(0.29)	0.673
Decisions: all members served	49	18	4	0.49	(0.17)	0.159
Decisions without Alibaba	43	17	5	-0.20	(0.30)	0.572
<i>Panel B. Open-weight premium, production-scale universe, outcome $\ln \hat{w}$</i>						
Models	186	67	55	0.51	(0.18)	0.012
Decisions	159	67	55	0.45	(0.17)	0.018
Decisions, 2023 and later	123	60	52	0.63	(0.15)	0.001
<i>Panel C. Descriptive: median s (reference)</i>						
Served (model-level)	26	6		0.83	[0.63, 0.91]	
Not served	51	15		0.74	[0.60, 0.84]	
Served, without Alibaba	7	5		0.55	[0.21, 0.78]	
Served, in a tier window	18	5		0.74	[0.57, 0.86]	
On-device target	9	2		0.90	[0.84, 0.92]	
Server or unspecified	65	16		0.70	[0.57, 0.85]	
Decisions, served	11	6		0.68	[0.30, 0.79]	
Decisions, not served	38	14		0.74	[0.61, 0.84]	
<i>Panel D. Randomization inference for the serving coefficient of Panel A (two-sided)</i>						
	n	Strata	Treated	Coef.	RI p	RI- t p
Models, strata: year $\times$ compute tercile	77	9	6	0.56	<0.001	0.06
Models, strata: year $\times$ compute half	77	6	6	0.56	<0.001	0.07
Models, strata: year	77	3	6	0.56	<0.001	0.07
Models without Alibaba, year $\times$ tercile	58	9	5	-0.09	0.63	0.76
Decisions, year $\times$ tercile	49	9	6	0.28	0.17	0.38
Decisions, strata: year	49	3	6	0.28	0.11	0.40
Developers, slope weighted by models	77	18	6	0.56	0.10	
Developers, unweighted slope	77	18	6	0.24	0.28	
Developers without Alibaba, weighted	58	17	5	0.01	0.94	

Note: Panels A–B: restricted wild cluster bootstrap- t by developer, Webb weights, $B = 9,999$; CRV1 standard errors in parentheses; treated = developer clusters with the attribute (with three or fewer the p -value is not interpretable). Model-level serving code: this size served first-party (per-token API or consumer product) within 180 days of release; ambiguous: Meta AI “built with Llama 3” (8B, 70B) and 01.AI’s early-access API (Yi-34B). Decisions: members of a family that share one token budget enter once (family share; total compute; first release year); in Panel B, the clean sample’s common budgets enter once and other models of the production-scale universe are decisions of their own. Panel C: interquartile range in brackets; columns 2–3 give n and developers. Panel D: randomization inference (MacKinnon and Webb, 2020), 9,999 permutations; RI p for the coefficient, RI- t p for its CRV1 t -statistic. The rows for models and decisions permute the serving code across units within the strata shown (column 3: number of strata), which treats each size as a separate assignment although one developer decides for all its sizes; the developer rows permute the developers’ shares of served models across the 18 developers (column 3: developers) and use the developer-level slope of the residuals of $\ln \hat{w}$ on $\ln C$ and year effects.

identified set for the developer’s zero point. It gives $M^*(10^{24}) \in [5.7, 248]$. At 10^{25} FLOP the set is $[4.5, 468]$. With every path and model in total parameters, the set at 10^{24} FLOP is $[4.4, 391]$. Marin’s band sets the upper end of the set for 76 of the 77 models and Llama 3’s band the lower end for all of them.

The sign. By Proposition A6(iii), $w > 1$ is identified whenever $\ln M$ exceeds the upper end of the set, for any technology with single-peaked IsoFLOP profiles. Table E10 reports the share identified as a function of a tilt allowance τ on M^* that covers tokenizer differences and data-quality tilts. With no allowance the sign is identified for 78 percent of models, 69 percent of the 49 decisions and 71 percent of the 56 units of the earlier version, but for only 24 percent of training compute (14 percent over decisions). No model is identified as below one, which is mechanical: every clean model has $M \geq 19.5$, while the lower end of the set at the models’ compute runs from 4.0 to 11.4. The union also matters for that statement: Marin’s band alone would identify $w < 1$ for six flagships (Llama 2 70B, LLaMA 65B, Llama 3.1 405B, Falcon 180B, Falcon 40B and DeepSeek LLM 67B). The 17 models whose sign is not identified all have M below 250 (the largest is Qwen2.5 72B, at 247.6), and only one of them, LLaMA 13B, has fewer than 15 billion parameters. By year, the compute-weighted identified share is 12 percent in 2023, 19 percent in 2024 and 49 percent in 2025.

Straight paths. The paths are straight lines by assumption, and inside the designs they are not: a quadratic term in $\ln C$ has a t -statistic of 7.15 for Chinchilla (convex), 2.39 for Llama 3 (convex) and -3.11 for Marin (concave). At the anchors the local slopes are 0.40 for Chinchilla and 0.25 for Llama 3, against -0.05 for Marin, so Marin’s steep line comes from its small budgets and the flat lines of Chinchilla and Llama 3 average a fall and a rise. The bands cover sampling error around a line, not this misspecification. Continuing Chinchilla’s and Llama 3’s paths above their anchors at their local slopes, beside the bands, identifies 75 percent of models and 20 percent of compute, with an upper end of 339 for $M^*(10^{24})$; with one standard error added to each local slope, 71 and 17 percent, and 477.

Other identified sets. Table E10 also reports sets under weaker or different restrictions. An earlier version of this paper (PI-1) put each design’s level on a bootstrap- t interval at its largest bracketed budget and bounded the path slopes by the point estimates of the paths and published laws, $[-0.156, 0.162]$; it identified 87 percent of models and 48 percent of compute. It excluded Marin’s Comma corpus, whose five bracketed budgets give the interval $[0.2, 3,586]$ for its anchor, by a rule of at least six budgets that was chosen after that interval was seen; with Comma only 8 percent of models were identified. Levels on t -intervals with $k - 2$ degrees of freedom, with Comma admitted with its own slope of 0.294, identify 68 percent of models and 17 percent of compute; PI-1’s levels with slopes anywhere in the hull of the slopes’ 95 percent intervals, $[-0.175, 0.353]$, identify 71 and 17 percent. Every set that carries slope uncertainty or admits Comma with a valid interval lies between 68 and 78 percent of models and 17 and 24 percent of compute, except one that lets every slope reach its own t -interval, Comma’s

0.459 included: 44 and 9 percent. With normal inputs as the only restriction ($|e_p| \leq 1$), 13 percent of models are identified (14 with PI-1’s levels); admitting every in-set technology’s in-support M^* as an anchor, 23 percent. With anchors and models all in total parameters the share of models is 71 percent, and all in FLOP-effective parameters 75 percent.

Tilt allowances and the breakdown frontier. Figure E1 traces the share identified against τ . The share of models falls below one half at $\tau = 3.7$ (2.7 with PI-1’s levels and slopes at their 95 percent intervals), the share of models below 15 billion parameters at 7.5, and the share of the 49 decisions at 2.7; the compute share is below one half at every τ . At $\tau = 1.84$, DataDecide’s largest tilt under the reference exponents, the shares are 66 percent of models, 61 percent of decisions and 17 percent of compute; at 3.4, the same tilt under DataDecide’s own exponents, 53, 47 and 16 percent.

Magnitudes and the curvature at frontier compute. Magnitudes need a curvature (Proposition A6(iv)). With $1/\sigma^* - 1$ between 0.40 and 0.68, from the in-design estimates (0.40 to 0.52) to the top-budget value ($\sigma^* = 0.594$), the medians of the models’ lower and upper bounds on s are 0.41 and 0.95 (0.41 and 0.90 with 0.40 to 0.52); tilt allowances of 1.84 and 3.4 lower the median lower bound to 0.24 and 0.03. Over the 49 decisions, whose bounds are the compute-weighted harmonic means of their members’ bounds, the medians are 0.33 and 0.88 with 0.40 to 0.52, and the median lower bound is 0.15 at $\tau = 1.84$ and -0.12 at 3.4: at DataDecide’s own factor the median decision’s sign is not identified. The curvature itself falls with compute in the two IsoFLOP designs that reach 10^{21} FLOP (Online Appendix D3). Beyond the designs, $\sigma^*(C)$ is partially identified (Manski, 2003): it lies between the value at the largest budgets and a linear continuation of the Chinchilla–Llama 3 drift if it does not rise with compute and falls no faster than where it is observed (Section III; Table E8). With σ^* at the lower end of that set at each model’s compute, the median upper bound on s is 0.99. We also propagate constant values of σ^* into the wedges with each decision’s reference zero point held fixed. Because $\ln w$ rises with $1/\sigma^* - 1$ when $w > 1$, the reference levels are lower bounds if σ^* falls with compute: the median decision share is 0.73 at the reference, 0.87 at $\sigma^* = 0.60$, 0.85 at 0.62 and 0.96 with σ^* at the lower end of its set. The share with $w > 1$ does not depend on the curvature. A drifting σ^* also violates the quasi-homotheticity behind the closed forms, so these are scale propagations, not re-estimations (Table E8).

E8. A Second Output

A developer that optimizes a task metric rather than validation loss is misread by an econometrician who uses loss, by the factor $w_L/w_{L'}$ of Proposition A3(ii). The OLMo ladder’s 30 runs are scored both on C4 cross-entropy and on task bits per byte, so technologies fitted to each output on the same runs measure that factor (a test of the output critique of Bond et al., 2021). Table E11 reports the median over the 77 clean models of $\ln(w_{\text{task}}/w_{C4})$: -0.11 with $\kappa = 1$ and

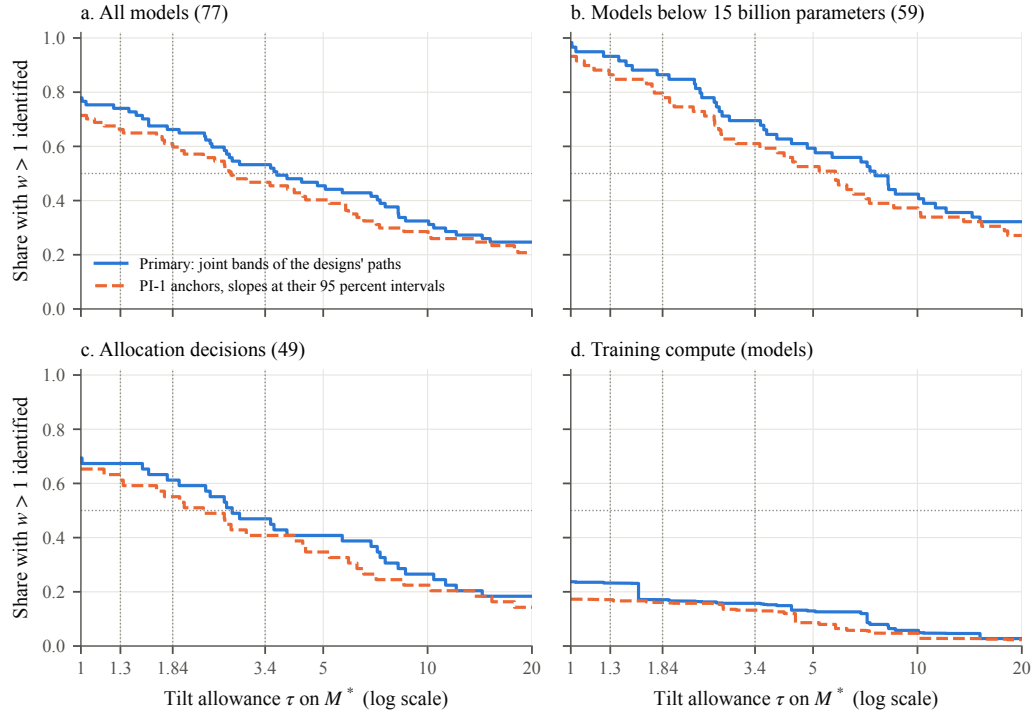


FIGURE E1. BREAKDOWN FRONTIER OF THE SIGN RESULT

Note: Share of the clean sample for which $w > 1$ is identified beyond the designs against the tilt allowance τ on M^* (log scale): all 77 models (panel a), the 59 below 15 billion parameters (b), the 49 allocation decisions (c) and training compute 6ND over models (d). Solid: the primary set of Table E10; dashed: PI-1's levels with path slopes anywhere in the hull of the designs' 95 percent slope intervals. Dotted vertical lines: $\tau = 1.3$ (tokenizer differences), 1.84 and 3.4 (DataDecide's largest tilt under the reference exponents and under its own); dotted horizontal line: one half. The share of models falls below one half at $\tau = 3.7$ (2.7 under the dashed set); the compute share is below one half at every τ . [TBD-m9: the experiment's implied factor for M^* at 10^{15} – 10^{17} FLOP (Amendment 1, Q2), marked as a small-scale magnitude beside DataDecide's.]

Source: Author's calculations.

TABLE E8. CURVATURE AT FRONTIER COMPUTE AND THE SIZE OF THE REVEALED WEDGES

<i>Panel A. $\sigma^*(C)$ beyond the designs and the wedge scale $\theta = 1/\sigma^* - 1$</i>						
Compute (FLOP)	10^{21}	10^{22}	10^{23}	10^{24}	10^{25}	10^{26}
σ^* , upper bound ^a	0.594	0.594	0.594	0.594	0.594	0.594
σ^* , lower bound, Chinchilla–Llama 3 drift (−0.058)	0.594	0.536	0.479	0.421	0.363	0.306
σ^* , lower bound, pooled drift (−0.033)	0.594	0.561	0.528	0.495	0.463	0.430
θ at the upper bound	0.684	0.684	0.684	0.684	0.684	0.684
θ at the lower bound (Chinchilla–Llama 3)	0.684	0.865	1.089	1.375	1.751	2.269
Memo: θ at $\sigma^* = 0.70$	0.429	0.429	0.429	0.429	0.429	0.429
<i>Panel B. Revealed wedges and shares on the clean sample (77 models)</i>						
Curvature	Median			Llama 3	OLMo 2	SmolLM2
	w	s	s , units ^b	herd s_f	7B w	1.7B w
Reference ($\sigma^* = 0.701$)	3.99	0.749	0.736	0.294	4.0	11.3
$\sigma^* = 0.70$	4.00	0.750	0.737	0.294	4.0	11.4
$\sigma^* = 0.65$	5.71	0.825	0.813	0.349	5.7	21.3
Top budgets, 95% upper ^c ($\sigma^* = 0.622$)	7.14	0.860	0.849	0.380	7.1	31.6
Top budgets ($\sigma^* = 0.594$)	9.13	0.890	0.881	0.412	9.1	48.6
$\sigma^* = 0.60$	8.64	0.884	0.874	0.405	8.6	44.2
$\sigma^* = 0.55$	14.11	0.929	0.922	0.464	14.1	104.4
$\sigma^*(C_i)$, pooled line	11.07	0.910	0.890	0.525	11.1	61.1
$\sigma^*(C_i)$, lower bound (Chin.–Llama)	45.50	0.978	0.963	0.750	41.8	526.0
$\sigma^*(C_i)$, lower bound (pooled)	19.80	0.949	0.935	0.594	19.8	165.6

Note: Panel A: ^aPartial identification (Manski, 2003). For compute above the largest bracketed budgets ($\approx 10^{21}$ FLOP), $\sigma^*(C) \in [\sigma_{\text{lin}}(C), \sigma_{\text{top}}]$ if σ^* does not rise with compute beyond the designs (upper bound $\sigma_{\text{top}} = 0.594$: inverse-variance mean of Chinchilla’s and Llama 3’s estimates at 6×10^{20} FLOP and above, placed at 10^{21}) and falls no faster than the within-design drift measured where it is observed (lower bound: linear continuation from σ_{top} , truncated to $(0, 1)$). If the drift is not real (Marin shows none up to 3×10^{20}), the level measured within the designs applies (0.687 across studies, 0.701 for the reference technology); the drift-agnostic set is $[\sigma_{\text{lin}}(C), 0.70]$. Panel B: $\ln w = \theta \ln(M/M_{\text{ref}}^*(C))$, each model’s reference zero point held fixed; $s = (w - 1)/w$. $\sigma^*(C_i)$ rows evaluate the curvature at each model’s training compute (median 2×10^{23} FLOP; 76 of 77 models above 10^{21}). ^bDecision units: one family-level share per common- D family ($\bar{w}_H = [\sum_j h_j/w_j]^{-1}$, 11 families) plus the 45 members of size-specific families and singletons: the 56 family-label units of Online Appendix E, a robustness definition. ^cUpper end of the top-budget value’s modified Hartung–Knapp–Sidik–Jonkman interval with independent budgets; with the variance of the joint bootstrap draws (Table D6) it is 0.627. The share with $w > 1$ (0.974) does not depend on θ . For $w > 1$, $\ln w$ increases with θ : if σ^* falls with compute, the reference level is a lower bound on the wedge, not a midpoint. The lower bound on $\sigma^*(C)$ assumes a linear decline at the Chinchilla–Llama 3 rate; a hinge fit gives a steeper decline above 3×10^{20} FLOP, so that bound is not conservative against acceleration.

Source: Author’s calculations.

TABLE E9. PARAMETRIC EXTRAPOLATION AGAINST THE MODEL-FREE WEDGE IN THREE RECIPES

	$M = D/N$				
	<16	16–64	64–256	256–1,024	$\geq 1,024$
<i>Farseeer: non-embedding N, 404 runs; $b_2 = +0.020$ [0.019, 0.022]</i>					
Local $\ln w$	−0.69	0.06	0.68	1.37	2.01
κ free, all runs	−0.02	+0.05	+0.00	−0.14	−0.36
	[−0.03, −0.01]	[0.05, 0.06]	[0.00, 0.01]	[−0.15, −0.13]	[−0.47, −0.32]
$\kappa = 1$, $M \leq 100$	+0.08	−0.23	−0.55	−0.95	−1.36
	[0.07, 0.09]	[−0.24, −0.22]	[−0.55, −0.52]	[−0.96, −0.92]	[−1.47, −1.31]
Convex part of local $\ln w$	+0.09	+0.00	+0.02	+0.17	+0.37
	[0.08, 0.10]	[−0.00, 0.00]	[0.00, 0.02]	[0.15, 0.18]	[0.34, 0.46]
<i>Llama 3: 133 runs; $b_2 = -0.029$ [−0.035, −0.009]</i>					
Local $\ln w$ (points)	−0.80 (70)	0.29 (38)	0.97 (16)	1.70 (2)	—
κ free, all runs	+0.27	+0.04	−0.08	−0.38	—
	[0.18, 0.31]	[0.01, 0.06]	[−0.14, −0.04]	[−0.52, −0.18]	
$\kappa = 1$, $M \leq 100$	+0.37	−0.16	−0.46	−0.90	—
	[0.26, 0.40]	[−0.22, −0.14]	[−0.52, −0.45]	[−1.03, −0.74]	
Convex part of local $\ln w$	−0.05	−0.04	−0.13	−0.01	—
	[−0.07, −0.02]	[−0.05, −0.02]	[−0.16, −0.06]	[−0.13, 0.15]	
<i>Marin, three corpora pooled (equal weights); $b_2 = +0.027$ to $+0.032$ by corpus</i>					
κ free, all runs	−0.09 [†]	+0.07	−0.09	−0.43	−1.03
	[−0.09, −0.04]	[0.05, 0.08]	[−0.09, −0.04]	[−0.48, −0.36]	[−1.28, −0.95]
$\kappa = 1$, $M \leq 100$	−0.07 [†]	−0.21	−0.53	−1.03	−1.73
	[−0.07, −0.03]	[−0.22, −0.17]	[−0.54, −0.44]	[−1.08, −0.91]	[−1.95, −1.63]
Convex part of local $\ln w$	+0.14	−0.06	+0.09	+0.43	+1.02
	[0.10, 0.14]	[−0.08, −0.04]	[0.01, 0.09]	[0.32, 0.45]	[0.94, 1.21]
Largest- M runs left out	+0.09	−0.03	+0.09	+0.38	—
Runs with $M > 1,000$	+0.13	−0.05	+0.10	+0.46	—

Note: Entries: $\Delta = \ln(w_{\text{param}}/w_{\text{local}})$ averaged over the evaluation points in each bin of $M = D/N$, with 95% percentile bootstrap intervals below; negative values mean that the parametric form understates the wedge. Local wedge: $w = (f_c - f_x)/(f_c + f_x)$ for $f = \ln L$, $x = \ln M$ and $c = \ln C$, from a Gaussian-kernel local quadratic in (x, c) with leave-one-out cross-validated bandwidths (one for Marin’s three corpora); it needs no functional form and no loss units. Evaluation points: 14 log-spaced M on every budget line inside its observed range, kernel effective sample size at least 8 (number of points in parentheses). Comparators: the κ family with κ free fitted to all runs and the Chinchilla form ($\kappa = 1$) fitted to the runs with $M \leq 100$, by Huber loss on $\ln L$ in the same FLOP-implied convention. Intervals: wild cluster bootstrap by budget (Webb weights, 999 draws), every fit recomputed in each draw; they exclude smoothing bias. [†]The percentile interval excludes the estimate at full precision, because the bootstrap population’s statistic differs from the estimate by more than the draws’ spread (bias-corrected basic intervals in the replication files). b_2 : coefficient on u^2 in $\ln w_{\text{local}} = b_1 u + b_2 u^2$, $u = \ln(M/M_{\text{local}}^*(C))$, $M_{\text{local}}^*(C)$ the root of $\ln w_{\text{local}} = 0$ on each budget line; $b_2 > 0$ means $\ln w$ is convex. Convex part: $\ln w_{\text{local}} - b_1 u$, the gap between the local wedge and its linear extrapolation from the local slope at the path; a form’s Δ is minus the convex part plus a rotation, $\ln w_{\text{param}} - b_1 u$. Leave-corner-out rows (points only): without each budget’s largest- M run the local wedge is evaluated up to $M \approx 320$ (the primary row’s convex part on those points is 0.28), and without runs with $M > 1,000$ up to $M \approx 595$. Marin: 85, 84 and 88 runs on Comma, DCLM and Nemotron-CC (per-corpus rows in the replication files); a run whose loss exceeds that of the same configuration on fewer tokens is dropped (DCLM, 157M, 1.8×10^{20} FLOP). The local wedge is evaluated up to $M \approx 1,110$ on Marin (runs reach 3,706) and $M \approx 290$ on Llama 3 (runs reach 643). The $M \geq 1,024$ bin rests on 1–2 corner points per corpus and is fragile; Llama 3’s 256–1,024 bin is two points at its lower end. Farseeer (module ra1, a near-factorial grid): local quadratic in $(\ln N, \ln D)$ evaluated at its 25 sizes $\times$ 24 log-spaced M up to 2,570 inside each size’s observed range of D , kernel effective sample size at least 15; wild cluster bootstrap by model size. *Source:* Li et al. (2025b); Held and Marin Community (2026); Grattafiori et al. (2024); Czech (2026); author’s calculations.

TABLE E10. SIGN IDENTIFICATION BEYOND THE DESIGNS, BY TILT ALLOWANCE

Identified set	Share identified, tilt allowance τ						Share of compute, τ			
	1	1.3	1.84	3.4	4.4	10	1	1.84	3.4	10
<i>Panel A. Models (77)</i>										
Primary: joint bands of the paths	0.78	0.74	0.66	0.53	0.47	0.32	0.24	0.17	0.16	0.06
Models below 15B parameters ($n = 59$)	0.98	0.93	0.86	0.69	0.61	0.42	1.00	0.96	0.88	0.32
PI-1: point slopes, Comma excluded	0.87	0.86	0.82	0.69	0.68	0.48	0.48	0.34	0.23	0.15
PI-1 with Marin Comma	0.08	0.05	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00
t -interval levels, Comma admitted	0.68	0.65	0.53	0.43	0.39	0.25	0.17	0.16	0.13	0.03
PI-1, slopes at 95 percent intervals	0.71	0.66	0.60	0.47	0.42	0.29	0.17	0.16	0.13	0.05
Normal inputs only ($ e \leq 1$)	0.13	0.13	0.10	0.10	0.10	0.03	0.00	0.00	0.00	0.00
Every technology as an anchor	0.23	0.21	0.14	0.10	0.09	0.04	0.03	0.01	0.01	0.00
All anchors in total parameters	0.71	0.65	0.58	0.45	0.42	0.26	0.17	0.16	0.13	0.03
All in FLOP-effective parameters	0.75	0.70	0.65	0.48	0.44	0.27	0.24	0.17	0.15	0.05
<i>Panel B. Allocation decisions (49)</i>										
Primary: joint bands of the paths	0.69	0.67	0.61	0.47	0.41	0.27	0.14	0.11	0.10	0.01
PI-1: point slopes, Comma excluded	0.82	0.80	0.76	0.63	0.61	0.41	0.40	0.32	0.14	0.10
PI-1 with Marin Comma	0.06	0.04	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
t -interval levels, Comma admitted	0.63	0.59	0.47	0.39	0.33	0.18	0.12	0.10	0.10	0.00
PI-1, slopes at 95 percent intervals	0.65	0.61	0.55	0.41	0.37	0.22	0.12	0.11	0.10	0.01
Normal inputs only ($ e \leq 1$)	0.14	0.14	0.10	0.10	0.10	0.02	0.00	0.00	0.00	0.00
Every technology as an anchor	0.16	0.12	0.10	0.08	0.08	0.02	0.00	0.00	0.00	0.00
All anchors in total parameters	0.65	0.59	0.53	0.41	0.37	0.20	0.12	0.10	0.10	0.00
All in FLOP-effective parameters	0.67	0.65	0.59	0.41	0.39	0.20	0.14	0.11	0.10	0.00
Primary, 56 family-label units	0.71	0.68	0.61	0.48	0.43	0.29	0.15	0.12	0.11	0.01
<i>Panel C. Primary, by release year (models)</i>										
2023 ($n = 24$)	0.63	0.50	0.33	0.13	0.08	0.04	0.12	0.05	0.02	0.01
2024 ($n = 40$)	0.88	0.88	0.83	0.70	0.60	0.45	0.19	0.11	0.10	0.03
2025 ($n = 13$)	0.77	0.77	0.77	0.77	0.77	0.46	0.49	0.49	0.49	0.20

Note: Share of the clean sample for which $w > 1$ is identified beyond the designs (Proposition A6(iii), (vii) and (viii)): $\ln M$ exceeds the upper end of the identified set for $\ln M^*(C)$ at the model's compute by at least $\ln \tau$; no model is identified as $w < 1$ in any row. Tilt allowances τ : 1.3, tokenizer differences; 1.84, the largest tilt across DataDecide's recipes, 0.26, which corresponds to $\tau \approx 1.8$ under the reference exponents (we use 1.84); 3.4, the same tilt under DataDecide's own exponents; 4.4, both 1.3 and 3.4; Figure E1 traces τ up to 20. *Primary:* one path per study, a line through the logged IsoFLOP minima at its bracketed budgets, extrapolated with a simultaneous 95 percent band for its level and slope (wild bootstrap- t , Webb weights, 9,999 draws, within-study pooled residual variance, critical value at least the normal-theory one) over 4.8×10^{20} (Marin: from 3×10^{20} to 4.0×10^{25} FLOP; brackets give pointwise 95 percent intervals. Chinchilla: 9 budgets to 3×10^{21} FLOP, total parameters as digitized (the convention of the models' M), level 20.7 [13.8, 30.9] at 3×10^{21} FLOP, slope -0.003 [$-0.122, 0.115$]. Llama 3: 8 budgets to 10^{21} , FLOP-implied $N_F = C/(6D)$, level 15.1 [11.2, 20.3], slope 0.008 [$-0.086, 0.102$]. Marin: DCLM and Nemotron-CC (7 budgets each) and Comma (5) to 3×10^{20} , FLOP-implied, corpus levels 15.0, 15.5 and 20.8 and one slope, 0.198 [0.132, 0.265], with a pooled residual variance (15 degrees of freedom). DeepSeek's published law enters as a point path in its own convention. The identified set is the union of the bands widened by $\ln \tau$; for $M^*(10^{24})$ it is [5.7, 248] and for $M^*(10^{25})$ [4.5, 468]. *PI-1* (version 3): bootstrap- t intervals for each design's level, slopes bounded by the point estimates of the paths and published laws, [$-0.156, 0.162$]; Marin Comma (five budgets, interval [0.2, 3,586]) was excluded by a rule of at least six budgets chosen after that interval was seen. *t-interval levels:* levels on t -intervals with $k - 2$ degrees of freedom, Comma admitted with its own slope, 0.294; slopes in [$-0.156, 0.294$]. *Slopes at 95 percent intervals:* PI-1's levels, slopes in [$-0.175, 0.353$]. *Normal inputs only:* the primary levels, $|e| \leq 1$. *Every technology:* the primary bands plus every in-set technology's in-support M^* , slopes in [$-0.131, 0.311$]. *Conventions:* models' M is in total parameters except in the rows in FLOP-effective parameters. In total parameters, Marin's minima are converted with its measured ratio of FLOP-implied to configuration parameters (1.03 to 1.05 at the anchors), and Llama 3's band is shifted by the range of $\ln(N_F/N)$ at its anchor over three FLOP accountings, [$-0.25, 0.20$]. In FLOP-effective parameters, Chinchilla's minima and the models' parameters follow Hoffmann et al.'s count (attention at 4,096 tokens for the models). Compute-weighted: shares of training compute $6ND$. Allocation decisions: members of one family that share one token budget (within 10 percent) form one decision; a decision is identified only if every member is.

Source: Author's calculations.

TABLE E11. A SECOND OUTPUT ON THE SAME RUNS: TASK BITS PER BYTE VERSUS C4 LOSS

Output	Chinchilla form ($\kappa = 1$)				κ free			
	Log ratio	[95%]	s	σ^*	Log ratio	[95%]	s	σ^*
C4 loss (baseline)			0.22	0.825			0.75	0.544
Task BPB, all tasks	-0.11	[-0.78, 0.60]	0.13	0.763	-0.37	[-1.17, 1.26]	0.64	0.512
MMLU	0.01	[-0.81, 0.77]	0.22	0.789	0.14	[-0.66, 0.98]	0.78	0.498
ARC, OBQA	0.17	[-0.92, 1.00]	0.34	0.720	-0.42	[-1.51, 1.94]	0.62	0.570
Commonsense	-0.07	[-0.96, 0.62]	0.16	0.734	-0.34	[-1.10, 1.10]	0.64	0.482
BoolQ	-1.75	[-57.96, -0.20]	-3.47	0.865	0.89	[-169.01, 5.59]	0.90	0.251

Note: Log ratio: $\ln(w_{\text{task}}/w_{\text{C4}})$. Commonsense: HellaSwag, PIQA, SocialIQA, WinoGrande, CommonsenseQA. OLMo ladder (Bhagia et al., 2025): 30 runs, the same runs scored on C4 cross-entropy and on task bits per byte (validation splits). Technologies are fitted separately to each output. $\ln(w_{\text{task}}/w_{\text{C4}})$: median over the 77 clean-sample models (OLMo’s parameter convention) of the log ratio of the wedges implied by the task-output and the C4-output technologies; by Proposition A3(ii) it measures how a developer optimizing task performance would be misread by an econometrician using loss. Brackets: 95 percent interval of the cross-model median from a pairs bootstrap by (size, multiplier) cell with the same resample for both outputs ($B = 399$). Across models (10th–90th percentile) the all-task ratio runs from -0.40 to 0.21 ($\kappa = 1$) and from -0.75 to 0.05 (κ free). s : median share under that output’s technology. BoolQ’s $\kappa = 1$ fit is degenerate ($M^*(10^{21})$ in the millions).

-0.37 with κ free for all tasks. Every 95 percent interval of the cross-model median includes zero except BoolQ’s under $\kappa = 1$, whose fit is degenerate; the effect varies by task family, and neither its sign nor its size is stable across forms. The intervals are wide: with κ free, the all-task interval allows a task-valuing developer’s wedge from a third of the loss-based one to three and a half times it. This is one recipe at small scale (190M to 3.2B parameters, $M \leq 200$): the design has no power, and the output critique remains unbounded.

E9. The Cost Side and Post-Training Compute

FLOPs are not costs. Table E12 lets the price of a training FLOP vary with model size (elasticity δ) and serving cost per token scale as N^ϕ , with a price ϱ of a serving FLOP relative to a training FLOP (Proposition A3(iii)(f)). The share $s = (w - 1 - \delta)/(w - 1 - \delta + \phi)$ does not depend on ϱ and moves little: the median is 0.72 to 0.77 for δ between -0.1 and 0.1 with $\phi = 1 + \delta$, and 0.70 to 0.86 for ϕ between 0.5 and 1.25 with $\delta = 0$. Token counts depend on ϱ : under the serving reading, planned serving tokens for the median model are 9.0 times training tokens at $\lambda\varrho = 1$ and 0.9 times at $\lambda\varrho = 10$, and for the median decision, $T/D = 3(w - 1)/(\lambda\varrho)$, 8.3 and 0.8.

Post-training compute scales with N at given post-training tokens and does not scale with D , so it enters the first-order condition like $\delta > 0$: $w = 1 + x_P + m_N$, with x_P post-training compute relative to pre-training compute. Disclosed budgets put x_P at 0.0019 for DeepSeek-V3 (DeepSeek-AI et al., 2024a) and at 0.0004 and 0.0007 for the final supervised stage of Tülu 3 on Llama 3.1 8B and 70B (Lambert et al., 2025), converting disclosed GPU-hours at an assumed 40 percent utilization. For Olmo 3 Think 32B (Team Olmo, Ettinger et al., 2025),

TABLE E12. SENSITIVITY OF THE REVEALED SHARE TO THE COST SIDE

Case	δ	ϕ	Share s		Median T/D		Llama 3 8B T/D	
			median	Llama 3 8B	$\varrho = 1$	$\varrho = 10$	$\varrho = 1$	$\varrho = 10$
A	-0.1	0.90	0.77	0.87	10.3	1.0	19.4	1.9
A	0.0	1.00	0.75	0.85	9.0	0.9	17.2	1.7
A	0.1	1.10	0.72	0.84	7.9	0.8	15.3	1.5
B	0.0	0.50	0.86	0.92	17.9	1.8	34.4	3.4
B	0.0	0.75	0.80	0.88	11.9	1.2	22.9	2.3
B	0.0	1.00	0.75	0.85	9.0	0.9	17.2	1.7
B	0.0	1.25	0.70	0.82	7.2	0.7	13.7	1.4

Note: Lifetime cost $6N^{1+\delta}D + 2\varrho N^\phi T$ in training-FLOP units at the model's own size: δ is the elasticity of the training cost of a FLOP with respect to N (model FLOPs utilization falling or rising with size), ϕ the elasticity of serving cost per token with respect to N , and ϱ the price of a serving FLOP relative to a training FLOP. The first-order condition is $w = 1 + \delta + \phi X_S/X_T$, with X_S and X_T serving and training expenditure, so $X_S/X_T = (w - 1 - \delta)/\phi$, $s = (w - 1 - \delta)/(w - 1 - \delta + \phi)$ and $T/D = 3(w - 1 - \delta)/(\varrho\phi)$. Case A: $\phi = 1 + \delta$, so that $w = (1 + \delta)(1 + \varrho T/3D)$. Case B: $\delta = 0$. The share s does not depend on ϱ . Clean sample, reference technology; T/D is the serving reading of the wedge (Proposition A3(iii)(a)), which requires that the developer bears the serving cost.

Source: Author's calculations.

reinforcement learning took 0.026 of pre-training compute at the initial release and 0.135 with the 21-day continuation of Olmo 3.1; all post-training is at most 0.21 of pre-training before the initial release and at most 0.32 including the continuation. At $x_P = 0.32$ for every model the median share falls from 0.749 to 0.727 over models and from 0.735 to 0.710 over decision units, and the 2025 compute-weighted share from 0.83 to 0.82 (to 0.77 if post-training compute is also counted in the denominator, $m_N/(1 + m_N + x_P)$). Post-training compute is second order for the median and trims the 2025 aggregate by about 0.01, or 0.06 under the wider definition.

E10. Trends and Validation

The trend from 2023. Table E13 applies one set of rules to the clean sample in 2023, 2024 and 2025, and Table E14 adds sampling uncertainty and composition. Under the reference, the compute-weighted share over decisions is 0.24, 0.54 and 0.81, and over models 0.27, 0.58 and 0.83. It rises in both steps under every one of the 32 technologies, over models and over decisions, and in all 18 samples that leave one developer out. A bootstrap that resamples the 18 developers (9,999 draws) gives a 95 percent interval of [0.19, 0.72] for the change over decisions from 2023 to 2025. For the change from 2023 to 2024 the interval is [0.02, 0.65]. For the step from 2024 to 2025 it is [-0.12, 0.44], so that step is not established. The medians behave differently: the median decision share is 0.49, 0.77 and 0.74 and rises in both steps under only 1 of the 32 technologies, while the median model share, 0.55, 0.82 and 0.87, rises in both under 31. Without any one of Alibaba, Hugging Face or Marin the median decision share falls from 2024 to 2025 by more, to 0.65 in each case. The trend largely restates the rise in M , whose median rose from 146 in 2023 to 2,438 in 2025, since at given compute every technology ranks

models by M . Llama 3.1 405B, whose sign is not identified, carries 55 percent of 2024 compute; the compute-weighted identified share is 12, 19 and 49 percent in the three years, so part of the weighted rise is a shift in composition; and the 2025 cohort has six decision units, all dense.

Composition, flags and rationales. Tier windows hold 47, 44 and 92 percent of clean-sample compute in 2023, 2024 and 2025. Adding the excluded mixture-of-experts models (six in 2024 and six in 2025) as decisions of their own moves the 2025 compute-weighted share over decisions from 0.81 to 0.82 at active parameters and to 0.73 at total parameters (2024: 0.56 and 0.52, against 0.54); once added, they carry 53 percent of 2025 compute. Within developers, a regression of the log decision wedge on the release date with developer fixed effects gives growth of 1.59-fold a year, with a 95 percent interval of $[0.91, 2.61]$ from the bootstrap over developers (2.70 given log compute), against 1.86 pooled; the slope is identified by the nine developers with decisions at more than one release date, six of them in more than one year. The wild cluster bootstrap- t that Section V uses gives wider intervals for the same regressions, $[0.47, 5.59]$ with developer fixed effects and $[0.96, 3.54]$ pooled (Online Appendix Table F2). The protocol’s synthetic-data rule flags every 2025 decision, so no 2025 share without flagged decisions exists under it; dropping SmoLLM 1–3, phi-1.5, phi-2, Qwen2.5, Qwen3 and OLMo 2 instead leaves three 2025 decisions, with a compute-weighted share of 0.67. Table E15 reports the stated sizing rationales, coded under the protocol by both coders. The share of decisions citing a compute-optimal target falls from 0.12 in 2023 to 0.04 in 2024 and zero in 2025 (weighted by compute, 0.45, 0.65 and zero, as the Llama 3 herd’s flagship dominates 2024); the three decisions that cite one (the Llama 3 herd, Falcon 40B and 180B) have a median share of 0.04. Among decisions with a deployment rationale (inference cost, or a device or memory target) the median share rises from 0.48 (six decisions) to 0.83 (18) and 0.89 (three); among decisions that state no rationale it is 0.55, 0.75 and 0.65.

Before 2023. The eight open-weight production-scale models of 2019–2022 have a compute-weighted share of 0.04 under the reference technology, but two Kaplan-era flagships carry 92 percent of the period’s compute. Under the belief that developers planned with the allocation law of Kaplan et al. (2020) (Proposition A3(ii)), the share is 0.19 to 0.29 (0.43 to 0.57 with Kaplan’s curvature as well), and OPT-175B, a replication of GPT-3, lies on the believed path by construction. The low pre-2023 value is therefore a belief regime, not a measured absence of value, and the paper states the trend from 2023.

Token volumes. Per-model token volumes on OpenRouter cannot be collected under its terms, and the developer-level totals of Aubakirova et al. (2026) show that one year of that traffic is a small part of the lifetime tokens that the releases imply under the serving reading at $\lambda\varrho = 1$ (4.6 percent for DeepSeek, 0.04 percent for Alibaba), which says only that the platform carries a small part of a model’s use; a test of levels is not possible with public data.

What is not done. A test on a second margin in the spirit of Raval (2023), such

as the sparsity of mixture-of-experts models, the number of key-value heads or the vocabulary, has not been run. Token counts are not normalized by bytes, and Qwen2.5’s per-member token counts are not audited. Durability, the obsolescence of models that shortens the horizon over which serving demand is planned, is part of T but is not modeled. Loss is not capability: the second-output test covers one recipe, and the effect of quantization (Kumar et al., 2025), which changes deployed quality without changing training, is signed through the output concept of Proposition 1(iii) but not measured.

TABLE E13. THE TREND FROM 2023 UNDER ONE SET OF RULES

	n	Ref.	Lab-own	32 technologies (range)	Leave one developer out	Rise under 2023–25
<i>Panel A. Median s</i>						
Models, 2023	24	0.55	0.56	[−0.19, 0.66]	[0.52, 0.57]	31/32
2024	40	0.82	0.82	[0.30, 0.93]	[0.81, 0.84]	LODO 17/18
2025	13	0.87	0.85	[0.31, 0.95]	[0.81, 0.89]	
Decision units, 2023	16	0.49	0.51	[−0.27, 0.64]	[0.46, 0.53]	1/32
2024	27	0.77	0.77	[0.21, 0.89]	[0.76, 0.82]	LODO 2/18
2025	6	0.74	0.73	[0.08, 0.80]	[0.65, 0.86]	
<i>Panel B. Compute-weighted s</i>						
Models, 2023	24	0.27	0.30	[0.01, 0.58]	[0.23, 0.40]	32/32
2024	40	0.58	0.61	[0.05, 0.74]	[0.48, 0.75]	LODO 18/18
2025	13	0.83	0.82	[0.24, 0.96]	[0.70, 0.87]	
Decision units, 2023	16	0.24	0.27	[0.00, 0.57]	[0.21, 0.36]	32/32
2024	27	0.54	0.56	[0.02, 0.70]	[0.44, 0.72]	LODO 18/18
2025	6	0.81	0.81	[0.23, 0.92]	[0.68, 0.86]	

Median M (models): 2023 146; 2024 1,217; 2025 2,438

Panel C. Before 2023 (robustness): 2019–2022 open-weight production-scale universe

	n	Median s	Weighted s
All eight models			
reference	8	0.22	0.04
Kaplan belief	8	0.52	0.19
Kaplan belief, $\sigma^* = 0.535$	8	0.77	0.43
OPT-175B and BLOOM-1.7B dropped (suites)			
reference	6	0.22	0.06
Kaplan belief	6	0.52	0.29
Kaplan belief, $\sigma^* = 0.535$	6	0.77	0.57
Contrast: clean sample 2023–2025			
Kaplan belief	77	0.91	0.93

Note: Clean sample, dense open-weight base models. Ref.: reference technology; Lab-own: lab’s own path with the common σ^* where available; 32 technologies: range over the technologies; leave one developer out: range when each developer is dropped in turn; Rise: number of the 32 technologies under which the statistic rises from 2023 to 2024 and from 2024 to 2025 (LODO: number of leave-one-developer-out samples, reference technology, in which it does). Compute-weighted $s = \sum (w^+ - 1)C / \sum w^+ C$, $w^+ = \max(w, 1)$. Decision units are dated by their first release. At given compute every technology ranks models by M , so the rise largely restates the rise in median M . Panel C: Kaplan belief, the developer planned with the Kaplan et al. (2020) allocation $N \propto C^{0.73}$ anchored at GPT-3 (a believed technology, Online Appendix Proposition A3(ii)), with the reference curvature or with the curvature of Kaplan’s joint law ($\sigma^* = 0.535$); OPT-175B, a GPT-3 replication, lies on that path by construction.

TABLE E14. THE TREND FROM 2023: SAMPLING UNCERTAINTY AND COMPOSITION

	2023		2024		2025	
	Point	[95%]	Point	[95%]	Point	[95%]
<i>Panel A. By release year</i>						
Weighted s , decision units	0.24	[0.09, 0.51]	0.54	[0.34, 0.79]	0.81	[0.65, 0.89]
Median s , decision units	0.49	[0.24, 0.59]	0.77	[0.73, 0.85]	0.74	[0.63, 0.89]
Weighted s , models	0.27	[0.10, 0.53]	0.58	[0.40, 0.80]	0.83	[0.67, 0.90]
Median s , models	0.55	[0.41, 0.62]	0.82	[0.75, 0.85]	0.87	[0.74, 0.92]
<i>Panel B. Changes</i>						
	2023 to 2024		2024 to 2025		2023 to 2025	
Weighted s , decision units	0.30	[0.02, 0.65]	0.28	[-0.12, 0.44]	0.57	[0.19, 0.72]
Median s , decision units	0.29	[0.19, 0.57]	-0.03	[-0.19, 0.13]	0.26	[0.07, 0.59]
Weighted s , models	0.31	[0.04, 0.65]	0.24	[-0.10, 0.39]	0.56	[0.18, 0.71]
Median s , models	0.27	[0.18, 0.39]	0.05	[-0.05, 0.10]	0.31	[0.19, 0.44]
<i>Panel C. Compute-weighted s over decision units: checks</i>						
	2024		2025			
Dense decision units (Panel A)	0.54		0.81			
MoE models added, active N	0.56		0.82			
MoE models added, total N	0.52		0.73			
Without synthetic-data units (protocol)	0.39		—			
Without synthetic-data units (family list)	0.43		0.67			
<i>Panel D. Annual growth of the decision-unit wedge</i>						
Pooled OLS	1.86	[1.25, 2.80]				
Pooled OLS, ln C	2.14	[1.68, 3.04]				
Developer FE	1.59	[0.91, 2.61]				
Developer FE, ln C	2.70	[1.24, 4.54]				

Note: Weighted: compute-weighted. Clean sample, 49 budget-level decision units (16, 27 and 6 in 2023, 2024 and 2025; 24, 40 and 13 models), reference technology. Compute-weighted $s = \sum(w^+ - 1)C / \sum w^+ C$, $w^+ = \max(w, 1)$. Intervals: percentile intervals from a bootstrap that resamples the 18 developers with replacement ($B = 9,999$; 4,999 in Panel D). Intervals in Panels A, B and D use this bootstrap. Panel C: the mixture-of-experts (MoE) models excluded from the clean sample enter as decision units of their own, with the reference wedge at active or at total parameters and weight $6N_{active}D$; synthetic-data flags coded under the reading protocol (Online Appendix E5); the family list drops SmolLM 1–3, phi-1.5, phi-2, Qwen2.5, Qwen3 and OLMo 2. Panel D: OLS of $\ln \hat{W}$ on the release date over decision units, pooled or with developer fixed effects, which identify the trend from the nine developers with decisions at more than one date; the wild cluster bootstrap- t of Online Appendix Table F2 gives wider intervals ([0.96, 3.54] pooled, [0.47, 5.59] with fixed effects). —: under the protocol's rule every 2025 decision is flagged.

TABLE E15. STATED SIZING RATIONALES BY RELEASE YEAR

Year	<i>n</i>	Share of decision units stating					Deployment rationale			None		
		C-opt.	C-opt.,	Inference	Device or	Deploy-	None	<i>n</i>	Median	Weighted	Median	Weighted
			weighted	cost	memory	ment			<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>
2023	16	0.12	0.45	0.38	0.31	0.38	0.50	6	0.48	0.35	0.55	0.37
2024	27	0.04	0.65	0.56	0.63	0.67	0.33	18	0.83	0.53	0.75	0.72
2025	6	0.00	0.00	0.17	0.33	0.50	0.50	3	0.89	0.88	0.65	0.67
2023–2025	49	0.06	0.52	0.45	0.49	0.55	0.41	27	0.77	0.63	0.69	0.65

Note: Stated sizing rationale of each of the 49 decision units, coded from the developers’ saved reports, release posts and model cards under the written protocol (Online Appendix E5): C-opt., the size or token budget was set to be compute-optimal or by a scaling-law allocation; inference cost, the model was sized or trained longer to lower inference or serving cost, or its size is said to make inference cheaper or faster; device or memory, the size targets a device, GPU or memory limit, or on-device, local or edge use; deployment, either of the two; none, no rationale stated. A unit can state several. Weighted: by training compute. Median and weighted *s* under the reference technology, among units with a deployment rationale and among units that state none of the three. Codes resolved from two coders, the second blind to the wedges (agreement in the replication package).

APPENDIX F. ECONOMIC IMPLICATIONS: ADDITIONAL RESULTS

This appendix supports Section V. Every estimate is regenerated by one deterministic script in the replication package from the clean sample and decision units of Section IV, the technologies and IsoFLOP minima of Section III, and two vintages of the Epoch AI database; values taken from the literature are cited where they appear.

Compute growth and the vintage of the database. Table F1 applies one estimator, the running top ten of training compute at release with a wild cluster bootstrap by developer, to Epoch AI’s public file as captured on May 31, 2024 (Epoch AI, 2024) and to the snapshot of September 23, 2026 (Epoch AI, 2026). On the 2024 file the estimator gives 4.13-fold growth, consistent with the published 4.2-fold rate of Sevilla and Roldán (2024) ($p = 0.73$). A two-factor Shapley decomposition attributes 64 percent of the gap to the 2026 file’s 5.23-fold rate to revised compute estimates and 36 percent to entries added after May 2024.

The trend in the revealed wedge. Table F2, panel A, gives the trend under alternative estimators: within developers, above 10^{24} FLOP, weighted by compute, for the four disclosed flagship runs above 10^{25} FLOP, and on version 3’s 56 family-label units. Over the decision units every interval includes no growth, except that of the within-developer trend conditional on log compute (2.70-fold a year). Panel B gives the compute-weighted wedge by release year, and panel C the five-year multiples of the scenario of Table 3, panel B, when the frontier’s wedge keeps rising.

The data wall at observed allocations. A developer with wedge w minimizes lifetime cost $6ND + \Xi N$, where ΞN is its value of compactness, subject to reaching the loss of its unconstrained choice; Ξ is set so that the unconstrained choice has $\varepsilon_N/\varepsilon_D = w$. Effective data follow Muennighoff et al. (2023), and the technology is

the reference or a member of its observationally equivalent family with the same expansion path and frontier. All objects depend on w and scarcity only. Table F3 reports the extra lifetime cost by w and r (panel A), the measured wedge under a cap (panel B), today’s frontier under each technology and at the stock’s lower bound (panel C), and, for compute-optimal runs, the hard wall, the shadow value, the elasticity of substitution between compute and unique tokens, and the return to compute behind the wall (panel D). Figure F1 plots the main relations. Two sensitivities are worth stating. If the frontier’s wedge had followed the open-weight trend since the median release date of the four anchor runs (mid-2024), frontier runs would already process 3.5 to 8.9 times the compute-optimal tokens ($\sigma^* = 0.60$ to 0.74 ; 6.2 at 0.70). They would have reached the 100-trillion-token stock between late 2025 and early 2026, so starting the rise in September 2026, as Section V does, is the conservative choice. And in the full repetition model the decay of excess parameters acts as a size penalty, so that measured wedges of models near the expansion path rise rather than fall as the wall binds (for the frontier with its wedge held, $\hat{w}/w$ is 1.04 in 2029 and 1.11 in 2030, against 0.94 and 0.87 under data decay only). Because data are nonrival (Jones and Tonetti, 2020), the private shadow values of unique tokens in Section V.B understate their social value when several developers use the same tokens.

The fleet share of AI compute. Read as planned serving expenditure, an interpretation that the serving evidence of Section IV does not support, the 2025 compute-weighted wedge of the decision units ($w - 1 = 4.4$) implies a share of AI compute spent on inference. Suppose that each model is served for one year while training compute grows 5.06-fold a year, that a FLOP costs the same in serving as in training, and that final training runs are 10 to 23 percent of developers’ research compute (Denain and Wu, 2026). The implied inference share is then 0.17 to 0.33 (0.68 for one-year lives if all AI compute were final training runs). Longer service lives lower it, because the models in service are then older vintages trained with less compute: it is 0.11 to 0.23 with two-year lives and 0.08 to 0.17 with three-year lives (Table F4, panel C). Without Alibaba the one-year range is 0.09 to 0.19. Disclosures are higher: inference took about three-fifths of Google’s machine-learning energy in 2019–2021 (Patterson et al., 2022) and 70 percent of Meta’s AI power capacity (Wu et al., 2022), and it was about 40 percent of Nvidia’s data-center revenue in fiscal 2024 (NVIDIA Corporation, 2024). The comparison is a loose check, and the 2025 wedge rests largely on one family, Alibaba’s Qwen3.

Growth calibration, fleet shares and synthetic tokens. Table F4 reports γ and the compute needed to halve reducible loss for every large design (panel A); profiles of the IsoFLOP frontier in the irreducible loss E (panel B), which bound E tightly for Chinchilla and Llama 3 but loosely or not at all for Marin’s two-decade ladders, and of which only Chinchilla’s is in known loss units and contains its design’s joint-fit E ; the fleet share of inference implied by the 2025 wedge for service lives of one to three years, R&D multiples of final training runs and

TABLE F1. FRONTIER COMPUTE GROWTH BY DATABASE VINTAGE

Growth of the running top-10 training runs, January 2018 to May 2024 (times per year)

Sample	Runs	Developers	Growth	95% CI	p ($4.2\times$)
Epoch AI, published estimate	—	—	4.2	[3.6, 4.9] ^a	—
Database of 31 May 2024	58	22	4.13	[3.75, 4.55]	0.732
notable models only	52	18	4.16	[3.73, 4.63]	0.855
+ compute revisions to Sep. 2026	66	23	4.76	[3.69, 6.13]	0.278
notable models only	65	19	4.66	[3.45, 6.38]	0.561
+ entries added after May 2024	63	25	4.46	[3.77, 5.29]	0.578
notable models only	58	21	4.45	[3.73, 5.33]	0.646
Database of 23 Sep. 2026	78	26	5.23	[4.02, 6.80]	0.038
notable models only	67	23	4.95	[4.07, 6.01]	0.080
January 2018 to September 2026 (Section V)	92	27	5.06	[4.27, 6.02]	0.041

Decomposition of $\ln(5.23/4.13)$: compute revisions 0.64, entries added 0.36 (Shapley)

Notes: OLS of $\log_{10} C$ on release date for models in the running top 10 of training compute at release (all domains); 95 percent wild cluster bootstrap- t intervals by developer (Webb weights, $B = 9,999$); p : restricted wild cluster bootstrap test of 4.2-fold growth. The database of 31 May 2024 is Epoch AI’s public file as captured by the Internet Archive three days after the published estimate (Epoch AI, 2024); the database of 23 September 2026 is Epoch AI (2026). Compute revisions: models present in both vintages, with the 2026 estimate (145 revised by more than 5 percent, 54 given an estimate, 21 whose estimate was withdrawn); additions: 978 entries released before June 2024 but added later. ^aSevilla and Roldán (2024): 90 percent interval of a piecewise model with an estimated break near 2018.

relative prices of an inference FLOP (panel C); and the break-even scarcities at which generating a token costs its shadow value (panel D).

TABLE F2. THE TREND IN THE REVEALED WEDGE AND FRONTIER DATA DEMAND

Panel A. Growth per year of the revealed wedge and the data multiple, 2023–2025

Estimator	n	w per year	95% CI	D/D^* per year at $\sigma^* = 0.70$ at $\sigma^* =$	
				(from M/M^*)	0.60; 0.74
Decision units, OLS (primary)	49	1.86	[0.96, 3.54]	2.06	1.59; 2.41
developer fixed effects	49	1.59	[0.47, 5.59]	1.72	1.42; 1.94
fixed effects and log compute	49	2.70	[1.39, 5.21]	3.19	2.10; 4.10
above 10^{24} FLOP (6 developers)	11	1.76	[0.78, 4.15]	1.93	1.53; 2.23
compute-weighted	49	2.21	[0.38, 13.1]	2.53	1.81; 3.09
Flagship runs above 10^{25} FLOP ^a	4	1.70	—	1.86	1.49; 2.12
Models, OLS	77	1.99	[0.95, 4.33]	2.24	1.68; 2.67
Models, compute-weighted	77	1.96	[1.01, 3.85]	2.19	1.66; 2.60
Aggregate wedge, decision units	49	2.02	—	2.28	1.69; 2.72
Median wedge, decision units	49	1.53	—	1.64	1.37; 1.83
Family-label units (version 3), OLS	56	1.91	[1.03, 3.48]	2.13	1.62; 2.51
Leave one developer out (units, OLS)	—	1.66–2.26	—	—	—
32 technologies (models, OLS)	77	1.17–4.28	—	1.97–2.38	—

Panel B. The compute-weighted wedge by release year

Release year	Units	s	w	D/D^* at this w	
				$\sigma^* = 0.70$	0.60; 0.74
2023	16	0.24	1.32	1.38	1.23; 1.48
2024	27	0.54	2.17	2.46	1.79; 3.00
2025	6	0.81	5.38	7.13	3.53; 10.98

Panel C. The frontier adopts an open-weight trend without a stop

Trend	σ^*	D/D^*		D over 5 years		Years to 2025 w
		per year		4.2×	5.06×	
Decision units	0.60	1.59	359	571		2.4
	0.70	2.06	1,305	2,074		2.4
	0.74	2.41	2,887	4,588		2.4
Within developers	0.70	1.72	529	841		3.2
Units above 10^{24} FLOP	0.70	1.93	945	1,502		2.6
Compute-weighted units	0.70	2.52	3,580	5,690		1.9
Flagships above 10^{25} FLOP ^a	0.70	1.86	774	1,230		2.8

Notes: Panel A: growth factor $\exp(\hat{b})$ of the slope of $\ln w$ (reference technology) on release date; decision units as in Section IV; 95 percent wild cluster bootstrap- t intervals by developer (Webb weights, $B = 9,999$; 18 developers; 6 above 10^{24} FLOP, so that interval is indicative); developer fixed effects: the slope is identified by the 9 developers with units at more than one date; ^athe four disclosed dense runs above 10^{25} FLOP in 2024–2025, descriptive (OLS, no interval; 1.31 without Pangu Ultra); aggregate and median wedge: growth between the 2023 and 2025 values, with the aggregate $w = 1/(1 - s)$ at the compute-weighted s ; D/D^* from M/M^* : $\exp(\hat{b}/2\theta)$, $\theta = 1/\sigma^* - 1$ of the reference, which needs no curvature given $M^*(C)$ and equals the conversion at the reference $\sigma^* = 0.70$; $\sigma^* = 0.60/0.74$: w growth converted with $D/D^* = w^{\sigma^*/[2(1-\sigma^*)]}$; 32 technologies: the ex-ante technologies of Section IV, each in its own parameter convention, on the audited token counts (the reference gives the models row). Panel B: $s = \sum(w^+ - 1)C / \sum w^+ C$ over decision units, $w^+ = \max(w, 1)$, reference technology; $w = 1/(1 - s)$; D/D^* that a model with this wedge would use at given compute; over models, s is 0.27, 0.58, 0.83. Panel C: the scenario of Table 3, panel B, without the stop at the 2025 wedge; D over 5 years: frontier tokens in 2031 relative to September 2026; years until the frontier's wedge reaches the 2025 compute-weighted wedge of the decision units ($w = 5.38$).

TABLE F3. THE DATA WALL AT OBSERVED ALLOCATIONS

Wedge w (D/D^* at $\sigma^* = 0.70$)	$r = 0.5$	$r = 1$	$r = 2$	$r = 4$	$r = 8$	$r = 16$
<i>Data decay only, $R_D^* = 15.4$</i>						
$w = 1$ (1.00)	0.0	0.0	1.6	7.4	22.2	63.4
$w = 2$ (2.25)	<0.1	1.1	4.4	11.9	29.6	76.4
$w = 5.38$ (7.16)	1.1	3.1	7.3	16.1	34.4	74.8
<i>Data-only fit, $R_D^* = 2.9$</i>						
$w = 1$ (1.00)	0.0	0.0	7.0	39.8	192.1	5,076.1
$w = 2$ (2.25)	0.1	4.9	21.1	67.5	246.5	2,733.9
$w = 5.38$ (7.16)	4.8	13.8	33.1	75.8	186.9	636.0
<i>Panel B. The measured wedge: $\hat{w}/(1+m_N) / \hat{w}/w$</i>						
Repetition model	$D/U = 1.5$	2	4	8	16	32
Hard cap (token cap)	0.58/0.81	0.38/0.69	0.14/0.47	0.04/0.30	0.01/0.18	0.00/0.08
Data-only fit, $R_D^* = 2.9$	0.90/0.96	0.83/0.93	0.67/0.85	0.49/0.74	0.31/0.60	0.17/0.45
Data decay only, $R_D^* = 15.4$	0.98/0.99	0.96/0.98	0.92/0.96	0.85/0.93	0.74/0.87	0.57/0.78
Full model	0.98/0.99	0.96/0.98	0.92/0.96	0.85/0.93	0.74/0.87	0.57/0.78
<i>Panel C. The frontier in September 2026 at observed data use</i>						
Technology	100T stock		22T stock (lower 95% bound)			
	D^*/U	D/U	D^*/U	D/U	Extra cost (%) ^b	
Chinchilla, κ free (reference)	0.50	0.62	2.26	2.82	3.1 / 7.4	
Chinchilla, $\kappa = 1$	0.42	0.60	1.90	2.71	2.8 / 6.6	
Llama 3 (Meta's law)	0.79	0.72	3.59	3.29	4.2 / 11.6	
DeepSeek LLM	0.35	0.57	1.57	2.61	2.6 / 5.8	
Farseer, $\kappa = 1$	0.38	0.57	1.73	2.59	2.5 / 5.6	
Farseer, own form	3.67	1.09	16.68	4.94	8.6 / 31.6	
Gadre et al., RefinedWeb	0.08	0.49	0.37	2.24	1.8 / 3.3	
OLMo ladder	4.54	1.03	20.65	4.70	7.9 / 28.1	
Marin, DCLM	1.05	0.84	4.79	3.83	5.6 / 17.1	
<i>Panel D. Compute-optimal allocation: hard wall, shadow value, substitution</i>						
Technology	σ^*	$r_{\max}$	Shadow value		σ_{CU}	$\gamma_{\text{eff}}/\gamma$
			$r = 4$	$r = 16$	$r = 4$	$r = 16$
Chinchilla, κ free (reference)	0.70	83	0.45	8.61	0.48	0.80
Equivalent, $\sigma^* = 0.60$	0.60	47	0.51	13.81	0.47	0.77
Equivalent, $\sigma^* = 0.74$	0.74	119	0.43	6.95	0.49	0.82
Reference, full model ($R_N^* = 5.3$)	0.70	9	3.76	—	0.49	—
Reference, data-only fit ($R_D^* = 2.9$)	0.70	20	1.83	117.85	0.53	0.50

Notes: Reference technology (κ free, $\sigma^* = 0.70$) with the repetition model of [Muennighoff et al. \(2023\)](#). A developer with wedge w minimizes $6ND + \Xi N$, where ΞN is the value of compactness, at the loss of its uncapped choice, which has $\varepsilon_N/\varepsilon_D = w$ and processes $D = D^*(C)w^{\sigma^*/[2(1-\sigma^*)]}$ tokens. Panel A: extra lifetime cost when the unique stock is $U = D^*(C)/r$, C the model's training compute; at $w = 1$ it is the extra compute of Table 3. The objects depend on (w, r) only (checked at 10^{25} and 10^{27} FLOP). Panel B: the wedge computed from the capped choice, treating tokens processed as fresh, relative to $1 + m_N = 1 + \Xi/(6D)$ at that choice and to the no-wall wedge $w = 5.38$ (D/U : no-wall tokens over the stock); $1 + m_N > w$ because the capped developer processes fewer tokens; $\hat{w}/w$ is the shortfall of the measured trend from its no-wall path. Token-cap case of Proposition 1: $\hat{w} = (1 + m_N)/(1 + \mu)$; under soft repetition $\mu = 1/\eta - 1$, $\eta = d \ln D'/d \ln D$ (checked). In the full model excess parameters also decay, a size penalty that raises the measured wedge once N exceeds the compute-optimal size for U ; for over-trained models it equals data decay only. Panel C: frontier compute of September 2026, stock grown 5 percent a year from 2024; D/U at the observed data multiple of the disclosed dense frontier runs; ^bextra lifetime cost at 22T, data decay only / full model (at 100T it is below 0.1 percent for every technology). Panel D: $r_{\max}$, hard wall; shadow value in units of $6N$; σ_{CU} , elasticity of substitution between compute and unique tokens; $\gamma_{\text{eff}}/\gamma$, relative return to compute behind the wall.

TABLE F4. GROWTH CALIBRATION, FLEET SHARES AND SYNTHETIC TOKENS

Technology	E from	γ	(s.e.)	$2^{1/\gamma}$ [95% CI]	Years at		Total elast.
					4.2 $\times$	5.06 $\times$	
Chinchilla, κ free (reference)	joint fit	0.165	(0.006)	66 [50, 90]	2.9	2.6	0.007
Chinchilla, $\kappa = 1$	joint fit	0.178	(0.006)	49 [39, 63]	2.7	2.4	0.006
Chinchilla, frontier	frontier	0.168	(0.007)	62 [44, 91]	2.9	2.5	–
Hoffmann et al. (published)	joint fit	0.155		88	3.1	2.8	–
Llama 3, frontier (Meta)	frontier	0.136	(0.006)	161 [108, 259]	3.5	3.1	–
Llama 3, runs, $\kappa = 1$	joint fit	0.149	(0.005)	104 [77, 146]	3.2	2.9	–
Farseer, $\kappa = 1$	joint fit	0.148	(0.009)	109 [66, 209]	3.3	2.9	0.009
Farseer, κ free	joint fit	0.096	(0.003)	1,351 [950, 1,996]	5.0	4.4	0.021

Panel B. E -profiles of the IsoFLOP frontier (model-free minima)

Design	Budgets	Decades	$\hat{E}$	95% set for E	$\hat{\gamma}$	γ over the set
Chinchilla	9	2.7	1.74	[1.68, 1.80]	0.168	[0.157, 0.180]
Llama 3	8	2.2	0.56	[0.52, 0.59]	0.127	[0.111, 0.143]
Marin, DCLM	7	2.0	1.65	[1.28, 1.90]	0.102	[0.083, 0.120]
Marin, Nemotron-CC	7	2.0	1.51	[1.37, 1.64]	0.100	[0.092, 0.107]
Marin, Comma	5	2.0	1.44	[0.00, 2.42]	0.083	[0.049, 0.161]

Chinchilla at $E = 1.76$: local γ 0.169 (low budgets), 0.173 (high); drift $-0.0007/\text{decade}$

Panel C. Fleet share of AI compute implied by the 2025 wedge, by service life

Service life	ϕ	$g = 1$			$g = 2$		
		$\Lambda = 1$	4.4	10.4	$\Lambda = 1$	4.4	10.4
1 year	0.49	0.68	0.33	0.17	0.81	0.50	0.29
2 years	0.30	0.57	0.23	0.11	0.72	0.37	0.20
3 years	0.20	0.47	0.17	0.08	0.64	0.29	0.15

Panel D. The shadow value of a unique token beside the cost of a synthetic token

Compute (FLOP)	N^*	Shadow value (\$/M)			Break-even r : generation		
		$r = 2$	$r = 4$	$r = 8$	own	\$0.60	\$3.96
1.0×10^{26}	0.9T	0.5	2.8	14	3.5–4.8	2.1	4.6
9.4×10^{26}	2.8T	1.6	8.8	45	3.5–4.8	1.5	2.9

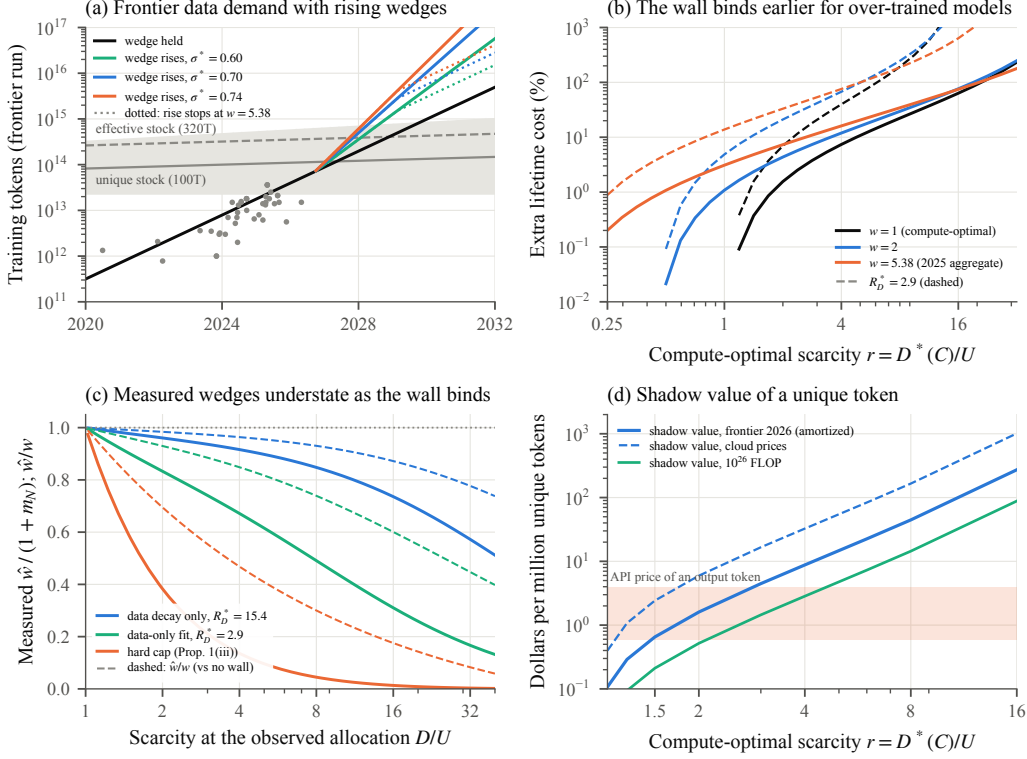


FIGURE F1. DATA DEMAND WITH RISING WEDGES, THE DATA WALL AT OBSERVED ALLOCATIONS, AND THE SHADOW VALUE OF DATA

Note: Reference technology (κ free, $\sigma^* = 0.70$) unless stated. Panel a: training tokens of frontier runs at 5.06-fold compute growth, anchored at the median data multiple of the four disclosed dense runs above 10^{25} FLOP; black, wedge held; colored, wedge rising 1.86-fold a year from September 2026 (the trend over the decision units) with $D/D^* = w^{\sigma^*/[2(1-\sigma^*)]}$; dotted, the rise stops at the 2025 aggregate wedge ($w = 5.38$). Gray dots: dense models in the verified sample above 10^{24} FLOP. Solid and dashed gray lines: the quality-adjusted (100T) and repetition-adjusted (320T) stocks of public text of Villalobos et al. (2024), growing 5 percent a year; shading: the 95 percent interval of the unique stock with stock growth between 0 and 10 percent a year. Panel b: extra lifetime cost against compute-optimal scarcity $r = D^*(C)/U$ for $w = 1, 2$ and 5.38 ; solid, data decay only ($R_D^* = 15.4$); dashed, data-only fit ($R_D^* = 2.9$). Panel c: the wedge measured from the capped choice, relative to one plus the value of compactness at that choice (solid) and to the no-wall wedge (dashed), for $w = 5.38$, against the no-wall tokens over the stock; the hard-cap curve is the token-cap case of Proposition 1. Panel d: shadow value of a unique token for a compute-optimal run at the frontier compute of 2026 (at $\$10^{-18}$ per FLOP, amortized, and at cloud prices) and at 10^{26} FLOP; shading: posted API prices of DeepSeek-V4.1-Flash (off-peak) to DeepSeek-V4-Pro (peak) per million output tokens on September 24, 2026 (DeepSeek, 2026).